

# MTH-RA, Real Analysis

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## General Information About the Course

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## §1 10 September 2026

The classical motivation for measure theory goes back more than two thousand years: the Greeks wanted to measure the area of a circle, the volume of a ball, and other regions in the plane and in space. This eventually evolved into an abstract theory: instead of working only in  $\mathbb{R}^d$ , we work with an arbitrary set  $\Omega$  and ask how to assign a consistent notion of “size” to certain of its subsets.

A key obstruction appears immediately:

- It is **not possible**, in general, to consistently measure *every* subset of a set (even of  $\mathbb{R}^d$ ).
- Instead, we single out a suitable collection of “good” subsets (we call them the **measurable sets**) and only attempt to measure those.

This leads to the first fundamental definition of the course: the notion of a  $\sigma$ -algebra.

**Definition 1.1** ( $\sigma$ -algebra). Let  $\Omega$  be a set. A collection  $\Sigma$  of subsets of  $\Omega$  is called a  $\sigma$ -algebra (or  $\sigma$ -field) on  $\Omega$  if:

- (i)  $\Sigma \neq \emptyset$  (equivalently,  $\Omega \in \Sigma$ );
- (ii) (closed under complements) if  $A \in \Sigma$ , then  $A^c := \Omega \setminus A \in \Sigma$ ;
- (iii) (closed under countable unions) if  $A_1, A_2, A_3, \dots \in \Sigma$ , then  $\bigcup_{n=1}^{\infty} A_n \in \Sigma$ .

The pair  $(\Omega, \Sigma)$  is called a **measurable space**, and the sets  $A \in \Sigma$  are called **measurable sets**.

**Remark 1.1.** Condition (i) simply rules out the degenerate empty collection, which would vacuously satisfy (ii) and (iii) but is of no interest. Some textbooks instead require  $\emptyset \in \Sigma$  directly; this is equivalent given (ii)–(iii) and a nonempty  $\Sigma$ .

### Proposition 1.1

Every  $\sigma$ -algebra  $\Sigma$  on  $\Omega$  satisfies:

- (a)  $\emptyset \in \Sigma$  and  $\Omega \in \Sigma$ ;
- (b)  $\Sigma$  is closed under countable intersections: if  $A_1, A_2, \dots \in \Sigma$  then  $\bigcap_{n=1}^{\infty} A_n \in \Sigma$ ;
- (c)  $\Sigma$  is closed under finite unions and finite intersections (in particular under set differences).

*Proof.* (a) Since  $\Sigma \neq \emptyset$ , pick  $A \in \Sigma$ . By (ii),  $A^c \in \Sigma$ , and by (iii) (a union of the two, or of  $A$  and infinitely many copies of it),  $A \cup A^c = \Omega \in \Sigma$ . Applying (ii) again,  $\Omega^c = \emptyset \in \Sigma$ .

(b) By De Morgan's law,

$$\bigcap_{n=1}^{\infty} A_n = \left( \bigcup_{n=1}^{\infty} A_n^c \right)^c. \quad (1.1)$$

Each  $A_n^c \in \Sigma$  by (ii), the union is in  $\Sigma$  by (iii), and its complement is in  $\Sigma$  by (ii) again.

(c) A finite union  $A_1 \cup \dots \cup A_k$  is a countable union with  $A_n = \emptyset$  for  $n > k$  (using  $\emptyset \in \Sigma$  from (a)), so it lies in  $\Sigma$  by (iii). Finite intersections follow similarly from (b), and  $A \setminus B = A \cap B^c \in \Sigma$ . ■

**Remark 1.2.** The definition does *not* require the sets  $A_n$  in condition (iii) to be pairwise disjoint; the countable union of *any* sequence of measurable sets is measurable.

**Example 1.1** (Trivial and power set  $\sigma$ -algebras)

For any  $\Omega$ :

- $\{\emptyset, \Omega\}$  is the smallest possible  $\sigma$ -algebra on  $\Omega$ .
- The power set  $\mathcal{P}(\Omega)$  (all subsets of  $\Omega$ ) is the largest possible  $\sigma$ -algebra on  $\Omega$ ; if  $|\Omega| = n < \infty$  then  $|\mathcal{P}(\Omega)| = 2^n$ .

**Example 1.2** (Generated  $\sigma$ -algebra)

Let  $X$  be an arbitrary collection of subsets of  $\Omega$ . There is a **smallest**  $\sigma$ -algebra  $\sigma(X)$  containing  $X$ , called the  $\sigma$ -algebra **generated by**  $X$ . It can be constructed as

$$\sigma(X) = \bigcap \{ \Sigma : \Sigma \text{ is a } \sigma\text{-algebra on } \Omega, X \subseteq \Sigma \}, \quad (1.2)$$

i.e. the intersection of *all*  $\sigma$ -algebras containing  $X$  (an arbitrary intersection of  $\sigma$ -algebras is again a  $\sigma$ -algebra, and  $\mathcal{P}(\Omega)$  is one such  $\sigma$ -algebra containing  $X$ , so this intersection is nonempty and well defined).

**Lemma 1.2**

Let  $\Omega$  be a set and  $\{\Sigma_\alpha\}_{\alpha \in J}$  an arbitrary collection of  $\sigma$ -algebras on  $\Omega$ . Then  $\bigcap_{\alpha \in J} \Sigma_\alpha$  is also a  $\sigma$ -algebra on  $\Omega$ .

*Proof.* Every  $\Sigma_\alpha$  contains  $\emptyset$  and  $\Omega$ , so  $\Sigma := \bigcap_{\alpha \in J} \Sigma_\alpha$  is nonempty.

Suppose  $A \in \Sigma$ . Then  $A \in \Sigma_\alpha$  for every  $\alpha$ . Since  $\Sigma_\alpha$  is a  $\sigma$ -algebra, it contains  $A^c$ , for every  $\alpha$ . Therefore,  $\Sigma := \bigcap_{\alpha \in J} \Sigma_\alpha$  also contains  $A^c$ .

Suppose  $A_n \in \Sigma$  for  $n \in \mathbb{N}$ . Then  $A_n \in \Sigma_\alpha$  for every  $\alpha$ , for every  $n$ . Since  $\Sigma_\alpha$  is a  $\sigma$ -algebra, it contains  $\bigcup_{n \in \mathbb{N}} A_n$ , for every  $\alpha$ . Therefore,  $\Sigma := \bigcap_{\alpha \in J} \Sigma_\alpha$  also contains  $\bigcup_{n \in \mathbb{N}} A_n$ . ■

**Lemma 1.2** shows that  $\sigma(X)$  from (1.2) is a  $\sigma$ -algebra.

**Example 1.3** (A concrete generated  $\sigma$ -algebra)

Let  $\Omega = \{1, 2, 3, 4\}$  and  $X = \{\{1, 2\}, \{1, 4\}\}$ . We compute  $\sigma(X)$  by closing under complements and unions:

$$\{1, 2\}, \{3, 4\}, \{1, 4\}, \{2, 3\}, \{1, 2, 4\} \rightarrow \{1, 2\} \cup \{1, 4\}, \{3\}, \{2\}, \{1\}, \{4\}, \dots \quad (1.3)$$

Continuing this process eventually generates every subset of  $\{1, 2, 3, 4\}$ , so in this case  $\sigma(X) = \mathcal{P}(\Omega)$ , with 16 elements.

**Example 1.4** (A proper subalgebra)

Let  $\Omega = \{1, 2, 3, 4\}$  and  $X = \{\{1, 2\}\}$ . Closing under complement and union gives

$$\sigma(X) = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}, \quad (1.4)$$

which has only  $4 = 2^2$  elements, which is a proper subset of  $\mathcal{P}(\Omega)$  (which has 16 elements). This shows that a  $\sigma$ -algebra need not be the full power set.

**Exercise 1.1** (Possible cardinalities). Let  $\Omega = \{1, \dots, n\}$ .

- (a) Show that every  $\sigma$ -algebra on a finite set  $\Omega$  has cardinality equal to a power of 2.
- (b) For a fixed achievable cardinality  $2^k$ , is the  $\sigma$ -algebra with that cardinality unique? Investigate how many distinct  $\sigma$ -algebras of cardinality  $2^k$  exist on  $\{1, \dots, n\}$ .

*Solution.* Let  $\Sigma$  be a  $\sigma$ -algebra on  $\Omega$ . We define a relation on  $\Omega$  as follows:

$$x \sim y \iff \text{for every } A \in \Sigma, x \in A \text{ if and only if } y \in A. \quad (1.5)$$

This is clearly an equivalence relation. Hence, the equivalence classes partition  $\Omega$ . Now, for a given  $x \in \Omega$ , let  $C_x$  be the equivalence class containing  $x$ . Then,

$$C_x = \bigcap_{A \in \Sigma, x \in A} A \in \Sigma, \quad (1.6)$$

as it's the intersection of sets in  $\Sigma$ . Furthermore, for  $A \in \Sigma$ ,

$$A = \bigcup_{x \in A} C_x. \quad (1.7)$$

Hence, every element of  $\Sigma$  is the disjoint union of members of  $\mathcal{C} = \{C_x : x \in \Omega\}$ . Furthermore, **every** disjoint union of members of  $\mathcal{C}$  is in  $\Sigma$ , since  $\mathcal{C} \subseteq \Sigma$ . Hence, the elements of  $\Sigma$  are in a bijection with the subsets of  $\mathcal{C}$ . Therefore,

$$|\Sigma| = 2^{|\mathcal{C}|}. \quad (1.8)$$

For the second part,  $|\Sigma| = 2^k$  corresponds to a partition of  $\Omega$  into  $k$  disjoint parts, which is the Stirling number of second kind, denoted as  $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ . ■

**Example 1.5 (Borel  $\sigma$ -algebra)**

Let  $\Omega = \mathbb{R}^d$  with its usual topology, and let  $X$  be the collection of all **open** subsets of  $\mathbb{R}^d$ . The generated  $\sigma$ -algebra

$$\mathcal{B}(\mathbb{R}^d) := \sigma(X) \quad (1.9)$$

is called the **Borel  $\sigma$ -algebra**. It contains every open set, every closed set (as complements of open sets), and all sets obtainable from these by countably many unions, intersections, and complements. In particular, on  $\mathbb{R}$ , it contains every interval (open, closed, or half-open).

It is useful to contrast a  $\sigma$ -algebra with a **topology**, since both are collections of subsets satisfying closure axioms.

**Definition 1.2 (Topology).** A **topology**  $\tau$  on a set  $\Omega$  is a collection of subsets of  $\Omega$  (called **open sets**) such that:

- (i)  $\emptyset, \Omega \in \tau$ ;
- (ii)  $\tau$  is closed under **arbitrary** unions: if  $\{O_j\}_{j \in J}$  is any family (not necessarily countable) of sets in  $\tau$ , then  $\bigcup_{j \in J} O_j \in \tau$ ;
- (iii)  $\tau$  is closed under **finite** intersections: if  $O_1, \dots, O_n \in \tau$ , then  $\bigcap_{i=1}^n O_i \in \tau$ .

**Remark 1.3 (Key differences).** • A topology allows *arbitrary* (possibly uncountable) unions but only *finite* intersections.

- A  $\sigma$ -algebra allows only *countable* unions (and, as a consequence, countable intersections), but requires closure under **complements**, which a topology does not.

These are genuinely different axiom systems, even though both start from “a set together with a distinguished collection of subsets.”

Having fixed a measurable space  $(\Omega, \Sigma)$ , we now assign a size to each measurable set.

**Definition 1.3 (Measure).** A **measure** on  $(\Omega, \Sigma)$  is a function  $\mu : \Sigma \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\} = [0, \infty]$  such that:

- (i) (countable additivity) if  $A_1, A_2, A_3, \dots \in \Sigma$  are pairwise disjoint, then

$$\mu \left( \bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mu(A_n); \quad (1.10)$$

(ii)  $\mu$  is not identically  $+\infty$  (i.e.  $\mu(A) < \infty$  for at least one  $A \in \Sigma$ ).

The triple  $(\Omega, \Sigma, \mu)$  is called a **measure space**.

**Remark 1.4** (Motivation for additivity). If a measure is meant to behave like a physical quantity, e.g. volume of water in a bucket, then combining two *disjoint* quantities should simply add their sizes: 3 liters and 5 liters combined give 8 liters, with nothing created or destroyed. Countable additivity is the natural axiomatization of this idea.

**Proposition 1.3** (The measure of the empty set)

Let  $\mu$  be a measure on  $(\Omega, \Sigma)$  satisfying countable additivity (condition (i) above), and suppose  $\mu$  is not identically  $+\infty$ . Then  $\mu(\emptyset) = 0$ .

*Proof.* Apply additivity to the disjoint sequence  $A, \emptyset, \emptyset, \dots$  for a fixed  $A$  with  $\mu(A) < \infty$ :

$$\mu(A) = \mu(A \cup \emptyset) = \mu(A) + \mu(\emptyset). \quad (1.11)$$

Since  $\mu(A) < \infty$ , we may subtract it from both sides to get  $\mu(\emptyset) = 0$ .

(If instead  $\mu(\emptyset) = c > 0$  were allowed, one could show  $\mu(\emptyset) = \mu(\emptyset \cup \emptyset) = 2c$ , forcing  $c \in \{0, \infty\}$ ; and if  $\mu(\emptyset) = \infty$  then, applying additivity with  $A$  and  $\emptyset$ ,  $\mu(A) = \mu(A) + \infty = \infty$  for every  $A$ , i.e.  $\mu$  is identically  $\infty$ . This is exactly the degenerate case excluded by (ii).) ■

**Remark 1.5.** Some textbooks instead simply impose  $\mu(\emptyset) = 0$  directly as part of the definition, in place of the “not identically  $\infty$ ” condition; the two conventions agree on all non-degenerate measures. This mirrors the analogous convention issue for topologies (where  $\emptyset, \Omega \in \tau$  is imposed directly).

**Definition 1.4.** Let  $(\Omega, \Sigma, \mu)$  be a measure space.

- $\mu$  is a **finite measure** if  $\mu(\Omega) < \infty$ .
- If  $\mu$  is finite and  $\mu(\Omega) = 1$ , then  $\mu$  is called a **probability measure**.
- $\mu$  is  **$\sigma$ -finite** if there exists a sequence  $A_1, A_2, A_3, \dots \in \Sigma$  such that

$$\mu(A_n) < \infty \text{ for every } n, \quad \text{and} \quad \Omega = \bigcup_{n=1}^{\infty} A_n. \quad (1.12)$$

**Remark 1.6.** If  $\mu(\Omega) < \infty$ , one can always **normalize**  $\mu$  by setting  $\tilde{\mu}(A) = \mu(A)/\mu(\Omega)$  (assuming  $\mu(\Omega) > 0$ ), producing a probability measure with the same measurable sets of measure zero.

**Proposition 1.4**

Let  $(M, \Sigma, \mu)$  be a measure space.

(a) If  $A_1 \subseteq A_2 \subseteq \cdots$  is an increasing sequence of measurable sets, then

$$\mu \left( \bigcup_{n=1}^{\infty} A_n \right) = \lim_{n \rightarrow \infty} \mu(A_n). \quad (1.13)$$

(b) If  $A_1 \supseteq A_2 \supseteq \cdots$  is a decreasing sequence of measurable sets and  $\mu(A_1) < \infty$ , then

$$\mu \left( \bigcap_{n=1}^{\infty} A_n \right) = \lim_{n \rightarrow \infty} \mu(A_n). \quad (1.14)$$

*Proof.* (a) We define  $B_1 = A_1$  and  $B_n = A_n - A_{n-1}$  for  $n \geq 2$ . Then the  $B_n$ 's are pairwise disjoint, and  $\bigcup_n B_n = \bigcup_n A_n$ . As a result,

$$\mu \left( \bigcup_{n=1}^{\infty} A_n \right) = \mu \left( \bigcup_{n=1}^{\infty} B_n \right) = \sum_{n=1}^{\infty} \mu(B_n) \quad (1.15)$$

$$= \lim_{N \rightarrow \infty} \sum_{n=1}^N \mu(B_n) = \lim_{N \rightarrow \infty} \mu \left( \bigcup_{n=1}^N B_n \right) \quad (1.16)$$

$$= \lim_{N \rightarrow \infty} \mu(A_N), \quad (1.17)$$

since  $\bigcup_{n=1}^N B_n = A_N$ .

(b) Define  $B_n = A_1 - A_n$ . Then  $B_n$  is an increasing sequence of measurable sets. Furthermore,

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (A_1 - A_n) = A_1 - \bigcap_{n=1}^{\infty} A_n. \quad (1.18)$$

**Claim 1.1:** If  $A, B$  are measurable sets with  $A \subseteq B$  and  $\mu(A) < \infty$ , then  $\mu(B - A) = \mu(B) - \mu(A)$ .

*Proof.*  $B = A \cup (B - A)$  and this is a disjoint decomposition. Therefore,

$$\mu(B) = \mu(A) + \mu(B - A). \quad (1.19)$$

Rearranging gives the desired result.  $\square$

By applying part (a), we get

$$\mu \left( \bigcup_{n=1}^{\infty} B_n \right) = \lim_{n \rightarrow \infty} \mu(B_n). \quad (1.20)$$

Applying [Claim 1.1](#) into both sides of the equation, we get

$$\mu(A_1) - \mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \mu(A_1) - \lim_{n \rightarrow \infty} \mu(A_n). \quad (1.21)$$

■

**Remark 1.7.** The condition that  $\mu(A_1) < \infty$  (or at least  $\mu(A_n) < \infty$  for some  $n$ ) is important here. For example, consider

$$\Omega = \mathbb{N}, \quad \Sigma = \mathcal{P}(\mathbb{N}), \quad \mu = \text{counting measure}.$$

Here,  $A_n = \{n, n+1, n+2, \dots\}$  is a decreasing sequence with  $\mu(A_n) = \infty$  for every  $n$ . But

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \mu(\emptyset) = 0 \text{ and } \lim_{n \rightarrow \infty} \mu(A_n) = \infty. \quad (1.22)$$

#### Example 1.6

Lebesgue measure on  $\mathbb{R}^d$  is not finite (the whole space has infinite volume), but it *is*  $\sigma$ -finite: taking  $A_n = B(0, n)$ , the ball of radius  $n$  centered at the origin, each  $A_n$  has finite volume and  $\mathbb{R}^d = \bigcup_{n=1}^{\infty} A_n$ .

**Remark 1.8.** The choice of  $\Omega$ ,  $\Sigma$ , and  $\mu$  is entirely up to the application at hand. There is no canonical choice in general. However, certain combinations are used so pervasively that they become the default setting for the theory; the main example developed in this course will be  $\mathbb{R}^d$  equipped with the Borel (or Lebesgue)  $\sigma$ -algebra and Lebesgue measure.

**Definition 1.5** (Measurable function). Let  $(\Omega, \Sigma)$  be a measurable space. A function  $f : \Omega \rightarrow \mathbb{R}$  is called **measurable** (with respect to  $\Sigma$ ) if, for every interval  $I \subseteq \mathbb{R}$ ,

$$f^{-1}(I) := \{x \in \Omega : f(x) \in I\} \in \Sigma. \quad (1.23)$$

**Remark 1.9.** It suffices to check this condition for intervals of the special form  $I = (a, \infty)$ ,  $a \in \mathbb{R}$ : since  $\Sigma$  is closed under complements, countable unions, and countable intersections, every interval (open, closed, half-open, bounded, or unbounded) can be built from intervals of this one type. Because,

$$\begin{aligned} (-\infty, a] &= (a, \infty)^c, \quad [a, \infty) = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, \infty\right), \quad (-\infty, a) = [a, \infty)^c, \\ (a, b) &= (a, \infty) \cap (-\infty, b), \quad [a, b] = [a, \infty) \cap (-\infty, b], \\ [a, b) &= [a, \infty) \cap (-\infty, b), \quad (a, b] = (a, \infty) \cap (-\infty, b]; \end{aligned}$$

and preimage distributes union, intersections and complements. That's why many texts simply *define* measurability by requiring  $f^{-1}((a, \infty)) \in \Sigma$  for all  $a \in \mathbb{R}$ .

**Remark 1.10** (Measurability depends on the choice of  $\Sigma$ ). Whether a given function is measurable depends entirely on the  $\sigma$ -algebra chosen on the domain. It is not an intrinsic property of the function alone.

**Example 1.7**

Let  $\Omega = \mathbb{R}$  and  $\Sigma = \{\emptyset, \mathbb{R}\}$  (the trivial  $\sigma$ -algebra).

- The constant function  $f(x) = 1$  is measurable: for any interval  $I$ ,  $f^{-1}(I)$  is either  $\mathbb{R}$  (if  $1 \in I$ ) or  $\emptyset$  (if  $1 \notin I$ ), and both lie in  $\Sigma$ .
- The identity function  $f(x) = x$  is **not** measurable: for a bounded interval such as  $I = (0, 1)$ ,  $f^{-1}(I) = (0, 1) \notin \{\emptyset, \mathbb{R}\}$ .

This illustrates that measurability is relative to  $\Sigma$ : the same function can be measurable for one  $\sigma$ -algebra and not for another.

**Example 1.8** (Basic examples of measurable functions)

The following are basic examples of measurable functions (with respect to the Borel  $\sigma$ -algebra, unless otherwise specified):

- Any constant function  $f \equiv c$  is measurable.
- For a measurable set  $A \subseteq \Omega$ , the **characteristic function** of  $A$  is defined by

$$\chi_A(x) := \begin{cases} 1, & \text{if } x \in A; \\ 0, & \text{if } x \notin A. \end{cases} \quad (1.24)$$

This function is measurable: for  $a \in \mathbb{R}$ , the following sets are all measurable

$$\{x \in \Omega : \chi_A(x) > a\} = \begin{cases} \Omega, & a < 0; \\ A, & 0 \leq a < 1; \\ \emptyset, & a \geq 1. \end{cases} \quad (1.25)$$

- Every continuous function  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  is measurable with respect to the Borel  $\sigma$ -algebra: for any  $a \in \mathbb{R}$ , the set  $\{x : f(x) > a\} = f^{-1}((a, \infty))$  is open (being the preimage of an open set under a continuous map), hence Borel measurable.
- Every monotone function  $f : \mathbb{R} \rightarrow \mathbb{R}$  (increasing or decreasing) is measurable with respect to the Borel  $\sigma$ -algebra: for any  $a \in \mathbb{R}$ , the set  $\{x : f(x) > a\}$  is always an interval (a ray, of one of the forms  $(c, \infty)$ ,  $[c, \infty)$ ,  $(-\infty, c)$ , or  $(-\infty, c]$ , depending on monotonicity and endpoint conventions), and every interval is Borel measurable.

**Remark 1.11** (Measurable maps between measure spaces). More generally, if  $(\Omega_1, \Sigma_1)$  and  $(\Omega_2, \Sigma_2)$  are two measurable spaces, a function  $f : \Omega_1 \rightarrow \Omega_2$  is called **measurable** if  $f^{-1}(B) \in \Sigma_1$  for every  $B \in \Sigma_2$ . The definition above is the special case  $\Omega_2 = \mathbb{R}$  with  $\Sigma_2 = \mathcal{B}(\mathbb{R})$ , the Borel  $\sigma$ -algebra.

It will be convenient for us to also allow our functions to take values in the extended real line  $\overline{\mathbb{R}}$ . The definition is the same as before: a function  $f : \Omega \rightarrow \overline{\mathbb{R}}$  is **measurable** if the preimage of any extended interval is measurable, or, alternatively, if the sets  $\{x \in \Omega : f(x) > a\}$  are measurable for each  $a \in \mathbb{R}$ . In this case, observe that the sets

$$\{x \in \Omega : f(x) = +\infty\} = \bigcap_{n=1}^{\infty} \{x \in \Omega : f(x) > n\}, \quad (1.26)$$

$$\{x \in \Omega : f(x) = -\infty\} = \left( \bigcup_{n=1}^{\infty} \{x \in \Omega : f(x) > -n\} \right)^c \quad (1.27)$$

are both measurable.

We shall adopt the convention that

$$0 \cdot (\pm\infty) = 0 \text{ and } (\pm\infty) + \pm(\mp\infty) = 0. \quad (1.28)$$

Let  $\mathcal{M} = \mathcal{M}(\Omega, \Sigma)$  denote the set of measurable functions  $f : \Omega \rightarrow \mathbb{R}$  (or  $\overline{\mathbb{R}}$ ).

### Proposition 1.5

Let  $f, g \in \mathcal{M}$  and  $c \in \mathbb{R}$ . Then each of the following functions is also measurable:

- (a)  $cf$ ;
- (b)  $f \pm g$ ;
- (c)  $fg$ ;
- (d)  $\frac{f}{g}$ , if  $g \neq 0$  everywhere;
- (e)  $|f|$ .

*Proof.* We are going to use the trick that  $f$  is measurable if the sets  $\{x \in \Omega : f(x) > a\}$  are all measurable.

- (a) If  $c = 0$ , then  $cf = 0$ , which is a constant function, so it's measurable. For  $c > 0$ ,

$$\{x \in \Omega : c \cdot f(x) > a\} = \left\{x \in \Omega : f(x) > \frac{a}{c}\right\} \quad (1.29)$$

is measurable since  $f$  is a measurable function. For  $c < 0$ ,

$$\{x \in \Omega : c \cdot f(x) > a\} = \left\{x \in \Omega : f(x) < \frac{a}{c}\right\} \quad (1.30)$$

is measurable since  $f$  is a measurable function.

(b) Note that (assuming  $f(x), g(x)$  are finite)

$$\begin{aligned} f(x) + g(x) > a &\iff f(x) > a - g(x) \\ &\iff \exists q \in \mathbb{Q} \text{ such that } f(x) > q > a - g(x) \\ &\iff \exists q \in \mathbb{Q} \text{ such that } f(x) > q \text{ and } g(x) > a - q. \end{aligned}$$

$$\begin{aligned} &\{x \in \Omega : f(x) + g(x) > a\} \\ &= \{x \in \Omega : \text{both } f(x) \text{ and } g(x) \text{ are finite, and } f(x) + g(x) > a\} \\ &\quad \cup (\{x \in \Omega : f(x) = \infty\} \cap \{x \in \Omega : g(x) > -\infty\}) \\ &\quad \cup (\{x \in \Omega : g(x) = \infty\} \cap \{x \in \Omega : f(x) > -\infty\}) \\ &\quad \cup X, \end{aligned} \tag{1.31}$$

where  $X$  is

$$X = \begin{cases} \{x \in \Omega : f(x) = +\infty, g(x) = -\infty\} \cup \{x \in \Omega : g(x) = +\infty, f(x) = -\infty\}, & a < 0, \\ \emptyset, & a \geq 0. \end{cases} \tag{1.32}$$

Therefore,

$$\begin{aligned} &\{x \in \Omega : f(x) + g(x) > a\} \\ &= \bigcup_{q \in \mathbb{Q}} (\{x \in \Omega : f(x) > q\} \cap \{x \in \Omega : g(x) > a - q\} \cap \{x \in \Omega : f(x), g(x) \in \mathbb{R}\}) \\ &\quad \cup (\{x \in \Omega : f(x) = \infty\} \cap \{x \in \Omega : g(x) > -\infty\}) \\ &\quad \cup (\{x \in \Omega : g(x) = \infty\} \cap \{x \in \Omega : f(x) > -\infty\}) \\ &\quad \cup X. \end{aligned} \tag{1.33}$$

This is a measurable set since it's made up of measurable sets by taking union and intersection.

(c) We're gonna prove an intermediate lemma first.

**Claim 1.2:** If  $f$  is measurable, then so is  $f^2$ .

*Proof.* If  $a < 0$ , then  $\{x \in \Omega : f(x)^2 > a\} = \Omega$ , is measurable. For  $a \geq 0$ ,

$$\{x \in \Omega : f(x)^2 > a\} = \{x \in \Omega : f(x) > \sqrt{a}\} \cup \{x \in \Omega : f(x) < -\sqrt{a}\} \tag{1.34}$$

is measurable. □

Now notice that

$$fg = \frac{1}{4} [(f+g)^2 - (f-g)^2], \tag{1.35}$$

is indeed measurable.

(d) It suffices to show that  $\frac{1}{g}$  is measurable.

$$\left\{x \in \Omega : \frac{1}{g(x)} > a\right\} = \begin{cases} \{x \in \Omega : 0 < g(x) < \frac{1}{a}\}, & a > 0, \\ \{x \in \Omega : g(x) > 0\}, & a = 0, \\ \{x \in \Omega : \frac{1}{a} < g(x) < 0\}, & a < 0. \end{cases} \quad (1.36)$$

These are all measurable since  $g$  is measurable.

(e) If  $a < 0$ , then  $\{x \in \Omega : |f(x)| > a\} = \Omega$ , is measurable. For  $a \geq 0$ ,

$$\{x \in \Omega : |f(x)| > a\} = \{x \in \Omega : f(x) > a\} \cup \{x \in \Omega : f(x) < -a\} \quad (1.37)$$

is measurable. ■

### Proposition 1.6

Let  $(f_n)_{n=1}^\infty$  be a countable sequence of measurable functions  $\Omega \rightarrow \overline{\mathbb{R}}$ . Then each of the following functions is measurable:

- (a)  $f(x) = \sup_n f_n(x)$ ;
- (b)  $f(x) = \inf_n f_n(x)$ ;
- (c)  $f(x) = \limsup_{n \rightarrow \infty} f_n(x)$ ;
- (d)  $f(x) = \liminf_{n \rightarrow \infty} f_n(x)$ .

Consequently, if  $f_n \rightarrow f$  pointwise, then  $f$  is measurable.

*Proof.* (a) Note that

$$\sup_n f_n(x) > a \iff f_n(x) > a \text{ for some } n. \quad (1.38)$$

Therefore,

$$\left\{x \in \Omega : \sup_n f_n(x) > a\right\} = \bigcup_n \{x \in \Omega : f_n(x) > a\}, \quad (1.39)$$

is measurable.

(b) Note that

$$\inf_n f_n(x) = - \left( \sup_n (-f_n(x)) \right). \quad (1.40)$$

(c) Follows from the fact that

$$\limsup_{n \rightarrow \infty} f_n(x) = \inf_m \sup_{n \geq m} f_n(x). \quad (1.41)$$

(d) Follows from the fact that

$$\liminf_{n \rightarrow \infty} f_n(x) = \sup_m \inf_{n \geq m} f_n(x). \quad (1.42)$$

When  $f_n \rightarrow f$  pointwise, then the limit, limsup and liminf coincides. ■

**Remark 1.12.** In the case of complex-valued functions, we say that  $f : \Omega \rightarrow \mathbb{C}$  is measurable if its real part and imaginary parts are both measurable.

**Definition 1.6** (Almost everywhere). Let  $(\Omega, \Sigma, \mu)$  be a measure space. We say that a property (P) holds  $\mu$ -**almost everywhere** (or simply **almost everywhere**, when the measure  $\mu$  is clear from context), abbreviated  $\mu$ -a.e. or a.e., if there exists a set  $N \in \Sigma$  with  $\mu(N) = 0$  such that (P) holds on  $N^c$ . Such a set  $N$  is called a **null set**.

### Example 1.9

Typical almost-everywhere statements include:

- $f = g$  almost everywhere, meaning  $\{x \in \Omega : f(x) \neq g(x)\}$  is contained in a null set;
- $\lim_{n \rightarrow \infty} f_n(x) = f(x)$  almost everywhere, meaning the set of  $x$  for which this limit fails to hold is contained in a null set.

**Remark 1.13** (Subsets of null sets need not be measurable). If  $\mu(N) = 0$  and  $B \subseteq N$ , one would like to conclude that  $\mu(B) = 0$  as well. However, this conclusion presupposes  $B \in \Sigma$ , which need not be the case, since  $\Sigma$  may fail to contain every subset of a null set.

**Definition 1.7** (Complete measure). A measure  $\mu$  on  $(\Omega, \Sigma)$  is called **complete** if its domain  $\Sigma$  contains every subset of every  $\mu$ -null set; that is, whenever  $N \in \Sigma$  satisfies  $\mu(N) = 0$  and  $B \subseteq N$ , then  $B \in \Sigma$  (and consequently  $\mu(B) = 0$ ).

**Remark 1.14.** Every measure can be made complete by slightly enlarging its  $\sigma$ -algebra, as follows. Let  $(\Omega, \Sigma, \mu)$  be a measure space. Let

$$\mathcal{N} := \{N \in \Sigma : \mu(N) = 0\}, \quad \bar{\Sigma} := \{A \cup B : A \in \Sigma \text{ and } B \subseteq N \text{ for some } N \in \mathcal{N}\}. \quad (1.43)$$

$\bar{\Sigma}$  is a  $\sigma$ -algebra. Indeed:

1. Take  $A \cup B \in \bar{\Sigma}$ , where  $A \in \Sigma$  and  $B \subseteq N$  for some  $N \in \mathcal{N}$ . Then

$$(A \cup B)^c = A^c \cap B^c = A^c - (B \cap A^c). \quad (1.44)$$

We're gonna use  $W - U = (W - V) \cup (V - U)$  for  $U \subseteq V \subseteq W$ , with  $U = B \cap A^c$ ,  $V = N \cap A^c$ ,  $W = A^c$ . Then the RHS of (1.44) becomes,

$$[A^c - (N \cap A^c)] \cup [(N \cap A^c) - (B \cap A^c)]. \quad (1.45)$$

$A^c - (N \cap A^c) \in \Sigma$  since it's made up of elements of  $\Sigma$  using complementation and finite intersection.  $(N \cap A^c) - (B \cap A^c) \subseteq N$ . Therefore,  $(A \cup B)^c \in \bar{\Sigma}$ .

2. Let  $A_i \cup B_i \in \bar{\Sigma}$  be a countable sequence of sets, with  $A_i \in \Sigma$  and  $B_i \subseteq N_i$  with  $\mu(N_i) = 0$ . Then

$$\mu\left(\bigcup_{i=1}^{\infty} N_i\right) \leq \sum_{i=1}^{\infty} \mu(N_i) = 0, \quad (1.46)$$

implying that  $\bigcup_{i=1}^{\infty} N_i \in \mathcal{N}$ . Now,

$$\bigcup_{i=1}^{\infty} (A_i \cup B_i) = \left(\bigcup_{i=1}^{\infty} A_i\right) \cup \left(\bigcup_{i=1}^{\infty} B_i\right), \quad (1.47)$$

where  $\bigcup_{i=1}^{\infty} A_i \in \Sigma$ , and  $\bigcup_{i=1}^{\infty} B_i \subseteq \bigcup_{i=1}^{\infty} N_i \in \mathcal{N}$ .

There is a unique extension  $\bar{\mu}$  of  $\mu$  to a complete measure on  $\bar{\Sigma}$ , given by

$$\bar{\mu}(A \cup B) = \mu(A), \quad (1.48)$$

where  $A \in \Sigma$  and  $B \subseteq N$  for some  $N \in \mathcal{N}$ . The measure space  $(\Omega, \bar{\Sigma}, \bar{\mu})$  is called the **completion** of  $(\Omega, \Sigma, \mu)$ .

### Proposition 1.7

$\bar{\mu}$  is a complete measure, and it's the unique measure on  $\bar{\Sigma}$  that extends  $\mu$ .

*Proof.* Suppose  $\mu(A \cup B) = 0$  for some  $A \cup B \in \bar{\Sigma}$ , where  $A \in \Sigma$  and  $B \subseteq N$  with  $\mu(N) = 0$ . Choose any subset  $C \subseteq A \cup B$ . We can write it as

$$C = \emptyset \cup C, \text{ for } C \subseteq A \cup N, \quad (1.49)$$

where  $A \cup N \in \Sigma$  with  $\mu(A \cup N) = 0$ . Therefore,  $C \in \bar{\Sigma}$ , and hence  $\bar{\mu}$  is a complete measure.

Assume for the sake of contradiction that  $\mu'$  is a measure on  $\bar{\Sigma}$  that extends  $\mu$ , i.e.  $\mu'|_{\Sigma} = \mu$ . For  $A \cup B \in \bar{\Sigma}$ , where  $A \in \Sigma$  and  $B \subseteq N$  with  $\mu(N) = 0$ , we have

$$\bar{\mu}(A \cup B) = \mu(A) = \mu'(A) \leq \mu'(A \cup B) \leq \mu'(A \cup N) = \mu(A \cup N) = \mu(A) = \bar{\mu}(A \cup B). \quad (1.50)$$

Therefore,  $\mu' = \bar{\mu}$ . ■

### Lemma 1.8

The following implications hold **if and only if** the measure  $\mu$  is complete:

- (i) if  $f$  is measurable and  $f = g$   $\mu$ -a.e., then  $g$  is measurable;
- (ii) if  $(f_n)_{n \geq 1}$  are measurable and  $f_n \rightarrow f$   $\mu$ -a.e., then  $f$  is measurable.

*Proof.* (i) Suppose  $\mu$  is complete; and  $f$  is measurable and  $f = g$  a.e. Then  $f \neq g$  on  $B \subseteq N$ , where  $\mu(N) = 0$ . Since  $\mu$  is complete,  $B \in \Sigma$ . Now,

$$\{x \in \Omega : g(x) > a\} = (\{x \in \Omega : g(x) > a\} \cap B) \cup (\{x \in \Omega : g(x) > a\} \cap B^c) \quad (1.51)$$

$$= (\text{subset of a null-set}) \cup (\{x \in \Omega : f(x) > a\} \cap B^c), \quad (1.52)$$

which is a measurable set.

Conversely, suppose that whenever  $f$  is measurable and  $f = g$  a.e., we have that  $g$  is measurable. Now, let  $N \in \mathcal{N}$  and  $B \subseteq N$ . Take

$$f \equiv 0, \quad g = \chi_B \quad (1.53)$$

They agree on  $N^c$ , so  $f = g$  a.e.  $f$  is a constant function so it's measurable. Therefore,  $g$  is also measurable, which forces

$$g^{-1}\{1\} = B \in \Sigma. \quad (1.54)$$

(ii) Suppose  $f_n \rightarrow g$  (everywhere). Then  $g$  is measurable, and  $f = g$  a.e. Therefore,  $f$  is measurable if and only if  $\mu$  is complete. ■

#### Lemma 1.9 (Borel–Cantelli lemma)

Let  $(\Omega, \Sigma, \mu)$  be a measure space. Let  $(E_k)_{k=1}^\infty$  be a countable family of measurable sets of  $\Omega$  such that

$$\sum_{k=1}^{\infty} \mu(E_k) < \infty. \quad (1.55)$$

Let  $E := \{x \in \Omega : x \in E_k \text{ for infinitely many } k\}$ . Then  $E$  is measurable and  $\mu(E) = 0$ .

*Proof.* Note that

$$\begin{aligned} x \in E_k \text{ for infinitely many } k &\iff \forall n \exists m \geq n \text{ such that } x \in E_m \\ &\iff \forall n, x \in \bigcup_{m \geq n} E_m \\ &\iff x \in \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} E_m. \end{aligned}$$

Therefore,  $E = \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} E_m$  is measurable as it's made up of measurable sets by countable unions and intersections. Now,

$$\mu\left(\bigcup_{m \geq 1} E_m\right) \leq \sum_{m=1}^{\infty} \mu(E_m) < \infty. \quad (1.56)$$

By continuity, since  $\left\{\bigcup_{m \geq n} E_m\right\}_n$  is a decreasing sequence,

$$\mu(E) = \lim_{n \rightarrow \infty} \mu\left(\bigcup_{m \geq n} E_m\right) \leq \lim_{n \rightarrow \infty} \sum_{m=n}^{\infty} \mu(E_m) = 0. \quad (1.57)$$

■

**Exercise 1.2.** Let  $(f_n)_{n \geq 1}$  be a sequence of (measurable) functions on a measure space  $(\Omega, \Sigma, \mu)$ . Suppose that for every  $x \in \Omega$ , the sequence  $(f_n(x))_{n \geq 1}$  is bounded. Let

$$E := \{x \in \Omega : (f_n(x))_{n \geq 1} \text{ converges}\}. \quad (1.58)$$

Show that  $E \in \Sigma$ .

*Solution.*  $(f_n(x))_{n \geq 1}$  converges if and only if  $\limsup_n f_n(x) = \liminf_n f_n(x)$ . We have already shown that  $\limsup_n f_n$  and  $\liminf_n f_n$  are measurable. Therefore, so is  $h := \limsup_n f_n - \liminf_n f_n$ . Now,

$$E = \{x \in \Omega : (f_n(x))_{n \geq 1} \text{ converges}\} = h^{-1}\{0\}, \quad (1.59)$$

which is measurable since  $h$  is measurable. ■

**Exercise 1.3.** For each  $n \geq 1$ , let  $f_n : [0, 1] \rightarrow \mathbb{R}$  be a continuous function. Suppose that:

- (i)  $0 \leq f_n(x)$  for all  $n$  and all  $x \in [0, 1]$ ;
- (ii) the sequence of partial sums  $\sum_{n=1}^N f_n(x)$  converges uniformly on  $[0, 1]$  as  $N \rightarrow \infty$ ;
- (iii) the sum  $\sum_{n=1}^{\infty} (\sup\{f_n(x) : 0 \leq x \leq 1\})$  diverges.

Either give an example of such a sequence of functions, or prove that no such sequence exists.

*Solution.* One such sequence of functions is

$$f_n(x) = \text{triangular bump function on } \left[\frac{1}{n+1}, \frac{1}{n}\right] \text{ of height } \frac{1}{n}. \quad (1.60)$$

Then

$$\sum_{n=1}^{\infty} (\sup\{f_n(x) : 0 \leq x \leq 1\}) = \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges.} \quad (1.61)$$

$$\sum_{n=1}^N f_n(x) = \text{triangular bump functions of height } \frac{1}{n} \text{ on each } \left[\frac{1}{n+1}, \frac{1}{n}\right], \text{ for } n = 1, 2, \dots, N. \quad (1.62)$$

which converges uniformly to

$$f(x) = \begin{cases} \text{triangular bump functions of height } \frac{1}{n} & \text{if } x \in \left[\frac{1}{n+1}, \frac{1}{n}\right] \text{ for } n \geq 1, \\ 0 & x = 0. \end{cases} \quad (1.63)$$

Indeed,

$$\sup_{x \in [0,1]} \left| f(x) - \sum_{n=1}^N f_n(x) \right| = \frac{1}{N+1} \rightarrow 0. \quad (1.64)$$

■

## §2 11 September 2026

Having built the machinery of  $\sigma$ -algebras, measures, and measurable functions, we now turn to the central question of this part of the course:

*Given a measure space  $(\Omega, \Sigma, \mu)$  and a measurable function  $f$ , how do we define  $\int_{\Omega} f \, d\mu$ ?*

The construction proceeds in three stages: first for characteristic functions, then for simple functions, and finally for general non-negative measurable functions (by approximation). Integration of signed or complex-valued functions is deferred to a later lecture.

**Definition 2.1** (Characteristic function). Let  $A \in \Sigma$  be a measurable set. The **characteristic function** (or **indicator function**) of  $A$  is the function  $\chi_A : \Omega \rightarrow \mathbb{R}$  (also written  $\mathbf{1}_A$ ) defined by

$$\chi_A(x) := \begin{cases} 1, & x \in A; \\ 0, & x \notin A. \end{cases} \quad (2.1)$$

**Remark 2.1.** As noted already,  $\chi_A$  is always measurable when  $A \in \Sigma$ : for an interval  $I \subseteq \mathbb{R}$ ,

$$\chi_A^{-1}(I) = \begin{cases} \Omega, & 0, 1 \in I; \\ A, & 1 \in I, 0 \notin I; \\ A^c, & 0 \in I, 1 \notin I; \\ \emptyset, & 0, 1 \notin I, \end{cases} \quad (2.2)$$

and each of  $\Omega, A, A^c, \emptyset$  lies in  $\Sigma$ .

The entire theory of integration is built by insisting on one basic axiom, motivated by the idea that the integral should compute the “volume” below the graph of a function: for a measurable set  $A$  and a constant height  $c \geq 0$ , the volume of the block of height  $c$  over the base  $A$  should be  $c$  times the size of the base. This gives the defining relation

$$\int_{\Omega} c \chi_A \, d\mu := c \mu(A), \quad c \geq 0, \, A \in \Sigma, \quad (2.3)$$

and, in particular (taking  $c = 1$ ),

$$\int_{\Omega} \chi_A \, d\mu := \mu(A). \quad (2.4)$$

**Definition 2.2** (Simple function). A measurable function  $\varphi : \Omega \rightarrow \mathbb{R}$  is called **simple** if it takes only finitely many values.

**Remark 2.2** (Standard representation). If  $\varphi$  is simple and  $a_1, \dots, a_N$  are its distinct *non-zero* values, set

$$A_i := \varphi^{-1}(\{a_i\}), \quad i = 1, \dots, N. \quad (2.5)$$

Each  $A_i$  is measurable (since  $\varphi$  is measurable), the sets  $A_1, \dots, A_N$  are pairwise disjoint, and

$$\varphi = \sum_{i=1}^N a_i \chi_{A_i}. \quad (2.6)$$

This expression is called the **standard representation** of  $\varphi$ .

**Remark 2.3** (Non-standard representations). A simple function can typically be written as a finite linear combination of characteristic functions in more than one way, using sets that are *not* disjoint and coefficients that are *not* distinct. For example, on disjoint sets  $A, B, C$  with

$$\varphi = 5 \chi_A + 8 \chi_B + 3 \chi_C \quad (2.7)$$

(the standard representation), one may check directly that

$$\varphi = 5 \chi_{A \cup B} + 3 \chi_{B \cup C} \quad (2.8)$$

as well, since on  $A$  this gives 5, on  $B$  it gives  $5 + 3 = 8$ , and on  $C$  it gives 3 — matching the standard representation, even though  $A \cup B$  and  $B \cup C$  overlap.

**Definition 2.3** (Integral of a non-negative simple function). Let  $\varphi$  be a non-negative simple function with standard representation  $\varphi = \sum_{i=1}^N a_i \chi_{A_i}$  (so  $a_i > 0$  and the  $A_i$  are pairwise disjoint, possibly with  $\mu(A_i) = \infty$ ). We define

$$\int_{\Omega} \varphi \, d\mu := \sum_{i=1}^N a_i \mu(A_i) \in [0, \infty]. \quad (2.9)$$

**Lemma 2.1** (Independence of the representation)

Let  $\varphi \geq 0$  be a simple function, and suppose

$$\varphi = \sum_{n=1}^N a_n \chi_{A_n} = \sum_{m=1}^M b_m \chi_{B_m} \quad (2.10)$$

are two representations of  $\varphi$  as finite linear combinations of characteristic functions, where the sets  $A_n$  (resp.  $B_m$ ) need *not* be pairwise disjoint, and the coefficients  $a_n, b_m \in \mathbb{R}$  need *not* be distinct or non-negative (only  $\varphi$  itself is assumed non-negative; we adopt the convention  $0 \times \infty = 0$  if some  $\mu(A_n) = \infty$  with  $a_n = 0$ ). Then

$$\sum_{n=1}^N a_n \mu(A_n) = \sum_{m=1}^M b_m \mu(B_m). \quad (2.11)$$

*Proof.* We'll show that  $\sum_{n=1}^N a_n \mu(A_n) = \sum_{i=1}^K r_i \mu(E_i)$ , where  $\varphi = \sum_{i=1}^K r_i \chi_{E_i}$  is the standard representation.

**Case 1:**  $A_i$ 's are pairwise disjoint, but  $a_i$ 's need not be distinct. In this case, for each  $a \in \{a_1, \dots, a_N\}$ , set  $B_a = \bigcup_{a_i=a} A_i$ . Then it's clear that  $B_a \cap B_{a'} = \emptyset$ , since  $A_i$ 's are pairwise disjoint. Furthermore,  $\varphi^{-1}(a) = B_a$ . Therefore,  $\varphi = \sum_a a \chi_{B_a}$  is the standard representation.

$$\sum_a a \mu(B_a) = \sum_a a \sum_{a_i=a} \mu(A_i) = \sum_{i=1}^N a_i \mu(A_i). \quad (2.12)$$

**Case 2:** Now let's assume that  $A_i$ 's are not necessarily pairwise disjoint.

**Claim 2.1:** There exists a refinement of the set  $\bigcup_{i=1}^N A_i$  with sets  $B_1, \dots, B_M$  such that  $\bigcup_{i=1}^N A_i = \bigcup_{j=1}^M B_j$ , and  $B_j$ 's are pairwise disjoint, and for each  $i$ ,  $A_i = \bigcup_{B_j \subseteq A_i} B_j$ .

*Proof.* For each nonempty subset  $S$  of  $\{1, \dots, N\}$ , we set

$$B_S = \left( \bigcap_{i \in S} A_i \right) \cap \left( \bigcap_{i \notin S} A_i^c \right). \quad (2.13)$$

This gives a collection of  $2^N - 1$  pairwise disjoint sets. Now, we'll show that

$$A_i = \bigcup_{B_S \subseteq A_i} B_S. \quad (2.14)$$

Clearly,  $\bigcup_{B_S \subseteq A_i} B_S \subseteq A_i$  holds. Now, take  $x \in A_i$ . Let  $S' = \{i : x \in A_i\}$ , which is a nonempty set. Then  $x \in B_{S'} \subseteq A_i$  since  $S'$  contains  $i$ . Therefore,  $\bigcup_{B_S \subseteq A_i} B_S \supseteq A_i$ .

This proves  $\bigcup_{S \neq \emptyset} B_S \supseteq \bigcup_{i=1}^N A_i$ . The reverse containment holds since  $B_S \subseteq \bigcup_{i=1}^N A_i$ .  $\square$

Now, for each  $S$ , we define  $b_S = \sum_{A_i \supseteq B_S} a_i$ . Then

$$\sum_{i=1}^N a_i \mu(A_i) = \sum_{i=1}^N a_i \sum_{B_S \subseteq A_i} \mu(B_S) = \sum_S \sum_{B_S \subseteq A_i} a_i \mu(B_S) = \sum_S b_S \mu(B_S). \quad (2.15)$$

This boils down to case 1, and we know that this is equal to the sum in the standard representation.  $\blacksquare$

**Remark 2.4.** The integral of a non-negative simple function may be computed using *any* representation of  $\varphi$  as a finite linear combination of characteristic functions of measurable sets, not only the standard one. Equivalently, a non-negative simple function may be described as any finite non-negative linear combination of characteristic functions of measurable sets, and its integral is unambiguously

$$\int_{\Omega} \varphi d\mu = \sum_n a_n \mu(A_n) \quad (2.16)$$

for any such representation  $\varphi = \sum_n a_n \chi_{A_n}$ .

### Lemma 2.2

Let  $\varphi$  and  $\psi$  be simple functions in  $\mathcal{M}^+$ .

- (a) If  $c \in \mathbb{R}$ , then  $\int c\varphi \, d\mu = c \int \varphi \, d\mu$ .
- (b)  $\int (\varphi + \psi) \, d\mu = \int \varphi \, d\mu + \int \psi \, d\mu$ .
- (c) If  $\varphi \leq \psi$ , then  $\int \varphi \, d\mu \leq \int \psi \, d\mu$ .
- (d)  $A \mapsto \int_A \varphi \, d\mu := \int_{\Omega} \varphi \cdot \chi_A \, d\mu$  is a measure on  $\Sigma$ .

*Proof.* (a) Let  $\varphi = \sum_{i=1}^n a_i \chi_{A_i}$ . Then  $c\varphi = \sum_{i=1}^n ca_i \chi_{A_i}$ .

$$\int_{\Omega} c\varphi \, d\mu = \sum_{i=1}^n ca_i \mu(A_i) = c \sum_{i=1}^n a_i \mu(A_i) = c \int_{\Omega} \varphi \, d\mu. \quad (2.17)$$

(b) Let  $\varphi = \sum_{i=1}^n a_i \chi_{A_i}$  and  $\psi = \sum_{j=1}^m b_j \chi_{B_j}$ . Then  $\varphi + \psi = \sum_{i=1}^n a_i \chi_{A_i} + \sum_{j=1}^m b_j \chi_{B_j}$ .

$$\int_{\Omega} (\varphi + \psi) \, d\mu = \sum_{i=1}^n a_i \mu(A_i) + \sum_{j=1}^m b_j \mu(B_j) = \int_{\Omega} \varphi \, d\mu + \int_{\Omega} \psi \, d\mu. \quad (2.18)$$

(c)  $\psi - \varphi$  is a non-negative simple function, whose integral is non-negative, by definition.

(d) Let  $\varphi = \sum_{i=1}^n a_i \chi_{E_i}$  be the standard representation.

$$\nu(A) := \int_{\Omega} \varphi \cdot \chi_A \, d\mu = \int_{\Omega} \sum_{i=1}^n a_i \chi_{E_i} \cdot \chi_A \, d\mu = \int_{\Omega} \sum_{i=1}^n a_i \chi_{E_i \cap A} \, d\mu = \sum_{i=1}^n a_i \mu(E_i \cap A). \quad (2.19)$$

Clearly,  $\nu(A) \geq 0$  and  $\nu(\emptyset) = 0$ . Now, if  $\{A_n\}_n$  is a countable collection of pairwise disjoint sets,

$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^n a_i \mu\left(E_i \cap \bigcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^n a_i \mu\left(\bigcup_{n=1}^{\infty} (E_i \cap A_n)\right) \quad (2.20)$$

$E_i \cap A_n$ 's are all pairwise disjoint, so

$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^n a_i \sum_{n=1}^{\infty} \mu(E_i \cap A_n) = \sum_{n=1}^{\infty} \sum_{i=1}^n a_i \mu(E_i \cap A_n) = \sum_{n=1}^{\infty} \nu(E_i \cap A_n). \quad (2.21)$$

■

We now turn to general non-negative measurable functions. The key technical tool, used repeatedly throughout analysis, is to approximate an object in a large class (here, non-negative measurable functions) by objects in a smaller class we already understand (here, simple functions).

**Theorem 2.3** (Approximation by simple functions)

Let  $f : \Omega \rightarrow [0, \infty]$  be measurable. Then there exists a sequence of simple functions  $(\varphi_n)_{n \geq 1}$ , each satisfying  $0 \leq \varphi_n \leq \varphi_{n+1}$ , such that

$$\varphi_n(x) \longrightarrow f(x) \quad \text{for every } x \in \Omega. \quad (2.22)$$

*Proof.* For each  $n \geq 1$  and each integer  $k$  with  $0 \leq k \leq n2^n - 1$ , define the measurable set

$$A_{k,n} := \left\{ x \in \Omega : \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} \right\}, \quad (2.23)$$

and, to handle the top of the range, set

$$A_{n2^n,n} := \{x \in \Omega : f(x) \geq n\}. \quad (2.24)$$

These sets are measurable (being preimages of intervals under  $f$ ) and, for each fixed  $n$ , pairwise disjoint, with

$$\Omega = \bigcup_{k=0}^{n2^n} A_{k,n}. \quad (2.25)$$

Define

$$\varphi_n := \sum_{k=0}^{n2^n} \frac{k}{2^n} \chi_{A_{k,n}}, \quad (2.26)$$

a non-negative simple function. In words:  $\varphi_n$  rounds  $f$  down to the nearest multiple of  $2^{-n}$ , on the region where  $f < n$ , and sets  $\varphi_n = n$  where  $f \geq n$ .

*Monotonicity.* Passing from level  $n$  to level  $n+1$  refines each interval  $[k/2^n, (k+1)/2^n)$  into

$$\left[ \frac{k}{2^n}, \frac{k+1}{2^n} \right) = \left[ \frac{2k}{2^{n+1}}, \frac{2k+1}{2^{n+1}} \right) \cup \left[ \frac{2k+1}{2^{n+1}}, \frac{2k+2}{2^{n+1}} \right). \quad (2.27)$$

So  $A_{k,n} = A_{2k,n+1} \cup A_{2k+1,n+1}$ ; on either piece, the assigned value at level  $n+1$  is at least the value  $k/2^n$  assigned at level  $n$ . Likewise, the top set  $A_{n2^n,n} = \{f \geq n\}$  splits, at level  $n+1$ , into the part where  $n \leq f < n+1$  (now assigned a value  $\geq n$ , refined further) and the part where  $f \geq n+1$  (assigned the value  $n+1 > n$ ). Hence  $\varphi_n \leq \varphi_{n+1}$  pointwise.

*Pointwise convergence.* Fix  $x \in \Omega$ . If  $f(x) = \infty$ , then  $\varphi_n(x) = n \rightarrow \infty = f(x)$ . If  $f(x) < \infty$ , then once  $n > f(x)$ , we have  $x \in A_{k,n}$  for the unique  $k$  with

$$\frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n}, \quad (2.28)$$

so

$$0 \leq f(x) - \varphi_n(x) < \frac{1}{2^n} \longrightarrow 0. \quad (2.29)$$

Hence  $\varphi_n(x) \rightarrow f(x)$  for every  $x \in \Omega$ . ■

**Remark 2.5.** The convergence in this proposition holds **everywhere**, not merely almost everywhere, and the approximating sequence is **monotonically increasing**. Both features will be essential in the proof of the Monotone Convergence Theorem below.

**Definition 2.4** (Integral of a non-negative measurable function). Let  $f : \Omega \rightarrow [0, \infty]$  be measurable. We define

$$\int_{\Omega} f \, d\mu := \sup \left\{ \int_{\Omega} \varphi \, d\mu : \varphi \text{ simple, } 0 \leq \varphi \leq f \right\} \in [0, \infty]. \quad (2.30)$$

**Remark 2.6.** This definition could have been adopted directly, without first proving the approximation proposition above; the proposition simply confirms that the class of simple functions lying below  $f$  is rich enough to be useful (e.g. in proving the Monotone Convergence Theorem).

Notation for the integral varies across the literature and is otherwise a matter of convention: one may write  $\int_{\Omega} f \, d\mu$ ,  $\int_{\Omega} f(x) \, d\mu(x)$ ,  $\int f \, d\mu$  (when  $\Omega$  is clear), or  $\int f$  (when  $\mu$  is also clear); some authors (especially in physics) write the measure before the function. Any of these is acceptable as long as the underlying set,  $\sigma$ -algebra, and measure are unambiguous.

The following two properties follow immediately from the definition as a supremum.

**Proposition 2.4** (Monotonicity)

If  $f, g : \Omega \rightarrow [0, \infty]$  are measurable and  $f \leq g$ , then

$$\int_{\Omega} f \, d\mu \leq \int_{\Omega} g \, d\mu. \quad (2.31)$$

*Proof.* Every simple function  $\varphi$  with  $0 \leq \varphi \leq f$  also satisfies  $0 \leq \varphi \leq g$ , so the set over which the supremum defining  $\int_{\Omega} f \, d\mu$  is taken is contained in the set over which the supremum defining  $\int_{\Omega} g \, d\mu$  is taken. A supremum over a smaller set cannot exceed a supremum over a larger set. ■

**Proposition 2.5** (Positive homogeneity)

If  $f : \Omega \rightarrow [0, \infty]$  is measurable and  $c \geq 0$ , then

$$\int_{\Omega} cf \, d\mu = c \int_{\Omega} f \, d\mu. \quad (2.32)$$

*Proof.* This conclusion clearly holds if  $c = 0$ . For  $c > 0$ , note that  $\varphi \leq cf \iff \frac{\varphi}{c} \leq f$ . ■

**Remark 2.7.** Additivity of the integral,  $\int_{\Omega} (f + g) \, d\mu = \int_{\Omega} f \, d\mu + \int_{\Omega} g \, d\mu$  for non-negative measurable  $f, g$ , is *not* immediate from the definition and requires a genuine proof; this will follow once the Monotone Convergence Theorem is available (approximate  $f$  and  $g$  by increasing sequences of simple functions, for which additivity is straightforward, and pass to the limit).

**Definition 2.5** (Integral over a subset). Let  $A \in \Sigma$  and let  $g : \Omega \rightarrow [0, \infty]$  be measurable. The integral of  $g$  over  $A$  is defined by

$$\int_A g \, d\mu := \int_{\Omega} g \chi_A \, d\mu. \quad (2.33)$$

We now come to the first of the two fundamental convergence theorems of integration theory (the second being the Dominated Convergence Theorem, to be treated in a later lecture). Both address the same basic question: when can a limit be passed inside an integral?

**Theorem 2.6** (Monotone Convergence Theorem)

Let  $(f_n)_{n \geq 1}$  be a sequence of measurable functions  $\Omega \rightarrow [0, \infty]$  such that

$$0 \leq f_n \leq f_{n+1} \quad \text{for every } n, \quad (2.34)$$

and let  $f(x) := \lim_{n \rightarrow \infty} f_n(x)$  (which exists in  $[0, \infty]$  for every  $x$ , by monotonicity). Then  $f$  is measurable, and

$$\int_{\Omega} f \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (2.35)$$

*Proof.* Since  $(f_n(x))_n$  is non-decreasing for each  $x$ , its limit equals  $\limsup_n f_n(x) = \sup_n f_n(x)$ , which is measurable by .

For each  $n$ , since  $f_n \leq f$ , monotonicity of the integral gives

$$\int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu \quad \text{for all } n. \quad (\star)$$

The left-hand side is non-decreasing in  $n$  (again by monotonicity, since  $f_n \leq f_{n+1}$ ) and bounded above by the fixed quantity  $\int_{\Omega} f \, d\mu$ , so the limit exists and

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu. \quad (2.36)$$

Fix a simple function  $\varphi$  with  $0 \leq \varphi \leq f$ , and fix a constant  $\alpha \in (0, 1)$ . Define

$$A_n := \{x \in \Omega : f_n(x) \geq \alpha \varphi(x)\}. \quad (2.37)$$

Each  $A_n$  is measurable (as  $f_n - \alpha \varphi$  is measurable).

**Claim 2.2:**  $A_n \subseteq A_{n+1}$  for every  $n$ .

*Proof.* If  $f_n(x) \geq \alpha \varphi(x)$ , then since  $f_{n+1}(x) \geq f_n(x)$ , also  $f_{n+1}(x) \geq \alpha \varphi(x)$ .  $\square$

**Claim 2.3:**  $\bigcup_{n=1}^{\infty} A_n = \Omega$ .

*Proof.* Fix  $x \in \Omega$ . If  $\varphi(x) = 0$ , then  $x \in A_1$  trivially (as  $f_1(x) \geq 0 = \alpha\varphi(x)$ ). If  $\varphi(x) > 0$ , then

$$\alpha\varphi(x) < \varphi(x) \leq f(x). \quad (2.38)$$

Since  $f_n(x) \rightarrow f(x)$ , eventually  $f_n(x) > \alpha\varphi(x)$ , i.e.  $x \in A_n$  for  $n$  large.  $\square$

By monotonicity and the definition of the integral over a subset,

$$\int_{\Omega} f_n \, d\mu \geq \int_{A_n} f_n \, d\mu \geq \int_{A_n} \alpha\varphi \, d\mu = \alpha \int_{A_n} \varphi \, d\mu. \quad (2.39)$$

Since  $\varphi$  is simple, the set function  $B \mapsto \int_B \varphi \, d\mu$  is a measure. Since  $(A_n)$  is an increasing sequence of measurable sets with  $\bigcup_n A_n = \Omega$ , by [Proposition 1.4](#),

$$\lim_{n \rightarrow \infty} \int_{A_n} \varphi \, d\mu = \int_{\Omega} \varphi \, d\mu. \quad (2.40)$$

Taking  $n \rightarrow \infty$  in (2.39),

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \alpha \int_{\Omega} \varphi \, d\mu. \quad (2.41)$$

This holds for every  $\alpha \in (0, 1)$ ; letting  $\alpha \rightarrow 1^-$  gives

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \int_{\Omega} \varphi \, d\mu. \quad (2.42)$$

Finally, taking the supremum over all simple  $\varphi$  with  $0 \leq \varphi \leq f$ ,

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \int_{\Omega} f \, d\mu. \quad (2.43)$$

Combining both inequalities gives the result.  $\blacksquare$

**Remark 2.8.** The restriction  $\alpha < 1$  (rather than simply taking  $\alpha = 1$ , i.e.  $A_n = \{f_n \geq \varphi\}$ ) is essential: without it, claim (b) above can fail. It is possible for a simple function  $\varphi \leq f$  to touch  $f$  exactly at some point  $x$ , while the approximating sequence  $f_n(x)$  increases to  $f(x)$  without ever reaching it, so that  $f_n(x) < \varphi(x)$  for every finite  $n$ .

**Remark 2.9.** The [Monotone Convergence Theorem](#) is, in essence, already implicit in the construction of  $\int_{\Omega} f \, d\mu$ . The integral of  $f$  was defined as a supremum of integrals of simple functions below  $f$ , and we exhibited an increasing sequence of simple functions converging to  $f$  everywhere. The [Monotone Convergence Theorem](#) shows that this phenomenon persists for *any* increasing sequence of non-negative measurable functions converging to  $f$ , not only sequences of simple functions.

**Lemma 2.7**

If  $f_1, f_2 \in \mathcal{M}^+$ , then  $\int_{\Omega} (f_1 + f_2) d\mu = \int_{\Omega} f_1 d\mu + \int_{\Omega} f_2 d\mu$ . By induction,

$$\int_{\Omega} \sum_{i=1}^n f_i d\mu = \sum_{i=1}^n \int_{\Omega} f_i d\mu. \quad (2.44)$$

*Proof.* By [Theorem 2.3](#), we can find increasing sequence of simple functions  $\varphi_n \rightarrow f_1, \psi_n \rightarrow f_2$ . Then  $\varphi_n + \psi_n$  is an increasing sequence of simple function that monotonically converges to  $f_1 + f_2$ . By [Monotone Convergence Theorem](#),

$$\int_{\Omega} (f_1 + f_2) d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} (\varphi_n + \psi_n) d\mu \quad (2.45)$$

$$= \lim_{n \rightarrow \infty} \int_{\Omega} \varphi_n d\mu + \lim_{n \rightarrow \infty} \int_{\Omega} \psi_n d\mu \quad (2.46)$$

$$= \int_{\Omega} f_1 d\mu + \int_{\Omega} f_2 d\mu. \quad (2.47)$$

■

**Proposition 2.8**

If  $\{f_n\}_n$  is a sequence in  $\mathcal{M}^+$ , then

$$\int_{\Omega} \sum_{i=1}^{\infty} f_i d\mu = \sum_{i=1}^{\infty} \int_{\Omega} f_i d\mu. \quad (2.48)$$

*Proof.*  $g_n = \sum_{i=1}^n f_i$  is a monotone increasing sequence converging to  $f = \sum_{i=1}^{\infty} f_i$ . Therefore, by [Monotone Convergence Theorem](#),

$$\int_{\Omega} \sum_{i=1}^{\infty} f_i d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} g_n d\mu = \lim_{n \rightarrow \infty} \sum_{i=1}^n \int_{\Omega} f_i d\mu = \sum_{i=1}^{\infty} \int_{\Omega} f_i d\mu. \quad (2.49)$$

■

**Proposition 2.9**

If  $f \in \mathcal{M}^+$  and  $\int_{\Omega} f d\mu = 0$ , then  $f = 0$   $\mu$ -a.e.

*Proof.* Suppose  $f = 0$  a.e. Let  $E$  be the set where  $f$  is nonzero, then  $\mu(E) = 0$  by hypothesis. Then consider a simple function in its standard form  $\varphi = \sum_{i=1}^n a_i \chi_{A_i} \leq f$ , with  $a_i > 0$  and  $A_i$ 's pairwise disjoint. Since  $\varphi \leq f$ , hence each  $A_i \subseteq E$ , and consequently,  $\mu(A_i) = 0$ . Then

$$\int_{\Omega} \varphi d\mu = \sum_{i=1}^n a_i \mu(A_i) = 0 \implies \int_{\Omega} f d\mu = \sup_{\substack{0 \leq \varphi \leq f \\ \varphi \text{ simple}}} \int_{\Omega} \varphi d\mu = 0. \quad (2.50)$$

Conversely, suppose  $\int_{\Omega} f \, d\mu = 0$ . Consider the set

$$E_n = \left\{ x \in \Omega : f(x) > \frac{1}{n} \right\}. \quad (2.51)$$

Then we have

$$E := \{x \in \Omega : f(x) > 0\} = \bigcup_{n=1}^{\infty} E_n. \quad (2.52)$$

We need to show that  $\mu(E) = 0$ . Assume for the sake of contradiction that it's strictly greater than 0. By continuity ([Proposition 1.4](#)),

$$0 < \mu(E) = \lim_{n \rightarrow \infty} \mu(E_n), \quad (2.53)$$

implies that there exists some  $m$  for which  $\mu(E_m) > 0$ . Then  $0 \leq \frac{1}{m} \chi_{E_m} \leq f$ , but then

$$\int_{\Omega} f \, d\mu \geq \int_{\Omega} \frac{1}{m} \chi_{E_m} \, d\mu = \frac{1}{m} \mu(E_m) > 0. \quad (2.54)$$

■

### Theorem 2.10

If  $\{f_n\}_{n \geq 1}$  and  $f$  are functions in  $\mathcal{M}^+$  such that  $\{f_n\}_{n \geq 1}$  converges monotonically increasing to  $f$  for  $\mu$ -a.e.  $x \in \Omega$ . Then conclusion of [Monotone Convergence Theorem](#) still holds, i.e.

$$\int_{\Omega} f \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (2.55)$$

*Proof.* Suppose  $f_n(x) \rightarrow f(x)$  monotonically increasing on  $E \subseteq \Omega$ , then  $\mu(E^c) = 0$ . Then  $f - f\chi_E = 0$  and  $f_n - f_n\chi_E = 0$  a.e. Also,  $f_n\chi_E$  converges monotonically increasing to  $f\chi_E$  everywhere. As a result, by [Monotone Convergence Theorem](#),

$$\int_{\Omega} f \, d\mu = \int_{\Omega} f\chi_E \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n\chi_E \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (2.56)$$

■

### Proposition 2.11

If  $f \in \mathcal{M}^+$ , then the map

$$A \mapsto \int_A f \, d\mu = \int_{\Omega} f \cdot \chi_A \, d\mu \quad (2.57)$$

is a measure on  $\Sigma$ .

*Proof.* Suppose  $\nu(A) = \int_{\Omega} f \cdot \chi_A \, d\mu$ . Clearly,  $\nu(A) \geq 0$ . Also

$$\nu(\emptyset) = \int_{\Omega} f \cdot \chi_{\emptyset} \, d\mu = \int_{\Omega} 0 \, d\mu = 0. \quad (2.58)$$

Now, suppose  $\{A_n\}_{n \geq 1}$  is a countable collection of pairwise disjoint sets. We consider

$$f_n = \sum_{i=1}^n (f \chi_{A_i}) = f \chi_{\bigcup_{i=1}^n A_i}, \quad (2.59)$$

which is a monotonically increasing sequence of measurable functions, that converge pointwise to  $f \chi_{\bigcup_{i=1}^{\infty} A_i}$ . Then by [Monotone Convergence Theorem](#),

$$\nu \left( \bigcup_{i=1}^{\infty} A_i \right) = \int_{\Omega} f \chi_{\bigcup_{i=1}^{\infty} A_i} d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n d\mu \quad (2.60)$$

$$= \lim_{n \rightarrow \infty} \int_{\Omega} \sum_{i=1}^n (f \chi_{A_i}) d\mu = \lim_{n \rightarrow \infty} \sum_{i=1}^n \int_{\Omega} (f \chi_{A_i}) d\mu \quad (2.61)$$

$$= \sum_{i=1}^{\infty} \nu(A_i). \quad (2.62)$$

■

#### Corollary 2.12

If  $A_1 \subseteq A_2 \subseteq \dots$  is an increasing sequence of measurable sets and  $f \in \mathcal{M}^+$ , then

$$\int_{\bigcup_{i=1}^{\infty} A_i} f d\mu = \lim_{n \rightarrow \infty} \int_{A_n} f d\mu. \quad (2.63)$$

*Proof.*  $A \mapsto \nu(A) = \int_A f dx$  is a measure on  $\Sigma$ , by [Proposition 2.11](#).  $\{A_n\}_n$  is an increasing sequence of measurable sets. Therefore, by [Proposition 1.4](#),

$$\nu \left( \bigcup_{i=1}^{\infty} A_i \right) = \lim_{n \rightarrow \infty} \nu(A_n) \implies \int_{\bigcup_{i=1}^{\infty} A_i} f d\mu = \lim_{n \rightarrow \infty} \int_{A_n} f d\mu. \quad (2.64)$$

■

## §3 14 September 2026

In our last lecture, we proved that if  $f_n$  is a monotonically increasing sequence of measurable functions, then we can essentially swap limit and integral:

$$\int_{\Omega} \lim_{n \rightarrow \infty} f_n \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.1)$$

The hypothesis that  $(f_n)$  is an *increasing* sequence is essential, as the following examples show.

### Example 3.1

Consider  $f_n = \chi_{(n, n+1)}$ . Then  $f_n(x) \rightarrow 0$  pointwise for every  $x \in \mathbb{R}$ , yet

$$\int_{\mathbb{R}} f_n \, d\mu = 1 \quad \text{for every } n, \quad \text{while} \quad \int_{\mathbb{R}} \left( \lim_{n \rightarrow \infty} f_n \right) \, d\mu = 0. \quad (3.2)$$

So the conclusion of [Monotone Convergence Theorem](#) fails without monotonicity.

### Example 3.2

Define, for  $n \geq 1$ ,

$$f_n = n \chi_{(0, \frac{1}{n})}, \quad (3.3)$$

or, to make it continuous, a triangular spike on  $[0, 1/n]$  of height  $2n$  so that its graph is a triangle of area 1. Then

$$\int_0^1 f_n(x) \, dx = 1 \quad \text{for every } n, \quad (3.4)$$

but for every fixed  $x > 0$  there is  $N$  such that  $1/n < x$  for all  $n \geq N$ , so  $f_n(x) \rightarrow 0$  pointwise (and  $f_n(0) = 0$  for all  $n$ , by construction). Hence  $f_n \rightarrow 0$  pointwise while  $\int_0^1 f_n \, d\mu = 1$ , again showing that monotonicity cannot be dropped.

This type of example is the classical illustration that pointwise convergence alone does not let one pass the limit inside the integral; one really needs either monotonicity or a dominating integrable function ([Dominated Convergence Theorem](#), see below).

### Theorem 3.1 (Fatou's Lemma)

Let  $(f_n)$  be a sequence of nonnegative measurable functions on  $\Omega$ . Then

$$\int_{\Omega} \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.5)$$

*Proof.* For each  $m$ , define

$$g_m(x) = \inf_{n \geq m} f_n(x). \quad (3.6)$$

For every  $n \geq m$  we have  $g_m \leq f_n$ , and since the integral is monotone,

$$\int g_m \, d\mu \leq \int f_n \, d\mu \quad \text{for all } n \geq m. \quad (3.7)$$

Taking the infimum over  $n \geq m$  on the right-hand side gives

$$\int g_m \, d\mu \leq \inf_{n \geq m} \int f_n \, d\mu. \quad (3.8)$$

The sequence  $(g_m)$  is increasing in  $m$  (an infimum over a smaller index set is larger), and

$$\lim_{m \rightarrow \infty} g_m(x) = \sup_m \inf_{n \geq m} f_n(x) = \liminf_{n \rightarrow \infty} f_n(x). \quad (3.9)$$

By the [Monotone Convergence Theorem](#) applied to  $(g_m)$ ,

$$\int \left( \liminf_{n \rightarrow \infty} f_n \right) d\mu = \lim_{m \rightarrow \infty} \int g_m \, d\mu \leq \lim_{m \rightarrow \infty} \inf_{n \geq m} \int f_n \, d\mu = \liminf_{n \rightarrow \infty} \int f_n \, d\mu. \quad (3.10)$$

■

### Proposition 3.2

If  $\{f_n\}_{n \geq 1}$  and  $f$  are functions in  $\mathcal{M}^+$  such that  $f_n(x)$  converges to  $f(x)$  for  $\mu$ -a.e.  $x \in \Omega$ , then the conclusion of [Fatou's Lemma](#) still holds, i.e.

$$\int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.11)$$

*Proof.* Suppose  $f_n(x) \rightarrow f(x)$  on  $E \subseteq \Omega$ , then  $\mu(E^c) = 0$ . Then  $f - f\chi_E = 0$  and  $f_n - f_n\chi_E = 0$  a.e. Also,  $f_n\chi_E$  converges to  $f\chi_E$  everywhere. As a result,

$$\int_{\Omega} f \, d\mu = \int_{\Omega} f\chi_E \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n\chi_E \, d\mu = \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.12)$$

■

### Proposition 3.3

If  $f \in \mathcal{M}^+$  and  $\int_{\Omega} f \, d\mu < \infty$ , then  $\{x \in \Omega : f(x) = \infty\}$  is a null set and  $\{x \in \Omega : f(x) > 0\}$  is  $\sigma$ -finite.

*Proof.* Suppose  $E := \{x \in \Omega : f(x) = \infty\}$ . Assume for the sake of contradiction that  $\mu(E) > 0$ . Then  $f \geq f \cdot \chi_E$ . So

$$\int_{\Omega} f \, d\mu \geq \int_{\Omega} f \cdot \chi_E \, d\mu = \int_{\Omega} \infty \cdot \chi_E \, d\mu = \infty \cdot \mu(E) = \infty, \quad (3.13)$$

which is a contradiction! Now, suppose  $F = \{x \in \Omega : f(x) > 0\}$ . Observe that

$$F = \bigcup_{n=1}^{\infty} \left\{ x \in \Omega : f(x) \geq \frac{1}{n} \right\} =: \bigcup_{n=1}^{\infty} F_n. \quad (3.14)$$

So it suffices to prove that each  $F_n$  has finite measure. Note that  $f(x) \geq \frac{1}{n}$  on  $F_n$ , so that  $n \cdot f(x) \geq 1$ . In other words,  $\chi_{F_n} \leq n \cdot f$ . By monotonicity of integrals,

$$\mu(F_n) = \int_{\Omega} \chi_{F_n} d\mu \leq \int_{\Omega} n \cdot f d\mu = n \int_{\Omega} f d\mu < \infty. \quad (3.15)$$

■

**Definition 3.1** (Positive and negative part). For a measurable function  $f : \Omega \rightarrow \mathbb{R}$  define

$$f^+(x) = \max\{f(x), 0\}, \quad f^-(x) = \max\{-f(x), 0\}, \quad (3.16)$$

to be the **positive part** and **negative part** of  $f$ , respectively.

**Remark 3.1.** It's clear that  $f^+, f^- \geq 0$  are measurable, and

$$f = f^+ - f^-, \quad |f| = f^+ + f^-. \quad (3.17)$$

This decomposition is not unique as a way of writing  $f$  as a difference of two nonnegative functions (one may add the same nonnegative function to both  $f^+$  and  $f^-$ ), but it is the canonical choice, and it is the one used to define the integral of a general function.

**Definition 3.2** (The space  $L^1$ ). Let  $(\Omega, \mu)$  be a measure space. Define

$$L^1(\Omega, \mu) = \left\{ f : \Omega \rightarrow \mathbb{R} \text{ measurable} : \int_{\Omega} |f| d\mu < \infty \right\}. \quad (3.18)$$

For  $f \in L^1(\Omega, \mu)$ , both  $\int f^+ d\mu$  and  $\int f^- d\mu$  are finite (since their sum is  $\int |f| d\mu < \infty$ ), and we define

$$\int_{\Omega} f d\mu := \int_{\Omega} f^+ d\mu - \int_{\Omega} f^- d\mu. \quad (3.19)$$

**Remark 3.2.** The integral is first defined for simple functions, then extended to nonnegative measurable functions via the supremum definition, and finally extended to  $L^1$  via this difference, precisely so as to avoid ever having to make sense of  $\infty - \infty$ .

**Proposition 3.4**

If  $f \in L^1$ , then

$$\left| \int_{\Omega} f d\mu \right| \leq \int_{\Omega} |f| d\mu. \quad (3.20)$$

*Proof.*

$$\left| \int_{\Omega} f \, d\mu \right| = \left| \int_{\Omega} f^+ \, d\mu - \int_{\Omega} f^- \, d\mu \right| \leq \int_{\Omega} f^+ \, d\mu + \int_{\Omega} f^- \, d\mu = \int_{\Omega} |f| \, d\mu. \quad (3.21)$$

■

### Proposition 3.5

If  $f, g \in L^1(\Omega, \mu)$  and  $c \in \mathbb{R}$ , then  $f + cg \in L^1(\Omega, \mu)$  and

$$\int_{\Omega} (f + cg) \, d\mu = \int_{\Omega} f \, d\mu + c \int_{\Omega} g \, d\mu. \quad (3.22)$$

*Proof.* First notice that

$$|f + cg| \leq |f| + |c| |g|, \quad (3.23)$$

which implies using monotonicity of integral of non-negative functions that

$$\int_{\Omega} |f + cg| \, d\mu \leq \int_{\Omega} |f| + |c| |g| \, d\mu = \int_{\Omega} |f| \, d\mu + |c| \int_{\Omega} |g| \, d\mu < \infty. \quad (3.24)$$

We'll now show that  $\int_{\Omega} cf \, d\mu = c \int_{\Omega} f \, d\mu$ . For  $c \geq 0$ ,

$$(cf)^+ = cf^+, (cf)^- = cf^-. \quad (3.25)$$

Then

$$\int_{\Omega} cf \, d\mu = \int_{\Omega} (cf)^+ \, d\mu - \int_{\Omega} (cf)^- \, d\mu = \int_{\Omega} cf^+ \, d\mu - \int_{\Omega} cf^- \, d\mu = c \int_{\Omega} f \, d\mu. \quad (3.26)$$

For  $c < 0$ ,

$$(cf)^+ = -cf^-, (cf)^- = -cf^+. \quad (3.27)$$

Then

$$\int_{\Omega} cf \, d\mu = \int_{\Omega} (cf)^+ \, d\mu - \int_{\Omega} (cf)^- \, d\mu = \int_{\Omega} -cf^- \, d\mu - \int_{\Omega} -cf^+ \, d\mu = c \int_{\Omega} f \, d\mu. \quad (3.28)$$

Now let's show that  $\int_{\Omega} f + g \, d\mu = \int_{\Omega} f \, d\mu + \int_{\Omega} g \, d\mu$ . Suppose  $h = f + g$ . Then

$$h^+ - h^- = f^+ - f^- + g^+ - g^- \implies h^+ + f^- + g^- = h^- + f^+ + g^+. \quad (3.29)$$

Using the additivity of integral of non-negative functions, we get

$$\int_{\Omega} h^+ \, d\mu + \int_{\Omega} f^- \, d\mu + \int_{\Omega} g^- \, d\mu = \int_{\Omega} h^- \, d\mu + \int_{\Omega} f^+ \, d\mu + \int_{\Omega} g^+ \, d\mu. \quad (3.30)$$

Regrouping gives us the desired result:

$$\int_{\Omega} h^+ \, d\mu - \int_{\Omega} h^- \, d\mu = + \int_{\Omega} f^+ \, d\mu - \int_{\Omega} f^- \, d\mu + \int_{\Omega} g^+ \, d\mu - \int_{\Omega} g^- \, d\mu, \quad (3.31)$$

i.e.  $\int_{\Omega} h \, d\mu = \int_{\Omega} f \, d\mu + \int_{\Omega} g \, d\mu$ . ■

### Lemma 3.6

If  $f \leq g$  a.e. then  $\int_{\Omega} f \, d\mu \leq \int_{\Omega} g \, d\mu$ .

*Proof.* Suppose  $f \leq g$  on  $E$  with  $\mu(E^c) = 0$ . Then  $(g - f)\chi_E \geq 0$ . The integral of a non-negative function is non-negative, so

$$\int_{\Omega} g \, d\mu - \int_{\Omega} f \, d\mu = \int_{\Omega} (g - f) \, d\mu = \int_{\Omega} (g - f)\chi_E \, d\mu \geq 0. \quad (3.32)$$
■

### Proposition 3.7

If  $f \in M$  and  $g \in L^1(\Omega, \mu)$  with  $|f| \leq g$  a.e., then  $f \in L^1(\Omega, \mu)$ . (A function dominated by an integrable function is itself integrable.)

*Proof.* By monotonicity of integrals,

$$\int_{\Omega} |f| \, d\mu \leq \int_{\Omega} g \, d\mu < \infty. \quad (3.33)$$
■

### Proposition 3.8

If  $f, g \in L^1$ , then the following are equivalent:

- (a)  $\int_A f \, d\mu = \int_A g \, d\mu$  for every  $A \in \Sigma$ ;
- (b)  $\int_{\Omega} |f - g| \, d\mu = 0$ ;
- (c)  $f = g$  a.e.

*Proof.* The equivalence of (b) and (c) follows from [Proposition 2.9](#):

$$\int_{\Omega} |f - g| \, d\mu = 0 \iff |f - g| = 0 \text{ a.e.} \iff f = g \text{ a.e.} \quad (3.34)$$

**(b) $\Rightarrow$ (a)** Take any  $A \in \Sigma$ . Then

$$\left| \int_A f \, d\mu - \int_A g \, d\mu \right| = \left| \int_{\Omega} (f - g)\chi_A \, d\mu \right| \leq \int_{\Omega} |f - g|\chi_A \, d\mu \leq \int_{\Omega} |f - g| \, d\mu = 0. \quad (3.35)$$

**(a) $\Rightarrow$ (c)** Suppose  $A = \{x \in \Omega : f(x) > g(x)\}$ , which is a measurable set. On  $A$ ,  $f - g > 0$ . If  $\mu(A) > 0$ , then  $\int_A (f - g) > 0$  by the following claim, which contradicts with the hypothesis.

**Claim 3.1:** If  $\mu(E) > 0$  and  $h > 0$  on  $E$ , then  $\int_E h \, d\mu > 0$ .

*Proof.* Let  $E_n = \{x \in E : h(x) \geq \frac{1}{n}\}$ . Then  $\{E_n\}_n$  is an increasing sequence with

$$E = \bigcup_{i=1}^{\infty} E_n. \quad (3.36)$$

By continuity (Proposition 1.4),

$$0 < \mu(E) = \lim_{n \rightarrow \infty} \mu(E_n), \quad (3.37)$$

implies that there exists some  $m$  for which  $\mu(E_m) > 0$ . Then  $0 \leq \frac{1}{m} \chi_{E_m} \leq h$ , but then

$$\int_E h \, d\mu \geq \int_{\Omega} \frac{1}{m} \chi_{E_m} \, d\mu = \frac{1}{m} \mu(E_m) > 0. \quad (3.38)$$

□

Therefore,  $\mu(A) = 0$ . Similarly, if  $B = \{x \in \Omega : f(x) < g(x)\}$ , then  $\mu(B) = 0$ . Now,

$$\{x \in \Omega : f(x) \neq g(x)\} = A \cup B, \quad (3.39)$$

is also a null set, since it's the union of two null sets. ■

**Remark 3.3.** Note that  $\int_{\Omega} f \, d\mu = \int_{\Omega} g \, d\mu$  does not necessarily imply that  $f = g$  a.e. For example, consider  $\Omega = [0, 2\pi]$  and  $f(x) = \sin x$ ,  $g(x) = \cos x$ . Then

$$\int_0^{2\pi} \sin x \, dx = \int_0^{2\pi} \cos x \, dx = 0, \quad (3.40)$$

but it's **NOT** true that  $\sin x = \cos x$  a.e.

### Theorem 3.9 (Dominated Convergence Theorem)

Let  $(f_n)$  be a sequence of measurable functions on  $\Omega$  such that  $f_n \rightarrow f$  pointwise a.e., and suppose there is a function  $g \in L^1(\Omega, \mu)$  such that

$$|f_n(x)| \leq g(x) \text{ a.e. for all } n. \quad (3.41)$$

Then  $f \in L^1(\Omega, \mu)$  and

$$\int_{\Omega} f \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.42)$$

*Proof.* Since  $|f_n| \leq g$  for every  $n$  and  $f_n \rightarrow f$  pointwise, also  $|f| \leq g$ , so  $f \in L^1(\Omega, \mu)$  Proposition 3.7.

Since  $g + f_n \geq 0$  (because  $|f_n| \leq g$ ) and  $g + f_n \rightarrow g + f$  pointwise a.e., [Fatou's Lemma](#) gives

$$\int_{\Omega} (g + f) \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g + f_n) \, d\mu = \int_{\Omega} g \, d\mu + \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.43)$$

Since  $\int_{\Omega} g \, d\mu$  is finite, it may be cancelled from both sides:

$$\int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.44)$$

Likewise  $g - f_n \geq 0$  and  $g - f_n \rightarrow g - f$  pointwise a.e., so [Fatou's Lemma](#) gives

$$\int_{\Omega} (g - f) \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g - f_n) \, d\mu = \int_{\Omega} g \, d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu, \quad (3.45)$$

using  $\liminf(-a_n) = -\limsup(a_n)$ . Cancelling  $\int_{\Omega} g \, d\mu$  (finite) from both sides:

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu. \quad (3.46)$$

Combining (3.44) and (3.46),

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu, \quad (3.47)$$

and since always  $\liminf \leq \limsup$ , all three quantities coincide. Hence  $\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu$  exists and equals  $\int_{\Omega} f \, d\mu$ . ■

**Remark 3.4.** The [Monotone Convergence Theorem](#), [Fatou's Lemma](#), and the [Dominated Convergence Theorem](#) are the three fundamental convergence theorems needed to justify moving a limit inside an integral: either the sequence is monotone (Monotone Convergence), or it is dominated by a single integrable function (Dominated Convergence); [Fatou's Lemma](#) always applies to nonnegative sequences and is the tool used to prove the other two.

### Theorem 3.10 (Generalized Dominated Convergence Theorem)

If  $f_n, g_n, g \in L^1$ ,  $f_n \rightarrow f$  and  $g_n \rightarrow g$  a.e.,

$$|f_n| \leq g_n, \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu = \int_{\Omega} g \, d\mu, \quad (3.48)$$

then  $f \in L^1$ , and

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu = \int_{\Omega} f \, d\mu. \quad (3.49)$$

*Proof.* Since  $|f_n| \leq g_n$  for every  $n$  and  $f_n \rightarrow f$  pointwise a.e. and  $g_n \rightarrow g$  a.e., we have that  $|f| \leq g$  a.e. So  $f \in L^1(\Omega, \mu)$  [Proposition 3.7](#).

Since  $g_n + f_n \geq 0$  (because  $|f_n| \leq g_n$ ) and  $g_n + f_n \rightarrow g + f$  pointwise a.e., [Fatou's Lemma](#) gives

$$\begin{aligned}
\int_{\Omega} g \, d\mu + \int_{\Omega} f \, d\mu &= \int_{\Omega} (g + f) \, d\mu \\
&\leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g_n + f_n) \, d\mu \\
&= \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu + \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \\
&= \int_{\Omega} g \, d\mu + \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu.
\end{aligned} \tag{3.50}$$

Since  $\int_{\Omega} g \, d\mu$  is finite, it may be cancelled from both sides:

$$\int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \tag{3.51}$$

Likewise  $g_n - f_n \geq 0$  and  $g_n - f_n \rightarrow g - f$  pointwise a.e., so [Fatou's Lemma](#) gives

$$\begin{aligned}
\int_{\Omega} g \, d\mu - \int_{\Omega} f \, d\mu &= \int_{\Omega} (g - f) \, d\mu \\
&\leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g_n - f_n) \, d\mu \\
&= \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \\
&= \int_{\Omega} g \, d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu.
\end{aligned} \tag{3.52}$$

Cancelling  $\int_{\Omega} g \, d\mu$  (finite) from both sides:

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu. \tag{3.53}$$

Combining [\(3.51\)](#) and [\(3.53\)](#),

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu, \tag{3.54}$$

and since always  $\liminf \leq \limsup$ , all three quantities coincide. Hence  $\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu$  exists and equals  $\int_{\Omega} f \, d\mu$ . ■

**Remark 3.5.** In the proof of [Generalized Dominated Convergence Theorem](#), we used the fact that if  $a_n, b_n$  are two sequences such that  $a_n \rightarrow a$ , then

$$\liminf_{n \rightarrow \infty} (a_n + b_n) = a + \liminf_{n \rightarrow \infty} b_n, \tag{3.55}$$

$$\liminf_{n \rightarrow \infty} (a_n - b_n) = a - \limsup_{n \rightarrow \infty} b_n. \tag{3.56}$$

To prove the first one, notice that

$$\inf_{m \geq n} (a_m + b_m) \geq \inf_{m \geq n} a_m + \inf_{m \geq n} b_m. \tag{3.57}$$

Taking  $n \rightarrow \infty$ , we get

$$\liminf_{n \rightarrow \infty} (a_n + b_n) \geq \liminf_{n \rightarrow \infty} a_n + \liminf_{n \rightarrow \infty} b_n \quad (3.58)$$

$$= a + \liminf_{n \rightarrow \infty} b_n. \quad (3.59)$$

On the other hand, writing  $b_n = (a_n + b_n) + (-a_n)$  and applying (3.58), we get

$$\liminf_{n \rightarrow \infty} b_n = \liminf_{n \rightarrow \infty} [(a_n + b_n) + (-a_n)] \quad (3.60)$$

$$\geq \liminf_{n \rightarrow \infty} (a_n + b_n) + \liminf_{n \rightarrow \infty} (-a_n) \quad (3.61)$$

$$= \liminf_{n \rightarrow \infty} (a_n + b_n) - a. \quad (3.62)$$

Therefore, combining (3.59) and (3.62), we have established (3.55). To see (3.56),

$$\liminf_{n \rightarrow \infty} (a_n - b_n) = a + \liminf_{n \rightarrow \infty} (-b_n) = a - \limsup_{n \rightarrow \infty} b_n.$$

### Example 3.3

Let

$$f_n(x) = n \chi_{(0,1/n)}(x), \quad x \in [0, 1]. \quad (3.63)$$

Then  $f_n(x) \rightarrow 0$  for every  $x \in [0, 1]$  (for  $x = 0$  every  $f_n(0) = 0$ ; for  $x > 0$ , eventually  $\frac{1}{n} < x$ ), so  $f_n \rightarrow 0$  pointwise. However,

$$\int_0^1 f_n dx = 1 \quad \text{for every } n, \quad \text{while } \int_0^1 \left( \lim_{n \rightarrow \infty} f_n \right) dx = 0. \quad (3.64)$$

The conclusion of the Dominated Convergence Theorem fails here because no single integrable function  $g$  dominates all the  $f_n$ . The pointwise supremum of the family is essentially  $g(x) = \frac{1}{x}$ , and

$$\int_0^1 \frac{1}{x} dx = \infty, \quad (3.65)$$

so  $g \notin L^1([0, 1])$ . This also shows the Monotone Convergence Theorem does not apply, since the sequence  $(f_n)$  is not pointwise increasing.

We now show that one cannot hope to define a measure on *all* subsets of  $\mathbb{R}$  that

- (i) is translation invariant,
- (ii) is monotone (if  $A \subseteq B$  then  $\mu(A) \leq \mu(B)$ ), and
- (iii) and that extends the usual length of intervals.

Consequently the  $\sigma$ -algebra of measurable sets must be a strict subset of the power set  $\mathcal{P}(\mathbb{R})$ . For this, we illustrate the following example:

On  $[0, 1]$ , declare  $x \sim y$  if  $x - y \in \mathbb{Q}$ . This is an equivalence relation, partitioning  $[0, 1]$  into equivalence classes. By the Axiom of Choice, choose exactly one representative  $x_I$  from each

equivalence class  $I$ , and let

$$A = \{x_I : I \in [0, 1] / \sim\} \subseteq [0, 1], \quad (3.66)$$

where  $I$  indexes the set of equivalence classes. The set  $A$  is called a *Vitali set*.

### Theorem 3.11

There is no measure  $\mu$  on  $\mathcal{P}(\mathbb{R})$  that is translation invariant, monotone, and assigns to every interval its usual length.

*Proof.* Suppose such a  $\mu$  exists. For  $r \in \mathbb{Q} \cap [-1, 1]$  set

$$Y_r = A + r = \{a + r : a \in A\}. \quad (3.67)$$

If  $Y_r \cap Y_s \neq \emptyset$  for  $r \neq s \in \mathbb{Q}$ , then  $x_i + r = x_j + s$  for some representatives, so

$$x_i - x_j = s - r \in \mathbb{Q}, \quad (3.68)$$

forcing  $x_i \sim x_j$ ; since  $A$  contains only one representative per class,  $x_i = x_j$ , hence  $r = s$ , a contradiction. So the sets  $\{Y_r\}_{r \in \mathbb{Q} \cap [-1, 1]}$  are pairwise disjoint.

Since  $A \subseteq [0, 1]$  and  $r \in [-1, 1]$ ,

$$Y_r \subseteq [-1, 2] \quad \text{for every } r \in \mathbb{Q} \cap [-1, 1]. \quad (3.69)$$

For any  $x \in [0, 1]$  there is a representative  $x_i \sim x$  with  $x - x_i \in \mathbb{Q} \cap [-1, 1]$ , so  $x \in Y_{x-x_i}$ . Hence

$$[0, 1] \subseteq \bigcup_{r \in \mathbb{Q} \cap [-1, 1]} Y_r \subseteq [-1, 2]. \quad (3.70)$$

By translation invariance,  $\mu(Y_r) = \mu(A)$  for every  $r$ . By monotonicity and countable additivity on the disjoint countable union,

$$\mu([0, 1]) \leq \mu\left(\bigcup_{r \in \mathbb{Q} \cap [-1, 1]} Y_r\right) = \sum_{r \in \mathbb{Q} \cap [-1, 1]} \mu(Y_r) = \sum_{r \in \mathbb{Q} \cap [-1, 1]} \mu(A) \leq \mu([-1, 2]). \quad (3.71)$$

Now  $\mu([0, 1]) = 1$  and  $\mu([-1, 2]) = 3$  (extending the usual length), so

$$1 \leq \sum_{r \in \mathbb{Q} \cap [-1, 1]} \mu(A) \leq 3. \quad (3.72)$$

But  $\mathbb{Q} \cap [-1, 1]$  is countably infinite, so the middle sum is a countable sum of a single fixed value  $\mu(A) \in [0, \infty]$ :

- if  $\mu(A) = 0$ , the sum is 0, contradicting  $1 \leq 0$ ;
- if  $\mu(A) > 0$ , the sum is  $+\infty$ , contradicting  $\infty \leq 3$ .

Either way we reach a contradiction. Hence no such measure  $\mu$  can exist on all of  $\mathcal{P}(\mathbb{R})$ . ■

**Remark 3.6.** This is the price one pays for wanting a translation-invariant, monotone measure extending length: it cannot be defined on *every* subset of  $\mathbb{R}$ . This motivates restricting attention to a suitable  $\sigma$ -algebra (the Lebesgue measurable sets), which is the object to be constructed next.

## §4 16 September 2026

Having seen that no measure on *all* subsets of  $\mathbb{R}$  can extend length while remaining translation invariant, we now begin the actual construction of Lebesgue measure. The construction proceeds geometrically, starting from rectangles (objects simple enough that their “volume” is unambiguous) and builds up to general open, and eventually general measurable, sets. This lecture develops the purely geometric facts about rectangles and open sets that the construction will rely on.

**Definition 4.1** (Rectangle). A (closed) **rectangle**  $R \subseteq \mathbb{R}^d$  is a product of  $d$  closed, bounded intervals:

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_d, b_d], \quad a_i \leq b_i. \quad (4.1)$$

The sides of  $R$  are parallel to the coordinate axes; rotated rectangles are not considered. The associated **open rectangle** is

$$R^\circ := (a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_d, b_d). \quad (4.2)$$

**Definition 4.2** (Volume). The **volume** of the rectangle  $R = \prod_{i=1}^d [a_i, b_i]$  is

$$|R| := \prod_{i=1}^d (b_i - a_i). \quad (4.3)$$

(The same formula, with the same value, defines the volume  $|R^\circ|$  of the associated open rectangle.)

**Definition 4.3** (Almost disjoint collection). A finite (or countable) collection of rectangles  $\{R_k\}$  is **almost disjoint** if the open rectangles  $R_k^\circ$  are pairwise disjoint; that is, the rectangles may touch along their boundaries (share a face, edge, or corner), but their interiors do not overlap.

### Proposition 4.1

Suppose a rectangle  $R$  is the almost disjoint union of finitely many rectangles  $R_1, \dots, R_n$ , i.e.  $R = \bigcup_{k=1}^n R_k$  with  $\{R_k\}$  almost disjoint. Then

$$|R| = \sum_{k=1}^n |R_k|. \quad (4.4)$$

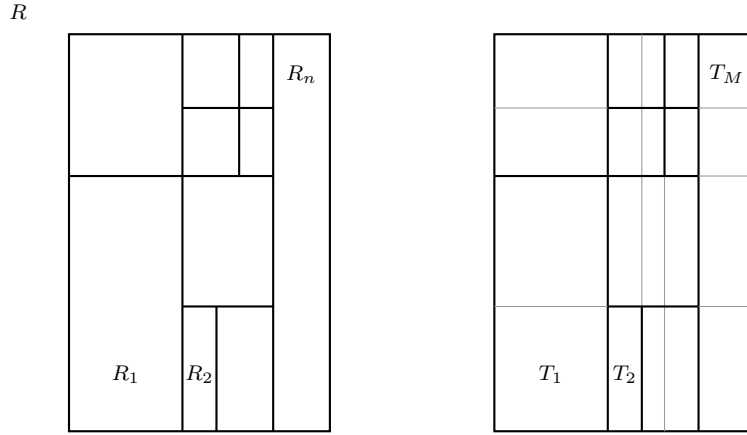
*Proof. Case 1:* Suppose first that each side of  $R = \prod_{i=1}^d [a_i, b_i]$  has been partitioned into finitely many subintervals. Say the  $i$ -th side  $[a_i, b_i]$  is cut into pieces of lengths  $\ell_{i,1}, \dots, \ell_{i,m_i}$  with  $\sum_j \ell_{i,j} = b_i - a_i$ , and the  $R_k$  are exactly the resulting grid cells (one for each choice of a

subinterval in each coordinate). Then, using distributivity of multiplication over addition in each coordinate,

$$|R| = \prod_{i=1}^d (b_i - a_i) = \prod_{i=1}^d \left( \sum_{j=1}^{m_i} \ell_{i,j} \right) = \sum_{j_1=1}^{m_1} \cdots \sum_{j_d=1}^{m_d} \ell_{1,j_1} \cdots \ell_{d,j_d} = \sum_k |R_k|, \quad (4.5)$$

since each term in the expanded product is exactly the volume of one grid cell, and every grid cell arises exactly once.

**Case 2:** Given an arbitrary almost disjoint decomposition  $R = \bigcup_{k=1}^n R_k$ , extend every face (bounding hyperplane) of every  $R_k$  across the entirety of  $R$ , in every coordinate direction.



This produces a common refining grid on  $R$ : each original rectangle  $R_k$  is itself partitioned by this grid into finitely many grid cells (an almost disjoint union of sub-rectangles), and by Case 1 applied to  $R_k$ ,

$$|R_k| = \sum_{\substack{\text{grid cells } T \\ T \subseteq R_k}} |T|. \quad (4.6)$$

On the other hand, since  $\{R_k\}$  is almost disjoint and covers  $R$ , every grid cell  $T$  of the common refinement lies in exactly one  $R_k$  (up to boundary overlaps, which do not affect volume), so summing over  $k$ ,

$$\sum_{k=1}^n |R_k| = \sum_{k=1}^n \sum_{T \subseteq R_k} |T| = \sum_{\text{all grid cells } T \text{ of } R} |T| = |R|, \quad (4.7)$$

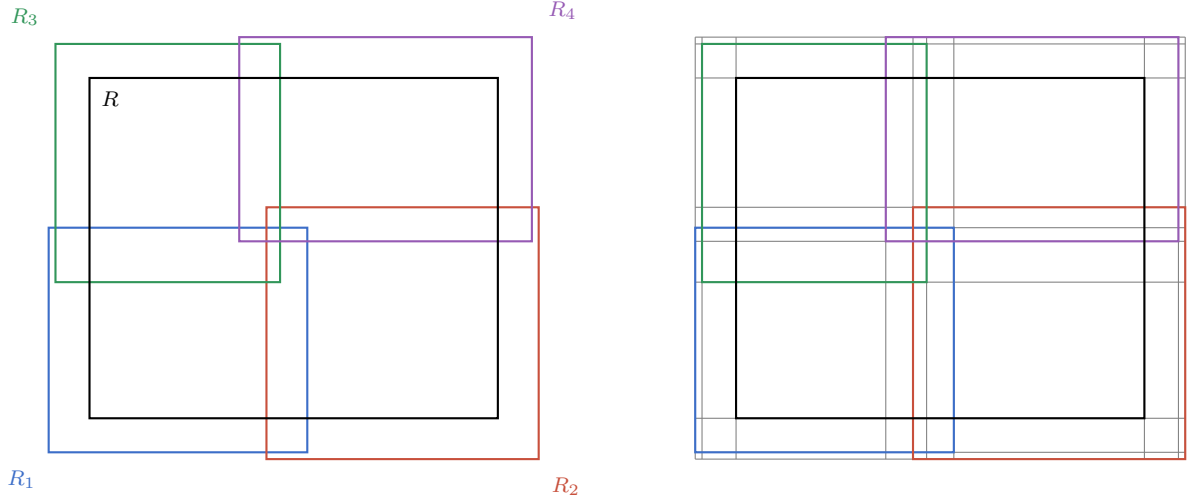
the last equality again by Case 1, applied this time to  $R$  itself with the common refining grid. ■

### Proposition 4.2

Suppose a rectangle  $R$  is covered by finitely many rectangles  $R_1, \dots, R_n$  (i.e.  $R \subseteq \bigcup_{k=1}^n R_k$ ), with no assumption that the  $R_k$  are disjoint or almost disjoint, and no assumption that  $R_k \subseteq R$ . Then

$$|R| \leq \sum_{k=1}^n |R_k|. \quad (4.8)$$

*Proof.* As in the proof of [Proposition 4.1](#), extend every face of every  $R_k$  and  $R$  to form a refining grid. This may create grid rectangles that are **NOT** in any of the  $R_i$ 's or  $R$ .



This decomposes  $R$  into finitely many almost disjoint grid cells  $T_1, \dots, T_m$ , so that  $R = \bigcup_j T_j$  almost disjointly, and by [Proposition 4.1](#),

$$|R| = \sum_{j=1}^m |T_j|. \quad (4.9)$$

Since  $R \subseteq \bigcup_k R_k$ , every grid cell  $T_j$  lies in at least one  $R_k$ ; assign each  $T_j$  to one such  $R_k$  (breaking ties arbitrarily).

$$|R| = \sum_{j=1}^m |T_j| = \sum_{k=1}^n \left( \sum_{\substack{T_j \text{ assigned} \\ \text{to } R_k}} |T_j| \right). \quad (4.10)$$

For a fixed  $k$ , the cells  $T_j$  assigned to  $R_k$  are almost disjoint and all contained in  $R_k \cap R \subseteq R_k$ . Note that they may not exhaust all of  $R_k$ . In fact, they are a subset of all the grid rectangles that are contained in  $R_k$ . Therefore,

$$\sum_{\substack{T_j \text{ assigned} \\ \text{to } R_k}} |T_j| \leq \sum_{\substack{T \text{ grid rectangle,} \\ T \subseteq R_k}} |T| = |R_k|, \quad (4.11)$$

where the last equality follows from [Proposition 4.1](#). Now combining (4.10) and (4.11), we are done! ■

**Remark 4.1.** [Proposition 4.2](#) is the geometric heart of **countable subadditivity** for the outer measure to be constructed in the next lecture: it says that volume cannot be “cheated” by covering a rectangle with other rectangles of smaller total volume.

**Proposition 4.3** (Structure of open subsets of  $\mathbb{R}$ )

Every open set  $O \subseteq \mathbb{R}$  can be written as a countable union of pairwise disjoint open intervals. Moreover, this decomposition is unique: it is exactly the decomposition of  $O$  into its connected components.

*Proof.* For  $x \in O$ , let

$$I_x := (a_x, b_x), \text{ where } a_x = \inf \{a : (a, x) \subseteq O\}, \text{ and } b_x = \sup \{b : (x, b) \subseteq O\}, \quad (4.12)$$

be the *largest* open interval contained in  $O$  that contains  $x$ . Since  $O$  is open,  $I_x$  is nonempty (it contains some small interval around  $x$ ), and clearly  $O = \bigcup_{x \in O} I_x$ .

If  $x, y \in O$  and  $I_x \cap I_y \neq \emptyset$ , then  $I_x \cup I_y$  is itself an open interval (a union of two overlapping intervals) contained in  $O$  and containing both  $x$  and  $y$ ; by maximality of  $I_x$  (as the *largest* such interval containing  $x$ ), we must have

$$I_x \cup I_y \subseteq I_x \implies I_y \subseteq I_x. \quad (4.13)$$

Similarly,  $I_x \subseteq I_y$ , so  $I_x = I_y$ . Hence, the collection  $\{I_x : x \in O\}$  consists of intervals that are pairwise either identical or disjoint; selecting one representative from each distinct interval gives a collection of pairwise disjoint open intervals  $\{I_j\}_{j \in J}$  with  $O = \bigsqcup_{j \in J} I_j$ .

Each  $I_j$  is a nonempty open interval, hence contains a rational number  $q_j$ ; since the  $I_j$  are pairwise disjoint, distinct intervals contain distinct rationals, giving an injection

$$\begin{aligned} J &\rightarrow \mathbb{Q} \\ j &\mapsto q_j. \end{aligned} \quad (4.14)$$

An injection into a countable set shows the index set  $J$  is countable.

Now about the uniqueness, suppose

$$O = \bigsqcup_k J_k, \quad (4.15)$$

is another decomposition into pairwise disjoint open intervals. Each  $J_k$ , being a connected subset of  $O$ , is contained in the maximal interval through any of its points, i.e.  $J_k \subseteq I_j$  for a **unique**  $j$ .

Conversely,  $I_j$  is covered by those  $J_k$ 's contained in it. But  $I_j$  is connected and the  $J_k$ 's are open and pairwise disjoint, so  $I_j$  cannot be covered by two or more nonempty disjoint open pieces (else  $I_j$  would be disconnected).

Hence exactly one  $J_k$  satisfies  $J_k \subseteq I_j$  with the covering forcing  $J_k = I_j$ . This gives a bijection between the two decompositions, proving uniqueness. ■

**Proposition 4.3** is special to  $\mathbb{R}$ : it relies essentially on the fact that connected open subsets of  $\mathbb{R}$  are exactly the open intervals. In  $\mathbb{R}^d$  for  $d \geq 2$ , connected open sets can have arbitrarily complicated shapes (e.g. an open disk), so an open set in  $\mathbb{R}^d$  cannot in general be written as a

disjoint union of open *rectangles*. The right substitute uses **closed rectangles**, allowing *almost* disjointness in place of disjointness.

**Remark 4.2.** In  $d = 1$ , consider the open set  $(0, 1) \subseteq \mathbb{R}$ . Then we can write  $(0, 1)$  as a countable union of almost-disjoint closed intervals:

$$(0, 1) = \bigcup_{n=2}^{\infty} \left[ \frac{1}{n+1}, \frac{1}{n} \right] \cup \bigcup_{n=2}^{\infty} \left[ 1 - \frac{1}{n}, 1 - \frac{1}{n+1} \right]. \quad (4.16)$$

**Definition 4.4** (Dyadic cubes). For  $n \in \mathbb{Z}$ , a **dyadic cube of generation  $n$**  is a set of the form

$$Q = \prod_{i=1}^d \left[ \frac{k_i}{2^n}, \frac{k_i+1}{2^n} \right], \quad k_1, \dots, k_d \in \mathbb{Z}. \quad (4.17)$$

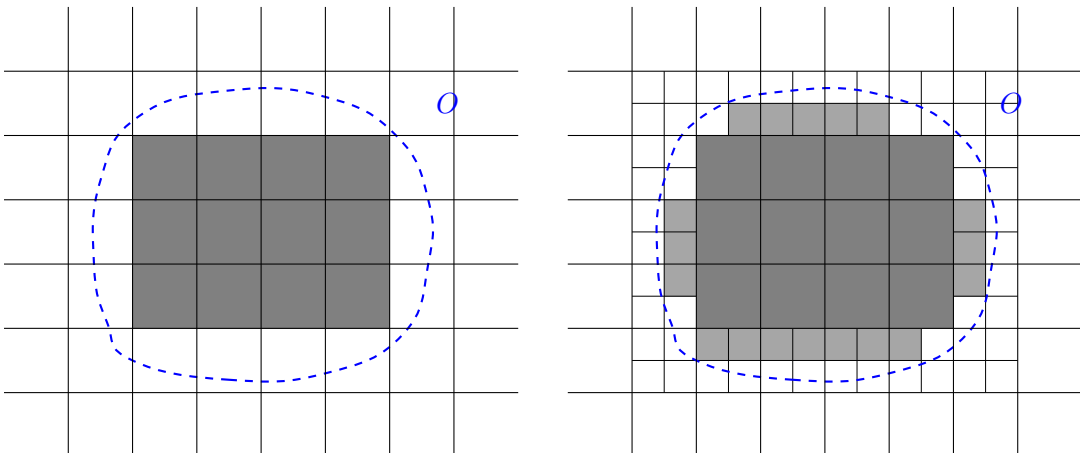
For fixed  $n$ , the dyadic cubes of generation  $n$  have disjoint interiors and their union is all of  $\mathbb{R}^d$  (they form a grid of cubes of side length  $2^{-n}$ ). Every dyadic cube  $Q$  of generation  $n$  is contained in a unique dyadic cube  $\tilde{Q}$  of generation  $n-1$  (its **parent**, of side length  $2^{-(n-1)}$ ), obtained by halving the grid spacing; conversely, each dyadic cube of generation  $n-1$  is the almost disjoint union of exactly  $2^d$  dyadic cubes of generation  $n$  (its **children**).

**Remark 4.3.** Two dyadic cubes (of possibly different generations) are always either almost disjoint (disjoint interiors) or one is contained in the other. This nesting property is the key structural fact that makes the dyadic grid useful.

**Proposition 4.4** (Structure of open subsets of  $\mathbb{R}^d$ )

Every open set  $O \subseteq \mathbb{R}^d$  can be written as a countable, almost disjoint union of closed rectangles.

*Proof.* Call a dyadic cube  $Q$  **selected** if  $Q \subseteq O$  but its parent  $\tilde{Q}$  is *not* contained in  $O$  (i.e.  $Q$  is a maximal dyadic cube contained in  $O$ , in the sense that enlarging it to its parent would exit  $O$ ).



Define

$$\mathcal{Q} := \{Q : Q \text{ is a selected dyadic cube}\}, \quad U := \bigcup_{Q \in \mathcal{Q}} Q. \quad (4.18)$$

We claim  $O = U$ .  $U \subseteq O$  is immediate, since every selected  $Q$  satisfies  $Q \subseteq O$  by definition.

Let  $x \in O$ . Since  $O$  is open, there is  $\varepsilon > 0$  with the open ball  $B(x, \varepsilon) \subseteq O$ . Choose  $n$  large enough that a dyadic cube of generation  $n$  has diameter

$$2^{-n}\sqrt{d} < \varepsilon. \quad (4.19)$$

Then there is a generation- $n$  dyadic cube  $Q_n$  containing  $x$  that satisfies

$$Q_n \subseteq B(x, \varepsilon) \subseteq O. \quad (4.20)$$

Since  $Q_n \subseteq O$ , that means  $Q_n$  has been selected at step  $n$ , or one of its previous generation parents has been selected at an earlier step. Therefore,  $x \in U = \bigcup_{Q \in \mathcal{Q}} Q$ .

Now we show that  $\mathcal{Q}$  is an almost-disjoint collection. Suppose  $Q, Q' \in \mathcal{Q}$  are distinct selected cubes with overlapping interiors. By the nesting property, one is contained in the other; say

$$Q' \subsetneq Q, \quad (4.21)$$

so that  $Q'$  is a proper descendant of  $Q$ , of strictly higher generation. Let  $Q''$  be the immediate parent of  $Q'$ . Then

$$Q' \subseteq Q'' \subseteq Q. \quad (4.22)$$

Since  $Q \subseteq O$  (as  $Q$  is selected), also  $Q'' \subseteq O$ . But this contradicts  $Q'$  being selected, which requires the immediate parent of  $Q'$  to *not* be contained in  $O$ . Hence no such containment can occur, and the interiors of distinct selected cubes are disjoint.

Finally, we'll now show that  $\mathcal{Q}$  is a countable collection. For each fixed generation  $n \in \mathbb{Z}$ , the dyadic cubes of generation  $n$  are indexed by  $\mathbb{Z}^d$ , a countable set.

$$\mathcal{Q} \subseteq \bigcup_{n \in \mathbb{Z}} \{\text{generation-}n \text{ dyadic cubes}\}, \quad (4.23)$$

which is a countable union of countable sets, hence countable. ■

**Remark 4.4** (Comparison with the one-dimensional case). For  $d = 1$ , the construction in [Proposition 4.4](#) produces a decomposition into closed dyadic *intervals*, which is generally coarser and less canonical than the connected-components decomposition of [Proposition 4.3](#) (dyadic intervals need not align with the true connected components of  $O$ , and distinct choices of the underlying grid (e.g. shifting or rescaling it) give different, equally valid, decompositions). This is consistent with the failure of *uniqueness* in higher dimensions: [Proposition 4.4](#) asserts existence of an almost disjoint decomposition into closed rectangle, not that it is the unique or canonical one.

**Remark 4.5.** The decomposition of [Proposition 4.4](#) cannot in general be made into a genuinely disjoint union of *closed* cubes (closed cubes covering a connected open set must share boundary faces where they meet), nor into a disjoint union of *open* rectangles when  $d \geq 2$  (e.g. an open disk is not a union of finitely or countably many disjoint open rectangles, since removing the boundary of any inscribed rectangle from a disk leaves a connected leftover region that cannot itself be a union of open rectangles disjoint from the first without gaps). Allowing cubes to touch along boundaries (“almost” disjoint) is what makes the construction possible in every dimension.

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We now begin the construction proper. The strategy has two stages: first, define a set function  $m^*$ , the **exterior measure** (or **outer measure**), on *every* subset of  $\mathbb{R}^d$  by covering from outside with cubes; second, single out a sub-collection of “good” sets (we’ll call them the **Lebesgue measurable sets**) on which  $m^*$  will turn out to be countably additive, i.e. a genuine measure in the sense of Section 3. [Theorem 3.11](#) tells us this sub-collection cannot be all of  $\mathcal{P}(\mathbb{R}^d)$ .

**Definition 5.1** (Exterior measure). For a set  $E \subseteq \mathbb{R}^d$ , the **exterior measure** of  $E$  is

$$m^*(E) := \inf \left\{ \sum_{j=1}^{\infty} |Q_j| : E \subseteq \bigcup_{j=1}^{\infty} Q_j, \quad Q_j \text{ closed cubes} \right\} \in [0, \infty]. \quad (5.1)$$

That is,  $m^*(E)$  is the infimum, over all countable covers of  $E$  by closed cubes (not required to be disjoint or almost disjoint), of the total volume of the cubes used.

**Remark 5.1.** Every set  $E \subseteq \mathbb{R}^d$  admits at least one such cover (e.g. by cubes tiling all of  $\mathbb{R}^d$ ), so  $m^*(E)$  is always a well-defined element of  $[0, \infty]$ .

**Remark 5.2** (Why cover from outside). One might instead try to define an “interior” measure by taking the *supremum* of the total volume of cubes contained *inside*  $E$ . This alternative fails badly. For example, a set with empty interior but which is, intuitively, “large” (such as the set of irrational numbers in  $[0, 1]$ , or more strikingly a dense set with no interior points at all) would have interior measure 0, since no cube of positive volume can fit inside it. Covering from *outside* does not suffer from this defect, which is why the construction proceeds this way.

**Remark 5.3** (Cubes versus rectangles). The definition uses closed cubes rather than closed rectangles purely for concretion. Using rectangles in place of cubes gives an identical set function (the two notions of covering are interchangeable, since every rectangle can be covered by finitely many cubes, and vice versa). Open cubes could equally be used in place of closed cubes, again without changing the value of  $m^*$ .

### Example 5.1

$m^*({x}) = 0$  for any single point  $x \in \mathbb{R}^d$ : cover  $\{x\}$  by a closed cube of side length  $\varepsilon$  centered at  $x$ , of volume  $\varepsilon^d$ . Since  $\varepsilon > 0$  is arbitrary, the infimum over such covers is 0.

**Remark 5.4.** Since  $m^*(E) = \inf_{E \subseteq \bigcup_{j=1}^{\infty} Q_j} \sum_{j=1}^{\infty} |Q_j|$ , by the definition of infimum, for every  $\varepsilon > 0$ , we can find a cover using closed cubes  $E \subseteq \bigcup_{j=1}^{\infty} Q_j$  such that

$$\sum_{j=1}^{\infty} |Q_j| \leq m^*(E) + \varepsilon. \quad (5.2)$$

This will turn out to be helpful while proving some identities.

We record some basic properties of  $m^*$ . These would guide us toward the properties we ultimately want a measure to have.

**Proposition 5.1 (Monotonicity)**

If  $E \subseteq F \subseteq \mathbb{R}^d$ , then  $m^*(E) \leq m^*(F)$ .

*Proof.* Every countable cover of  $F$  by closed cubes is also a cover of  $E$  (since  $E \subseteq F$ ), so the collection of covers admissible for computing  $m^*(F)$  is a subset of those admissible for  $m^*(E)$ . An infimum taken over a smaller collection of covers cannot be less than one taken over a larger collection, i.e.  $m^*(E) \leq m^*(F)$ . ■

**Proposition 5.2 (Countable subadditivity)**

For any countable collection of sets  $E_1, E_2, \dots \subseteq \mathbb{R}^d$ ,

$$m^*\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{\infty} m^*(E_j). \quad (5.3)$$

*Proof.* If the right side is infinite there is nothing to prove, so assume it is finite. Fix  $\varepsilon > 0$ . For each  $j$ , by definition of  $m^*(E_j)$  as an infimum, there is a countable cover of  $E_j$  by closed cubes  $\{Q_{j,i}\}_{i \geq 1}$  with

$$\sum_{i=1}^{\infty} |Q_{j,i}| \leq m^*(E_j) + \frac{\varepsilon}{2^j}. \quad (5.4)$$

The combined collection  $\{Q_{j,i}\}_{j,i \geq 1}$  is a countable cover of  $\bigcup_j E_j$  by closed cubes, so by definition of  $m^*\left(\bigcup_j E_j\right)$  as an infimum,

$$m^*\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} |Q_{j,i}| \leq \sum_{j=1}^{\infty} \left(m^*(E_j) + \frac{\varepsilon}{2^j}\right) = \sum_{j=1}^{\infty} m^*(E_j) + \varepsilon. \quad (5.5)$$

Since  $\varepsilon > 0$  was arbitrary, the claimed inequality follows. ■

**Corollary 5.3**

If  $E$  is countable, then  $m^*(E) = 0$ .

*Proof.* Suppose  $E = \{x_1, x_2, x_3, \dots\}$ . Then

$$E = \bigcup_{i=1}^{\infty} \{x_i\} \implies 0 \leq m^*(E) \leq \sum_{i=1}^{\infty} m^*(\{x_i\}) = 0. \quad (5.6)$$

■

**Proposition 5.4** (Exterior measure of a cube equals its volume)

If  $Q$  is a closed cube, then  $m^*(Q) = |Q|$ .

*Proof.* The single-cube cover  $\{Q\}$  is admissible, giving  $m^*(Q) \leq |Q|$  directly from the definition.

Now we shall prove that  $m^*(Q) \geq |Q|$ . Let  $\{Q_j\}_{j \geq 1}$  be any countable cover of  $Q$  by closed cubes; we must show  $\sum_{j=1}^{\infty} |Q_j| \geq |Q|$ .

Fix  $\varepsilon > 0$ , and for each  $j$  let  $\tilde{Q}_j^\circ$  be the open cube with the same center as  $Q_j$ , dilated by a factor  $(1 + \varepsilon)$ , i.e.

$$\tilde{Q}_j^\circ = (1 + \varepsilon) Q_j^\circ, \quad (5.7)$$

so that  $Q_j \subset \tilde{Q}_j^\circ$  and

$$|\tilde{Q}_j^\circ| = |\tilde{Q}_j| = (1 + \varepsilon)^d |Q_j|. \quad (5.8)$$

Then  $\{\tilde{Q}_j^\circ\}$  is a cover of  $Q$  by open sets, and since  $Q$  is compact, finitely many of them already cover  $Q$ , say  $Q \subseteq \bigcup_{j=1}^n \tilde{Q}_j^\circ$ . By [Proposition 4.2](#) (subadditivity of volume for coverings of a rectangle),

$$|Q| \leq \sum_{j=1}^n |\tilde{Q}_j^\circ| = (1 + \varepsilon)^d \sum_{j=1}^n |Q_j| \leq (1 + \varepsilon)^d \sum_{j=1}^{\infty} |Q_j|. \quad (5.9)$$

Since  $\varepsilon > 0$  is arbitrary, letting  $\varepsilon \rightarrow 0^+$  gives  $|Q| \leq \sum_{j=1}^{\infty} |Q_j|$ . As this holds for every admissible cover  $\{Q_j\}$ , taking the infimum over all such covers gives  $|Q| \leq m^*(Q)$ . ■

**Proposition 5.5**

If  $Q^\circ$  is an open cube, then  $m^*(Q^\circ) = |Q^\circ|$ .

*Proof.* Let  $Q$  be the closure of  $Q^\circ$ . Then  $Q^\circ \subseteq Q$  and monotonicity ([Proposition 5.1](#)) gives us

$$m^*(Q^\circ) \leq m^*(Q) = |Q| = |Q^\circ|. \quad (5.10)$$

Now note that  $(1 - \varepsilon)Q \subseteq Q^\circ$ , and hence, monotonicity will give us

$$m^*(Q^\circ) \geq m^*((1 - \varepsilon)Q) = |(1 - \varepsilon)Q| = (1 - \varepsilon)^d |Q| = (1 - \varepsilon)^d |Q^\circ|. \quad (5.11)$$

Letting  $\varepsilon \rightarrow 0^+$  forces  $m^*(Q^\circ) \geq |Q^\circ|$ . ■

The same argument (covering by cubes and using the volume-subadditivity [Proposition 4.2](#)) shows more generally that  $m^*(R) = |R|$  for any rectangle  $R$ .

**Proposition 5.6**

For any rectangle  $R$ ,  $m^*(R) = |R|$ .

*Proof.* Let  $\{Q_j\}_{j \geq 1}$  be any countable cover of  $R$  by closed cubes. Fix  $\varepsilon > 0$ , and for each  $j$  let

$$\tilde{Q}_j^\circ = (1 + \varepsilon) Q_j^\circ, \quad (5.12)$$

so that  $R \subseteq \bigcup_{j=1}^\infty Q_j \subseteq \bigcup_{j=1}^\infty \tilde{Q}_j^\circ$ . By the compactness of  $R$ , we have  $R \subseteq \bigcup_{j=1}^n \tilde{Q}_j^\circ$ . Using [Proposition 4.2](#), we get

$$|R| \leq \sum_{j=1}^n |\tilde{Q}_j^\circ| = (1 + \varepsilon)^d \sum_{j=1}^n |Q_j| \leq (1 + \varepsilon)^d \sum_{j=1}^\infty |Q_j|. \quad (5.13)$$

Since  $\varepsilon > 0$  is arbitrary, letting  $\varepsilon \rightarrow 0^+$  gives  $|R| \leq \sum_{j=1}^\infty |Q_j|$ . As this holds for every admissible cover  $\{Q_j\}$ , taking the infimum over all such covers gives  $|R| \leq m^*(R)$ .

Now we'll show that  $m^*(R) \leq |R|$ . Consider a grid in  $\mathbb{R}^d$  formed by cubes of size  $\frac{1}{k}$ .

$$\mathcal{Q} := \{Q : Q \text{ is a cube contained in } R\}, \quad (5.14)$$

$$\mathcal{Q}' := \{Q : Q \text{ is a cube that intersect both } R \text{ and } R^c\}. \quad (5.15)$$

Then we have

$$\bigcup_{Q \in \mathcal{Q}} Q \subseteq R \subseteq \bigcup_{Q \in \mathcal{Q} \cup \mathcal{Q}'} Q. \quad (5.16)$$

$R$  is made out of  $\bigcup_{Q \in \mathcal{Q}} Q$  (which itself is a collection of almost-disjoint rectangles) and some more almost-disjoint rectangles. Therefore,

$$|R| = \sum_{Q \in \mathcal{Q}} |Q| + \text{possibly some more} \geq \sum_{Q \in \mathcal{Q}} |Q|. \quad (5.17)$$

**Claim 5.1:** The size of  $\mathcal{Q}'$  is at most  $Ck^{d-1}$ , for some constant  $C$  (not depending on  $k$ ).

*Proof.* The cubes in  $\mathcal{Q}'$  intersect both  $R$  and  $R^c$ , i.e. they cover the  $(d-1)$ -dimensional faces of  $R$ . There are exactly  $2d$  of them.

Let  $a$  be the largest edge-length of  $R$ . To cover the edge of length  $a$ , we would need around  $\frac{a}{k-1} = ak$  cubes. But this would be the case if the cubes fit perfectly into the edge. There may be a little space leftover, so the number of cubes we need is no more than  $ak + 2$  (adding one for each endpoint of the edge). Therefore, to cover a  $(d-1)$ -dimensional face of  $R$ , the number of cubes needed is at most

$$(ak + 2)^{d-1} = (a + 1)^{d-1} k^{d-1}. \quad (5.18)$$

There are  $2d$  faces, so the number of cubes needed to cover the faces is at most

$$2d \cdot (a + 1)^{d-1} k^{d-1}. \quad (5.19)$$

□

Then we have  $R \subseteq \bigcup_{Q \in \mathcal{Q} \cup \mathcal{Q}'} Q$ , and each cube has volume  $\frac{1}{k^d}$ . Therefore,

$$\sum_{Q \in \mathcal{Q} \cup \mathcal{Q}'} |Q| = \sum_{Q \in \mathcal{Q}} |Q| + \sum_{Q \in \mathcal{Q}'} |Q| \leq |R| + Ck^{d-1} \cdot \frac{1}{k^d} = |R| + \frac{C}{k}. \quad (5.20)$$

Taking  $k \rightarrow \infty$ , we get  $m^*(R) \leq |R|$ . ■

### Proposition 5.7

For any  $E \subseteq \mathbb{R}^d$ ,

$$m^*(E) = \inf\{m^*(O) : E \subseteq O, O \text{ open}\}. \quad (5.21)$$

*Proof.* For any open  $O \supseteq E$ , monotonicity (Proposition 5.1) gives  $m^*(E) \leq m^*(O)$ ; taking the infimum over such  $O$  preserves the inequality:

$$m^*(E) \leq \inf_{\substack{E \subseteq O, \\ O \text{ open}}} m^*(O). \quad (5.22)$$

Now, for the other inequality: fix  $\varepsilon > 0$ . By definition of  $m^*(E)$ , there is a countable cover of  $E$  by closed cubes  $\{Q_j\}$  with

$$\sum_{j=1}^{\infty} |Q_j| \leq m^*(E) + \varepsilon. \quad (5.23)$$

For each  $j$ , let

$$\tilde{Q}_j^\circ = (1 + \varepsilon) Q_j^\circ, \quad \text{and} \quad O := \bigcup_j \tilde{Q}_j^\circ, \quad (5.24)$$

which is an open set containing  $E$  (since  $E \subseteq \bigcup_j Q_j \subseteq \bigcup_j \tilde{Q}_j^\circ$ ). By countable subadditivity (Proposition 5.2) and Proposition 5.4,

$$m^*(O) \leq \sum_{j=1}^{\infty} m^*(\tilde{Q}_j^\circ) = \sum_{j=1}^{\infty} |\tilde{Q}_j^\circ| \leq (1 + \varepsilon)^d \sum_{j=1}^{\infty} |Q_j| \leq (1 + \varepsilon)^d m^*(E) + \varepsilon. \quad (5.25)$$

Taking infimum on the LHS gives

$$\inf_{\substack{E \subseteq O, \\ O \text{ open}}} m^*(O) \leq (1 + \varepsilon)^d m^*(E) + \varepsilon, \quad (5.26)$$

for every  $\varepsilon > 0$ . Now taking  $\varepsilon \rightarrow 0^+$  gives us the desired bound. ■

**Definition 5.2.** The diameter of a set  $X \subseteq \mathbb{R}^d$  is

$$\text{diam}(X) = \sup\{|x - y| : x, y \in X\}. \quad (5.27)$$

The distance between two sets  $X, Y \subseteq \mathbb{R}^d$  is

$$d(X, Y) := \inf\{|x - y| : x \in X, y \in Y\}. \quad (5.28)$$

**Proposition 5.8** (Additivity for positively separated sets)

If  $d(E_1, E_2) > 0$ , then

$$m^*(E_1 \cup E_2) = m^*(E_1) + m^*(E_2). \quad (5.29)$$

*Proof.* Applying subadditivity (Proposition 5.2) to the two-set collection  $\{E_1, E_2\}$ , we get

$$m^*(E_1 \cup E_2) \leq m^*(E_1) + m^*(E_2). \quad (5.30)$$

For the other inequality, we may assume  $m^*(E_1 \cup E_2) < \infty$  (else there is nothing to prove). Fix  $\varepsilon > 0$  and let  $\{Q_j\}$  be a cover of  $E_1 \cup E_2$  by closed cubes with

$$\sum_{j=1}^{\infty} |Q_j| \leq m^*(E_1 \cup E_2) + \varepsilon. \quad (5.31)$$

By subdividing each  $Q_j$  into finitely many smaller closed cubes if necessary (which does not change the total volume, by Proposition 4.1), we may further assume every cube in the cover has diameter less than  $\delta$ , with

$$0 < \delta < d(E_1, E_2). \quad (5.32)$$

Then no single cube  $Q_j$  can intersect both  $E_1$  and  $E_2$ ; indeed if  $Q_i$  intersects both  $E_1$  and  $E_2$  at points  $e_1$  and  $e_2$ , respectively, then

$$|e_1 - e_2| \leq \text{diam}(Q_j) \leq \delta < d(E_1, E_2) \leq |e_1 - e_2|, \quad (5.33)$$

which is not possible. Now, we partition the cover into  $\{Q_j : Q_j \cap E_1 \neq \emptyset\}$ , which covers  $E_1$ , and  $\{Q_j : Q_j \cap E_2 \neq \emptyset\}$ , which covers  $E_2$  (every cube meets at most one of the two sets, so this is a genuine partition of the index set). Then

$$m^*(E_1) \leq \sum_{Q_j \cap E_1 \neq \emptyset} |Q_j| \quad \text{and} \quad m^*(E_2) \leq \sum_{Q_j \cap E_2 \neq \emptyset} |Q_j|. \quad (5.34)$$

Adding these, we get

$$m^*(E_1) + m^*(E_2) \leq \sum_{Q_j \cap E_1 \neq \emptyset} |Q_j| + \sum_{Q_j \cap E_2 \neq \emptyset} |Q_j| = \sum_{j=1}^{\infty} |Q_j| \leq m^*(E_1 \cup E_2) + \varepsilon. \quad (5.35)$$

Since  $\varepsilon > 0$  is arbitrary,  $m^*(E_1) + m^*(E_2) \leq m^*(E_1 \cup E_2)$ . ■

**Corollary 5.9**

If  $E_1, E_2, \dots, E_n \subseteq \mathbb{R}^d$  with  $d(E_i, E_j) > 0$  for every  $i \neq j$ , then

$$m^*\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n m^*(E_i). \quad (5.36)$$

**Proposition 5.10** (Exterior measure of an almost disjoint union of cubes)

If  $E = \bigcup_{j=1}^{\infty} Q_j$  is a countable, almost disjoint union of closed cubes, then

$$m^*(E) = \sum_{j=1}^{\infty} |Q_j|. \quad (5.37)$$

*Proof.*  $m^*(E) \leq \sum_{j=1}^{\infty} |Q_j|$  directly follows from the definition, since  $\bigcup_{j=1}^{\infty} Q_j$  is countable collection of closed cubes that cover  $E$ . For the other direction, fix  $N \geq 1$  and  $\varepsilon > 0$ , and let

$$\tilde{Q}_j = (1 - \varepsilon) Q_j. \quad (5.38)$$

Since  $Q_1, \dots, Q_N$  are almost disjoint (disjoint interiors) and there are only finitely many of them, shrinking creates a positive gap between each pair, i.e. for  $i \neq j$ ,

$$d(Q'_i, Q'_j) > 0. \quad (5.39)$$

Corollary 5.9 together with Proposition 5.4 gives

$$m^*\left(\bigcup_{j=1}^N \tilde{Q}_j\right) = \sum_{j=1}^N m^*(\tilde{Q}_j) = \sum_{j=1}^N |\tilde{Q}_j| = (1 - \varepsilon)^d \sum_{j=1}^N |Q_j|. \quad (5.40)$$

Since  $\bigcup_{j=1}^N Q'_j \subseteq E$ , monotonicity (Proposition 5.1) gives

$$m^*(E) \geq m^*\left(\bigcup_{j=1}^N \tilde{Q}_j\right) = (1 - \varepsilon)^d \sum_{j=1}^N |Q_j|. \quad (5.41)$$

Letting  $\varepsilon \rightarrow 0^+$  (with  $N$  fixed), so  $m^*(E) \geq \sum_{j=1}^N |Q_j|$ . Since this holds for every  $N$ , letting  $N \rightarrow \infty$  gives  $m^*(E) \geq \sum_{j=1}^{\infty} |Q_j|$ . ■

**Remark 5.5.** Proposition 5.10 combined with Proposition 4.4 (every open set is a countable almost disjoint union of dyadic cubes), gives a concrete way of computing  $m^*(O)$  for an open set  $O$ . That is, decompose  $O$  into almost disjoint dyadic cubes and sum their volumes.

We now single out the sets on which  $m^*$  will be shown (in the next lecture) to be countably additive.

**Definition 5.3** (Lebesgue measurable set). A set  $E \subseteq \mathbb{R}^d$  is **Lebesgue measurable** (or simply **measurable**, in this context) if, for every  $\varepsilon > 0$ , there exists an open set  $O \supseteq E$  such that

$$m^*(O - E) \leq \varepsilon. \quad (5.42)$$

**Remark 5.6.** Informally,  $E$  is measurable if it can be approximated from outside by open sets, in the sense that the “error”  $O - E$  can be made to have arbitrarily small exterior measure.

This definition makes no direct reference to a  $\sigma$ -algebra. We will show below that the collection of measurable sets does in fact form a  $\sigma$ -algebra.

**Remark 5.7.** [Proposition 5.7](#) says that for every  $\varepsilon > 0$ , we can find an open set  $O \supseteq E$  such that

$$m^*(O) \leq m^*(E) + \varepsilon. \quad (5.43)$$

Note that this is quite different from (5.42), since it's **NOT** always true that  $m^*(O - E) = m^*(O) - m^*(E)$ .

We now establish a few basic properties of Lebesgue measurable sets, culminating in the fact that they form a  $\sigma$ -algebra.

**Proposition 5.11** (Open sets are measurable)

Every open set  $O \subseteq \mathbb{R}^d$  is measurable.

*Proof.* Take the open set to be  $O$  itself. Then  $m^*(O - O) = m^*(\emptyset) = 0 < \varepsilon$  for every  $\varepsilon > 0$ . ■

**Proposition 5.12** (Sets of exterior measure zero are measurable)

If  $m^*(E) = 0$ , then  $E$  is measurable. (In particular,  $m^*(E) = 0$  forces  $E$  to be measurable with  $m^*$  restricting to a genuine value 0 on  $E$ ; such sets are called **null sets**.)

*Proof.* Fix  $\varepsilon > 0$ . By [Proposition 5.7](#) applied to  $E$ , since  $m^*(E) = 0$ , there is an open set  $O \supseteq E$  with  $m^*(O) \leq \varepsilon$ . Since  $O - E \subseteq O$ , monotonicity gives

$$m^*(O - E) \leq m^*(O) \leq \varepsilon. \quad (5.44)$$

■

**Proposition 5.13** (Countable unions of measurable sets are measurable)

If  $E_1, E_2, \dots$  are measurable, then  $E := \bigcup_{n=1}^{\infty} E_n$  is measurable.

*Proof.* Fix  $\varepsilon > 0$ . For each  $n$ , since  $E_n$  is measurable, choose an open set  $O_n \supseteq E_n$  with

$$m^*(O_n - E_n) \leq \frac{\varepsilon}{2^n}. \quad (5.45)$$

Let  $O := \bigcup_{n=1}^{\infty} O_n$ , which is an open set containing  $E$ . Since

$$O - E \subseteq \bigcup_{n=1}^{\infty} (O_n - E_n) \quad (5.46)$$

(a point in  $O$  not in  $E$  lies in some  $O_n$ , and since it is not in  $E \supseteq E_n$ , it is not in  $E_n$  either, so it lies in  $O_n - E_n$ ), countable subadditivity ([Proposition 5.2](#)) gives

$$m^*(O - E) \leq \sum_{n=1}^{\infty} m^*(O_n - E_n) \leq \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} = \varepsilon. \quad (5.47)$$

**Proposition 5.14** (Closed sets are measurable)

Every closed set  $F \subseteq \mathbb{R}^d$  is measurable.

*Proof. Case 1: Compact Sets.* Let  $K$  be compact (closed and bounded), we'll show that it's measurable. Fix  $\varepsilon > 0$ . By [Proposition 5.7](#), there is an open set  $O \supseteq K$  with

$$m^*(O) \leq m^*(K) + \varepsilon. \quad (5.48)$$

Since  $K$  is bounded,  $m^*(K) < \infty$ , so this is a genuine (finite) bound. The set  $O - K$  is open (as  $K$  is closed), so by [Proposition 4.4](#) it may be written as a countable almost disjoint union of closed cubes,

$$O - K = \bigcup_{n=1}^{\infty} Q_n. \quad (5.49)$$

For each fixed  $N$ , the finite union  $S_N := \bigcup_{n=1}^N Q_n$  is compact (a finite union of closed bounded cubes), disjoint from  $K$  (since each  $Q_n \subseteq O - K$ ), and two disjoint compact sets have positive distance; hence  $d(K, S_N) > 0$  (see the following claim).

**Claim 5.2:** If  $X, Y \subseteq \mathbb{R}^d$  are disjoint sets with  $X$  being compact and  $Y$  closed, then  $d(X, Y) > 0$ .

*Proof.* Assume for the sake of contradiction that there exists two sequences  $x_n \in X$  and  $y_n \in Y$  such that  $|x_n - y_n| \rightarrow 0$ .  $X$  is compact, so there exists a convergent subsequence  $x_{n_i} \rightarrow x \in X$ . Then

$$|x - y_{n_i}| \leq |x - x_{n_i}| + |x_{n_i} - y_{n_i}|. \quad (5.50)$$

Both terms of RHS converges to 0. Therefore  $|x - y_{n_i}| \rightarrow 0$ . As  $Y$  is closed, it contains all the limit points, so  $x \in Y$ . But we are given that  $X$  and  $Y$  are disjoint. Contradiction!  $\square$

By [Proposition 5.8](#) and [Proposition 5.10](#),

$$m^*\left(K \cup \bigcup_{n=1}^N Q_n\right) = m^*(K) + m^*\left(\bigcup_{n=1}^N Q_n\right) = m^*(K) + \sum_{n=1}^N |Q_n|. \quad (5.51)$$

Since  $K \cup \bigcup_{n=1}^N Q_n \subseteq O$ , monotonicity gives

$$m^*(K) + \sum_{n=1}^N |Q_n| \leq m^*(O) \leq m^*(K) + \varepsilon \implies \sum_{n=1}^N |Q_n| \leq \varepsilon, \quad (5.52)$$

for every  $N$ , and letting  $N \rightarrow \infty$ ,

$$\sum_{n=1}^{\infty} |Q_n| \leq \varepsilon. \quad (5.53)$$

By [Proposition 5.10](#) again,

$$m^*(O - K) = m^*\left(\bigcup_{n=1}^{\infty} Q_n\right) = \sum_{n=1}^{\infty} |Q_n| \leq \varepsilon. \quad (5.54)$$

Since  $\varepsilon > 0$  was arbitrary,  $K$  is measurable.

**Case 2: General closed sets.** Let  $F$  be closed (not necessarily bounded). Write

$$F = \bigcup_{n=1}^{\infty} \left( F \cap \overline{B(\mathbf{0}, n)} \right), \quad (5.55)$$

a countable union of compact sets (each  $F \cap \overline{B(\mathbf{0}, n)}$  is closed and bounded). By Case 1, each is measurable, and by [Proposition 5.13](#), their countable union  $F$  is measurable. ■

**Proposition 5.15** (Complements of measurable sets are measurable)

If  $E \subseteq \mathbb{R}^d$  is measurable, then  $E^c = \mathbb{R}^d - E$  is measurable.

*Proof.* For each  $n \geq 1$ , since  $E$  is measurable, choose an open set  $O_n \supseteq E$  with

$$m^*(O_n - E) \leq \frac{1}{n}. \quad (5.56)$$

Each  $O_n^c$  is closed, hence measurable by [Proposition 5.14](#). Moreover,  $O_n^c \subseteq E^c$ , since  $E \subseteq O_n$ . Let

$$S := \bigcup_{n=1}^{\infty} O_n^c, \quad (5.57)$$

a countable union of measurable sets, hence measurable by [Proposition 5.13](#). Since each  $O_n^c \subseteq E^c$ , also  $S \subseteq E^c$ . We claim  $m^*(E^c - S) = 0$ . Indeed,

$$E^c - S = E^c \cap S^c = E^c \cap \bigcap_{n=1}^{\infty} O_n = \bigcap_{n=1}^{\infty} (E^c \cap O_n) \subseteq E^c \cap O_n = O_n - E, \quad (5.58)$$

for every  $n$ . so

$$m^*(E^c - S) \leq m^*(O_n - E) \leq \frac{1}{n}, \quad (5.59)$$

for every  $n$ , forcing  $m^*(E^c - S) = 0$ . By [Proposition 5.12](#),  $E^c - S$  is measurable. Finally,

$$E^c = S \cup (E^c - S) \quad (5.60)$$

is a union of two measurable sets, hence measurable by [Proposition 5.13](#). ■

**Theorem 5.16** (The Lebesgue  $\sigma$ -algebra)

The collection  $\mathcal{L}$  of Lebesgue measurable subsets of  $\mathbb{R}^d$  is a  $\sigma$ -algebra.

*Proof.* [Proposition 5.13](#) gives closure of  $\mathcal{L}$  under countable unions, and [Proposition 5.15](#) gives closure under complements. Furthermore,  $\emptyset \in \mathcal{L}$  as it's a null-set. These are exactly the defining properties of a  $\sigma$ -algebra ([Definition 1.1](#)). ■

**Remark 5.8.**  $\mathcal{L}$  contains every open set and hence contains the Borel  $\sigma$ -algebra  $\mathcal{B}(\mathbb{R}^d)$ . So in general

$$\mathcal{B}(\mathbb{R}^d) \subseteq \mathcal{L} \subsetneq \mathcal{P}(\mathbb{R}^d), \quad (5.61)$$

with the last containment strict by Vitali's theorem ([Theorem 3.11](#)).

**Definition 5.4** (Lebesgue measure). The **Lebesgue measure** of a set  $E \in \mathcal{L}$  is defined as

$$m(E) := m^*(E), \quad (5.62)$$

i.e.  $m$  is the restriction of the exterior measure  $m^*$  to the  $\sigma$ -algebra  $\mathcal{L}$  of measurable sets.

It remains to verify that  $m$  is in fact countably additive on  $\mathcal{L}$ , i.e. that  $(\mathbb{R}^d, \mathcal{L}, m)$  is genuinely a measure space in the sense of [Definition 1.3](#).

## §6 18 September 2026

The first order of business is to show that the set function  $m$  just defined is in fact a genuine *measure*: that it is countably additive on disjoint measurable sets.

### Lemma 6.1

If  $E$  is measurable, then for every  $\varepsilon > 0$ , there exists a closed set  $F \subseteq E$  such that  $m^*(E - F) \leq \varepsilon$ .

*Proof.*  $E^c$  is measurable, so there is an open set  $O \supseteq E^c$  with  $m(O - E^c) \leq \varepsilon$ . Let  $F = O^c$ ; then  $F$  is closed, and  $F = O^c \subseteq (E^c)^c = E$ , while

$$E - F = E \cap O = O - E^c, \quad (6.1)$$

so  $m(E - F) = m(O - E^c) \leq \varepsilon$ . ■

### Theorem 6.2 (Countable additivity)

If  $E_1, E_2, E_3, \dots$  is a countable collection of pairwise disjoint measurable sets, and  $E = \bigcup_{j=1}^{\infty} E_j$ , then  $E$  is measurable and

$$m(E) = \sum_{j=1}^{\infty} m(E_j). \quad (6.2)$$

*Proof.* Measurability of  $E = \bigcup_j E_j$  is immediate, since  $\mathcal{L}$  is already known to be a  $\sigma$ -algebra (closed under countable unions). Also, since the exterior measure is already known to be countably *subadditive*,

$$m(E) \leq \sum_{j=1}^{\infty} m(E_j). \quad (6.3)$$

So we'll now prove the other inequality. First, assume each  $E_j$  is bounded. Fix  $\varepsilon > 0$ . Since each  $E_j$  is measurable, there is a closed set  $F_j \subseteq E_j$  with

$$m^*(E_j - F_j) \leq \frac{\varepsilon}{2^j}. \quad (6.4)$$

Since  $E_j - F_j$  and  $F_j$  are disjoint with union  $E_j$ , using subadditivity,

$$m^*(E_j) \leq m^*(F_j) + m^*(E_j - F_j) \leq m^*(F_j) + \frac{\varepsilon}{2^j} \implies m(F_j) \geq m(E_j) - \frac{\varepsilon}{2^j}. \quad (6.5)$$

Now fix  $N$  and consider  $F_1, \dots, F_N$ . Since  $E_1, \dots, E_N$  are pairwise disjoint, so are  $F_1, \dots, F_N$ . Moreover, each  $F_j$  is closed and bounded (as  $F_j \subseteq E_j$  and  $E_j$  is bounded), hence compact.

Applying [Corollary 5.9](#) and [Claim 5.2](#),

$$m^*\left(\bigcup_{j=1}^N F_j\right) = \sum_{j=1}^N m(F_j). \quad (6.6)$$

Since  $\bigcup_{j=1}^N F_j \subseteq E$ , monotonicity of outer measure gives

$$m(E) \geq m^*\left(\bigcup_{j=1}^N F_j\right) = \sum_{j=1}^N m(F_j) \geq \sum_{j=1}^N \left(m(E_j) - \frac{\varepsilon}{2^j}\right) \geq \sum_{j=1}^N m(E_j) - \varepsilon. \quad (6.7)$$

Letting  $N \rightarrow \infty$  and then  $\varepsilon \rightarrow 0$  (which was arbitrary) yields

$$m(E) \geq \sum_{j=1}^{\infty} m(E_j), \quad (6.8)$$

as desired. Now we move to the general scenario of when  $E_j$ 's need not be bounded. Write  $\mathbb{R}^d$  as an increasing union of closed balls: let  $B_k$  be the closed ball of radius  $k$  centered at the origin, and set

$$S_1 = B_1, \quad S_k = B_k - B_{k-1} \quad (k \geq 2), \quad (6.9)$$

so that the  $S_k$  are pairwise disjoint and  $\mathbb{R}^d = \bigcup_{k=1}^{\infty} S_k$ . For each  $j$ , decompose  $E_j$  according to this partition:

$$E_{j,k} = E_j \cap S_k, \quad E_j = \bigcup_{k=1}^{\infty} E_{j,k}, \quad (6.10)$$

a disjoint union of *bounded* measurable sets. By applying the bounded case to the sets  $\{E_{j,k}\}_k$  (disjoint and bounded, for fixed  $j$ ),

$$m(E_j) = \sum_{k=1}^{\infty} m(E_{j,k}). \quad (6.11)$$

Now  $E = \bigcup_j E_j = \bigcup_{j,k} E_{j,k}$ , and all the sets  $E_{j,k}$  (over all  $j$  and  $k$ ) are pairwise disjoint and bounded (each being contained in  $S_k$ ), so the bounded case applies again to this doubly-indexed countable disjoint bounded family:

$$m(E) = \sum_{j,k} m(E_{j,k}) = \sum_{j=1}^{\infty} \left( \sum_{k=1}^{\infty} m(E_{j,k}) \right) = \sum_{j=1}^{\infty} m(E_j). \quad (6.12)$$

■

**Remark 6.1.** The key point used in this argument was that each  $E_j$  could be approximated from inside by a *compact* set  $F_j$ . Disjointness of the  $E_j$  then translates into a positive distance between the compact pieces  $F_j$ , which is what makes the finite additivity of outer measure on compact sets available.

**Corollary 6.3**

If  $F \subseteq E$  are both measurable with  $m(F) < \infty$ , then

$$m(E - F) = m(E) - m(F). \quad (6.13)$$

*Proof.*  $E = F \cup (E - F)$ , and they are disjoint. So by additivity,

$$m(E) = m(F) + m(E - F). \quad (6.14)$$

Since  $m(F) < \infty$ , we can subtract it from both sides and the desired equality follows.  $\blacksquare$

**Proposition 6.4**

Let  $E \subseteq \mathbb{R}^d$  be Lebesgue measurable. Then the following hold.

- (i) If  $m(E) < \infty$ , then for every  $\varepsilon > 0$  there exists a compact set  $K \subseteq E$  with

$$m(E - K) \leq \varepsilon. \quad (6.15)$$

- (ii) If  $m(E) < \infty$ , then for every  $\varepsilon > 0$  there exists a finite union  $F = \bigcup_{i=1}^N Q_i$  of almost disjoint closed cubes such that

$$m(E \triangle F) \leq \varepsilon, \quad (6.16)$$

where  $E \triangle F = (E - F) \cup (F - E)$  denotes the symmetric difference.

*Proof.* (i) Write  $E$  as the disjoint union of its intersections with the shells  $S_k$  from earlier:

$$E = \bigsqcup_{k=1}^{\infty} E_k, \quad E_k = E \cap S_k. \quad (6.17)$$

By countable additivity,  $m(E) = \sum_{k=1}^{\infty} m(E_k)$ . Since  $m(E) < \infty$  this series converges, so there is  $N$  such that

$$\sum_{k=1}^N m(E_k) \geq m(E) - \frac{\varepsilon}{2}. \quad (6.18)$$

Since  $\bigcup_{k=1}^N S_k = B_n$  (the closed ball of radius  $N$ ),

$$m(E \cap B_n) = \sum_{k=1}^n m(E_k) \geq m(E) - \frac{\varepsilon}{2}. \quad (6.19)$$

Now, by applying [Lemma 6.1](#) to  $E \cap B_n$ , there is a closed set  $K \subseteq E \cap B_n$  with

$$m((E \cap B_n) - K) \leq \frac{\varepsilon}{2}. \quad (6.20)$$

Since  $K \subseteq B_n$ ,  $K$  is closed and bounded, hence compact, and  $K \subseteq E$ . Finally

$$E - K = (E - (E \cap B_n)) \cup ((E \cap B_n) - K), \quad (6.21)$$

a disjoint union, so

$$m(E - K) = m(E) - m(E \cap B_n) + m((E \cap B_n) - K) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad (6.22)$$

- (ii) Fix an open set  $O \supseteq E$  with  $m(O - E) < \varepsilon/2$ . In particular,  $m(O) < \infty$ . Every open subset of  $\mathbb{R}^d$  can be written as a countable union of almost disjoint closed cubes,

$$O = \bigcup_{j=1}^{\infty} Q_j, \quad \{Q_j\} \text{ almost disjoint}, \quad (6.23)$$

and by countable additivity of almost-disjoint cubes ([Proposition 5.10](#)),

$$m(O) = \sum_{j=1}^{\infty} |Q_j| < \infty. \quad (6.24)$$

Choose  $N$  so that

$$\sum_{j=1}^N |Q_j| \geq m(O) - \frac{\varepsilon}{2}, \quad (6.25)$$

and set  $F = \bigcup_{j=1}^N Q_j$ , a finite union of almost disjoint closed cubes with  $F \subseteq O$  and

$$m(O - F) = m(O) - m(F) \leq \frac{\varepsilon}{2}. \quad (6.26)$$

Now estimate the symmetric difference  $E \triangle F = (E - F) \cup (F - E)$ : Since  $E \subseteq O$ , we have  $E - F \subseteq O - F$ , so

$$m(E - F) \leq m(O - F) \leq \frac{\varepsilon}{2}. \quad (6.27)$$

Since  $F \subseteq O$ , we have  $F - E \subseteq O - E$ , so

$$m(F - E) \leq m(O - E) \leq \frac{\varepsilon}{2}. \quad (6.28)$$

Adding,  $m(E \triangle F) \leq m(E - F) + m(F - E) \leq \varepsilon$ . ■

**Remark 6.2.** These statements express the same intuitive fact in different guises. A measurable set can be approximated from outside by an open set, from inside by a closed set, from inside by a compact set (when it has finite measure), and by a finite union of almost-disjoint cubes, up to a set of arbitrarily small measure. One chooses whichever form of approximation is most convenient for the problem at hand.

One can show that  $(\mathbb{R}^d, \mathcal{L}, m)$  is a complete measure space.

**Theorem 6.5**

$m$  is a complete measure.

*Proof.* Suppose  $E \in \mathcal{L}$  with  $m(E) = 0$ . Then for any subset  $A \subseteq E$ , by the monotonicity of exterior measure,  $m^*(A) \leq m^*(E) = 0$ , implying that  $A$  is also a null-set. Since null-sets are measurable,  $A \in \mathcal{L}$ . So  $m$  is a complete measure. ■

The Lebesgue measurable sets arise as the completion of the  $\sigma$ -algebra of Borel sets, that is by adjoining all subsets of Borel sets of measure zero.

**Definition 6.1.** A set is called  $G_\delta$  set if it is the intersection of a countable collection of open sets. A set is called  $F_\sigma$  set if it is the union of a countable collection of closed sets.

**Proposition 6.6**

Let  $E \subseteq \mathbb{R}^d$ . The following are equivalent:

- (a)  $E$  is measurable.
- (b) There is a  $G_\delta$ -set  $G \supseteq E$  such that  $m^*(G - E) = 0$ .
- (c) There is a  $F_\sigma$ -set  $F \subseteq E$  such that  $m^*(E - F) = 0$ .

*Proof.* **(b) $\Rightarrow$ (a)**  $G$  is measurable, since it's the intersection of countably many measurable sets. So is  $G - E$ , since it's a null-set. Therefore,  $E = G - (G - E)$  is a measurable set.

**(c) $\Rightarrow$ (a)**  $F$  is measurable, since it's the union of countably many measurable sets. So is  $E - F$ , since it's a null-set. Therefore,  $E = F \cup (E - F)$  is a measurable set.

**(a) $\Rightarrow$ (b)** Since  $E$  is measurable, for every  $n$ , there exists an open set  $O_n \supseteq E$  such that

$$m^*(O_n - E) \leq \frac{1}{n}. \quad (6.29)$$

Choose  $G = \bigcap_{n=1}^{\infty} O_n \supseteq E$ . Then,

$$G - E = \bigcap_{n=1}^{\infty} O_n - E \subseteq O_n - E \quad (6.30)$$

$$\implies m^*(G - E) \leq m^*(O_n - E) \leq \frac{1}{n}, \quad (6.31)$$

for every  $n$ . Therefore,  $m^*(G - E) = 0$ .

**(a) $\Rightarrow$ (c)** Since  $E$  is measurable, for every  $n$ , there exists an open set  $F_n \subseteq E$  such that

$$m^*(E - F_n) \leq \frac{1}{n}. \quad (6.32)$$

Choose  $F = \bigcup_{n=1}^{\infty} F_n \subseteq E$ . Then,

$$E - F = E - \bigcap_{n=1}^{\infty} F_n \subseteq E - F_n \quad (6.33)$$

$$\implies m^*(E - F) \leq m^*(E - F_n) \leq \frac{1}{n}, \quad (6.34)$$

for every  $n$ . Therefore,  $m^*(E - F) = 0$ . ■

### Proposition 6.7

$E \subseteq \mathbb{R}^d$  is measurable if and only if for every  $\varepsilon > 0$ , there exists  $F \subseteq E \subseteq U$  with  $U$  open and  $F$  closed such that  $m^*(U - F) \leq \varepsilon$ .

*Proof.* Suppose  $E$  is measurable. Then there exists  $F \subseteq E \subseteq U$  with  $U$  open and  $F$  closed such that

$$m^*(U - E) \leq \frac{\varepsilon}{2} \text{ and } m^*(E - F) \leq \frac{\varepsilon}{2}. \quad (6.35)$$

Then  $U - F = (U - E) \cup (E - F)$  is a disjoint union, and hence

$$m^*(U - F) \leq m^*(U - E) + m^*(E - F) \leq \varepsilon. \quad (6.36)$$

Conversely, suppose such  $F \subseteq E \subseteq U$  exists for every  $\varepsilon$ . In particular, for every  $n$ , we can choose  $F_n \subseteq E \subseteq U_n$  with  $F_n$  closed and  $U_n$  open, and

$$m^*(U_n - F_n) \leq \frac{1}{n}. \quad (6.37)$$

Choose  $F = \bigcup_{n=1}^{\infty} F_n$  is a  $F_\sigma$  set. Then we have  $F \subseteq E$ . Furthermore,

$$E - F = E - \bigcup_{n=1}^{\infty} F_n \subseteq E - F_n \subseteq U_n - F_n. \quad (6.38)$$

By monotonicity, we have

$$m^*(E - F) \leq m^*(U_n - F_n) \leq \frac{1}{n}, \quad (6.39)$$

which is true for every  $n$ . Therefore,  $m^*(E - F) = 0$ . This gives  $E$  is measurable by [Proposition 6.6](#). ■

Now we'll discuss some invariance properties of Lebesgue measure.

### Theorem 6.8 (Translation invariance)

Let  $E \subseteq \mathbb{R}^d$  be measurable and  $x_0 \in \mathbb{R}^d$ . Then  $E + x_0 := \{x + x_0 : x \in E\}$  is measurable and

$$m(E + x_0) = m(E). \quad (6.40)$$

*Proof.* We divide the proof into several steps.

**Claim 6.1:** Outer measure is translation invariant.

*Proof.* Every covering of  $E$  by cubes  $\{Q_j\}$  gives, upon translating each cube by  $x_0$ , a covering of  $E + x_0$  by cubes  $\{Q_j + x_0\}$  of the *same* volumes (translating a cube does not change its side lengths). Hence

$$m^*(E + x_0) = m^*(E) \quad (6.41)$$

for every set  $E$ , since the coverings of  $E$  and of  $E + x_0$  by cubes are in volume-preserving bijection with each other.  $\square$

**Claim 6.2:** If  $E$  is measurable, so is  $E + x_0$ .

*Proof.* If  $E$  is measurable, for  $\varepsilon > 0$  choose an open  $O \supseteq E$  with  $m^*(O - E) \leq \varepsilon$ . Then  $O + x_0$  is open, contains  $E + x_0$ , and

$$(O + x_0) - (E + x_0) = (O - E) + x_0, \quad (6.42)$$

so by [Claim 6.1](#),

$$m^*((O + x_0) - (E + x_0)) = m^*(O - E) \leq \varepsilon. \quad (6.43)$$

$\square$

Hence  $E + x_0$  is measurable, and its measure equals  $m^*(E + x_0) = m^*(E) = m(E)$ .  $\blacksquare$

**Theorem 6.9** (Scaling/dilation invariance)

Let  $E \subseteq \mathbb{R}^d$  be measurable and  $\lambda > 0$ . Then  $\lambda E := \{\lambda x : x \in E\}$  is measurable and

$$m(\lambda E) = \lambda^d m(E). \quad (6.44)$$

*Proof.* As in the translation case, we divide the proof into several steps.

**Claim 6.3:**  $m^*(\lambda E) = \lambda^d m^*(E)$  for every  $E \subseteq \mathbb{R}^d$ .

*Proof.* If  $\{Q_j\}$  is a covering of  $E$  by cubes, then  $\{\lambda Q_j\}$  covers  $\lambda E$ , and a cube of side length  $\ell$  scales to a cube of side length  $\lambda\ell$ , so its volume scales by  $\lambda^d$ . Hence

$$m^*(\lambda E) = \lambda^d m^*(E), \quad (6.45)$$

for every set  $E$ , since the coverings of  $E$  and of  $\lambda E$  by cubes are in volume-scaling-by- $\lambda^d$  bijection with each other.  $\square$

**Claim 6.4:** If  $E$  is measurable, so is  $\lambda E$ .

*Proof.* If  $E$  is measurable, choosing an open  $O \supseteq E$  with  $m^*(O - E) < \frac{\varepsilon}{\lambda^d}$ . The set  $\lambda O$  is open, contains  $\lambda E$ , and  $(\lambda O) - (\lambda E) = \lambda(O - E)$ , so

$$m^*(\lambda O - \lambda E) = \lambda^d m^*(O - E) \leq \varepsilon, \quad (6.46)$$

as desired.  $\square$

Hence,  $\lambda E$  is measurable with  $m(\lambda E) = \lambda^d m(E)$ .  $\blacksquare$

One can similarly prove a rotation invariance. However one can prove a more general fact.

### Theorem 6.10

Let  $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$  be a linear map. If  $E \subseteq \mathbb{R}^d$  is measurable, so is  $T(E)$ . Furthermore,

$$m(T(E)) = |\det T| m(E). \quad (6.47)$$

We shall prove it in several steps.

### Proposition 6.11

Let  $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$  be a linear map with  $\det T = 0$ . Then  $m^*(T(\mathbb{R}^d)) = 0$ .

*Proof.* Let  $H = T(\mathbb{R}^d)$ . We can write

$$H = \bigcup_{R=1}^{\infty} (H \cap [-R, R]^d) =: \bigcup_{R=1}^{\infty} H_R. \quad (6.48)$$

By countable additivity,

$$m^*(H) \leq \sum_{R=1}^{\infty} m^*(H \cap [-R, R]^d). \quad (6.49)$$

So we will prove that  $m^*(H \cap [-R, R]^d) = 0$  for every  $R$ . Fix  $N$ , and let's divide each side of  $[-R, R]^d$  into  $N$  equal parts, i.e. divide the cube into  $N^d$  small cubes, each of side length  $\frac{2R}{N}$ . Let

$$\mathcal{Q} = \left\{ Q : Q \text{ is a small cube that intersects } H \cap [-R, R]^d \right\}. \quad (6.50)$$

Then  $\mathcal{Q}$  covers  $H \cap [-R, R]^d$ , so

$$m^*(H \cap [-R, R]^d) \leq \sum_{Q \in \mathcal{Q}} |Q| = |\mathcal{Q}| \left( \frac{2R}{N} \right)^d, \quad (6.51)$$

where  $|\mathcal{Q}|$  denotes the size of the set  $\mathcal{Q}$ . <sup>1</sup>

<sup>1</sup>**Abuse of notation:** For a cube  $Q$ ,  $|Q|$  is its volume, and for the set  $\mathcal{Q}$ ,  $|\mathcal{Q}|$  is its cardinality.

**Claim 6.5:**  $|\mathcal{Q}| \leq C \cdot N^{d-1}$  for some constant  $C$  (not depending on  $N$ ).

*Proof.* Since  $H$  is a strictly lower-dimensional hyperplane through the origin, there exists some nonzero vector  $\mathbf{v} \in \mathbb{R}^d$  such that  $\mathbf{x} \cdot \mathbf{v} = 0$  for every  $\mathbf{x} \in H$ . WLOG we can assume  $v_d \neq 0$  (well, there is a coordinate  $v_{i_0} \neq 0$ , then  $d$  can be replaced with  $i_0$  the following argument). Consider a “column”

$$C = I_1 \times I_2 \times \cdots \times I_{d-1} \times [-R, R], \quad (6.52)$$

where  $I_i$  are intervals of length  $\frac{2R}{N}$  in each direction. There are  $N^{d-1}$  such columns. Therefore,

$$|\mathcal{Q}| \leq N^{d-1} \times (\text{maximum number of cubes in a column that intersect } H_R). \quad (6.53)$$

If  $\mathbf{y}, \mathbf{z} \in H_R$  that belong to the same column, then

$$\mathbf{y} \cdot \mathbf{v} = 0 \implies y_d = - \sum_{i=1}^{d-1} \frac{v_i}{v_d} y_i, \quad (6.54)$$

$$\mathbf{z} \cdot \mathbf{v} = 0 \implies z_d = - \sum_{i=1}^{d-1} \frac{v_i}{v_d} z_i, \quad (6.55)$$

$$\implies |y_d - z_d| \leq \left| \sum_{i=1}^{d-1} \frac{v_i}{v_d} (y_i - z_i) \right| \leq \sum_{i=1}^{d-1} \left| \frac{v_i}{v_d} \right| \frac{2R}{N}, \quad (6.56)$$

since  $\mathbf{y}$  and  $\mathbf{z}$ 's  $i$ -th coordinate differs by at most  $\frac{2R}{N}$ . Hence, if  $\mathbf{y}, \mathbf{z} \in H_R$  that belong to the same column, then their  $d$ -th coordinate differs by at most  $M \cdot \frac{2R}{N}$ , for some constant  $M$  that doesn't depend on  $N$  (but  $M$  may depend on  $T$ ).

Therefore, we can argue that there are **at most**  $M + 2$  cubes in a column that intersect  $H_R$ . Otherwise, we can find two cubes in a column, both of them intersect  $H_R$ , and there are at least  $M + 1$  cubes in between them. This way, the distance between the  $d$ -th coordinate of those two points (at which the said cubes intersect  $H_R$ ) would be at least  $(M + 1) \frac{2R}{N}$ .

Hence,  $|\mathcal{Q}| \leq (M + 2) N^{d-1}$ . □

Now, using (6.51),

$$m^* \left( H \cap [-R, R]^d \right) \leq |\mathcal{Q}| \left( \frac{2R}{N} \right)^d = C \cdot N^{d-1} \frac{(2R)^d}{N^d} = \frac{C'}{N}. \quad (6.57)$$

This is true for every  $N$ . By taking  $N \rightarrow \infty$ , we get  $m^*(H_R) = 0$ . By using subadditivity (6.49), we get that  $m^*(H) = 0$ . ■

This proves the  $\det T = 0$  case of Theorem 6.10. Now, we focus on the  $\det T \neq 0$  case. A standard linear algebra fact says that any non-singular linear map can be written as a composition of

“elementary” linear maps of the following types:

- (a)  $T_1(x_1, \dots, x_i, \dots, x_d) = (x_1, \dots, cx_i, \dots, x_d)$  for  $c \neq 0$ ;
- (b)  $T_2(x_1, \dots, x_i, \dots, x_j, \dots, x_d) = (x_1, \dots, x_j, \dots, x_i, \dots, x_d)$ .
- (c)  $T_3(x_1, x_2, \dots, x_d) = (x_1 + x_2, x_2, \dots, x_d)$ .

Also, note that

$$\det T_1 = c, \quad \det T_2 = -1, \quad \det T_3 = 1. \quad (6.58)$$

### Lemma 6.12

Let  $Q = [0, 1]^d$  be the unit cube. Then

- (a)  $m^*(T_1(Q)) = |c|$ ;
- (b)  $m^*(T_2(Q)) = 1$ ;
- (c)  $m^*(T_3(Q)) \leq 1$ .

In other words,  $m^*(T_i(Q)) \leq |\det T_i|$  for each  $i = 1, 2, 3$ .

*Proof.* For  $T_1$ , the image  $T_1(Q)$  is the rectangle

$$[0, 1] \times \cdots \times [0, 1] \times [0, c] \times [0, 1] \times \cdots \times [0, 1], \quad (6.59)$$

or  $[c, 0]$  instead of  $[0, c]$ . We know that the exterior measure of a rectangle is its volume (Proposition 5.6), hence,

$$m^*(T_1(Q)) = |c| = |\det T_1|. \quad (6.60)$$

For  $T_2$ , the image  $T_2(Q) = Q$ , so

$$m^*(T_2(Q)) = m^*(Q) = 1 = |\det T_2|. \quad (6.61)$$

For  $T_3$ , note that

$$T_3(Q) = \{(x_1, \dots, x_d) : 0 \leq x_i \leq 1 \text{ for } i \geq 2; \text{ and } x_2 \leq x_1 \leq x_2 + 1\}. \quad (6.62)$$

Note that  $\tilde{Q} := T_3(Q)$  is contained in the rectangle  $[0, 2] \times [0, 1]^{d-1}$ . We cover this rectangle using cubes of side length  $\frac{1}{N}$ . Let

$$\mathcal{Q} = \{Q' : Q' \text{ is a small cube that intersects } \tilde{Q}\}. \quad (6.63)$$

Then  $\mathcal{Q}$  covers  $\tilde{Q}$ , so

$$m^*(\tilde{Q}) \leq \sum_{Q' \in \mathcal{Q}} |Q'| = |\mathcal{Q}| \cdot \frac{1}{N^d}. \quad (6.64)$$

Again,  $|\mathcal{Q}|$  denotes the size of the set  $\mathcal{Q}$ .

**Claim 6.6:**  $|\mathcal{Q}| = (1 + N) N^{d-1}$ .

*Proof.* Fix a “row”

$$R = [0, 2] \times I_2 \times I_3 \times \cdots \times I_d, \quad (6.65)$$

where each  $I_i$  is an interval of length  $\frac{1}{N}$ . There are  $N^{d-1}$  such rows. Now let's see how many small cubes of this row intersect  $\tilde{Q}$ .

Suppose  $I_2 = [\frac{k}{N}, \frac{k+1}{N}]$  for some  $k$ . Now, take  $\mathbf{x} \in \tilde{Q} \cap R$ . Then

$$x_2 \leq x_1 \leq x_2 + 1 \implies x_1 \in \left[ \frac{k}{N}, \frac{k + (N + 1)}{N} \right]. \quad (6.66)$$

Therefore, a small cube  $Q' = I_1 \times I_2 \times \cdots \times I_d$  intersects  $\tilde{Q}$  if and only if  $I_1 \subseteq [\frac{k}{N}, \frac{k + (N + 1)}{N}]$ . There are  $N + 1$  choices for  $I_1$  then. As a result, there are  $(N + 1)$  cubes in  $R$  that intersect  $\tilde{Q}$ . Since this is the same for all rows, we have  $|\mathcal{Q}| = (1 + N) N^{d-1}$ .  $\square$

Now, (6.64) gives us

$$m^*(T_3(Q)) \leq \frac{(1 + N) N^{d-1}}{N^d} = 1 + \frac{1}{N}. \quad (6.67)$$

Since this is true for every  $N$ , by taking  $N \rightarrow \infty$ , we get  $m^*(T_3(Q)) \leq 1$ .  $\blacksquare$

So we have shown that under elementary linear maps  $T$ , we have

$$m^*(T(Q)) \leq |\det T| m^*(Q), \quad (6.68)$$

where  $Q = [0, 1]^d$  is the unit cube.

### Lemma 6.13

For **every** cube  $Q'$  and elementary linear map  $T$ ,

$$m^*(T(Q')) \leq |\det T| m^*(Q'). \quad (6.69)$$

*Proof.* If  $Q'$  is another cube, then it's just a dilation and translation of  $Q$ , i.e.  $Q' = \delta Q + s$ . Then

$$T(Q') = T(\delta Q + s) = \delta T(Q) + T(s). \quad (6.70)$$

By the dilation and translation invariance of outer measure,

$$m^*(T(Q')) = m^*(\delta T(Q) + T(s)) = \delta^d m^*(T(Q)) \quad (6.71)$$

$$\leq |\det T| \delta^d m^*(Q) = |\det T| m^*(\delta Q + s) \quad (6.72)$$

$$= |\det T| m^*(Q'). \quad (6.73)$$

$\blacksquare$

**Lemma 6.14**

For **every** open set  $O \subseteq \mathbb{R}^d$  and elementary linear map  $T$ ,

$$m^*(T(O)) \leq |\det T| m^*(O). \quad (6.74)$$

*Proof.* Write  $O$  as a almost-disjoint countable union of cubes:  $O = \bigcup_{i=1}^{\infty} Q_i$ . Then

$$m^*(T(O)) = m^*\left(T\left(\bigcup_{i=1}^{\infty} Q_i\right)\right) = m^*\left(\bigcup_{i=1}^{\infty} T(Q_i)\right) \quad (6.75)$$

$$\leq \sum_{i=1}^{\infty} m^*(T(Q_i)) \quad (6.76)$$

$$\leq |\det T| \sum_{i=1}^{\infty} m^*(Q_i) = |\det T| m^*(O), \quad (6.77)$$

where the last equality follows from [Proposition 5.10](#). ■

**Lemma 6.15**

Let  $G \subseteq \text{GL}(d, \mathbb{R})$  be the set

$$G = \left\{ T \in \text{GL}(d, \mathbb{R}) : m^*(T(O)) \leq |\det T| m^*(O) \text{ for every open } O \subseteq \mathbb{R}^d \right\}. \quad (6.78)$$

Then  $G$  is closed under composition.

*Proof.* A linear map is continuous, and hence, an invertible linear map is a homeomorphism  $\mathbb{R}^d \rightarrow \mathbb{R}^d$ . Therefore, if  $O$  is open and  $S, T \in G$ , then both  $T(O)$  and  $ST(O)$  are open.

Now, since  $T(O)$  is open and  $S \in G$ , we have

$$m^*(ST(O)) \leq |\det S| m^*(T(O)). \quad (6.79)$$

Since  $T \in G$  and  $O$  is open, we have

$$m^*(T(O)) \leq |\det T| m^*(O). \quad (6.80)$$

Combining these, we have

$$m^*(ST(O)) \leq |\det S| |\det T| m^*(O) = |\det(ST)| m^*(O), \quad (6.81)$$

implying that  $ST \in G$ . ■

**Proposition 6.16**

$G = \text{GL}(d, \mathbb{R})$ .

*Proof.* Every invertible linear map  $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$  can be written as a composition of the elementary maps. By [Lemma 6.14](#), elementary maps are in  $G$ . Since  $G$  is closed under composition, we have  $T \in G$  for every  $T$ . ■

In other words, for **every**  $T \in \text{GL}(d, \mathbb{R})$ , we have

$$m^*(T(O)) \leq |\det T| m^*(O). \quad (6.82)$$

Actually, we can prove the equality in (6.82).

### Proposition 6.17

For every open set  $O \subseteq \mathbb{R}^d$  and  $T \in \text{GL}(d, \mathbb{R})$ , we have

$$m^*(T(O)) = |\det T| m^*(O). \quad (6.83)$$

*Proof.* We already have  $m^*(T(O)) \leq |\det T| m^*(O)$ . Now note that

$$m^*(O) = m^*(T^{-1}(T(O))). \quad (6.84)$$

Since  $T(O)$  is also open, and  $T^{-1} \in \text{GL}(d, \mathbb{R})$ , using (6.82),

$$m^*(O) = m^*(T^{-1}(T(O))) \leq |\det(T^{-1})| m^*(T(O)) = \frac{1}{|\det T|} m^*(T(O)). \quad (6.85)$$

Rearranging, we get

$$m^*(T(O)) \geq |\det T| m^*(O). \quad (6.86)$$

■

### Proposition 6.18

For every  $E \subseteq \mathbb{R}^d$ ,

$$m^*(T(E)) = |\det T| m^*(E). \quad (6.87)$$

*Proof.* We know from [Proposition 5.7](#) that

$$m^*(E) = \inf_{O \supseteq E, O \text{ open}} m^*(O). \quad (6.88)$$

Since  $T$  is a homeomorphism,  $O \supseteq T(E)$  if and only if  $O = T(U)$  for some open  $U \supseteq E$ .

$$m^*(T(E)) = \inf_{\substack{O \supseteq T(E), \\ O \text{ open}}} m^*(O) = \inf_{\substack{U \supseteq E, \\ U \text{ open}}} m^*(T(U)) \quad (6.89)$$

$$= |\det T| \inf_{\substack{U \supseteq E, \\ U \text{ open}}} m^*(U) = |\det T| m^*(E). \quad (6.90)$$

■

Now we are pretty much ready to finish the proof of [Theorem 6.10](#).

*Proof of Theorem 6.10.* Suppose  $E$  is measurable, and  $\varepsilon > 0$ . Then we can find an open set  $O \supseteq E$  such that

$$m^*(O - E) \leq \frac{\varepsilon}{\det T}. \quad (6.91)$$

Since  $T$  is a homeomorphism,  $T(O)$  is an open set containing  $T(E)$ , and  $T(O) - T(E) = T(O - E)$ . As a result,

$$m^*(T(O) - T(E)) = m^*(T(O - E)) = |\det T| m^*(O - E) \leq \varepsilon. \quad (6.92)$$

So  $T(E)$  is measurable. Then  $m(T(E)) = |\det T| m(E)$  follows from [Proposition 6.18](#). ■

**Remark 6.3.** For a linear map  $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ , measurability of  $E$  implies the measurability of  $T(E)$  because we are working in the same dimension. If  $T : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$  for  $d' < d$ , then measurability of  $E$  does not always imply the measurability of  $T(E)$ . For instance, suppose  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^1$  be the projection onto the first component.

Consider  $E = A \times \{0\}$ , where  $A$  is the non-measurable set constructed in [Theorem 3.11](#). Then  $E$  has outer measure 0 as it can be covered using the 0-measure rectangle  $[0, 1] \times [0, 0]$ . So  $E \subseteq \mathbb{R}^2$  is measurable. But  $T(E) = A \subset \mathbb{R}$  is **NOT** measurable.

**Exercise 6.1.** Find a sequence of real-valued non-negative functions  $\{f_k\}_{k \geq 1}$  on  $[0, 1]$  so that

$$\limsup_{k \rightarrow \infty} f_k(x) = \infty \quad \forall x \in [0, 1] \quad \text{and} \quad \int_0^1 f_k(x) dx \rightarrow 0. \quad (6.93)$$

*Solution.* Define

$$\begin{aligned} f_1 &= \chi_{[0,1]} \\ f_2 &= 2\chi_{[0, \frac{1}{2}]}, \quad f_3 = 2\chi_{[\frac{1}{2}, 1]} \\ f_4 &= 3\chi_{[0, \frac{1}{4}]}, \quad f_5 = 3\chi_{[\frac{1}{4}, \frac{1}{2}]}, \quad f_6 = 3\chi_{[\frac{1}{2}, \frac{3}{4}]}, \quad f_7 = 3\chi_{[\frac{3}{4}, 1]} \\ &\dots \end{aligned}$$

In other words, for  $n = 2^k + m$ , with  $k$  being the unique number satisfying  $2^k \leq n < 2^{k+1}$ , and  $0 \leq m < 2^k$ , we define

$$f_n = (k+1) \chi_{[\frac{m}{2^k}, \frac{m+1}{2^k}]}. \quad (6.94)$$

Then

$$\int_0^1 f_n(x) dx = \frac{k+1}{2^k} \rightarrow 0. \quad (6.95)$$

Now, given any  $x \in [0, 1]$ , and for every  $M > 0$ , there are non-negative integers  $k, m$  such that

$$k > M \quad \text{and} \quad m \leq 2^k x \leq m+1. \quad (6.96)$$

Now,  $f_{2^k+m}(x) = k + 1 > M$ . Therefore,  $\limsup_{k \rightarrow \infty} f_k(x) = \infty$ . ■

**Exercise 6.2.** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a Lebesgue measurable function. Assume that there exists a positive constant  $c$  with the following property: If  $A \subseteq \mathbb{R}$  is bounded and measurable, and  $f$  is bounded on  $A$ , then

$$\left| \int_A f(x) dx \right| \leq c. \quad (6.97)$$

Prove or disprove:  $f$  is Lebesgue integrable.

*Solution.* Yes,  $f$  is Lebesgue integrable. We shall prove it by showing that  $\int_{\mathbb{R}} f^+ dm \leq c$  and  $\int_{\mathbb{R}} f^- dm \leq c$ . Given  $n, M \geq 1$ , we define

$$A_{n,M} = \{x \in [-n, n] : 0 \leq f(x) \leq M\}. \quad (6.98)$$

Then  $A$  is bounded, measurable (since it's the intersection of  $[-n, n]$  and  $\{0 \leq f \leq M\}$ ), and  $f$  is bounded on  $A$ . Therefore,

$$\left| \int_{A_{n,M}} f(x) dx \right| \leq c \implies \int_{A_{n,M}} f^+(x) dx \leq c. \quad (6.99)$$

By [Corollary 2.12](#), since  $\{A_{n,M}\}$  is an increasing sequence of measurable sets, and

$$A_M = \bigcup_{n=1}^{\infty} A_{n,M} = \{x \in \mathbb{R} : 0 \leq f(x) \leq M\}. \quad (6.100)$$

Therefore,

$$\int_{A_M} f^+(x) dx = \lim_{n \rightarrow \infty} \int_{A_{n,M}} f^+(x) dx \leq c. \quad (6.101)$$

Now we again apply [Corollary 2.12](#) on the increasing sequence  $\{A_M\}_M$ .

$$\bigcup_{M=1}^{\infty} A_M = \{x \in \mathbb{R} : f(x) \geq 0\}. \quad (6.102)$$

Now,

$$\int_{\mathbb{R}} f^+(x) dx = \int_{\bigcup_{M=1}^{\infty} A_M} f^+(x) dx = \lim_{M \rightarrow \infty} \int_{A_M} f^+(x) dx \leq c. \quad (6.103)$$

Similarly, doing the same procedure on the function  $-f$ , we will get  $\int_{\mathbb{R}} f^-(x) dx \leq c$ . Therefore,  $f$  is Lebesgue-integrable. ■

**Exercise 6.3.** Let  $A \subset [0, 1]$  be a closed set. For each positive integer  $N$  let  $b(N)$  be the number of integers  $n$  such that  $0 \leq n \leq N - 1$  and

$$\left[ \frac{n}{N}, \frac{n+1}{N} \right] \cap A \neq \emptyset. \quad (6.104)$$

Show that

$$\lim_{N \rightarrow \infty} \frac{b(N)}{N} = m(A), \quad (6.105)$$

where  $m$  denotes the Lebesgue measure.

*Solution.* Let  $B_N$  be the union of the intervals  $\left[\frac{n}{N}, \frac{n+1}{N}\right]$  that intersect  $A$ .

$$B_N = \bigcup_{\left[\frac{n}{N}, \frac{n+1}{N}\right] \cap A \neq \emptyset} \left[\frac{n}{N}, \frac{n+1}{N}\right] \quad (6.106)$$

Since  $A \subset [0, 1]$ ,  $A \subseteq B_N$ . Therefore,

$$m(A) \leq m(B_N) \leq \sum_{\left[\frac{n}{N}, \frac{n+1}{N}\right] \cap A \neq \emptyset} m\left(\left[\frac{n}{N}, \frac{n+1}{N}\right]\right) = \frac{b(N)}{N}. \quad (6.107)$$

Therefore,

$$m(A) \leq \liminf_{n \rightarrow \infty} \frac{b(N)}{N}. \quad (6.108)$$

Now, let  $c(N) = N - b(N)$  be the number of intervals  $\left[\frac{n}{N}, \frac{n+1}{N}\right]$  that don't intersect  $A$ , i.e. intervals that are contained in  $A^c = [0, 1] - A$ . Given  $\varepsilon > 0$ , we can choose a compact set  $K \subseteq A^c$  such that

$$m(A^c) - m(K) = m(A^c - K) \leq \varepsilon. \quad (6.109)$$

Since  $K$  is compact and  $A$  is closed, and they are disjoint, by [Claim 5.2](#),  $d(A, K) = \delta > 0$ . Choose  $N$  such that  $\delta N > 1$ . Now, let

$$C_N = \bigcup_{\left[\frac{n}{N}, \frac{n+1}{N}\right] \subseteq A^c} \left[\frac{n}{N}, \frac{n+1}{N}\right]. \quad (6.110)$$

**Claim 6.7:**  $K \subseteq C_N$ .

*Proof.* Take  $x \in K$ . Then  $x$  is in some interval of the form  $\left[\frac{n}{N}, \frac{n+1}{N}\right]$ . If this interval intersects  $A$  at  $a$ , then

$$|a - x| \leq \frac{1}{N} < \delta = d(A, K), \quad (6.111)$$

which is a contradiction. Therefore,  $\left[\frac{n}{N}, \frac{n+1}{N}\right] \subseteq A^c$ . So this interval is in  $C_N$ . Hence, every  $x \in K$  is contained in  $C_N$ .  $\square$

Therefore,

$$m(K) \leq m(C_N) \leq \sum_{\left[\frac{n}{N}, \frac{n+1}{N}\right] \subseteq A^c} m\left(\left[\frac{n}{N}, \frac{n+1}{N}\right]\right) = \frac{c(N)}{N}. \quad (6.112)$$

Combining this with  $m(K) \geq m(A^c) - \varepsilon$ , we get

$$m(A^c) - \varepsilon \leq \frac{c(N)}{N} \implies 1 - m(A) - \varepsilon \leq 1 - \frac{b(N)}{N} \implies \frac{b(N)}{N} \leq m(A) + \varepsilon, \quad (6.113)$$

for every  $\varepsilon$  and for some sufficiently large  $N$ . Therefore, taking  $\varepsilon \rightarrow 0^+$ , we have

$$\limsup_{n \rightarrow \infty} \frac{b(N)}{N} \leq m(A). \quad (6.114)$$

Combining (6.108) and (6.114), we can conclude that  $\lim_{n \rightarrow \infty} \frac{b(N)}{N}$  exists, and it equals  $m(A)$ . ■

**Exercise 6.4.** For  $n = 1, 2, \dots$ , let  $f_n : \mathbb{R} \rightarrow \mathbb{R}$  be a Lebesgue measurable function. Assume that the set  $A_n = \{x \in \mathbb{R} : |f_n(x)| > n^{-2}\}$  has Lebesgue measure  $m(A_n) \leq 2^{-n}$ . Show that the series  $\sum_{n=1}^{\infty} f_n(x)$  is absolutely convergent for almost every  $x \in \mathbb{R}$ .

*Solution.* We have that

$$\sum_{n=1}^{\infty} m(A_n) \leq \sum_{n=1}^{\infty} 2^{-n} = 1 < \infty. \quad (6.115)$$

So we can apply [Borel–Cantelli lemma](#). Let  $E = \{x : x \in A_n \text{ for infinitely many } n\}$ . By [Borel–Cantelli lemma](#),  $m(E) = 0$ . We'll prove that  $\sum_{n=1}^{\infty} |f_n(x)|$  converges for  $x \notin E$ .

For  $x \notin E$ , there exists some  $N$  such that  $x \notin A_n$  for  $n > N$ . Therefore, for  $n > N$ ,  $|f_n(x)| \leq n^{-2}$ .

$$\sum_{n=1}^{\infty} |f_n(x)| = \sum_{n=1}^N |f_n(x)| + \sum_{n=N+1}^{\infty} |f_n(x)| \leq \sum_{n=1}^N |f_n(x)| + \sum_{n=N+1}^{\infty} \frac{1}{n^2}. \quad (6.116)$$

The first is a finite sum, and the second sum converges. Therefore,  $\sum_{n=1}^{\infty} |f_n(x)|$  converges. ■

**Exercise 6.5.** Let  $F = (f_1, f_2, \dots, f_d) : \mathbb{R}^d \rightarrow \mathbb{R}^d$  be a polynomial map, i.e., each  $f_i \in \mathbb{R}[x_1, x_2, \dots, x_d]$  for  $i = 1, 2, \dots, d$ . Show that if  $E \subset \mathbb{R}^d$  has Lebesgue measure zero, then  $F(E)$  also has Lebesgue measure zero.

(Hint: You may first try to prove that a Lipschitz map  $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$  takes sets of measure zero into sets of measure zero.)

*Solution.* Suppose  $G : U \subset \mathbb{R}^d \rightarrow \mathbb{R}^d$  is Lipschitz, i.e. there is a constant  $C > 0$  such that

$$\|G(\mathbf{x}) - G(\mathbf{y})\| \leq C \cdot \|\mathbf{x} - \mathbf{y}\|, \quad (6.117)$$

for every  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ . Let  $E \subseteq \mathbb{R}^d$  be a measure-0 set. Then given any  $\varepsilon > 0$ , there exists a countable collection of cubes  $\{Q_i\}_{i \geq 1}$  such that  $E \subseteq \bigcup_{i=1}^{\infty} Q_i$  and  $\sum_{i=1}^{\infty} |Q_i| \leq \frac{\varepsilon}{C^d d^{d/2}}$ .

Now let's examine  $G(Q_i)$ . Suppose  $Q_i$  has side length  $\ell_i$ . If we take any two points  $G(\mathbf{x}), G(\mathbf{y}) \in G(Q_i)$ ,

$$\left| (G(\mathbf{x}) - G(\mathbf{y}))_j \right| \leq \|G(\mathbf{x}) - G(\mathbf{y})\| \leq C \cdot \|\mathbf{x} - \mathbf{y}\| \leq C\sqrt{d}\ell_i. \quad (6.118)$$

Therefore,  $G(Q_i)$  is contained in a cube  $\tilde{Q}_i$  of side length  $C\sqrt{d}\ell_i$ . As a result,

$$G(E) \subseteq G\left(\bigcup_{i=1}^{\infty} Q_i\right) = \bigcup_{i=1}^{\infty} G(Q_i) \subseteq \bigcup_{i=1}^{\infty} \tilde{Q}_i. \quad (6.119)$$

Now,

$$\sum_{i=1}^{\infty} |\tilde{Q}_i| = \sum_{i=1}^{\infty} (C\sqrt{d}\ell_i)^d = C^d d^{d/2} \sum_{i=1}^{\infty} \ell_i^d = C^d d^{d/2} \sum_{i=1}^{\infty} |Q_i| \leq \varepsilon. \quad (6.120)$$

Therefore,  $G(E)$  has measure 0, i.e. Lipschitz function takes measure-0 sets to measure-0 sets.

Now we move back to our problem. Let  $E$  be a set of measure 0. We write

$$E = \bigcup_{n=1}^{\infty} (E \cap B_n) =: \bigcup_{n=1}^{\infty} E_n, \quad \text{where } B_n = \overline{B(\mathbf{0}, n)}. \quad (6.121)$$

Since  $E$  has measure 0, by monotonicity,  $m(E_n) = 0$ . Now,  $F$  is a polynomial map, so  $\|DF\|$  is continuous. Hence, it attains a maximum value on the compact set  $B_n$ , say  $K_n$ . As a result, on  $B_n$ , by Mean-Value inequality,

$$\|F(\mathbf{x}) - F(\mathbf{y})\| \leq K_n \|\mathbf{x} - \mathbf{y}\|. \quad (6.122)$$

Therefore,  $F$  is Lipschitz on  $B_n$ , and hence  $F(E_n)$  has measure 0. Now,

$$m(F(E)) = m\left(\bigcup_{n=1}^{\infty} F(E_n)\right) \leq \sum_{n=1}^{\infty} m(F(E_n)) = 0. \quad (6.123)$$

■

**Exercise 6.6.** Let  $E_k$  be a sequence of Lebesgue measurable sets in  $\mathbb{R}^d$  such that  $\sum_{k=1}^{\infty} m(E_k) < \infty$ . Let  $F_j$  be the subset of  $\mathbb{R}^d$  consisting of the points that belong to precisely  $j$  of the subsets  $\{E_k\}_{k=1}^{\infty}$ . Show that

$$\sum_{k=1}^{\infty} m(E_k) = \sum_{j=1}^{\infty} j m(F_j). \quad (6.124)$$

*Solution.* We shall use the fact that  $m(A) = \int_{\mathbb{R}^d} \chi_A dx$ . Then

$$\sum_{k=1}^{\infty} m(E_k) = \sum_{k=1}^{\infty} \int_{\mathbb{R}^d} \chi_{E_k} dx = \int_{\mathbb{R}^d} \left( \sum_{k=1}^{\infty} \chi_{E_k} \right) dx, \quad (6.125)$$

where the swapping of sum and integral is justified by [Proposition 2.8](#). Since  $\sum_{k=1}^{\infty} m(E_k) < \infty$ , by [Borel–Cantelli lemma](#), almost every element of  $\mathbb{R}^d$  belongs to only finitely many  $E_k$ ’s. In other words,  $\sum_{k=1}^{\infty} \chi_{E_k}$  is finite almost everywhere.

Now, if  $x \in F_j$ , then  $x$  belongs to  $j$ -many elements in  $\{E_k\}_{k \geq 1}$ . Therefore,

$$\sum_{k=1}^{\infty} \chi_{E_k}(x) = j, \quad \text{for } x \in F_j. \quad (6.126)$$

$\bigcup_{j=1}^{\infty} F_j$  is a disjoint union, which covers almost everywhere. Therefore,

$$\sum_{k=1}^{\infty} \chi_{E_k} = \sum_{j=1}^{\infty} j \chi_{F_j}, \quad \text{a.e.} \quad (6.127)$$

Therefore, their integrals are the same, i.e.

$$\begin{aligned} \sum_{k=1}^{\infty} m(E_k) &= \int_{\mathbb{R}^d} \left( \sum_{k=1}^{\infty} \chi_{E_k} \right) dx = \int_{\mathbb{R}^d} \left( \sum_{j=1}^{\infty} j \chi_{F_j} \right) dx \\ &= \sum_{j=1}^{\infty} j \int_{\mathbb{R}^d} \chi_{F_j} dx = \sum_{j=1}^{\infty} j m(F_j). \end{aligned} \quad (6.128)$$

Again, the swapping of sum and integral is facilitated by [Proposition 2.8](#). ■

**Exercise 6.7.** Construct a (non-measurable) function  $f : \mathbb{R} \rightarrow \mathbb{R}$  with the following property: for any  $g : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$|g(x) - f(x)| < 1 \quad \text{for all } x \in \mathbb{R}, \quad (6.129)$$

$g$  is not measurable.

*Solution.* Let  $V$  be a non-measurable set. We take

$$f = 2\chi_V. \quad (6.130)$$

Let  $g : \mathbb{R} \rightarrow \mathbb{R}$  such that  $|g(x) - f(x)| < 1$  for all  $x \in \mathbb{R}$ . Now, for  $x \in V$

$$|g(x) - 2| < 1 \implies g(x) > 1. \quad (6.131)$$

For  $x \notin V$ ,

$$|g(x) - 0| < 1 \implies g(x) < 1. \quad (6.132)$$

Now,  $g^{-1}(1, \infty) = V$  is not measurable. Therefore,  $g$  can't be measurable. ■

**Exercise 6.8.** We can construct generalized Cantor sets as follows. If  $I$  is a bounded interval and  $\alpha \in (0, 1)$ , let us call the open interval with the same midpoint of  $I$  and length equal to  $\alpha$  times the length of  $I$  the “open middle  $\alpha$ th” of  $I$ . If  $\{\alpha_j\}_{j \geq 1}$  is any sequence of numbers in  $(0, 1)$ , we can define a decreasing sequence  $\{K_j\}_{j \geq 0}$  of closed sets as follows. Let  $K_0 = [0, 1]$ , and, for  $j \geq 1$ ,  $K_j$  is obtained from  $K_{j-1}$  by removing the open middle  $\alpha_j$ -th of each of the intervals that make up  $K_{j-1}$ . The resulting limiting set

$$K = \bigcap_{j=0}^{\infty} K_j$$

is our generalized Cantor set (sometimes also called “fat” Cantor set). The usual Cantor set arises when taking  $\alpha_j = \frac{1}{3}$  for all  $j$ . Show that

- (a)  $K$  is compact, has empty interior, and is totally disconnected (i.e. the only connected subsets of  $K$  are single points). Moreover,  $K$  has no isolated points.
- (b)  $m(K) = \prod_{j=1}^{\infty} (1 - \alpha_j)$ .
- (c) Show that  $m(K) > 0$  if and only if  $\sum_{j=1}^{\infty} \alpha_j < \infty$ .
- (d) Given  $\beta \in (0, 1)$  exhibit a sequence  $\{\alpha_j\}_{j \geq 1}$  such that  $m(K) = \beta$ .

*Solution.* Note that,  $K_n$  consists of  $2^n$  intervals. Because at each step we are removing the middle  $\alpha_j$ -th of each interval, so the number of intervals gets doubled at each step.

Also, we claim that the intervals of  $K_n$  has length  $\ell_n = \prod_{j=1}^n \frac{1-\alpha_j}{2}$ . We shall prove this by induction on  $n$ . It clearly holds for  $n = 0$ . Suppose it holds for some  $n$ . Then at the next step, each interval's middle  $\alpha_{n+1}$ -th part gets removed, and we are left with the  $1 - \alpha_{n+1}$ , but that part gets divided into two parts. Therefore,

$$\ell_{n+1} = \frac{1 - \alpha_{n+1}}{2} \ell_n = \prod_{j=1}^{n+1} \frac{1 - \alpha_j}{2} \quad (6.133)$$

- (a)  $K$  is bounded, and also closed (as it's the intersection of closed sets). Hence, it's compact.

Suppose  $\text{Int } K \neq \emptyset$ . Then it contains an interval, say of size  $\varepsilon$ . Now,  $\ell_n \leq 2^{-n} \rightarrow 0$ , so we can find  $N$  such that  $\ell_N < \varepsilon$ . Since all the intervals in  $K_N$  has size  $\ell_N < \varepsilon$ , that means an interval of size  $\varepsilon$  can't be contained in  $K_N$ . Therefore, it can't be contained in  $K$  as it's the intersection of all  $K_n$ 's.

Take two distinct points  $x < y$ . Suppose  $\varepsilon = |x - y|$ . As before, choose  $N$  such that  $\ell_N < \varepsilon$ . Now,  $x, y \in K \subseteq K_N$ . Since  $\ell_N < \varepsilon$ ,  $x$  and  $y$  are in two different intervals in  $K_N$ . That means there was an interval  $(a, b)$  removed at one of the earlier stages. Then

$$K = \left[ K \cap \left( -\infty, \frac{a+b}{2} \right) \right] \cup \left[ K \cap \left( \frac{a+b}{2}, \infty \right) \right], \quad (6.134)$$

proving that  $x, y$  are separated. Therefore,  $K$  is totally separated, and in particular, totally disconnected.

Assume for the sake of contradiction that  $K$  has an isolated point  $x$ . Then there exists  $\varepsilon > 0$  with the property that  $B(x, \varepsilon)$  contain no point of  $K - \{x\}$ . As before, choose  $N$  such that  $\ell_N < \varepsilon$ . Since  $x \in K \subseteq K_N$ , so  $x$  is in one of the components of  $K_N$ , say  $[a, b]$ .

The length of the interval  $[a, b]$  is  $\ell_N$ , that means the distance from  $x$  to any of the points in this interval is at most  $\ell_N$ , which is strictly smaller than  $\varepsilon$ . Therefore,  $[a, b] \subseteq B(x, \varepsilon)$ .

If  $x \neq a$ ,  $a \in B(x, \varepsilon) \cap (K - \{x\})$ , since the endpoints are never removed. If  $x = a$ ,  $b \in B(x, \varepsilon) \cap (K - \{x\})$ . Both cases, contradict with the assumption that  $B(x, \varepsilon)$  contain no point of  $K - \{x\}$ .

- (b)  $K_n$  is the union of closed intervals, so it is measurable. Therefore,  $K$  is also measurable as it is the intersection of countably many measurable sets. Now,  $K_n$  has  $2^n$  disjoint intervals, each with length  $\ell_n = \prod_{j=1}^n \frac{1-\alpha_j}{2}$ . Therefore,

$$m(K_n) = 2^n \ell_n = \prod_{j=1}^n (1 - \alpha_j). \quad (6.135)$$

Now,  $K_n$  is a decreasing sequence of measurable sets, and  $K$  is their intersection, with  $m(K_0) = 1 < \infty$ . Therefore,

$$m(K) = \lim_{n \rightarrow \infty} m(K_n) = \lim_{n \rightarrow \infty} \prod_{j=1}^n (1 - \alpha_j) = \prod_{j=1}^{\infty} (1 - \alpha_j). \quad (6.136)$$

- (c) We are gonna use the bound  $1 - x \leq e^{-x}$  for  $x \in (0, 1)$  and  $1 - x \geq e^{-2x}$  for  $x \in (0, \frac{1}{2}]$ .

If  $\sum_{j=1}^{\infty} \alpha_j$  diverges, then we have

$$\prod_{j=1}^n (1 - \alpha_j) \leq \prod_{j=1}^n \exp(-\alpha_n) = \exp\left(-\sum_{j=1}^n \alpha_n\right). \quad (6.137)$$

As  $n \rightarrow \infty$ ,  $\sum_{j=1}^n \alpha_n \rightarrow \infty$ , so in that limit we have

$$m(K) = \lim_{n \rightarrow \infty} \prod_{j=1}^n (1 - \alpha_j) \leq \exp\left(-\lim_{n \rightarrow \infty} \sum_{j=1}^n \alpha_n\right) = 0. \quad (6.138)$$

Conversely, suppose  $\sum_{j=1}^{\infty} \alpha_j$  converges. Then there exists some  $N$  such that  $\alpha_n \leq \frac{1}{2}$  for  $n \geq N$ . Now,

$$\prod_{j=N}^M (1 - \alpha_j) \geq \prod_{j=N}^M \exp(-2\alpha_n) = \exp\left(-2 \sum_{j=N}^M \alpha_n\right). \quad (6.139)$$

As  $M \rightarrow \infty$ , the sum  $\sum_{j \geq N} \alpha_n$  converges to a finite value, say  $s$ . Hence,

$$\prod_{j=N}^{\infty} (1 - \alpha_j) \geq \exp\left(-2 \sum_{j=N}^{\infty} \alpha_n\right) = e^{-2s} > 0. \quad (6.140)$$

Therefore,  $\prod_{j=1}^{\infty} (1 - \alpha_j) > 0$ .

- (d) Given  $0 < \beta < 1$ , we need to have

$$\beta = \prod_{j=1}^{\infty} (1 - \alpha_j) \implies \log \beta = \sum_{j=1}^{\infty} \log(1 - \alpha_j). \quad (6.141)$$

Therefore, we need to have that  $\log(1 - \alpha_j)$  sums to  $\log \beta$ . Indeed,

$$\log \beta = \log \beta \sum_{j=1}^{\infty} 2^{-j} = \sum_{j=1}^{\infty} [2^{-j} \log \beta], \quad (6.142)$$

so we may try  $\log(1 - \alpha_j) = 2^{-j} \log \beta$ . Solving for  $\alpha_j$ , we have

$$\alpha_j = 1 - \beta^{2^{-j}}. \quad (6.143)$$

This is clearly within  $(0, 1)$ , and it satisfies  $m(K) = \beta$ . ■

**Exercise 6.9.** A measurable function is called *essentially bounded* if there exists a finite number  $\beta \geq 0$  such that the set  $\{x : |f(x)| > \beta\}$  has measure zero. Prove or disprove the following proposition:

“Let  $f : [0, 1] \rightarrow \mathbb{R}$  be Lebesgue integrable. Then there exists  $0 < a < b < 1$  such that  $f$  is essentially bounded in the interval  $(a, b)$ .”

*Solution.* Let  $q_1, q_2, q_3, \dots$  be an enumeration of  $\mathbb{Q} \cap [0, 1]$ . We define

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q} \cap [0, 1], \\ \sum_{k=1}^{\infty} \frac{1}{2^k \sqrt{|x - q_k|}} & \text{otherwise.} \end{cases} \quad (6.144)$$

$f$  is a non-negative function. So to show integrability, it suffices to show that  $\int_0^1 f(x) dx < \infty$ . Since the terms are all non-negative, by [Proposition 2.8](#),

$$\int_0^1 f(x) dx = \int_0^1 \sum_{k=1}^{\infty} \frac{1}{2^k \sqrt{|x - q_k|}} dx = \sum_{k=1}^{\infty} \frac{1}{2^k} \int_0^1 \frac{1}{\sqrt{|x - q_k|}} dx. \quad (6.145)$$

Let's now compute the integral:

$$\int_0^1 \frac{dx}{\sqrt{|x - q_k|}} = \int_0^{q_k} \frac{dx}{\sqrt{q_k - x}} + \int_{q_k}^1 \frac{dx}{\sqrt{x - q_k}} = 2\sqrt{q_k} + 2\sqrt{1 - q_k} \leq 4. \quad (6.146)$$

Therefore,

$$\int_0^1 f(x) dx \leq \sum_{k=1}^{\infty} \frac{4}{2^k} = 4 < \infty. \quad (6.147)$$

Therefore,  $f$  is integrable. Now, assume for the sake of contradiction that  $f$  is essentially bounded on  $(a, b)$ . Let's take a rational  $q_m \in (0, 1)$ . Then for irrational  $x$ ,

$$f(x) = \sum_{k=1}^{\infty} \frac{1}{2^k \sqrt{|x - q_k|}} \geq \frac{1}{2^m \sqrt{|x - q_m|}} \quad (6.148)$$

Suppose  $f$  is essentially bounded by  $\beta$  on  $(a, b)$ . Now, for irrational  $x \in \left(q_m - \frac{1}{4^m \beta^2}, q_m + \frac{1}{4^m \beta^2}\right)$ ,

$$|x - q_m| < \frac{1}{4^m \beta^2} \implies \frac{1}{2^m \sqrt{|x - q_m|}} > \beta. \quad (6.149)$$

Therefore,  $\{x : |f(x)| > \beta\}$  contains an interval (technically, all irrationals of an interval), so it has positive measure. Therefore,  $f$  can't be essentially bounded. ■

**Exercise 6.10.** Let  $E \subset [0, 1]$  be an open set, and let  $\chi_E$  be its characteristic function. Prove or disprove:  $\chi_E$  is Riemann integrable.

*Solution.* We shall use the following result:

A bounded function  $f : [0, 1] \rightarrow \mathbb{R}$  is Riemann-integrable if and only if the set of its discontinuities has measure 0.

Let  $q_n$  be an enumeration of the rational numbers in  $(0, 1)$ . For every  $q_n$ , we can choose open intervals  $(a_n, b_n) \subseteq (0, 1)$  containing  $q_n$  such that

$$\sum_{n=1}^{\infty} (b_n - a_n) < 1. \quad (6.150)$$

This is because  $\mathbb{Q} \cap (0, 1)$  has measure 0, so we can cover it with intervals of arbitrarily small total length. Then we choose  $E = \bigcup_{n=1}^{\infty} (a_n, b_n)$ . Then  $E$  is an open set, with boundary

$$\partial E = \overline{E} - E = [0, 1] - E. \quad (6.151)$$

Then  $[0, 1] = E \cup \partial E$ , and

$$1 = m([0, 1]) = m(E) + m(\partial E) \implies m(\partial E) = 1 - m(E) \geq 1 - \sum_{n=1}^{\infty} (b_n - a_n) > 0. \quad (6.152)$$

Therefore, the points of discontinuities of  $\chi_E$ , i.e. the boundary of  $E$ , has positive measure. Hence,  $\chi_E$  is **NOT** Riemann integrable. ■

## §7 Extra material: General outer measure

We show other important ideas to construct general measures starting from an outer measure. In particular, the discussion here leads us to an alternative way of constructing the Lebesgue measure.

**Definition 7.1** (Outer measure). An **outer measure**  $\mu^*$  on a set  $\Omega$  is a function  $\mu^* : \mathcal{P}(\Omega) \rightarrow [0, \infty]$  satisfying the following properties:

- (i)  $\mu^*(\emptyset) = 0$ ;
- (ii)  $\mu^*(A) \leq \mu^*(B)$  if  $A \subseteq B$ ;
- (iii)  $\mu^*$  is countably sub-additive, i.e.

$$\mu^* \left( \bigcup_{n=1}^{\infty} A_n \right) \leq \sum_{n=1}^{\infty} \mu^*(A_n). \quad (7.1)$$

We now describe a particularly useful way to generate outer measures. You may start with a collection of “elementary” sets  $\mathcal{E} \subseteq \mathcal{P}(\Omega)$  that you “know” how to measure (e.g. balls or rectangles in  $\mathbb{R}^d$ ) and then approximate generic sets from the outside by countable unions of such elementary sets.

**Proposition 7.1** (Generating outer measures)

Let  $\mathcal{E} \subseteq \mathcal{P}(\Omega)$  and  $\rho : \mathcal{E} \rightarrow [0, \infty]$  be such that  $\emptyset \in \mathcal{E}$ ,  $\Omega \in \mathcal{E}$  and  $\rho(\emptyset) = 0$ . For any  $A \subseteq \Omega$  define

$$\mu^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \rho(E_n) : E_n \in \mathcal{E} \text{ and } A \subseteq \bigcup_{n=1}^{\infty} E_n \right\}. \quad (7.2)$$

Then  $\mu^*$  is an outer measure.

*Proof.* From the definition it is clear that  $\mu^*(\emptyset) = 0$  and that  $\mu^*(A) \leq \mu^*(B)$  if  $A \subseteq B$ . To prove the countable sub-additivity we argue as follows. Suppose  $\{A_j\}_{j=1}^{\infty} \subseteq \mathcal{P}(\Omega)$  and let  $\varepsilon > 0$  be given. By definition, for each  $j$  there exists a collection  $\{E_j^n\}_{n=1}^{\infty} \subseteq \mathcal{E}$  such that  $A_j \subseteq \bigcup_{n=1}^{\infty} E_j^n$  and

$$\sum_{n=1}^{\infty} \rho(E_j^n) \leq \mu^*(A_j) + \frac{\varepsilon}{2^j}. \quad (7.3)$$

If  $A = \bigcup_{j=1}^{\infty} A_j$  we then have  $A \subseteq \bigcup_{j,n=1}^{\infty} E_j^n$  and hence

$$\mu^*(A) \leq \sum_{j,n=1}^{\infty} \rho(E_j^n) \leq \sum_{j=1}^{\infty} \mu^*(A_j) + \varepsilon. \quad (7.4)$$

Since  $\varepsilon > 0$  is arbitrary, we are done. ■

**Remark 7.1.** Note that, at this point, we are not claiming that  $\mu^*(E) = \rho(E)$  for a set  $E \in \mathcal{E}$ .

There is a beautiful and fundamental result that leads us from outer measures to measures as we now describe.

**Definition 7.2.** If  $\mu^*$  is an outer measure on  $\Omega$ , a set  $A \subseteq \Omega$  will be called  $\mu^*$ -measurable if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \quad (7.5)$$

for every  $E \subseteq \Omega$ .

**Remark 7.2.** From the definition of outer measure, note that we always have

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c), \quad (7.6)$$

so it is the other inequality

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c) \quad (7.7)$$

that needs to be verified in order for a set  $A$  to be  $\mu^*$ -measurable.

**Theorem 7.2 (Carathéodory's theorem)**

If  $\mu^*$  is an outer measure on  $\Omega$ , the collection  $\Sigma^*$  of  $\mu^*$ -measurable sets is a  $\sigma$ -algebra, and the restriction of  $\mu^*$  to  $\Sigma^*$  is a complete measure.

*Proof.* Let us divide this proof into its core steps.

*Step 1:  $\Sigma^*$  is closed under complements.* This follows from the fact that definition (7.5) is symmetric with respect to  $A$  and  $A^c$ .

*Step 2:  $\Sigma^*$  is closed under finite unions and  $\mu^*$  is finitely additive.* Let  $A, B \in \Sigma^*$  and  $E \subseteq \Omega$ . Then, from the definition

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c). \end{aligned} \quad (7.8)$$

Noting that  $A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$ , and using the sub-additivity, we have

$$\mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) \geq \mu^*(E \cap (A \cup B)). \quad (7.9)$$

From (7.8) and (7.9) we get

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c), \quad (7.10)$$

and it follows that  $A \cup B \in \Sigma^*$ . From this we get that  $\Sigma^*$  is closed under finite unions.

If  $A, B \in \Sigma^*$  and  $A \cap B = \emptyset$ , we have

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) = \mu^*(A) + \mu^*(B). \quad (7.11)$$

*Step 3:  $\Sigma^*$  is closed under countable unions and  $\mu^*$  is countably additive.* Now let  $\{A_j\}_{j=1}^\infty \subseteq \Sigma^*$ ; we want to show that  $\bigcup_{j=1}^\infty A_j \in \Sigma^*$ . Note that it suffices to prove this claim for  $A_j$ 's that are disjoint, since in the generic case we could consider

$$\tilde{A}_1 = A_1, \text{ and } \tilde{A}_j = A_j - (A_1 \cup \dots \cup A_{j-1}). \quad (7.12)$$

Now,  $\tilde{A}_j$ 's that are disjoint and

$$\bigcup_{j=1}^n \tilde{A}_j = \bigcup_{j=1}^n A_j \quad (7.13)$$

for each  $n$ , and

$$\bigcup_{j=1}^\infty \tilde{A}_j = \bigcup_{j=1}^\infty A_j. \quad (7.14)$$

So let us assume then that our original  $A_j$ 's are disjoint. Let

$$B_n = \bigcup_{j=1}^n A_j, \quad B = \bigcup_{j=1}^\infty A_j. \quad (7.15)$$

(set  $B_0 = \emptyset$ ). Then for any  $E \subseteq \Omega$ , we have

$$\begin{aligned} \mu^*(E \cap B_n) &= \mu^*(E \cap B_n \cap A_n) + \mu^*(E \cap B_n \cap A_n^c) \\ &= \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1}). \end{aligned} \quad (7.16)$$

From (7.16) we can easily show by induction that

$$\mu^*(E \cap B_n) = \sum_{j=1}^n \mu^*(E \cap A_j). \quad (7.17)$$

Therefore, since  $B_n \subseteq B$ ,  $B_n^c \supseteq B^c$ , and using monotonicity,

$$\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \geq \sum_{j=1}^n \mu^*(E \cap A_j) + \mu^*(E \cap B^c). \quad (7.18)$$

Letting  $n \rightarrow \infty$  in (7.18) we get

$$\begin{aligned} \mu^*(E) &\geq \sum_{j=1}^\infty \mu^*(E \cap A_j) + \mu^*(E \cap B^c) \\ &\geq \mu^*\left(\bigcup_{j=1}^\infty (E \cap A_j)\right) + \mu^*(E \cap B^c) \\ &= \mu^*(E \cap B) + \mu^*(E \cap B^c), \end{aligned} \quad (7.19)$$

and we conclude that  $B \in \Sigma^*$ . Moreover, all the inequalities in (7.19) are in fact equalities. Taking  $E = B$  in (7.19) we obtain

$$\mu^*(B) = \sum_{j=1}^{\infty} \mu^*(A_j) \quad (7.20)$$

which is the desired countable additivity.

*Step 4: The measure  $\mu^*$  is complete.* If  $\mu^*(A) = 0$ , for any  $E \subseteq \Omega$  we have

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c) = \mu^*(E \cap A^c) \leq \mu^*(E). \quad (7.21)$$

Hence  $A \in \Sigma^*$ . This shows that  $\mu^*$  restricted to the  $\sigma$ -algebra  $\Sigma^*$  is indeed a complete measure. This concludes the proof.  $\blacksquare$

At this point we cannot yet say too much about the  $\sigma$ -algebra  $\Sigma^*$  of  $\mu^*$ -measurable sets. It could be the case, for example, that we have  $\Sigma^* = \{\emptyset, \Omega\}$ . We need a little bit more structure in order to apply [Carathéodory's theorem](#) in a way to generate concrete  $\sigma$ -algebras and measures.

**Definition 7.3 (Algebra).** We say that a non-empty collection  $\mathcal{A}$  of subsets of  $\Omega$  is an **algebra** if it is closed under complements and finite unions, that is

- (i) If  $A$  belongs to  $\mathcal{A}$  then its complement  $A^c := \Omega - A$  also belongs to  $\mathcal{A}$ ;
- (ii) If  $\{A_n\}_{n=1}^N \subseteq \mathcal{A}$  then  $\bigcup_{n=1}^N A_n \in \mathcal{A}$ .

**Definition 7.4 (Pre-measure).** Let  $\mathcal{A}$  be an algebra. A function  $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$  is called a **pre-measure** if the following conditions are verified:

- (i)  $\mu_0(\emptyset) = 0$ ;
- (ii) If  $\{A_n\}_{n=1}^{\infty}$  is a sequence of disjoint sets in  $\mathcal{A}$  such that  $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$ , then

$$\mu_0\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_0(A_n). \quad (7.22)$$

In particular, the pre-measure is finitely additive since we can always take  $A_n = \emptyset$  for  $n > N$ . The notions of finite and  $\sigma$ -finite pre-measures are the same as we have discussed for measures.

If  $\mu_0$  is a pre-measure on the algebra  $\mathcal{A}$ , we have seen in [Proposition 7.1](#) that

$$\mu^*(E) = \inf \left\{ \sum_{n=1}^{\infty} \mu_0(A_n) : A_n \in \mathcal{A} \text{ and } E \subseteq \bigcup_{n=1}^{\infty} A_n \right\} \quad (7.23)$$

defines an outer measure on  $\Omega$ . In this setup we can prove the following.

### Proposition 7.3

If  $\mu_0$  is a pre-measure on the algebra  $\mathcal{A}$ , and  $\mu^*$  is the outer measure defined by (7.23), then

- (i)  $\mu^*|_{\mathcal{A}} = \mu_0$ .
- (ii) Every set in  $\mathcal{A}$  is  $\mu^*$ -measurable.

*Proof.* (i) Let  $E \in \mathcal{A}$ . Clearly, since  $E$  covers itself (i.e. we may take  $A_1 = E$  and  $A_j = \emptyset$  for  $j \geq 2$ ), we have

$$\mu^*(E) \leq \mu_0(E). \quad (7.24)$$

If  $E \subseteq \bigcup_{n=1}^{\infty} A_n$  with  $A_n \in \mathcal{A}$ , we may consider

$$\tilde{A}_1 = E \cap A_1, \text{ and } \tilde{A}_j = E \cap \left( A_j - \bigcup_{i=1}^{j-1} A_i \right). \quad (7.25)$$

Then  $\tilde{A}_j$ 's are disjoint members of  $\mathcal{A}$  whose union is  $E$  and we get

$$\mu_0(E) = \sum_{n=1}^{\infty} \mu_0(\tilde{A}_j) \leq \sum_{n=1}^{\infty} \mu_0(A_j). \quad (7.26)$$

Hence it follows that  $\mu_0(E) \leq \mu^*(E)$ . Therefore,  $\mu_0(E) = \mu^*(E)$ .

- (ii) If  $A \in \mathcal{A}$  and  $E \subseteq \Omega$ , and  $\varepsilon > 0$  is given, there is a sequence  $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{A}$  such that

$$E \subseteq \bigcup_{n=1}^{\infty} A_n, \text{ and } \sum_{n=1}^{\infty} \mu_0(A_n) \leq \mu^*(E) + \varepsilon. \quad (7.27)$$

Since  $\mu_0$  is additive on  $\mathcal{A}$ , for each  $A_n$ ,

$$\mu_0(A_n) = \mu_0(A_n \cap A) + \mu_0(A_n \cap A^c). \quad (7.28)$$

As a result,

$$\mu^*(E) + \varepsilon \geq \sum_{n=1}^{\infty} \mu_0(A_n) = \sum_{n=1}^{\infty} \mu_0(A_n \cap A) + \sum_{n=1}^{\infty} \mu_0(A_n \cap A^c). \quad (7.29)$$

Now, since  $\{A_n \cap A\}_{n \geq 1} \subseteq \mathcal{A}$  covers  $E \cap A$ ,

$$\mu^*(E \cap A) \leq \sum_{n=1}^{\infty} \mu_0(A_n \cap A). \quad (7.30)$$

Similarly,

$$\mu^*(E \cap A^c) \leq \sum_{n=1}^{\infty} \mu_0(A_n \cap A^c). \quad (7.31)$$

Combining all these, we find

$$\mu^*(E) + \varepsilon \geq \sum_{n=1}^{\infty} \mu_0(A_n \cap A) + \sum_{n=1}^{\infty} \mu_0(A_n \cap A^c) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c). \quad (7.32)$$

Since  $\varepsilon > 0$  is arbitrary,  $\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ . This shows that  $A$  is  $\mu^*$ -measurable. ■

With this in mind, the construction given by [Carathéodory's theorem](#) applied to a pre-measure  $\mu_0$  will give us a powerful result.

#### Theorem 7.4

Let  $\mathcal{A} \subseteq \mathcal{P}(\Omega)$  be an algebra,  $\mu_0$  a pre-measure on  $\mathcal{A}$ , and  $\Sigma$  be the  $\sigma$ -algebra generated by  $\mathcal{A}$ . There exists a measure  $\mu$  on  $\Sigma$  whose restriction to  $\mathcal{A}$  is  $\mu_0$ , namely  $\mu = \mu^*|_{\Sigma}$ , with  $\mu^*$  given by (7.23).

If  $\nu$  is another measure on  $\Sigma$  that extends  $\mu_0$  then  $\nu(E) \leq \mu(E)$  for all  $E \in \Sigma$  with equality if  $\mu(E) < \infty$ . If  $\mu_0$  is  $\sigma$ -finite, then  $\mu$  is the unique extension of  $\mu_0$  to a measure on  $\Sigma$ .

*Proof.* The first assertion follows immediately from [Carathéodory's theorem](#) and [Proposition 7.3](#), since the  $\sigma$ -algebra  $\Sigma^*$  of  $\mu^*$ -measurable sets contains  $\mathcal{A}$  and hence the  $\sigma$ -algebra  $\Sigma$  generated by  $\mathcal{A}$ .

For the second assertion, if  $E \in \Sigma$  and  $E \subseteq \bigcup_{n=1}^{\infty} A_n$  with  $A_n \in \mathcal{A}$ , then

$$\nu(E) \leq \sum_{n=1}^{\infty} \nu(A_n) = \sum_{n=1}^{\infty} \mu_0(A_n). \quad (7.33)$$

Since  $\mu(E) = \inf \sum_{n=1}^{\infty} \mu_0(A_n)$ , we can take infimum on the RHS and we get

$$\nu(E) \leq \inf_{\substack{\{A_n\}_n \subseteq \mathcal{A}, \\ E \subseteq \bigcup_n A_n}} \sum_{n=1}^{\infty} \mu_0(A_n) = \mu(E). \quad (7.34)$$

If we set  $A = \bigcup_{n=1}^{\infty} A_n$  we have

$$\nu(A) = \lim_{N \rightarrow \infty} \nu\left(\bigcup_{n=1}^N A_n\right) = \lim_{N \rightarrow \infty} \mu\left(\bigcup_{n=1}^N A_n\right) = \mu(A). \quad (7.35)$$

If  $\mu(E) < \infty$  and  $\varepsilon > 0$  is given, we can choose  $A$  above such that

$$\mu(A) \leq \sum_{n=1}^{\infty} \mu(A_n) = \sum_{n=1}^{\infty} \mu_0(A_n) \leq \mu(E) + \varepsilon. \quad (7.36)$$

Since  $\mu(A) = \mu(A - E) + \mu(E)$  we get  $\mu(A - E) \leq \varepsilon$  and

$$\mu(E) \leq \mu(A) = \nu(A) = \nu(E) + \nu(A - E) \leq \nu(E) + \mu(A - E) \leq \nu(E) + \varepsilon. \quad (7.37)$$

Since  $\varepsilon > 0$  is arbitrary, we conclude that  $\nu(E) = \mu(E)$ .

Finally, assume that  $\Omega = \bigcup_{n=1}^{\infty} A_n$  with  $\mu_0(A_n) < \infty$ . We may also assume that all the  $A_n$ 's are disjoint. Then, for any  $E \in \Sigma$ , we have

$$\mu(E) = \sum_{n=1}^{\infty} \mu(E \cap A_n) = \sum_{n=1}^{\infty} \nu(E \cap A_n) = \nu(E). \quad (7.38)$$

This concludes the proof. ■

**Remark 7.3.** Observe that  $\mu_0$  can actually be extended to a measure  $\mu^*$  on the  $\sigma$ -algebra  $\Sigma^*$ .

**Exercise 7.1.** Let  $\mathcal{A} \subseteq \mathcal{P}(\Omega)$  be an algebra,  $\mathcal{A}_\sigma$  be the collection of countable unions of sets in  $\mathcal{A}$  and  $\mathcal{A}_{\sigma\delta}$  be the collection of countable intersections of sets in  $\mathcal{A}_\sigma$ . Let  $\mu_0$  be a pre-measure on  $\mathcal{A}$  and  $\mu^*$  be the induced outer measure. Show that:

- (i) For any  $E \subseteq \Omega$  and  $\varepsilon > 0$  there exists  $A \in \mathcal{A}_\sigma$  with  $E \subseteq A$  and  $\mu^*(A) \leq \mu^*(E) + \varepsilon$ .
- (ii) If  $\mu^*(E) < \infty$  then  $E$  is  $\mu^*$ -measurable if and only if there exists  $B \in \mathcal{A}_{\sigma\delta}$  with  $E \subseteq B$  and  $\mu^*(B - E) = 0$ .
- (iii) If  $\mu_0$  is  $\sigma$ -finite, show that the condition  $\mu^*(E) < \infty$  in (ii) is not necessary.

*Solution.* (i) By definition of outer measure as an infimum, there exists  $\{A_n\}_n \subseteq \mathcal{A}$  such that

$$\sum_{n=1}^{\infty} \mu_0(A_n) \leq \mu^*(E) + \varepsilon. \quad (7.39)$$

WLOG, we can assume  $A_n$ 's are pairwise disjoint. If not, we can do a similar construction as in (7.12). We claim that  $A = \bigcup_{n=1}^{\infty} A_n$  is the desired set. Notice that  $A_n \in \mathcal{A}$  implies they are  $\mu^*$ -measurable. Then so is  $A$ . Then by continuity,

$$\mu^*(A) = \mu^*\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{N \rightarrow \infty} \mu^*\left(\bigcup_{n=1}^N A_n\right) = \lim_{N \rightarrow \infty} \mu_0\left(\bigcup_{n=1}^N A_n\right). \quad (7.40)$$

Since  $\mathcal{A}$  is closed under finite union, and  $\mu_0$  is finitely additive,

$$\mu^*(A) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \mu_0(A_n) = \sum_{n=1}^{\infty} \mu_0(A_n) \leq \mu^*(E) + \varepsilon. \quad (7.41)$$

- (ii) Suppose  $E$  is measurable. By part (i), there exists  $A_n$  with  $E \subseteq A_n$  and

$$\mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}. \quad (7.42)$$

Consider  $B = \bigcap_{n=1}^{\infty} A_n$ . This contains  $E$ . Then

$$\mu^*(E) \leq \mu^*(B) \leq \mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}, \quad (7.43)$$

which is true for every  $n$ . Therefore,  $\mu^*(B) = \mu^*(E)$ . Since  $E$  is measurable,

$$\mu^*(B) = \mu^*(B \cap E) + \mu^*(B \cap E^c) \implies \mu^*(B - E) = \mu^*(B) - \mu^*(E) = 0. \quad (7.44)$$

Conversely, suppose there exists  $B \in \mathcal{A}_{\sigma\delta}$  such that  $E \subseteq B$  and  $\mu^*(B - E) = 0$ . Since  $\mu^*$  is a complete measure,  $B - E$  is measurable. Then so is

$$B - (B - E) = E. \quad (7.45)$$

- (iii) The if direction doesn't require  $\mu^*(E)$  to be finite. So we'll prove the other direction. Suppose  $E$  is measurable.

Since  $\mu_0$  is  $\sigma$ -finite, there's a sequence of sets  $X_1, X_2, \dots$ , each with finite  $\mu_0$ -measure, and  $\Omega = \bigcup_{n=1}^{\infty} X_n$ . WLOG, they are disjoint.  $X_n \in \mathcal{A}$ , so they are measurable. Therefore,  $E \cap X_n$  is also measurable.

Given  $\varepsilon > 0$ , we can find  $B_n \in \mathcal{A}_{\sigma}$  such that  $B_n \supseteq E \cap X_n$ , and

$$\mu^*(B_n) \leq \mu^*(E \cap X_n) + \frac{\varepsilon}{2^n}. \quad (7.46)$$

Since  $(E \cap X_n)$  has finite measure,

$$\mu^*(B_n - (E \cap X_n)) = \mu^*(B_n) - \mu^*(E \cap X_n) \leq \frac{\varepsilon}{2^n}. \quad (7.47)$$

By subadditivity,

$$\mu^*\left(\bigcup_{n=1}^{\infty} (B_n - (E \cap X_n))\right) \leq \sum_{n=1}^{\infty} \mu^*(B_n - (E \cap X_n)) \leq \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} = \varepsilon. \quad (7.48)$$

Now,

$$\left(\bigcup_{n=1}^{\infty} B_n\right) - E = \left(\bigcup_{n=1}^{\infty} B_n\right) - \left(\bigcup_{n=1}^{\infty} (E \cap X_n)\right) \subseteq \bigcup_{n=1}^{\infty} (B_n - (E \cap X_n)). \quad (7.49)$$

Therefore,  $\mu^*((\bigcup_{n=1}^{\infty} B_n) - E) \leq \varepsilon$ , where  $E \subseteq \bigcup_{n=1}^{\infty} B_n \in \mathcal{A}_{\sigma\sigma} = \mathcal{A}_{\sigma}$ . Therefore, for each  $n$ , we can find  $A_n \in \mathcal{A}_{\sigma}$  such that  $E \subseteq A_n$  and

$$\mu^*(A_n - E) \leq \frac{1}{n}. \quad (7.50)$$

Then we take

$$B = \bigcap_{n=1}^{\infty} A_n \in \mathcal{A}_{\sigma\delta}. \quad (7.51)$$

Clearly,  $E \subseteq B$ . Now,

$$\mu^*(B - E) = \mu^*\left(\left(\bigcap_{n=1}^{\infty} A_n\right) - E\right) \leq \mu^*(A_n - E) \leq \frac{1}{n}, \quad (7.52)$$

which is true for any  $n$ . Therefore,  $\mu^*(B - E) = 0$ . ■

**Exercise 7.2.** Let  $\mu^*$  be an outer measure on  $\Omega$ ,  $\Sigma^*$  be the  $\sigma$ -algebra of  $\mu^*$ -measurable sets,  $\mu = \mu^*|_{\Sigma^*}$ , and  $\mu^+$  be the outer measure induced by  $\mu$  as in (7.23) (with  $\mu$  and  $\Sigma^*$  replacing  $\mu_0$  and  $\mathcal{A}$  in that definition). Show that

- (i) If  $E \subseteq \Omega$ , we have  $\mu^*(E) \leq \mu^+(E)$ , with equality if and only if there exists  $A \in \Sigma^*$  with  $E \subseteq A$  and  $\mu^*(A) = \mu^*(E)$ .
- (ii) If  $\mu^*$  is induced by a pre-measure then  $\mu^* = \mu^+$ .
- (iii) If  $\Omega = \{0, 1\}$ , construct an example such that  $\mu^* \neq \mu^+$ .

*Solution.* (i) Let  $\{A_n\}_n \subseteq \Sigma^*$  such that  $E \subseteq \bigcup_{n=1}^{\infty} A_n$ . By monotonicity and subadditivity,

$$\mu^*(E) \leq \mu^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n) = \sum_{n=1}^{\infty} \mu(A_n), \quad (7.53)$$

since  $A_n \in \Sigma^*$  and  $\mu = \mu^*|_{\Sigma^*}$ . Now, taking infimum on the RHS over all  $\{A_n\}_n \subseteq \Sigma^*$  such that  $E \subseteq \bigcup_{n=1}^{\infty} A_n$ , we get

$$\mu^*(E) \leq \inf_{\substack{A_n \in \Sigma^*, \\ E \subseteq \bigcup_n A_n}} \sum_{n=1}^{\infty} \mu(A_n) = \mu^+(E). \quad (7.54)$$

If there exists  $A \in \Sigma^*$  with  $E \subseteq A$  and  $\mu^*(A) = \mu^*(E)$ , then  $A$  covers  $E$ , so

$$\mu^+(E) \leq \mu(A) = \mu^*(A) = \mu^*(E), \quad (7.55)$$

proving that  $\mu^*(E) = \mu^+(E)$ .

Now, conversely, suppose  $\mu^*(E) = \mu^+(E)$ . Given  $n$ , by the definition of  $\mu^+$ , there exists  $\{B_{n,m}\}_m \subseteq \Sigma^*$  such that  $E \subseteq \bigcup_{m=1}^{\infty} B_{n,m} =: A_n$  and

$$\sum_{m=1}^{\infty} \mu(B_{n,m}) \leq \mu^+(E) + \frac{1}{n} = \mu^*(E) + \frac{1}{n}. \quad (7.56)$$

The LHS is at least  $\mu(A_n)$ , by subadditivity. Now we take  $A = \bigcap_{n=1}^{\infty} A_n \in \Sigma^*$ . Then  $A$  contains  $E$ , so by monotonicity,  $\mu^*(E) \leq \mu^*(A)$ . Now,

$$\mu^*(A) \leq \mu^*(A_n) \leq \sum_{m=1}^{\infty} \mu(B_{n,m}) \leq \mu^*(E) + \frac{1}{n}, \quad (7.57)$$

which is true for every  $n$ , proving that  $\mu^*(A) = \mu^*(E)$ . Therefore,  $\mu^*(A) = \mu^*(E)$ .

- (ii) Suppose  $\mu^*$  is induced by a pre-measure  $\mu_0$  on  $\mathcal{A}$ . Take  $E \subseteq \Omega$ . By [Exercise 7.1\(i\)](#), we can find  $A_n \in \mathcal{A}_\sigma \subseteq \Sigma^*$  with  $E \subseteq A_n$  and

$$\mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}. \quad (7.58)$$

Now we take  $A = \bigcap_{n=1}^{\infty} A_n \in \mathcal{A}_{\sigma\delta} \subseteq \Sigma^*$ . Then  $A$  contains  $E$ , so by monotonicity,  $\mu^*(E) \leq \mu^*(A)$ . Now,

$$\mu^*(A) \leq \mu^*(A_n) \leq \sum_{m=1}^{\infty} \mu(B_{n,m}) \leq \mu^*(E) + \frac{1}{n}, \quad (7.59)$$

which is true for every  $n$ , proving that  $\mu^*(A) = \mu^*(E)$ . Therefore,  $\mu^*(A) = \mu^*(E)$ . Then by part (ii),  $\mu^*(E) = \mu^+(E)$ .

- (iii) We define the outer measure  $\mu^*$  on  $\mathcal{P}(\Omega)$  as follows:

$$\mu^*(\emptyset) = 0, \quad \mu^*(\{0\}) = 2, \quad \mu^*(\{1\}) = 3, \quad \mu^*(\{0, 1\}) = 4. \quad (7.60)$$

This is an outer measure, as it satisfies monotonicity, and subadditivity. Clearly,  $\emptyset, \Omega$  are  $\mu^*$ -measurable. We claim that these are the only measurable sets. Indeed, if  $\{0\}$  was measurable, then we needed to have

$$\mu^*(\{0, 1\}) = \mu^*(\{0, 1\} \cap \{0\}) + \mu^*(\{0, 1\} \cap \{0\}^c), \quad (7.61)$$

but which is **NOT** true as  $4 \neq 2 + 3$ . Therefore,  $\Sigma^* = \{\emptyset, \Omega\}$ .

Finally, to compute  $\mu^+(\{0\})$ , there's only one set in  $\Sigma^*$  that covers  $\{0\}$ , namely  $\{0, 1\}$ . As a result,  $\mu^+(\{0\}) = 4 \neq \mu^*(\{0\})$ . ■

**Exercise 7.3.** Let  $(\Omega, \Sigma, \mu)$  be a measure space,  $\mu^*$  be the outer measure induced by  $\mu$  as in [\(7.23\)](#),  $\Sigma^*$  be the  $\sigma$ -algebra of  $\mu^*$ -measurable sets and  $\bar{\mu} = \mu^*|_{\Sigma^*}$ . Show that if  $\mu$  is  $\sigma$ -finite then  $\bar{\mu}$  is the completion of  $\mu$ .

*Solution.*  $\bar{\mu}$  is a complete measure on  $\Sigma^*$ , by [Carathéodory's theorem](#). Let  $\hat{\mu}$  be the completion on  $\mu$ , whose domain is

$$\bar{\Sigma} = \{E \cup F : E \in \Sigma, F \subseteq N \text{ for some } N \in \Sigma \text{ with } \mu(N) = 0\}. \quad (7.62)$$

$\hat{\mu}$  is the unique completion of  $\mu$  on  $\bar{\Sigma}$ . In order to show that  $\bar{\mu} = \hat{\mu}$ , since both are complete measures and both of them extend  $\mu$ , it suffices to show that their domains are equal. So we'll show  $\Sigma^* = \bar{\Sigma}$ .

Take  $E \cup F \in \overline{\Sigma}$ , where  $E \in \Sigma$  and  $F \subseteq N$  for some  $N \in \Sigma$  with  $\mu(N) = 0$ . Then  $E \cup N \in \Sigma$ . Now,

$$(E \cup N) - (E \cup F) \subseteq N - F. \quad (7.63)$$

As a result,

$$\mu^*((E \cup N) - (E \cup F)) \leq \mu^*(N - F) \leq \mu^*(N) = 0, \quad (7.64)$$

proving that  $\mu^*((E \cup N) - (E \cup F)) = 0$ . Now,  $E \cup N \in \Sigma = \Sigma_{\sigma\delta}$ . Since  $\mu$  is  $\sigma$ -finite, by [Exercise 7.1\(iii\)](#),  $E \cup F$  is  $\mu^*$ -measurable, i.e.  $E \cup F \in \Sigma^*$ .

Now, suppose  $X \in \Sigma^*$ . By [Exercise 7.1\(iii\)](#), we can find  $B \in \Sigma_{\sigma\delta} = \Sigma$  such that

$$X \subseteq B, \text{ and } \mu^*(B - X) = 0. \quad (7.65)$$

This proves that  $B - X$  is  $\mu^*$ -measurable, since it's a complete measure. Then again by [Exercise 7.1\(iii\)](#), we can find  $E \in \Sigma_{\sigma\delta} = \Sigma$  such that

$$B - X \subseteq E, \text{ and } \mu^*(E - (B - X)) = 0. \quad (7.66)$$

Now, by subadditivity,

$$\mu(E) = \mu^*(E) \leq \mu^*(B - X) + \mu^*(E - (B - X)) = 0. \quad (7.67)$$

Now, we write

$$X^c = \Omega - X = (\Omega - B) \cup (B - X). \quad (7.68)$$

$B \in \Sigma$ , so  $\Omega - B \in \Sigma$ . Furthermore,  $B - X$  is contained in a  $\mu$ -measure zero set  $E$ . Therefore,  $X^c \in \overline{\Sigma}$ . Since  $\overline{\Sigma}$  is a  $\sigma$ -algebra,  $X \in \overline{\Sigma}$ . ■

Let  $\mathcal{B}_{\mathbb{R}}$  be the Borel  $\sigma$ -algebra on  $\mathbb{R}$ . A little bit of motivation for what we are going to do: suppose that  $\mu$  is a measure on  $\mathcal{B}_{\mathbb{R}}$  we can consider the distribution function  $F : \mathbb{R} \rightarrow \mathbb{R}$

$$F(x) = \mu((-\infty, x]). \quad (7.69)$$

One can see that this function is non-decreasing and right-continuous, that is  $\lim_{y \rightarrow x^+} F(y) = F(x)$ . Indeed, if  $x_n \rightarrow x$  decreasingly, then

$$(-\infty, x] = \bigcap_{n=1}^{\infty} (-\infty, x_n]. \quad (7.70)$$

By continuity of measure,

$$F(x) = \mu((-\infty, x]) = \lim_{n \rightarrow \infty} \mu((-\infty, x_n]) = \lim_{n \rightarrow \infty} F(x_n). \quad (7.71)$$

So  $F : \mathbb{R} \rightarrow \mathbb{R}$ , defined as  $F(x) = \mu((-\infty, x])$  is non-decreasing and right-continuous. We want to revert this process and construct a measure  $\mu$  starting from a non-decreasing and right-continuous function  $F : \mathbb{R} \rightarrow \mathbb{R}$ .

We will consider the collection of half-open intervals to the left, called here simply  $h$ -intervals. These will be intervals of the following form:  $(a, b]$ ,  $(a, \infty)$  or  $\emptyset$ , where  $-\infty \leq a < b < \infty$ .

**Lemma 7.5**

The collection  $\mathcal{A}$  of finite disjoint unions of  $h$ -intervals is an algebra. Furthermore, the  $\sigma$ -algebra generated by  $\mathcal{A}$  is  $\mathcal{B}_{\mathbb{R}}$ .

*Proof.* Let  $\mathcal{I}_h$  be the collection of “basic”  $h$ -intervals,

$$\mathcal{I}_h = \{(a, b], (a, \infty), \emptyset : -\infty \leq a < b < \infty\}, \quad (7.72)$$

and  $\mathcal{A}$  is the collection of finite disjoint unions of elements of  $\mathcal{I}_h$ . Clearly,  $\emptyset, \mathbb{R} \in \mathcal{A}$ , as they are in  $\mathcal{I}_h$ .

$\mathcal{A}$  is closed under finite intersection. Let  $I, J \in \mathcal{I}_h$ . Then we have the following cases:

- $I = (a, b], J = (c, d]$ . Then  $I \cap J = (\max\{a, c\}, \min\{b, d\}]$ .
- $I = (a, \infty), J = (c, \infty)$ . Then  $I \cap J = (\max\{a, c\}, \infty)$ .
- $I = (a, b], J = (c, \infty)$ . Then  $I \cap J = (\max\{a, c\}, b]$  or  $\emptyset$ .

Therefore, the intersection of basic  $h$ -intervals is again a basic  $h$ -interval. Now, pick  $A = \bigcup_{i=1}^n I_i, B = \bigcup_{j=1}^m J_j \in \mathcal{A}$ , where  $I_i$ 's are pairwise disjoint  $h$ -intervals and  $J_j$ 's are pairwise disjoint  $h$ -intervals.

$$A \cap B = \left( \bigcup_{i=1}^n I_i \right) \cap \left( \bigcup_{j=1}^m J_j \right) = \bigcup_{i=1}^n \bigcup_{j=1}^m (I_i \cap J_j). \quad (7.73)$$

$I_i \cap J_j$  is again a basic  $h$ -interval, and they are all pairwise disjoint. Therefore,  $A \cap B \in \mathcal{A}$ .

$\mathcal{A}$  is closed under complement. Let  $I \in \mathcal{I}_h$ . Then we have the following cases:

- $I = (a, b]$ . Then  $I^c = (-\infty, a] \cup (b, \infty) \in \mathcal{A}$ .
- $I = (a, \infty)$ . Then  $I^c = (-\infty, a] \in \mathcal{A}$ .

Therefore, the complement of basic  $h$ -intervals is in  $\mathcal{A}$ . Now, pick  $A = \bigcup_{i=1}^n I_i \in \mathcal{A}$ , where  $I_i$ 's are pairwise disjoint  $h$ -intervals.

$$A^c = \left( \bigcup_{i=1}^n I_i \right)^c = \bigcap_{i=1}^n I_i^c. \quad (7.74)$$

$I_i^c \in \mathcal{A}$  and  $\mathcal{A}$  is closed under finite intersection. Therefore,  $A^c \in \mathcal{A}$ . Hence,  $\mathcal{A}$  is an algebra.

$\mathcal{A}$  generates  $\mathcal{B}_{\mathbb{R}}$ . Clearly  $\sigma(\mathcal{A}) \subseteq \mathcal{B}_{\mathbb{R}}$ . Now, we show the reverse inclusion. Note that any open interval  $(a, b)$  can be written as

$$(a, b) = \bigcup_{n=1}^{\infty} \left( a, b - \frac{1}{n} \right] \in \sigma(\mathcal{A}). \quad (7.75)$$

Since any open set is the countable union of such intervals, we can say that any open set is in  $\sigma(\mathcal{A})$ . Hence,  $\mathcal{B}_{\mathbb{R}} \subseteq \sigma(\mathcal{A})$ . ■

### Lemma 7.6

Let  $F : \mathbb{R} \rightarrow \mathbb{R}$  be a non-decreasing and right-continuous function. If  $(a_j, b_j]$  are disjoint  $h$ -intervals ( $j = 1, 2, 3, \dots, n$ ) let

$$\mu_0 \left( \bigcup_{j=1}^n (a_j, b_j] \right) = \sum_{j=1}^n (F(b_j) - F(a_j)) \quad (7.76)$$

and let  $\mu_0(\emptyset) = 0$ . Then  $\mu_0$  is well-defined and defines a pre-measure on  $\mathcal{A}$ .

*Proof.* ■

The next result is a pay-off of our hard work in the previous subsections.

### Theorem 7.7

If  $F : \mathbb{R} \rightarrow \mathbb{R}$  is a non-decreasing and right-continuous function, there is a unique Borel measure  $\mu_F$  on  $\mathbb{R}$  such that  $\mu_F((a, b]) = F(b) - F(a)$  for all  $a, b$ . If  $G$  is another such function, then  $\mu_F = \mu_G$  if and only if  $F - G$  is constant.

Conversely, if  $\mu$  is a Borel measure on  $\mathbb{R}$  that is finite on all bounded Borel sets and we define

$$F(x) = \begin{cases} \mu((0, x]), & \text{if } x > 0; \\ 0, & \text{if } x = 0; \\ -\mu((x, 0]), & \text{if } x < 0; \end{cases} \quad (7.77)$$

then  $F$  is non-decreasing, right-continuous and  $\mu = \mu_F$ .

*Proof.* Each such  $F$  induces a pre-measure on  $\mathcal{A}$  (finite disjoint unions of  $h$ -intervals) by [Lemma 7.6](#). It is clear that  $F$  and  $G$  induce the same pre-measure if and only if  $F - G$  is constant. These pre-measures are  $\sigma$ -finite since

$$\mathbb{R} = \bigcup_{j=1}^{\infty} (j, j+1]. \quad (7.78)$$

The first assertion then follows from [Theorem 7.4](#) and [Lemma 7.5](#).

For the second part one can verify that  $F$  defined by (7.77) is non-decreasing and right-continuous, and that  $\mu = \mu_F$  on  $\mathcal{A}$ . Hence  $\mu = \mu_F$  on  $\mathcal{B}_{\mathbb{R}}$  by the uniqueness part of Theorem 7.4 ■

The theory developed in the previous subsections gives, for each non-decreasing and right-continuous  $F : \mathbb{R} \rightarrow \mathbb{R}$ , not only a Borel measure  $\mu_F$  but a complete measure  $\overline{\mu_F}$  on a  $\sigma$ -algebra that contains  $\mathcal{B}_{\mathbb{R}}$ . In fact,  $\overline{\mu_F}$  is just the completion of  $\mu$  (see Exercise 7.3). This complete measure is usually referred to as the *Lebesgue–Stieltjes measure* associated to  $F$ . The Lebesgue measure on  $\mathbb{R}$  is simply  $\mu_F$  in the particular case of  $F(x) = x$ .

## §8 21 September 2026

We now discuss three informal heuristics, due to Littlewood, that summarize the “philosophy” of Lebesgue measure and integration on  $\mathbb{R}^d$ . Each heuristic is deliberately vague (hence the scare quotes around “nearly”); the content of this lecture is to make each one precise and prove it.

### Littlewood’s three principles.

- (a) Every measurable set is “nearly” a finite union of cubes.
- (b) Every measurable function is “nearly” continuous.
- (c) Every pointwise convergent sequence of measurable functions is “nearly” uniformly convergent.

**Remark 8.1.** Principle (a) is exactly Proposition 6.4(ii), proved in the previous lecture: for  $E$  measurable with  $m(E) < \infty$  and any  $\varepsilon > 0$ , there is a finite almost-disjoint union of cubes  $F'$  with  $m(E \Delta F') \leq \varepsilon$ .

We now turn to Principles (c) and (b), in that order (as the proof of Principle (b) will make use of Principle (c)).

### Theorem 8.1 (Egorov’s theorem)

Let  $E \subseteq \mathbb{R}^d$  be measurable with  $m(E) < \infty$ , and let  $(f_k)_{k \geq 1}$  be a sequence of measurable functions on  $E$ , finite-valued a.e., such that  $f_k \rightarrow f$  pointwise a.e. on  $E$ , for some measurable  $f$  finite a.e. on  $E$ . Then for every  $\varepsilon > 0$  there is a closed set  $F_\varepsilon \subseteq E$  such that

$$m(E - F_\varepsilon) < \varepsilon \quad \text{and} \quad f_k \rightarrow f \text{ uniformly on } F_\varepsilon. \quad (8.1)$$

*Proof.* By discarding a null set from  $E$  (which affects no measure computation below), we may assume WLOG that  $f_k(x) \rightarrow f(x)$  for every  $x \in E$ . For  $n, k \geq 1$ , define

$$E_{n,k} := \left\{ x \in E : |f_j(x) - f(x)| \leq \frac{1}{n} \text{ for all } j \geq k \right\}. \quad (8.2)$$

For fixed  $n$ ,

- $E_{n,k}$  is increasing in  $k$  (a larger  $k$  is a weaker requirement, so  $E_{n,k} \subseteq E_{n,k+1}$ ),
- measurable (an intersection of preimages under measurable functions),
- and  $\bigcup_{k=1}^{\infty} E_{n,k} = E$ : since  $f_j(x) \rightarrow f(x)$  for every  $x \in E$ , so for any  $n$ , the condition  $|f_j(x) - f(x)| \leq \frac{1}{n}$  holds for all sufficiently large  $j$ .

Now, continuity of measure from below gives  $m(E_{n,k}) \rightarrow m(E)$  as  $k \rightarrow \infty$ ,

$$\lim_{k \rightarrow \infty} m(E_{n,k}) = m(E) \implies \lim_{k \rightarrow \infty} m(E) - m(E_{n,k}) = 0 \implies \lim_{k \rightarrow \infty} m(E - E_{n,k}), \quad (8.3)$$

where the last implication follows since  $m(E) < \infty$  and [Corollary 6.3](#). Choose  $k_n$  such that

$$m(E - E_{n,k_n}) < \frac{\varepsilon}{2^{n+1}}. \quad (8.4)$$

Define

$$A_\varepsilon := \bigcap_{n=1}^{\infty} E_{n,k_n}. \quad (8.5)$$

Then  $E - A_\varepsilon = \bigcup_{n=1}^{\infty} (E - E_{n,k_n})$ , so by countable subadditivity,

$$m(E - A_\varepsilon) \leq \sum_{n=1}^{\infty} \frac{\varepsilon}{2^{n+1}} = \frac{\varepsilon}{2}. \quad (8.6)$$

Now, we'll show that  $f_j \rightarrow f$  uniformly on  $A_\varepsilon$ . Given  $\delta > 0$ , choose  $n$  with  $\frac{1}{n} < \delta$ . For every  $x \in A_\varepsilon$ ,  $x \in E_{n,k_n}$ , so

$$|f_j(x) - f(x)| \leq \frac{1}{n} < \delta \quad \text{for } j \geq k_n. \quad (8.7)$$

Since  $k_n$  does not depend on  $x$ , this is uniform convergence on  $A_\varepsilon$ .

Now, by [Lemma 6.1](#), choose a closed set  $F_\varepsilon \subseteq A_\varepsilon$  with  $m(A_\varepsilon - F_\varepsilon) \leq \varepsilon/2$ . Then

$$m(E - F_\varepsilon) \leq m(E - A_\varepsilon) + m(A_\varepsilon - F_\varepsilon) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \quad (8.8)$$

and  $f_k \rightarrow f$  uniformly on  $F_\varepsilon \subseteq A_\varepsilon$ , since uniform convergence on a set persists on every subset. ■

**Definition 8.1** (Step function). A **step function** on  $\mathbb{R}^d$  is a finite linear combination of characteristic functions of cubes:

$$\psi = \sum_{j=1}^N a_j \chi_{Q_j}, \quad (8.9)$$

where  $Q_1, \dots, Q_N$  are cubes (not necessarily disjoint) and  $a_1, \dots, a_N \in \mathbb{R}$ .

**Lemma 8.2** (Approximation of characteristic functions by step functions)

Let  $E \subseteq \mathbb{R}^d$  be measurable and  $m(E) < \infty$ . For every  $\varepsilon > 0$ , there is a step function  $\psi$  such that

$$m\left(\left\{x \in \mathbb{R}^d : \psi(x) \neq \chi_E(x)\right\}\right) \leq \varepsilon. \quad (8.10)$$

*Proof.* By [Proposition 6.4\(ii\)](#), choose a finite almost disjoint union of cubes

$$F' = \bigcup_{j=1}^N Q_j, \quad \text{with } m(E \triangle F') \leq \varepsilon. \quad (8.11)$$

Let  $\psi := \sum_{j=1}^N \chi_{Q_j}$ , a step function. For  $x \in \bigcup_{j=1}^N Q_j^\circ$ , since it's a disjoint union

$$\psi(x) = 1 = \chi_{F'}(x). \quad (8.12)$$

For  $x \notin F$ ,  $x \notin Q_j$  for any  $j$ , so

$$\psi(x) = 0 = \chi_{F'}(x). \quad (8.13)$$

Therefore,  $\psi = \chi_{F'}$  on  $B^c$ , where  $B = \bigcup_{j=1}^N (\partial Q_j)$  is the union of the boundaries of  $Q_j$ . Hence,

$$\left\{x \in \mathbb{R}^d : \chi_E(x) \neq \psi(x)\right\} \cap B^c = \left\{x \in \mathbb{R}^d : \chi_E(x) \neq \chi_{F'}(x)\right\} \cap B^c = (E \Delta F') \cap B^c. \quad (8.14)$$

Now, using the fact that  $m(X) = m(X \cap B) + m(X \cap B^c)$  (this is a consequence of additivity) for measurable sets  $X$  and  $B$ , and if  $B$  is a null-set

$$m(X) = m(X \cap B) + m(X \cap B^c) = m(X \cap B^c). \quad (8.15)$$

Therefore,

$$\begin{aligned} m\left(\left\{x \in \mathbb{R}^d : \chi_E(x) \neq \psi(x)\right\}\right) &= m\left(\left\{x \in \mathbb{R}^d : \chi_E(x) \neq \psi(x)\right\} \cap B^c\right) \\ &= m((E \Delta F') \cap B^c) \\ &= m(E \Delta F') \leq \varepsilon. \end{aligned} \quad (8.16)$$

■

**Proposition 8.3** (Approximation of measurable functions by step functions, a.e.)

Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  be measurable and finite-valued a.e. Then there is a sequence of step functions  $(\psi_k)_{k \geq 1}$  such that  $\psi_k \rightarrow f$  a.e. on  $\mathbb{R}^d$ .

*Proof.* By the approximation of measurable functions by simple functions (Theorem 2.3 applied to  $f^+$  and  $f^-$  separately), there is a sequence of simple functions  $\phi_n \rightarrow f$  everywhere. There is no loss of generality if we replace  $\phi_n$  with  $\varphi_n := \phi_n \cdot \chi_{B_n}$  (where  $B_n = \overline{B(\mathbf{0}, n)}$  is the closed ball of radius  $n$  centered at the origin). This does not affect pointwise convergence, as  $\chi_{B_n}(x) = 1$  for  $n \geq |x|$ .

Therefore, there is a sequence of simple functions  $\varphi_j \rightarrow f$  pointwise, each of the form

$$\varphi_j = \sum_{i=1}^{n_j} a_{ij} \chi_{E_{ij}} \quad (8.17)$$

with every  $E_{ij}$  bounded and measurable. By Proposition 8.2, for each  $i, j$  choose a step function  $\eta_{ij}$  with

$$m(\{\eta_{ij} \neq \chi_{E_{ij}}\}) \leq \frac{1}{2^j n_j (1 + |a_{ij}|)}, \quad (8.18)$$

and set

$$\psi_j = \sum_{i=1}^{n_j} a_{ij} \eta_{ij} \quad (8.19)$$

a step function. Let

$$Y_j := \bigcup_{i=1}^{n_j} \{\eta_{ij} \neq \chi_{E_{ij}}\}. \quad (8.20)$$

Then we have

$$m(Y_j) \leq \sum_{i=1}^{n_j} m(\{\eta_{ij} \neq \chi_{E_{ij}}\}) \leq \sum_{i=1}^{n_j} \frac{1}{2^j n_j (1 + |a_{ij}|)} \leq \frac{1}{2^j}. \quad (8.21)$$

Outside  $Y_j$ ,  $\eta_{ij} = \chi_{E_{ij}}$  for every  $i$ , so  $\psi_j(x) = \varphi_j(x)$  for  $x \notin Y_j$ . Since  $\sum_{j=1}^{\infty} m(Y_j) < \infty$ , the [Borel–Cantelli lemma](#) gives

$$m(Y^*) = m(\{x : x \in Y_j \text{ for infinitely many } j\}) = 0. \quad (8.22)$$

If  $x$  lies in only finitely many  $Y_j$ , so  $\psi_j(x) = \varphi_j(x)$  for all sufficiently large  $j$ . Since  $\varphi_j(x) \rightarrow f(x)$ , we can conclude that  $\psi_j(x) \rightarrow f(x)$  holds for every  $x \notin Y^*$ , i.e. almost everywhere, since  $m(Y^*) = 0$ . ■

#### **Theorem 8.4** (Lusin's theorem)

Let  $E \subseteq \mathbb{R}^d$  be measurable with  $m(E) < \infty$ , and let  $f : E \rightarrow \mathbb{R}$  be measurable and finite-valued. Then for every  $\varepsilon > 0$  there is a closed set  $F_\varepsilon \subseteq E$  such that

$$m(E - F_\varepsilon) < \varepsilon \quad \text{and} \quad f|_{F_\varepsilon} \text{ is continuous.} \quad (8.23)$$

**Remark 8.2.** The conclusion concerns the *restriction*  $f|_{F_\varepsilon}$ , viewed as a function  $F_\varepsilon \rightarrow \mathbb{R}$  in its own right (with the subspace topology); it does *not* assert that  $f$ , as a function on  $E$ , is continuous *at* the points of  $F_\varepsilon$ , which is a much stronger and generally false statement. For example,  $f = \chi_{\mathbb{Q} \cap [0,1]}$  on  $[0,1]$  is discontinuous at *every* point of  $[0,1]$ , yet  $f$  restricted to the set of irrationals (of full measure) is identically 0, hence (trivially) continuous.

*Proof.* Extend  $f$  to all of  $\mathbb{R}^d$  by setting  $f \equiv 0$  outside  $E$ ; this does not affect measurability. By [Proposition 8.3](#), choose step functions  $\psi_k \rightarrow f$  a.e. on  $\mathbb{R}^d$ . Each step function

$$\psi_k = \sum_{i=1}^{n_i} a_{ik} \chi_{Q_{ik}} \quad (8.24)$$

is continuous at every point outside of the boundary

$$B_k = \bigcup_i \partial Q_{ik}, \quad (8.25)$$

which a closed set with  $m(B_k) = 0$ .

By [Egorov's theorem](#) (applied on  $E$ , using  $m(E) < \infty$ ), choose a closed set  $\tilde{F} \subseteq E$  with

$m(E - \tilde{F}) \leq \varepsilon/2$  such that  $\psi_k \rightarrow f$  uniformly on  $\tilde{F}$ . Let

$$G := \tilde{F} - \bigcup_{k=1}^{\infty} B_k. \quad (8.26)$$

Since

$$m\left(\bigcup_{k=1}^{\infty} B_k\right) \leq \sum_{k=1}^{\infty} m(B_k) = 0, \quad (8.27)$$

we have  $m(\tilde{F} - G) = 0$ , so

$$m(E - G) \leq m(E - \tilde{F}) + m(\tilde{F} - G) \leq \frac{\varepsilon}{2}. \quad (8.28)$$

On  $G$ , every  $\psi_k$  is continuous (having removed all cube boundaries), and  $\psi_k \rightarrow f$  uniformly on  $G \subseteq \tilde{F}$ ; since a uniform limit of continuous functions is continuous,  $f|_G$  is continuous.

Finally,  $G$  need not be closed. By [Lemma 6.1](#), choose a closed set  $F_\varepsilon \subseteq G$  with  $m(G - F_\varepsilon) \leq \frac{\varepsilon}{2}$ ; then

$$m(E - F_\varepsilon) \leq m(E - G) + m(G - F_\varepsilon) \leq \varepsilon, \quad (8.29)$$

and  $f|_{F_\varepsilon}$ , being the restriction of the continuous function  $f|_G$  to a subset, is continuous.  $\blacksquare$

**Exercise 8.1.** Let  $f : [0, 1] \rightarrow [0, 1]$  be measurable, one-to-one, and onto, satisfying the following condition:

$$\forall N \subseteq [0, 1], \quad m(N) = 0 \implies m(f(N)) = 0. \quad (8.30)$$

Prove that  $f^{-1} : [0, 1] \rightarrow [0, 1]$  is measurable.

*Solution.* By [Lusin's theorem](#), there is a closed set  $F_n \subseteq [0, 1]$  such that

$$m([0, 1] - F_n) \leq \frac{1}{n}, \quad \text{and } f|_{F_n} \text{ is continuous.} \quad (8.31)$$

Since each  $F_n$  is closed and bounded, they are all compact. Consider

$$N := [0, 1] - \left(\bigcup_{n=1}^{\infty} F_n\right). \quad (8.32)$$

Then  $N \subseteq [0, 1] - F_n$  for each  $n$ . Hence,

$$m(N) \leq m([0, 1] - F_n) \leq \frac{1}{n}, \quad (8.33)$$

for each  $n$ , implying that  $m(N) = 0$ . By definition of measurable functions,  $f^{-1}$  is measurable if and only if

$$(f^{-1})^{-1}((a, \infty)) = f((a, \infty) \cap [0, 1]) \quad (8.34)$$

is Lebesgue measurable for every  $a \in \mathbb{R}$ . We write

$$(a, \infty) \cap [0, 1] = (a, \infty) \cap \left( \bigcup_{n=1}^{\infty} F_n \cup N \right) = \bigcup_{n=1}^{\infty} ((a, \infty) \cap F_n) \cup ((a, \infty) \cap N) \quad (8.35)$$

Therefore,

$$f((a, \infty) \cap [0, 1]) = \bigcup_{n=1}^{\infty} f((a, \infty) \cap F_n) \cup f((a, \infty) \cap N) \quad (8.36)$$

$(a, \infty) \cap N \subseteq N$ , so it's a nullset. Therefore,  $f((a, \infty) \cap N)$  has measure 0, by hypothesis. So it's Lebesgue measurable. For other's, we'll use the following result from topology:

If  $X$  is a compact space and  $Y$  is a Hausdorff space, and  $g : X \rightarrow Y$  is a bijective continuous map, then  $g$  is a homeomorphism.

Now,  $f|_{F_n} : F_n \rightarrow f(F_n)$  is a continuous bijective map from a compact space to a Hausdorff space, so it's a homeomorphism. Therefore, it maps (relative) open sets to (relative) open sets. This proves that  $f((a, \infty) \cap F_n)$  is open in  $f(F_n)$ , i.e. it's an open set in  $\mathbb{R}$  intersected with  $f(F_n)$  (which is compact). So it's also Lebesgue measurable.

Since all the parts in the RHS of (8.36) are Lebesgue measurable, we can conclude that  $f^{-1}$  is measurable. ■

Having developed general measure theory, the construction of Lebesgue measure, and the basic convergence theorems (culminating in Littlewood's three principles), we now turn to the theory of *function spaces*. The first, and most fundamental, family of examples is the  $L^p$  spaces.

Throughout,  $(\Omega, \mathcal{M}, \mu)$  denotes a measure space: a set  $\Omega$ , a  $\sigma$ -algebra  $\mathcal{M}$  of subsets of  $\Omega$ , and a measure  $\mu$  on  $\mathcal{M}$ . We consider measurable functions  $f : \Omega \rightarrow \mathbb{R}$  (or  $\Omega \rightarrow \mathbb{C}$ ).

**Definition 9.1** ( $L^p$  norm). For a measurable function  $f$  and  $1 \leq p < \infty$ , define

$$\|f\|_p = \left( \int_{\Omega} |f(x)|^p d\mu \right)^{\frac{1}{p}} \in [0, \infty]. \quad (9.1)$$

For  $p = \infty$ , define the *essential supremum* of  $f$ :

$$\|f\|_{\infty} = \inf \{ \alpha \geq 0 : |f(x)| \leq \alpha \text{ for } \mu\text{-a.e. } x \} =: \text{ess sup } f. \quad (9.2)$$

**Remark 9.1** (Why “essential”). The essential supremum is designed to be insensitive to changes of  $f$  on a set of measure zero, unlike the ordinary supremum. If  $f$  equals some bounded function except on a null set, where it is altered arbitrarily (even made unbounded there),  $\|f\|_{\infty}$  is unaffected: it is the smallest  $\alpha$  such that  $|f| \leq \alpha$  holds *almost everywhere*, not everywhere.

**Remark 9.2** (Equivalent formulation). One can equivalently write

$$\|f\|_{\infty} = \inf_{\substack{N \subseteq \Omega \\ \mu(N)=0}} \sup_{x \in \Omega - N} |f(x)|, \quad (9.3)$$

i.e. the infimum, over all null sets  $N$ , of the ordinary supremum of  $|f|$  on the complement of  $N$ .

**Remark 9.3.** The cases  $0 < p < 1$  can also be defined formally by the same integral formula, but  $\|\cdot\|_p$  fails to be a norm for such  $p$  (the triangle inequality fails), so these are not normed spaces in the usual sense; we will not use them.

**Proposition 9.1** (Homogeneity)

For any scalar  $c$  and  $1 \leq p \leq \infty$ ,

$$\|cf\|_p = |c| \|f\|_p. \quad (9.4)$$

*Proof.* For  $1 \leq p < \infty$ , pull  $|c|^p$  out of the integral and take the  $p$ -th root:

$$\|cf\|_p = \left( \int_{\Omega} |c|^p |f|^p d\mu \right)^{\frac{1}{p}} = |c| \left( \int_{\Omega} |f|^p d\mu \right)^{\frac{1}{p}} = |c| \|f\|_p. \quad (9.5)$$

For  $p = \infty$ ,  $|cf(x)| \leq \alpha$  a.e. if and only if  $|f(x)| \leq \alpha/|c|$  a.e. (for  $c \neq 0$ ), so the essential suprema scale accordingly. The case  $c = 0$  is immediate. ■

**Proposition 9.2** (Finiteness is preserved under addition)

If  $\|f\|_p < \infty$  and  $\|g\|_p < \infty$  for some  $1 \leq p \leq \infty$ , then  $\|f + g\|_p < \infty$ .

*Proof.* **Case 1:**  $p = \infty$ . If  $|f(x)| \leq \alpha$  a.e. and  $|g(x)| \leq \beta$  a.e., then  $|f(x) + g(x)| \leq \alpha + \beta$  a.e., so  $\|f + g\|_\infty \leq \alpha + \beta < \infty$ .

**Case**  $1 \leq p < \infty$ . Pointwise,

$$|f(x) + g(x)|^p \leq (2 \max\{|f(x)|, |g(x)|\})^p = 2^p \max\{|f(x)|, |g(x)|\}^p \leq 2^p (|f(x)|^p + |g(x)|^p). \quad (9.6)$$

Integrating,

$$\int_{\Omega} |f + g|^p \, d\mu \leq 2^p \left( \int_{\Omega} |f|^p \, d\mu + \int_{\Omega} |g|^p \, d\mu \right) < \infty. \quad (9.7)$$

■

**Remark 9.4.** This is only a finiteness statement, not yet the (much sharper) triangle inequality  $\|f + g\|_p \leq \|f\|_p + \|g\|_p$ , which will be proved later.

**Proposition 9.3** (Vanishing)

For  $1 \leq p < \infty$ ,  $\|f\|_p = 0$  if and only if  $f = 0$   $\mu$ -almost everywhere.

*Proof.* This follows immediately from [Proposition 2.9](#). ■

**Remark 9.5.** Homogeneity and finiteness-under-addition show that for fixed  $1 \leq p \leq \infty$ , the collection  $\{f : \|f\|_p < \infty\}$  is closed under scalar multiplication and addition, hence is a vector space. Once the triangle inequality is established,  $\|\cdot\|_p$  will be a genuine norm on the space of *equivalence classes* of functions equal a.e., since as it stands  $\|f\|_p = 0$  only forces  $f = 0$  a.e., not  $f \equiv 0$  (i.e. it is a priori only a *seminorm* before this identification).

**Definition 9.2** (Conjugate exponents). Two numbers  $1 \leq p, q \leq \infty$  are *conjugate exponents* if  $\frac{1}{p} + \frac{1}{q} = 1$  (with the convention  $\frac{1}{\infty} = 0$ , so  $p = 1$  is conjugate to  $q = \infty$ ).

**Lemma 9.4** (Young's inequality)

Let  $1 < p, q < \infty$  be conjugate exponents. For all real numbers  $a, b \geq 0$ ,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}, \quad (9.8)$$

with equality if and only if  $a^p = b^q$ .

*Proof.* If  $b = 0$  the inequality is trivial, so assume  $b > 0$ . Define, for  $x \geq 0$ ,

$$\varphi(x) = \frac{x^p}{p} + \frac{b^q}{q} - xb. \quad (9.9)$$

Then  $\varphi'(x) = x^{p-1} - b$ , which is negative for  $x < b^{\frac{1}{p-1}}$ , zero at  $x_0 = b^{\frac{1}{p-1}}$ , and positive for  $x > b^{\frac{1}{p-1}}$ ; so  $\varphi$  attains its global minimum on  $[0, \infty)$  at  $x_0$ . Compute, using  $q = \frac{p}{p-1}$  (equivalently  $(p-1)q = p$ ),

$$\varphi(x_0) = \frac{x_0^p}{p} + \frac{b^q}{q} - x_0 b = \frac{b^{\frac{p}{p-1}}}{p} + \frac{b^q}{q} - b^{\frac{1}{p-1}} b \quad (9.10)$$

$$= \frac{b^q}{p} + \frac{b^q}{q} - b^q \quad (9.11)$$

$$= b^q \left( \frac{1}{p} + \frac{1}{q} - 1 \right) = 0. \quad (9.12)$$

Hence  $\varphi(x) \geq \varphi(x_0) = 0$  for all  $x \geq 0$ , i.e.  $xb \leq \frac{x^p}{p} + \frac{b^q}{q}$  for all  $x \geq 0$ . Taking  $x = a$  gives the claimed inequality. Equality holds exactly when  $a = x_0 = b^{\frac{1}{p-1}}$ ,

$$a^{p-1} = b \iff a^p = ab = b^{\frac{1}{p-1}+1} = b^q. \quad (9.13)$$

■

#### Theorem 9.5 (Hölder's inequality)

Let  $1 \leq p, q, r \leq \infty$  satisfy

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{r}. \quad (9.14)$$

If  $f, g$  are measurable functions,

$$\|fg\|_r \leq \|f\|_p \|g\|_q. \quad (9.15)$$

When the RHS of (9.15) is finite, and  $1 < p, q < \infty$ , equality holds if and only if  $f = 0$  a.e., or  $g = 0$  a.e., or there is  $\alpha > 0$  with  $|g(x)|^q = \alpha |f(x)|^p$  for a.e.  $x$ .

**Remark 9.6.** The classical statement usually seen in textbooks is the case  $r = 1$ , where  $p$  and  $q$  are conjugate exponents. The general statement above (with an arbitrary target exponent  $r$ ) reduces to that case, as shown below.

**Remark 9.7** (Why this relation among exponents). The relation  $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$  is exactly what makes (9.15) consistent with respect to dilation of the space. Suppose we put no restriction on the exponents  $p, q, r$  yet. We want to have a universal inequality

$$\|fg\|_r \leq C \|f\|_p \|g\|_q, \quad (9.16)$$

where the constant  $C$  does not depend on  $f, g$ . Suppose we dilated the space  $\Omega = \mathbb{R}^d$  by a

positive factor  $\lambda > 0$ ; in other words, we replace  $f, g$  with

$$f_\lambda(x) = f(\lambda x), \quad g_\lambda(x) = g(\lambda x). \quad (9.17)$$

Then (9.16) should hold for  $f_\lambda, g_\lambda$  as well, i.e.

$$\|f_\lambda g_\lambda\|_r \leq C \|f_\lambda\|_p \|g_\lambda\|_q. \quad (9.18)$$

Now,

$$\|f_\lambda\|_p = \left( \int_{\mathbb{R}^d} |f(\lambda x)|^p dx \right)^{\frac{1}{p}} = \left( \int_{\mathbb{R}^d} |f(u)|^p \lambda^{-d} du \right)^{\frac{1}{p}} = \lambda^{-\frac{d}{p}} \|f\|_p, \quad (9.19)$$

where we have invoked the change of variables  $u = \lambda x$ , so that  $dx = \lambda^{-d} du$ . In a similar manner as (9.19), we get

$$\|g_\lambda\|_q = \lambda^{-\frac{d}{q}} \|g\|_q, \quad \|f_\lambda g_\lambda\|_r = \lambda^{-\frac{d}{r}} \|fg\|_r. \quad (9.20)$$

Plugging everything back to (9.18), we have

$$\lambda^{-\frac{d}{r}} \|fg\|_r \leq C \cdot \lambda^{-\frac{d}{p}} \|f\|_p \cdot \lambda^{-\frac{d}{q}} \|g\|_q. \quad (9.21)$$

After rearranging,

$$\|fg\|_r \leq C \cdot \lambda^{d\left(\frac{1}{r} - \frac{1}{p} - \frac{1}{q}\right)} \|f\|_p \|g\|_q, \quad (9.22)$$

which should hold for any  $\lambda > 0$ .

- If  $\frac{1}{r} - \frac{1}{p} - \frac{1}{q} > 0$ , then as  $\lambda \rightarrow 0$ , the RHS approaches 0. This implies  $\|fg\|_r \leq 0$ , which is obviously false for general measurable functions  $f, g$ .
- If  $\frac{1}{r} - \frac{1}{p} - \frac{1}{q} < 0$ , then as  $\lambda \rightarrow \infty$ , the RHS approaches 0. This also implies  $\|fg\|_r \leq 0$ .

Therefore, the only way an inequality like  $\|fg\|_r \leq C \|f\|_p \|g\|_q$  can make sense is if  $\frac{1}{r} - \frac{1}{p} - \frac{1}{q} = 0$ . In this case, the constant  $C$  is actually 1, as we'll prove now.

*Proof of Hölder's inequality.* If  $\|f\|_p = 0$  or  $\|g\|_q = 0$ , then (by the Proposition 9.3)  $f = 0$  or  $g = 0$  a.e., so  $fg = 0$  a.e. and both sides of (9.15) are 0. So assume  $\|f\|_p, \|g\|_q > 0$ .

If  $\|f\|_p = \infty$  or  $\|g\|_q = \infty$ , (9.15) is trivially true (right side infinite). So assume also  $0 < \|f\|_p, \|g\|_q < \infty$ .

**Case 1:**  $r = \infty$  (hence  $p = q = \infty$ ). If  $|f| \leq \alpha$  a.e. and  $|g| \leq \beta$  a.e., then  $|fg| \leq \alpha\beta$  a.e., so  $\|fg\|_\infty \leq \|f\|_\infty \|g\|_\infty$ .

**Case 2:**  $p = \infty$  or  $q = \infty$  with  $r < \infty$ , say  $q = \infty$  (so  $r = p$ ). Then  $|g(x)| \leq \|g\|_\infty$  a.e., so

$$\int_{\Omega} |fg|^p d\mu \leq \int_{\Omega} |f|^p \|g\|_\infty^p d\mu = \|g\|_\infty^p \int_{\Omega} |f|^p d\mu, \quad (9.23)$$

and taking  $p$ -th roots gives  $\|fg\|_p \leq \|f\|_p \|g\|_\infty$ .

**Case 3:**  $r = 1$ ,  $1 < p, q < \infty$  conjugate. It remains to prove

$$\int_{\Omega} |fg| \, d\mu \leq \|f\|_p \|g\|_q. \quad (9.24)$$

If  $\|f\|_p = 0$  or  $\|g\|_q = 0$  this is trivial as noted above, so assume both are finite and positive. Apply Young's inequality pointwise to  $a = \frac{|f(x)|}{\|f\|_p}$  and  $b = \frac{|g(x)|}{\|g\|_q}$ :

$$\frac{|f(x)|}{\|f\|_p} \cdot \frac{|g(x)|}{\|g\|_q} \leq \frac{1}{p} \left( \frac{|f(x)|}{\|f\|_p} \right)^p + \frac{1}{q} \left( \frac{|g(x)|}{\|g\|_q} \right)^q. \quad (9.25)$$

Integrating both sides over  $\Omega$ ,

$$\frac{1}{\|f\|_p \|g\|_q} \int_{\Omega} |fg| \, d\mu \leq \frac{1}{p} \cdot \frac{1}{\|f\|_p^p} \int_{\Omega} |f|^p \, d\mu + \frac{1}{q} \cdot \frac{1}{\|g\|_q^q} \int_{\Omega} |g|^q \, d\mu = \frac{1}{p} + \frac{1}{q} = 1, \quad (9.26)$$

using the definitions of  $\|f\|_p, \|g\|_q$ . Multiplying through by  $\|f\|_p \|g\|_q$  gives  $\int_{\Omega} |fg| \, d\mu \leq \|f\|_p \|g\|_q$ , as required.

**Case 4:**  $r > 1$ . Suppose  $1 < r < \infty$  and  $1 \leq p, q < \infty$ . Let

$$F = |f|^r, \quad G = |g|^r, \quad P = \frac{p}{r}, \quad Q = \frac{q}{r}. \quad (9.27)$$

Then

$$\frac{1}{P} + \frac{1}{Q} = \frac{r}{p} + \frac{r}{q} = r \left( \frac{1}{p} + \frac{1}{q} \right) = 1, \quad (9.28)$$

so  $P, Q$  are conjugate exponents. Applying the case  $r = 1$  applied to  $F, G, P, Q$ ,

$$\int_{\Omega} FG \, d\mu \leq \left( \int_{\Omega} F^P \, d\mu \right)^{\frac{1}{P}} \left( \int_{\Omega} G^Q \, d\mu \right)^{\frac{1}{Q}}, \quad (9.29)$$

i.e.

$$\int_{\Omega} |f|^r |g|^r \, d\mu \leq \left( \int_{\Omega} |f|^p \, d\mu \right)^{\frac{r}{p}} \left( \int_{\Omega} |g|^q \, d\mu \right)^{\frac{r}{q}}, \quad (9.30)$$

and raising both sides to the power  $\frac{1}{r}$  gives exactly  $\|fg\|_r \leq \|f\|_p \|g\|_q$ .

**Equality.** Equality in the integrated Young's inequality forces pointwise equality a.e. (since the difference of the two sides is a nonnegative function integrating to 0), i.e.

$$\left( \frac{|f(x)|}{\|f\|_p} \right)^p = \left( \frac{|g(x)|}{\|g\|_q} \right)^q \quad \text{a.e.}, \quad (9.31)$$

which is exactly  $|g(x)|^q = \alpha |f(x)|^p$  a.e. for  $\alpha = \|g\|_q^q / \|f\|_p^p > 0$ . ■

**Remark 9.8.**  $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$  is only a *necessary* condition, extracted purely from the dilation symmetry of Lebesgue measure on  $\mathbb{R}^d$  (that  $m(\lambda E) = \lambda^d m(E)$ ); it says nothing about whether the inequality actually holds once the relation is satisfied. Here it does hold (Hölder's inequality), but the scaling argument alone only rules out the impossible relations, narrowing down the

search before any real analytic work is done. The moral: any inequality intended to hold globally on  $\mathbb{R}^d$  must respect the symmetries (here, dilation invariance) of the underlying measure.

**Remark 9.9** (Contrast: finite measure spaces). On a measure space with  $\mu(\Omega) = 1$  (a probability space), there is no dilation symmetry to exploit, and indeed no relation among exponents is forced in the same way. In fact, on such a space the  $L^p$  norms are *nested*: if  $1 \leq a < b \leq \infty$ , then

$$\|f\|_a \leq \|f\|_b \quad \text{for every measurable } f. \quad (9.32)$$

Assume  $b < \infty$  (the case  $b = \infty$  is similar). Apply Hölder's inequality to  $|f|^a \cdot 1$  with exponents  $c = b/a$  (so  $c > 1$ ) and its conjugate  $c'$ , where  $1/c + 1/c' = 1$ :

$$\int_{\Omega} |f|^a \cdot 1 \, d\mu \leq \left( \int_{\Omega} |f|^{ac} \, d\mu \right)^{1/c} \left( \int_{\Omega} 1^{c'} \, d\mu \right)^{1/c'} = \|f\|_b^a \cdot \mu(\Omega)^{1/c'}. \quad (9.33)$$

Since  $\mu(\Omega) = 1$ , the second factor is 1, so  $\|f\|_a^a = \int |f|^a \, d\mu \leq \|f\|_b^a$ , i.e.  $\|f\|_a \leq \|f\|_b$ .

Therefore, if  $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$ , and  $q' > q$ , we have

$$\|fg\|_r \leq \|f\|_p \|g\|_q \leq \|f\|_p \|g\|_{q'}, \quad (9.34)$$

so  $\|fg\|_r \leq \|f\|_p \|g\|_{q'}$ , despite  $\frac{1}{p} + \frac{1}{q'} \neq \frac{1}{r}$ . Therefore, if we restrict our attention to a specific measure space,  $\|fg\|_r \leq \|f\|_p \|g\|_q$  may happen **without**  $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$ . But for  $\mathbb{R}^d$ , the condition  $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$  is necessary.

**Remark 9.10** (Another instance of the scaling heuristic). As a second illustration of the scaling method, consider inequalities of the form

$$\|f\|_q \leq C \|\nabla f\|_p, \quad f \in C_c^\infty(\mathbb{R}^d) \quad (9.35)$$

(smooth, compactly supported functions), where  $\nabla f$  is the gradient. The heuristic is: to make  $f$  large while keeping its support fixed, the function must vary quickly, forcing  $\nabla f$  to be large; so a bound on the size of the gradient should control the size of the function itself, *provided*  $f$  is compactly supported (so it cannot simply be a large constant, which has zero gradient but unbounded  $L^q$  norm if the domain were unbounded).

If such an inequality holds for all  $f \in C_c^\infty(\mathbb{R}^d)$  with  $C$  independent of  $f$ , then we need to have some condition on  $p, q, d$ . We argue similarly as before. For  $\lambda > 0$  let

$$f_\lambda(x) = f(\lambda x). \quad (9.36)$$

Then

$$\nabla f_\lambda(x) = \lambda (\nabla f)(\lambda x) \quad (9.37)$$

As in the previous scaling computation,

$$\|f_\lambda\|_q = \lambda^{-\frac{d}{q}} \|f\|_q, \quad \|\nabla f_\lambda\|_p = \lambda \cdot \lambda^{-\frac{d}{p}} \|\nabla f\|_p = \lambda^{1-\frac{d}{p}} \|\nabla f\|_p. \quad (9.38)$$

Applying the hypothesized inequality (9.35) to  $f_\lambda$ :

$$\lambda^{-\frac{d}{q}} \|f\|_q \leq C \lambda^{1-\frac{d}{p}} \|\nabla f\|_p \quad \text{for all } \lambda > 0. \quad (9.39)$$

After rearranging,

$$\|f\|_q \leq C \lambda^{d(\frac{1}{q} + \frac{1}{d} - \frac{1}{p})} \|\nabla f\|_p \quad \text{for all } \lambda > 0. \quad (9.40)$$

As before, the only way this can hold for every  $\lambda > 0$  is if the exponent of  $\lambda$  is 0, i.e.

$$\frac{1}{d} = \frac{1}{p} - \frac{1}{q}. \quad (9.41)$$

The exponent  $q = \frac{dp}{d-p}$  determined this way is called the *Sobolev conjugate* of  $p$  (defined when  $p < d$ ). That the Sobolev inequality **actually** holds with this exponent is a real theorem (the Sobolev embedding theorem), well beyond what the scaling argument alone establishes. The scaling argument only identifies the *unique candidate* exponent for which such an inequality could possibly be true, exactly as in the Hölder case above.

**Remark 9.11** (General moral). Every “global” inequality on  $\mathbb{R}^d$ , one asserted to hold for all functions in certain norms, with a universal constant, must be consistent with the dilation (and, more generally, translation and rotation) symmetries of Lebesgue measure. Checking this consistency via a one-parameter scaling substitution is a quick and standard first step, before attempting to prove the inequality itself, in identifying which exponent relations are even conceivable.

#### Theorem 9.6 (Minkowski's inequality)

Let  $1 \leq p \leq \infty$  and  $f, g$  be measurable functions. Then

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p. \quad (9.42)$$

For  $1 < p < \infty$ , equality holds if and only if  $g = \alpha f$  a.e. for some  $\alpha \geq 0$ , or  $f = 0$  a.e., or  $g = 0$  a.e. For  $p = 1$ , equality holds if and only if  $f(x)g(x) \geq 0$  a.e. (in the real case:  $f$  and  $g$  have the same sign a.e., allowing either to vanish).

*Proof.* Finiteness of  $\|f + g\|_p$  was already shown in Proposition 9.2.

**Case 1:**  $p = 1$ . Immediate from the pointwise triangle inequality  $|f(x) + g(x)| \leq |f(x)| + |g(x)|$ , integrated. Equality throughout requires pointwise equality a.e., which (for real or complex numbers) holds iff  $f(x)$  and  $g(x)$  are nonnegative multiples of one another at a.e.  $x$  (they point in the same direction in  $\mathbb{C}$ , equivalently have the same sign when real).

**Case 2:**  $p = \infty$ . If  $|f| \leq \alpha$  a.e. and  $|g| \leq \beta$  a.e., then  $|f + g| \leq \alpha + \beta$  a.e., so  $\|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty$ .

**Case 3:**  $1 < p < \infty$ . If  $\|f + g\|_p = 0$  the inequality is trivial, so assume  $\|f + g\|_p > 0$ ; we may also assume  $\|f + g\|_p < \infty$  (already shown). Pointwise, using the triangle inequality,

$$|f(x) + g(x)|^p = |f(x) + g(x)| \cdot |f(x) + g(x)|^{p-1} \leq (|f(x)| + |g(x)|) |f(x) + g(x)|^{p-1}. \quad (9.43)$$

Integrating,

$$\int_{\Omega} |f + g|^p \, d\mu \leq \int_{\Omega} |f| |f + g|^{p-1} \, d\mu + \int_{\Omega} |g| |f + g|^{p-1} \, d\mu. \quad (9.44)$$

Let  $q = \frac{p}{p-1}$ , the conjugate exponent to  $p$ . Apply [Hölder's inequality](#) (with exponents  $p, q$ ) to each term on the right of (9.44), treating  $|f + g|^{p-1}$  as the second factor:

$$\int_{\Omega} |f| |f + g|^{p-1} \, d\mu \leq \|f\|_p \left( \int_{\Omega} |f + g|^{(p-1)q} \, d\mu \right)^{\frac{1}{q}} \quad (9.45)$$

$$= \|f\|_p \left( \int_{\Omega} |f + g|^p \, d\mu \right)^{\frac{1}{p} \cdot \frac{p}{q}} \quad (9.46)$$

$$= \|f\|_p \|f + g\|_p^{p/q} \quad (9.47)$$

Similarly,

$$\int_{\Omega} |g| |f + g|^{p-1} \, d\mu \leq \|g\|_p \|f + g\|_p^{p/q}. \quad (9.48)$$

Substituting into (9.44),

$$\|f + g\|_p^p \leq (\|f\|_p + \|g\|_p) \|f + g\|_p^{p/q}. \quad (9.49)$$

Since  $0 < \|f + g\|_p < \infty$ , divide both sides by  $\|f + g\|_p^{p/q}$ , and use

$$p - \frac{p}{q} = p \left( 1 - \frac{1}{q} \right) = p \cdot \frac{1}{p} = 1, \quad (9.50)$$

to arrive at

$$\|f + g\|_p = \|f + g\|_p^{p-p/q} \leq \|f\|_p + \|g\|_p. \quad (9.51)$$

**Equality.** Tracing the argument back, equality in Minkowski's inequality for  $1 < p < \infty$  requires equality both in the pointwise triangle inequality  $|f + g| \leq |f| + |g|$  (a.e.) and in both applications of [Hölder's inequality](#). The former forces  $f, g$  to have the same sign (or one to vanish) a.e. The latter, via the equality case of Hölder, forces  $|f + g|^{(p-1)q} = |f + g|^p$  to be proportional to  $|f|^p$  and to  $|g|^p$  a.e., which (combined with the same-sign condition) reduces to  $g = \alpha f$  a.e. for some  $\alpha \geq 0$ , or one of  $f, g$  vanishing a.e. ■

We have now shown that for  $1 \leq p \leq \infty$ ,  $\|\cdot\|_p$  satisfies:

- $\|cf\|_p = |c| \|f\|_p$  (homogeneity);
- $\|f + g\|_p \leq \|f\|_p + \|g\|_p$  (Minkowski's inequality/the triangle inequality);
- $\|f\|_p = 0 \iff f = 0$  a.e.

Identifying functions that agree  $\mu$ -almost everywhere (working with equivalence classes  $[f]$  rather than individual functions  $f$ ) turns the third property into the genuine positive-definiteness required of a norm, and  $L^p(\Omega, \mu)$  becomes a normed vector space. The next goal is to show that this normed space is *complete*, i.e. a Banach space.

## §10 24 September 2026

We continue to work on a fixed measure space  $(\Omega, \Sigma, \mu)$ .

**Definition 10.1** (Equivalence class). For a measurable function  $f$ , let

$$[f] = \{g \text{ measurable} : g = f \text{ } \mu\text{-a.e.}\}. \quad (10.1)$$

Two functions with  $g \in [f]$  are said to be *identified*: they represent the same element of  $L^p$ .

**Remark 10.1** (The norm and operations are well defined on classes). Since altering a function on a null set does not change the value of its integral,  $\|g\|_p = \|f\|_p$  whenever  $g \in [f]$ ; so  $\|[f]\|_p := \|f\|_p$  is unambiguous. Likewise

$$[f] + [g] := [f + g], \quad c[f] := [cf] \quad (10.2)$$

are well defined, independent of the representatives chosen: changing  $f$  or  $g$  on a set of measure zero changes  $f + g$  only on a set of measure zero (a finite union of null sets is null), and similarly for scalar multiples.

**Abuse of Notation.** From now on we abuse notation and write  $f$  for the equivalence class  $[f]$ , rather than carrying the brackets throughout. Thus

$$L^p(\Omega, \mu) = \left\{ f \text{ (an equivalence class)} : \|f\|_p < \infty \right\}. \quad (10.3)$$

### Proposition 10.1

With this identification,  $L^p(\Omega, \mu)$  is a normed vector space:  $\|\cdot\|_p$  satisfies homogeneity, the triangle inequality (Minkowski), and

$$\|f\|_p = 0 \iff f = 0 \text{ in } L^p, \quad (10.4)$$

where “ $f = 0$  in  $L^p$ ” now means literally  $f$  is the zero equivalence class, i.e. any representative of  $f$  vanishes  $\mu$ -a.e.

**Remark 10.2.** Before this identification,  $\|\cdot\|_p$  was only a *seminorm* on the space of functions (it could vanish on a nonzero function). Passing to equivalence classes is exactly what upgrades it to a genuine norm.

**Definition 10.2** (Banach space). A normed vector space is a **Banach space** if it is *complete*: every Cauchy sequence converges to an element of the space. A sequence  $(f_n)$  in a normed space is *Cauchy* if for every  $\varepsilon > 0$  there is  $N$  such that  $\|f_m - f_n\| < \varepsilon$  whenever  $m, n \geq N$ .

**Example 10.1**

$\mathbb{Q}$  (with the usual absolute value) is not complete: a Cauchy sequence of rationals may converge to an irrational limit, which lies outside  $\mathbb{Q}$ . By contrast,  $\mathbb{R}$  is complete: every Cauchy sequence of reals converges to a real number.

**Theorem 10.2 (Riesz–Fischer)**

For  $1 \leq p \leq \infty$ ,  $L^p(\Omega, \mu)$  is a Banach space.

*Proof.* Having already shown  $L^p$  is a normed vector space, it remains only to prove completeness. We treat  $1 \leq p < \infty$  first, then  $p = \infty$  separately. Let  $(f_n)$  be a Cauchy sequence in  $L^p(\Omega, \mu)$ : for every  $\varepsilon > 0$  there is  $N(\varepsilon)$  such that  $\|f_m - f_n\|_p < \varepsilon$  whenever  $m, n \geq N(\varepsilon)$ .

**Case 1:**  $1 \leq p < \infty$ . For each  $k = 1, 2, 3, \dots$ , apply the Cauchy property with  $\varepsilon = 2^{-k}$  to choose an index  $N(2^{-k})$ , and select a subsequence  $g_k := f_{n_k}$  (with  $n_k$  chosen increasing and, at each stage, at least as large as  $N(2^{-k})$  and larger than the previously chosen index) so that

$$\|g_{k+1} - g_k\|_p \leq 2^{-k} \quad \text{for every } k \geq 1. \quad (10.5)$$

(This is possible because the tail of the original sequence is eventually within  $2^{-k}$  of any of its later terms, for each  $k$ .) Define

$$h_n(x) = |g_1(x)| + \sum_{k=1}^n |g_{k+1}(x) - g_k(x)|, \quad n \geq 1. \quad (10.6)$$

Each  $h_n \geq 0$ , and  $(h_n)$  is pointwise increasing in  $n$ . By the [Minkowski's inequality](#) in  $L^p$ ,

$$\|h_n\|_p \leq \|g_1\|_p + \sum_{k=1}^n \|g_{k+1} - g_k\|_p \leq \|g_1\|_p + \sum_{k=1}^{\infty} 2^{-k} = \|g_1\|_p + 1 =: M < \infty, \quad (10.7)$$

a bound independent of  $n$ . Let  $h(x) = \lim_{n \rightarrow \infty} h_n(x) \in [0, \infty]$  (the limit exists in  $[0, \infty]$  since  $(h_n)$  is increasing). By the Monotone Convergence Theorem applied to the monotone increasing sequence  $h_n^p \rightarrow h^p$ ,

$$\int_{\Omega} h^p d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} h_n^p d\mu = \lim_{n \rightarrow \infty} \|h_n\|_p^p \leq M^p < \infty, \quad (10.8)$$

so  $h \in L^p(\Omega, \mu)$ ; in particular  $h(x) < \infty$  for  $\mu$ -a.e.  $x$  ([Proposition 3.3](#)). For a.e.  $x$ ,  $h(x) < \infty$ , i.e. the series

$$\sum_{k=1}^{\infty} |g_{k+1}(x) - g_k(x)| \quad (10.9)$$

converges (absolutely) as a numerical series. Therefore, it converges without absolute value as well! Since

$$g_n(x) = g_1(x) + \sum_{k=1}^{n-1} (g_{k+1}(x) - g_k(x)) \quad (10.10)$$

is a telescoping sum, the sequence  $(g_n(x))$  converges, for  $\mu$ -a.e.  $x$ . Define

$$f(x) = \lim_{n \rightarrow \infty} g_n(x) \quad (10.11)$$

wherever this limit exists (a.e.), and  $f(x) = 0$  on the remaining null set; this  $f$  is our candidate limit. For every  $n$ ,

$$|g_n(x)| \leq h_n(x) \leq h(x) \quad \text{for a.e. } x, \quad (10.12)$$

so letting  $n \rightarrow \infty$ ,  $|f(x)| \leq h(x)$  a.e. as well. Since  $h \in L^p$ , this shows  $f \in L^p(\Omega, \mu)$ . Moreover

$$|f(x) - g_n(x)| \leq |f(x)| + |g_n(x)| \leq 2h(x) \quad \text{a.e.}, \quad (10.13)$$

and  $(2h)^p \in L^1$ , while  $|f - g_n|^p \rightarrow 0$  pointwise a.e. Therefore, [Dominated Convergence Theorem](#) applies to  $|f - g_n|^p$  (dominated by  $(2h)^p \in L^1$ ), giving

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f - g_n|^p d\mu = 0 \implies \lim_{n \rightarrow \infty} \|f - g_n\|_p = 0. \quad (10.14)$$

We have shown that the *subsequence*  $g_n = f_{n_k}$  converges to  $f$  in  $L^p$ .

**Claim 10.1:** If  $(x_n)_n$  is a Cauchy sequence in a metric space  $(X, d)$ , and it has a subsequence  $(x_{n_k})_k$  which converges to  $x \in X$ , then  $x_n \rightarrow x$ .

*Proof.* Fix  $\varepsilon > 0$ . There exists  $N_1$  such that

$$d(x_{n_k}, x) \leq \frac{\varepsilon}{2} \quad \text{for } n_k \geq N_1. \quad (10.15)$$

There exists  $N_2$  such that

$$d(x_n, x_m) \leq \frac{\varepsilon}{2} \quad \text{for } n, m \geq N_2. \quad (10.16)$$

Now, for all  $n, n_k \geq \max\{N_1, N_2\}$ ,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad (10.17)$$

Therefore,  $x_n \rightarrow x$ . □

Since  $f_n$  is a Cauchy sequence, and the subsequence  $g_n$  converges to  $f$ , therefore,  $f_n \rightarrow f$  in  $L^p$ .

**Case 2:**  $p = \infty$ . Let  $(f_n)$  be Cauchy in  $L^\infty(\Omega, \mu)$ . For each pair  $(m, n)$  and each  $k$ , the inequality  $|f_m(x) - f_n(x)| \leq \|f_m - f_n\|_\infty$  and  $|f_k(x)| \leq \|f_k\|_\infty$  holds for a.e.  $x$  by definition of the essential supremum. Let  $Z$  be the (countable) union, over all pairs  $(m, n)$  and all  $k$ , of the corresponding null exceptional sets. Then  $\mu(Z) = 0$ , and on  $\Omega - Z$ ,

$$|f_m(x) - f_n(x)| \leq \|f_m - f_n\|_\infty. \quad (10.18)$$

Given  $\varepsilon > 0$ , Cauchy-ness of  $(f_n)$  in  $L^\infty$  gives  $N(\varepsilon)$  with  $\|f_m - f_n\|_\infty < \varepsilon$  for  $m, n \geq N(\varepsilon)$ .

Combined with (10.18), for every  $x \in \Omega - Z$ , and  $m, n \geq N(\varepsilon)$ ,

$$|f_m(x) - f_n(x)| \leq \|f_m - f_n\|_\infty < \varepsilon. \quad (10.19)$$

So for each fixed  $x \in \Omega - Z$ ,  $(f_n(x))$  is a Cauchy sequence of real (or complex) numbers, hence converges. So we define

$$f(x) = \lim_{n \rightarrow \infty} f_n(x), \quad x \in \Omega - Z, \quad (10.20)$$

and  $f(x) = 0$  on  $Z$ . Letting  $n \rightarrow \infty$  in  $|f_m(x) - f_n(x)| < \varepsilon$  (for fixed  $m \geq N(\varepsilon)$ ,  $x \in \Omega - Z$ ) gives

$$|f_m(x) - f(x)| \leq \varepsilon, \quad (10.21)$$

for  $x \in \Omega - Z$ ,  $m \geq N(\varepsilon)$ . i.e. for  $m \geq N(\varepsilon)$ ,

$$\|f_m - f\|_\infty \leq \varepsilon. \quad (10.22)$$

This shows that  $f_m \rightarrow f$  in  $L^\infty$ . Now, taking  $\varepsilon = 1$  and for some  $m \geq N(1)$ ,

$$|f(x)| \leq |f(x) - f_m(x)| + |f_m(x)| \leq 1 + |f_m(x)| \leq 1 + \|f_m\|_\infty < \infty, \quad (10.23)$$

which is true a.e. Therefore,  $f \in L^\infty$ . ■

### Corollary 10.3

If  $f_n \rightarrow f$  in  $L^p(\Omega, \mu)$  ( $1 \leq p \leq \infty$ ), then there is a subsequence  $(f_{n_k})$  such that

$$f_{n_k}(x) \longrightarrow f(x) \quad \text{pointwise for } \mu\text{-a.e. } x. \quad (10.24)$$

**Remark 10.3** (This is *not* automatic). Convergence  $f_n \rightarrow f$  in the  $L^p$  norm is, a priori, a statement only about the size of  $\|f_n - f\|_p$ . It says nothing directly about the pointwise values  $f_n(x)$  versus  $f(x)$ . (Indeed the various  $L^p$  norms are generally not comparable to one another without extra hypotheses, e.g. finiteness of the measure.) It is a genuine theorem, not a triviality, that norm convergence forces a subsequence to converge pointwise a.e.

For example, consider the sequence of functions  $[0, 1] \rightarrow \mathbb{R}$  from [Exercise 6.1](#):

$$\begin{aligned} f_1 &= \chi_{[0,1]} \\ f_2 &= 2\chi_{[0, \frac{1}{2}]}, \quad f_3 = 2\chi_{[\frac{1}{2}, 1]} \\ f_4 &= 3\chi_{[0, \frac{1}{4}]}, \quad f_5 = 3\chi_{[\frac{1}{4}, \frac{1}{2}]}, \quad f_6 = 3\chi_{[\frac{1}{2}, \frac{3}{4}]}, \quad f_7 = 3\chi_{[\frac{3}{4}, 1]} \\ &\dots \end{aligned}$$

In other words, for  $n = 2^k + m$ , with  $k$  being the unique number satisfying  $2^k \leq n < 2^{k+1}$ , and  $0 \leq m < 2^k$ , we define

$$f_n = (k+1) \chi_{[\frac{m}{2^k}, \frac{m+1}{2^k}]}. \quad (10.25)$$

Then

$$\|f_n - 0\|_1 = \int_0^1 |f_n(x)| \, dx = \frac{k+1}{2^k} \rightarrow 0. \quad (10.26)$$

Now, given any  $x \in [0, 1]$ , and for every  $M > 0$ , there are non-negative integers  $k, m$  such that

$$k > M \text{ and } m \leq 2^k x \leq m+1. \quad (10.27)$$

Now,  $f_{2^k+m}(x) = k+1 > M$ . Therefore,  $\limsup_{k \rightarrow \infty} f_k(x) = \infty$ . In particular, the pointwise limit  $\lim_{k \rightarrow \infty} f_k(x)$  doesn't exist anywhere.

In order to do some of the exercises, we'll need to prove some density identities. To prove a property for all of  $L^p$ , it is often convenient to prove it first on a dense subclass and then extend.

#### Proposition 10.4

If  $f \in L^p(\Omega, \mu)$  for  $1 \leq p < \infty$ , then there exists a sequence  $\{\varphi_n\}$  of simple functions such that  $\varphi_n \rightarrow f$  in  $L^p$ .

*Proof.* By [Theorem 2.3](#) (applied on  $f^+$  and  $f^-$  and then adding), there exists an increasing sequence of simple functions  $\varphi_n \rightarrow f$  pointwise everywhere. Now,

$$|\varphi_n - f|^p \leq 2^p |f|^p \in L^1(\Omega, \mu). \quad (10.28)$$

Therefore, by [Dominated Convergence Theorem](#),

$$\lim_{n \rightarrow \infty} \|\varphi_n - f\|_p^p = \lim_{n \rightarrow \infty} \int_{\Omega} |\varphi_n - f|^p \, d\mu = \int_{\Omega} \lim_{n \rightarrow \infty} |\varphi_n - f|^p \, d\mu = 0. \quad (10.29)$$

■

Note that since  $f \in L^p$ , if we write  $\varphi_n = \sum_{i=1}^N a_i \chi_{E_i}$ , then we must have  $\mu(E_i) < \infty$  for all  $i$ . Because otherwise,  $\int_{\Omega} \varphi_n^p \, d\mu$  would be infinity. But then

$$\infty = \left| \int_{\Omega} \varphi_n^p \, d\mu \right| \leq \int_{\Omega} |\varphi_n|^p \, d\mu \leq \int_{\Omega} |f|^p \, d\mu < \infty, \quad (10.30)$$

we get a contradiction!

#### Lemma 10.5

Suppose  $1 \leq p < \infty$ . Let  $E \subseteq \mathbb{R}^d$  be measurable and  $m(E) < \infty$ . For every  $\varepsilon > 0$ , there is a step function  $\psi$  such that  $\|\psi - \chi_E\|_p \leq \varepsilon$ .

*Proof.* We proceed similarly as [Proposition 8.2](#). By [Proposition 6.4\(ii\)](#), choose a finite almost disjoint union of cubes

$$F' = \bigcup_{j=1}^N Q_j, \quad \text{with } m(E \Delta F') \leq \varepsilon^p. \quad (10.31)$$

Let  $\psi := \sum_{j=1}^N \chi_{Q_j}$ , a step function. For  $x \in \bigcup_{j=1}^N Q_j^\circ$ , since it's a disjoint union

$$\psi(x) = 1 = \chi_{F'}(x). \quad (10.32)$$

For  $x \notin F$ ,  $x \notin Q_j$  for any  $j$ , so

$$\psi(x) = 0 = \chi_{F'}(x). \quad (10.33)$$

Therefore,  $\psi = \chi_{F'}$  on  $B^c$ , where  $B = \bigcup_{j=1}^N (\partial Q_j)$  is the union of the boundaries of  $Q_j$ . Hence,

$$\left\{x \in \mathbb{R}^d : \chi_E(x) \neq \psi(x)\right\} \cap B^c = \left\{x \in \mathbb{R}^d : \chi_E(x) \neq \chi_{F'}(x)\right\} \cap B^c = (E \triangle F') \cap B^c. \quad (10.34)$$

Now, on  $B^c$ , the points where  $\chi_E \neq \psi$  are precisely  $(E \triangle F') \cap B^c$ . On  $B^c$ ,  $\psi$  is either 0 or 1. Hence,  $|\chi_E - \psi| \leq 1$  here. Therefore,

$$\begin{aligned} \|\psi - \chi_E\|_p^p &= \int_{\mathbb{R}} |\psi - \chi_E|^p \, dm = \int_{B^c} |\psi - \chi_E|^p \, dm \\ &= \int_{(E \triangle F') \cap B^c} |\psi - \chi_E|^p \, dm \\ &\leq \int_{(E \triangle F') \cap B^c} 1 \, dm \\ &= m((E \triangle F') \cap B^c) \leq \varepsilon^p. \end{aligned} \quad (10.35)$$

■

### Lemma 10.6

Suppose  $1 \leq p < \infty$ . If  $\varphi = \sum_{i=1}^N a_i \chi_{E_i}$  is a simple function in  $\mathbb{R}^d$ , with  $m(E_i) < \infty$ , then for every  $\varepsilon > 0$  there exists a step function  $\psi$  such that  $\|\varphi - \psi\|_p \leq \varepsilon$ .

*Proof.* For each  $i$ , choose a step function  $\psi_i$  such that

$$\|\psi_i - \chi_{E_i}\|_p \leq \frac{\varepsilon}{N(1 + |a_i|)}. \quad (10.36)$$

Then we define the step function

$$\psi = \sum_{i=1}^N a_i \psi_i. \quad (10.37)$$

As a result,

$$\begin{aligned} \|\varphi - \psi\|_p &= \left\| \sum_{i=1}^N a_i \chi_{E_i} - \sum_{i=1}^N a_i \psi_i \right\|_p \leq \sum_{i=1}^N |a_i| \|\chi_{E_i} - \psi_i\|_p \\ &\leq \sum_{i=1}^N |a_i| \frac{\varepsilon}{N(1 + |a_i|)} < \varepsilon. \end{aligned} \quad (10.38)$$

■

**Proposition 10.7**

If  $f \in L^p(\mathbb{R}^d)$  for  $1 \leq p < \infty$ , given any  $\varepsilon > 0$ , there exists a step function  $\psi$  such that  $\|f - \psi\|_p \leq \varepsilon$ .

*Proof.* By Proposition 10.4, we can find a sequence of simple functions  $\varphi_n$  such that  $\|\varphi_n - f\|_p \rightarrow 0$ . So we choose  $N$  large enough so that

$$\|f - \varphi_N\|_p \leq \frac{\varepsilon}{2}. \quad (10.39)$$

If we write  $\varphi_n = \sum_{i=1}^N a_i \chi_{E_i}$ , then we must have  $m(E_i) < \infty$  for all  $i$ , according to the remark after Proposition 10.4. By Lemma 10.6, we choose a step function  $\psi$  such that

$$\|\varphi_N - \psi\|_p \leq \frac{\varepsilon}{2}. \quad (10.40)$$

Therefore,

$$\|f - \psi\|_p \leq \|f - \varphi_N\|_p + \|\varphi_N - \psi\|_p \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad (10.41)$$

■

**Corollary 10.8**

If  $f \in L^p(\mathbb{R}^d)$  for  $1 \leq p < \infty$ , then there exists a sequence  $\{\psi_n\}$  of step functions such that  $\psi_n \rightarrow f$  in  $L^p$ .

*Proof.* Choose  $\psi_n$  with  $\|f - \psi_n\|_p \leq \frac{1}{n}$ . ■

**Lemma 10.9**

Suppose  $1 \leq p < \infty$ . Let  $Q$  be a cube in  $\mathbb{R}^d$ . For every  $\varepsilon > 0$ , there exists a compactly-supported continuous function  $g$  in  $\mathbb{R}^d$  such that  $\|\chi_Q - g\|_p \leq \varepsilon$ .

*Proof.* Suppose  $Q = \prod_{i=1}^d [a_i, b_i]$ . Enlarge the cube in all sides to form the cube

$$Q' = \prod_{i=1}^d [a_i - \delta, b_i + \delta], \quad (10.42)$$

in such a way that

$$|Q'| \leq |Q| + \varepsilon^p. \quad (10.43)$$

Now, we define the function  $g_i : \mathbb{R} \rightarrow \mathbb{R}$  as

$$g_i(x) = \begin{cases} 1 & \text{if } x \in [a, b], \\ 0 & \text{if } x \notin [a_i - \delta, b_i + \delta], \\ \text{linear} & \text{if } x \in (a_i - \delta, a_i) \text{ or } x \in [b_i, b_i + \delta]. \end{cases} \quad (10.44)$$

Then  $g_i$  is continuous. We define  $g : \mathbb{R}^d \rightarrow \mathbb{R}$  as

$$g(\mathbf{x}) = \prod_{i=1}^d g_i(x_i). \quad (10.45)$$

Then  $g$  is continuous (as product of continuous functions), with  $\text{supp } g = Q'$  which is compact. Then  $g$  agrees with  $\chi_Q$  on  $Q$ , and on  $\mathbb{R}^d - Q'$ . So they disagree only on  $Q' - Q$ , and on this region,  $|\chi_Q - g| \leq 1$ .

$$\|\chi_Q - g\|_p^p = \int_{\mathbb{R}^d} |\chi_Q - g|^p \, dm = \int_{Q' - Q} |\chi_Q - g|^p \, dm \leq \int_{Q' - Q} 1 \, dm = m(Q' - Q) \leq \varepsilon^p. \quad (10.46)$$

■

#### Lemma 10.10

Suppose  $1 \leq p < \infty$ . Let  $\psi = \sum_{i=1}^N a_i \chi_{Q_i}$  be a step function on  $\mathbb{R}^d$ . For every  $\varepsilon > 0$ , there exists a compactly-supported continuous function  $g$  in  $\mathbb{R}^d$  such that  $\|\psi - g\|_p \leq \varepsilon$ .

*Proof.* For each  $i$ , choose a compactly-supported continuous function  $g_i$  such that

$$\|g_i - \chi_{Q_i}\|_p \leq \frac{\varepsilon}{N(1 + |a_i|)}. \quad (10.47)$$

Then we define the compactly-supported continuous function

$$g = \sum_{i=1}^N a_i g_i. \quad (10.48)$$

As a result,

$$\|\psi - g\|_p = \left\| \sum_{i=1}^N a_i \chi_{Q_i} - \sum_{i=1}^N a_i g_i \right\|_p \leq \sum_{i=1}^N |a_i| \|\chi_{Q_i} - g_i\|_p \leq \sum_{i=1}^N |a_i| \frac{\varepsilon}{N(1 + |a_i|)} < \varepsilon. \quad (10.49)$$

■

#### Proposition 10.11

If  $f \in L^p(\mathbb{R}^d)$  for  $1 \leq p < \infty$ , given any  $\varepsilon > 0$ , there exists a compactly-supported continuous function  $g$  such that  $\|f - g\|_p \leq \varepsilon$ .

*Proof.* By [Proposition 10.7](#), we can find a step function  $\psi$  such that

$$\|f - \psi\|_p \leq \frac{\varepsilon}{2}. \quad (10.50)$$

By [Lemma 10.10](#) we choose a compactly-supported continuous function  $g$  such that

$$\|\psi - g\|_p \leq \frac{\varepsilon}{2}. \quad (10.51)$$

Therefore,

$$\|f - g\|_p \leq \|f - \psi\|_p + \|\psi - g\|_p \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad (10.52)$$

■

### Corollary 10.12

If  $f \in L^p(\mathbb{R}^d)$  for  $1 \leq p < \infty$ , then there exists a sequence  $\{g_n\}$  of compactly-supported continuous functions such that  $g_n \rightarrow f$  in  $L^p$ .

*Proof.* Choose  $g_n$  with  $\|f - g_n\|_p \leq \frac{1}{n}$ . ■

Therefore, the following classes are dense in  $L^p$  for  $1 \leq p < \infty$ :

- (a) simple functions;
- (b) step functions;
- (c) compactly-supported continuous functions;

In fact, we can even take  $g_i$  in [\(10.44\)](#) to be a smooth bump function which gives 1 at  $[a_i, b_i]$ , and whose support is  $[a_i - \delta_i, b_i + \delta_i]$ . Thus compactly-supported smooth functions are also dense in  $L^p$  for  $1 \leq p < \infty$ . But  $p = \infty$  is a different scenario altogether.

**Remark 10.4.** Let  $f \in L^\infty(\Omega, \mu)$ . Then  $|f|$  is bounded almost everywhere. Note that by the construction of [Theorem 2.3](#), for every non-negative function  $f \in L^\infty$ , we can construct a sequence of simple functions  $\varphi_n$  such that

$$0 \leq f(x) - \varphi_n(x) \leq \frac{1}{2^n} \quad \text{a.e.} \quad (10.53)$$

Hence,

$$\|f - \varphi_n\|_\infty \leq \frac{1}{2^n} \rightarrow 0. \quad (10.54)$$

For real functions,

$$\|(f^+ - f^-) - (\varphi_n^+ - \varphi_n^-)\|_\infty \leq \|f^+ - \varphi_n^+\|_\infty + \|f^- - \varphi_n^-\|_\infty \rightarrow 0. \quad (10.55)$$

For complex functions,

$$\|(f + ig) - (\varphi_n + i\psi_n)\| \leq \|f - \varphi_n\| + \|g - \psi_n\| \rightarrow 0. \quad (10.56)$$

Therefore, simple functions are dense in  $L^\infty$ .

**Remark 10.5.** The same is not true step functions. For instance, if we choose

$$f(x) = \sin\left(\frac{1}{x}\right) \quad \text{for } x > 0, \quad (10.57)$$

and  $f(x) = 0$  for  $x \leq 0$ . Then  $|f| \leq 1$ , everywhere, so that  $f \in L^\infty(\mathbb{R})$ . Choose any step function  $\psi$ . There exists some  $\delta$  such that  $\psi$  is constant on  $(0, \varepsilon)$ . But since  $\sin\left(\frac{1}{x}\right)$  is rapidly oscillating, it hits both  $+1$  and  $-1$  on  $(0, \varepsilon)$ . Therefore,

$$\|f - \psi\|_\infty = \text{ess sup } |f - \psi| \geq \sup_{x \in (0, \varepsilon)} |f(x) - \psi(x)| \geq \max\{|1 - c|, |-1 - c|\} \geq 1. \quad (10.58)$$

Therefore, step functions are not dense in  $L^\infty(\mathbb{R})$ .

**Remark 10.6.** Continuous functions are not dense in  $L^\infty(\mathbb{R})$ . Consider the function

$$f(x) = \chi_{[0, \infty)} \in L^\infty(\mathbb{R}). \quad (10.59)$$

Assume for the sake of contradiction that there exists a continuous function  $g$  such that  $\|f - g\|_\infty < \frac{1}{3}$ . In other words, for almost all  $x$ ,

$$f(x) - \frac{1}{3} \leq g(x) \leq f(x) + \frac{1}{3}. \quad (10.60)$$

Then  $g(x) \leq \frac{1}{3}$  for almost all  $x < 0$ , so by continuity,

$$g(0) \leq \frac{1}{3}. \quad (10.61)$$

On the other hand,  $g(x) \geq \frac{2}{3}$  for almost all  $x > 0$ , so by continuity,

$$g(0) \geq \frac{2}{3}, \quad (10.62)$$

which is a contradiction! Therefore, continuous functions are not dense in  $L^\infty(\mathbb{R})$ . As a result, the smaller class of compactly supported continuous functions also can't be dense in  $L^\infty(\mathbb{R})$ .

**Exercise 10.1.** A function  $\varphi : (a, b) \rightarrow \mathbb{R}$  is said to be convex if

$$\varphi((1 - \lambda)x + \lambda y) \leq (1 - \lambda)\varphi(x) + \lambda\varphi(y) \quad (10.63)$$

for all  $x, y \in (a, b)$  and  $0 \leq \lambda \leq 1$ . If  $\varphi : (a, b) \rightarrow \mathbb{R}$  is convex, prove that it is continuous.

*Solution.* Choose  $x_0 \in (a, b)$ . Then there exists  $R$  such that  $[x_0 - R, x_0 + R] \subseteq (a, b)$ . We shall prove that  $\varphi$  is Lipschitz on  $[x_0 - \frac{R}{2}, x_0 + \frac{R}{2}]$ , which will prove that  $\varphi$  is continuous at  $x_0$ .

First, we'll show that  $\varphi$  is bounded above on  $[x_0 - R, x_0 + R]$ . Indeed, any point in the interval  $[x_0 - R, x_0 + R]$  is a convex combination of its endpoints. Hence,

$$\varphi(x) \leq (1 - \lambda)\varphi(x_0 - R) + \lambda\varphi(x_0 + R) \leq \max\{\varphi(x_0 - R), \varphi(x_0 + R)\} =: M. \quad (10.64)$$

Next, we'll show that  $\varphi$  is bounded below on  $[x_0 - R, x_0 + R]$ . Take  $x \in [x_0 - R, x_0 + R]$ . Then take  $x' \in [x_0 - R, x_0 + R]$  such that  $x_0$  is the midpoint of  $x$  and  $x'$ . Then

$$\varphi(x_0) \leq \frac{1}{2}\varphi(x) + \frac{1}{2}\varphi(x') \implies \varphi(x) \geq 2\varphi(x_0) - \varphi(x') \geq 2\varphi(x_0) - M. \quad (10.65)$$

Therefore,  $\varphi$  is bounded on  $[x_0 - R, x_0 + R]$ , say  $|\varphi(x)| \leq K$ . Now take  $x, y \in [x_0 - \frac{R}{2}, x_0 + \frac{R}{2}]$ , and WLOG  $x < y$ . Take  $z = y + \frac{R}{2}$  and  $w = x - \frac{R}{2}$ . Then all  $w, x, y, z \in [x_0 - R, x_0 + R]$  in that order.

We know that for a convex function  $\varphi : (a, b) \rightarrow \mathbb{R}$  and  $a < s < t < u < b$ ,

$$\frac{\varphi(t) - \varphi(s)}{t - s} \leq \frac{\varphi(u) - \varphi(t)}{u - t}. \quad (10.66)$$

Now, applying it to  $w < x < y$  and  $x < y < z$ , we get

$$\frac{\varphi(x) - \varphi(w)}{R/2} \leq \frac{\varphi(y) - \varphi(x)}{y - x} \leq \frac{\varphi(z) - \varphi(y)}{R/2}. \quad (10.67)$$

Now,

$$\frac{\varphi(z) - \varphi(y)}{R/2} \leq \frac{K + K}{R/2} = \frac{4K}{R} \text{ and } \frac{\varphi(x) - \varphi(w)}{R/2} \geq \frac{-K - K}{R/2} = -\frac{4K}{R}. \quad (10.68)$$

Therefore,

$$\left| \frac{\varphi(y) - \varphi(x)}{y - x} \right| \leq \frac{4K}{R}, \quad (10.69)$$

a constant that doesn't depend on  $x, y$ . Therefore,  $\varphi$  is Lipschitz on  $[x_0 - \frac{R}{2}, x_0 + \frac{R}{2}]$ . ■

**Exercise 10.2.** Determine the values of  $a \in \mathbb{R}$  for which the function  $g : \mathbb{R} \rightarrow \mathbb{R}$  given by

$$g(x) = \frac{1}{|x|^a (1 + |x|)^{2a}} \quad (10.70)$$

belongs to  $L^1(\mathbb{R})$ .

*Solution.*  $g \in L^1(\mathbb{R})$  if and only if  $\int_{\mathbb{R}} g < \infty$ .

$$\int_{-\infty}^{\infty} \frac{1}{|x|^a (1 + |x|)^{2a}} dx = 2 \int_0^{\infty} \frac{1}{x^a (1 + x)^{2a}} dx \quad (10.71)$$

$$= 2 \int_0^1 \frac{1}{x^a (1 + x)^{2a}} dx + 2 \int_1^{\infty} \frac{1}{x^a (1 + x)^{2a}} dx. \quad (10.72)$$

So  $g \in L^1(\mathbb{R})$  if and only if both these integrals are finite. Now, on  $(0, 1]$ ,

$$\begin{aligned} 1 \leq 1+x \leq 2 &\implies 1 \leq (1+x)^{2a} \leq 2^{2a} = 4^a \\ &\implies 4^{-a} \leq \frac{1}{(1+x)^{2a}} \leq 1 \\ &\implies \frac{4^{-a}}{x^a} \leq \frac{1}{x^a(1+x)^{2a}} \leq \frac{1}{x^a}. \end{aligned}$$

(There is an implicit assumption that  $a \geq 0$ . But that's not an issue since the bounds are gonna just get swapped for the negative case.) Therefore,  $\int_0^1 \frac{1}{x^a(1+x)^{2a}} dx < \infty$  if and only if  $\int_0^1 \frac{1}{x^a} dx$  is finite. Clearly it's infinite for  $a = 1$ . For  $a \neq 1$ ,

$$\int_0^1 \frac{1}{x^a} dx = \left[ \frac{x^{-a+1}}{-a+1} \right]_0^1 = \frac{1}{1-a} - \frac{1}{1-a} \lim_{x \rightarrow 0^+} x^{1-a}. \quad (10.73)$$

If  $1-a < 0$ , the limit blows up. If  $1-a > 0$ , then this is finite. Therefore,  $\int_0^1 g(x) dx < \infty$  if and only if  $a < 1$ .

$$\frac{1}{x^a(1+x)^{2a}} = \frac{1}{x^a x^{2a} \left(\frac{1}{x} + 1\right)^{2a}} = x^{-3a} \left(1 + \frac{1}{x}\right)^{-2a}. \quad (10.74)$$

On  $[1, \infty)$ ,

$$1 \leq 1 + \frac{1}{x} \leq 2 \implies 4^{-a} \leq \frac{1}{\left(1 + \frac{1}{x}\right)^{2a}} \leq 1 \implies 4^{-a} x^{-3a} \leq \frac{x^{-3a}}{\left(1 + \frac{1}{x}\right)^{2a}} \leq x^{-3a}. \quad (10.75)$$

(Again, there is an implicit assumption that  $a \geq 0$ . But that's not an issue since the bounds are gonna just get swapped for the negative case.) Therefore,  $\int_1^\infty g(x) dx < \infty$  if and only if  $\int_1^\infty x^{-3a} dx$  is finite. Clearly it's infinite for  $a = \frac{1}{3}$ . For  $a \neq \frac{1}{3}$ ,

$$\int_1^\infty x^{-3a} dx = \left[ \frac{x^{1-3a}}{1-3a} \right]_1^\infty = \frac{1}{1-3a} \lim_{x \rightarrow \infty} x^{1-3a} - \frac{1}{1-3a}. \quad (10.76)$$

This will be finite if and only if  $1-3a < 0$ , i.e.  $a > \frac{1}{3}$ . Therefore,  $g \in L^1(\mathbb{R})$  if and only if  $\frac{1}{3} < a < 1$ . ■

**Exercise 10.3.** Let  $\alpha_1, \alpha_2, \dots, \alpha_n$  be positive numbers with  $\sum_{j=1}^n \alpha_j = 1$ . Let  $x_1, x_2, \dots, x_n$  be non-negative real numbers. Show that

$$x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} \leq \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n. \quad (10.77)$$

*Solution.* Note that  $f(x) = \log x$  is a concave function, i.e.

$$f((1-\lambda)x + \lambda y) \geq (1-\lambda)f(x) + \lambda f(y). \quad (10.78)$$

This is true because  $f'' < 0$ . We'll first show that for a concave function  $f$  and positive reals

$\alpha_1, \dots, \alpha_n$  with  $\sum_{j=1}^n \alpha_j = 1$ ,

$$f\left(\sum_{j=1}^n \alpha_j x_j\right) \geq \sum_{j=1}^n \alpha_j f(x_j). \quad (10.79)$$

This follows by induction. The base case  $n = 1$  is trivial. Suppose it's true for  $n$ -variables. Now, for  $\sum_{j=1}^{n+1} \alpha_j = 1$ ,

$$f\left(\sum_{j=1}^{n+1} \alpha_j x_j\right) = f\left((1 - \alpha_{n+1}) \sum_{j=1}^n \frac{\alpha_j}{1 - \alpha_{n+1}} x_j + \alpha_{n+1} x_{n+1}\right) \quad (10.80)$$

$$\geq (1 - \alpha_{n+1}) f\left(\sum_{j=1}^n \frac{\alpha_j}{1 - \alpha_{n+1}} x_j\right) + \alpha_{n+1} f(x_{n+1}) \quad (10.81)$$

By induction hypothesis, since  $\sum_{j=1}^n \frac{\alpha_j}{1 - \alpha_{n+1}} = 1$ ,

$$f\left(\sum_{j=1}^n \frac{\alpha_j}{1 - \alpha_{n+1}} x_j\right) \geq \sum_{j=1}^n \frac{\alpha_j}{1 - \alpha_{n+1}} f(x_j). \quad (10.82)$$

Combining (10.81) and (10.82), we are done!

Now that we have proved (10.79), we shall apply it on  $\log$ .

$$\log\left(\sum_{j=1}^n \alpha_j x_j\right) \geq \sum_{j=1}^n \alpha_j \log(x_j) = \log\left(\prod_{j=1}^n x_j^{\alpha_j}\right). \quad (10.83)$$

By exponentiating both sides, we get

$$\sum_{j=1}^n \alpha_j x_j \geq \prod_{j=1}^n x_j^{\alpha_j}. \quad (10.84)$$

■

**Exercise 10.4.** Assume that  $f \in L^\infty([0, 1])$  (with respect to the Lebesgue measure). Show that

$$\lim_{p \rightarrow \infty} \|f\|_p = \|f\|_\infty. \quad (10.85)$$

*Solution.* In fact, this works for any finite measure space  $(\Omega, \Sigma, \mu)$ . If  $\|f\|_\infty = 0$ , then  $f = 0$  a.e. So the statement is trivially true. So we assume  $\|f\|_\infty > 0$ . Since  $f \in L^\infty$ ,  $|f(x)| \leq \|f\|_\infty$  a.e. Therefore,

$$\|f\|_p = \left(\int_\Omega |f|^p d\mu\right)^{\frac{1}{p}} \leq \left(\int_\Omega \|f\|_\infty^p d\mu\right)^{\frac{1}{p}} = \mu(\Omega)^{\frac{1}{p}} \|f\|_\infty. \quad (10.86)$$

Taking  $p \rightarrow \infty$ , we have

$$\limsup_{p \rightarrow \infty} \|f\|_p \leq \|f\|_\infty. \quad (10.87)$$

Choose  $0 < \varepsilon < \|f\|_\infty$ . We define

$$E_\varepsilon = \{x \in \Omega : |f(x)| > \|f\|_\infty - \varepsilon\}. \quad (10.88)$$

By the definition of essential supremum,  $\mu(E_\varepsilon) > 0$ . Now,

$$\|f\|_p = \left( \int_\Omega |f|^p d\mu \right)^{\frac{1}{p}} \geq \left( \int_{E_\varepsilon} |f|^p d\mu \right)^{\frac{1}{p}} \quad (10.89)$$

$$\geq \left( \int_{E_\varepsilon} (\|f\|_\infty - \varepsilon)^p d\mu \right)^{\frac{1}{p}} = (\|f\|_\infty - \varepsilon) \mu(E_\varepsilon)^{\frac{1}{p}}. \quad (10.90)$$

Taking  $p \rightarrow \infty$ , we have

$$\liminf_{p \rightarrow \infty} \|f\|_p \geq \|f\|_\infty - \varepsilon. \quad (10.91)$$

Since  $\varepsilon > 0$  is arbitrarily small, we can take  $\varepsilon \rightarrow 0^+$ .

$$\liminf_{p \rightarrow \infty} \|f\|_p \geq \|f\|_\infty. \quad (10.92)$$

Combining (10.87) and (10.92), we have the desired limit  $\lim_{p \rightarrow \infty} \|f\|_p = \|f\|_\infty$ . ■

**Exercise 10.5.** If  $f \in L^p(\mathbb{R}^d)$  and  $v \in \mathbb{R}^d$ , we define the translation  $\tau_v f(x) := f(x - v)$ .

(i) If  $1 \leq p < \infty$ , show that translation is a continuous operator in  $L^p(\mathbb{R}^d)$ , that is, show that

$$\lim_{|v| \rightarrow 0} \|\tau_v f - f\|_p = 0. \quad (10.93)$$

(ii) Show that this is not true when  $p = \infty$ .

*Solution.* (i) Let's first consider the case when  $f$  is a compactly-supported continuous function, where  $\text{supp } f = K$  is compact. WLOG, assume  $|v| \leq 1$ . Then

$$[\tau_v f - f](x) = f(x - v) - f(x), \quad (10.94)$$

has the support contained in  $K' = \{x : d(x, K) \leq 1\}$ . This is bounded, since if  $K$  is bounded in a ball of radius  $R$ ,  $K'$  is bounded in a ball of radius  $R + 1$ . Furthermore, this is closed, as

$$K' = d(-, K)^{-1}[0, 1], \quad (10.95)$$

and  $d(-, K)$  is continuous. Now,

$$\|\tau_v f - f\|_p^p = \int_{\mathbb{R}^d} |\tau_v f - f|^p dm = \int_{K'} |\tau_v f - f|^p dm. \quad (10.96)$$

Since  $f$  is a continuous function on a compact set, it's uniformly continuous. Therefore,

given  $\varepsilon$ , we can find  $\delta > 0$  such that

$$|f(x-v) - f(x)| < \frac{\varepsilon}{m(K')^{\frac{1}{p}}} \text{ whenever } |v| < \delta. \quad (10.97)$$

Since this holds uniformly for all  $x \in K'$ ,

$$\|\tau_v f - f\|_p^p = \int_{K'} |\tau_v f - f|^p \, dm \leq \int_{K'} \frac{\varepsilon^p}{m(K')} \, dm = \frac{\varepsilon^p}{m(K')} m(K') = \varepsilon^p. \quad (10.98)$$

Hence,  $\|\tau_v f - f\|_p \leq \varepsilon$  whenever  $|v| < \delta$ , proving that

$$\lim_{|v| \rightarrow 0} \|\tau_v f - f\|_p = 0, \quad \text{for } f \in C_c(\mathbb{R}^d). \quad (10.99)$$

Now, we get back to the general case when  $f \in L^p(\mathbb{R}^d)$ . Given  $\varepsilon > 0$ , we can find a compactly-supported continuous function  $g$  such that

$$\|f - g\|_p \leq \frac{\varepsilon}{3}. \quad (10.100)$$

Since (10.99) holds for  $g$ , we can also find  $\delta > 0$  such that whenever  $|v| < \delta$ ,

$$\|\tau_v g - g\|_p \leq \frac{\varepsilon}{3}. \quad (10.101)$$

Now, notice that translation preserves norm, i.e. given  $h$ ,

$$\|\tau_v h\|_p = \|h\|_p, \quad (10.102)$$

which follows from the fact that integration is translation invariant. Therefore,

$$\begin{aligned} \|\tau_v f - f\|_p &\leq \|\tau_v f - \tau_v g\|_p + \|\tau_v g - g\|_p + \|g - f\|_p \\ &= \|\tau_v(f - g)\|_p + \|\tau_v g - g\|_p + \|g - f\|_p \\ &\leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon, \end{aligned} \quad (10.103)$$

whenever  $|v| < \delta$ . Therefore,

$$\lim_{|v| \rightarrow 0} \|\tau_v f - f\|_p = 0, \quad \text{for } f \in L^p(\mathbb{R}^d). \quad (10.104)$$

- (ii) Consider the function  $f = \chi_{[0, \infty)}$ . It's in  $L^\infty(\mathbb{R})$ , since its essential supremum is 1. For any  $v$ , the translation  $\tau_v f = \chi_{[v, \infty)}$ . Then

$$\tau_v f - f = \chi_{[v, \infty)} - \chi_{[0, \infty)} = \chi_{[0, v)} \text{ or } \chi_{[v, 0)}. \quad (10.105)$$

Both of these has essential supremum 1. Therefore,  $\lim_{|v| \rightarrow 0} \|\tau_v f - f\|_\infty = 1 \neq 0$ .

■

- Exercise 10.6.** (i) Does there exist a function  $f$  that belongs to  $L^2([0, 1])$  but does not belong to any other  $L^p([0, 1])$ , for  $1 \leq p \leq \infty$ ,  $p \neq 2$ ? If yes, give an example. If not, justify.
- (ii) Does there exist a function  $f$  that belongs to  $L^2(\mathbb{R})$  but does not belong to any other  $L^p(\mathbb{R})$ , for  $1 \leq p \leq \infty$ ,  $p \neq 2$ ? If yes, give an example. If not, justify.
- (iii) Does there exist a function  $f$  that belongs to  $L^1(\mathbb{R})$  but does not belong to any other  $L^p(I)$  for any open interval  $I \subset \mathbb{R}$  and any  $p > 1$ ? If yes, give an example. If not, justify.

*Solution.* (i) No. From [Remark 9.9](#), since  $m([0, 1]) = 1$ , any  $1 \leq p < 2$  has the property that  $\|f\|_p \leq \|f\|_2$ , so if  $\|f\|_2$  is finite, then so is  $\|f\|_p$  for  $1 \leq p < 2$ .

However, we can construct a function that is in  $L^2([0, 1])$  but not in  $L^p$  for  $p > 2$ . For instance, we define

$$f(x) = \frac{1}{\sqrt{x} \cdot (1 - \log x)}. \quad (10.106)$$

Then

$$\int_0^1 |f(x)|^2 dx = \int_0^1 \frac{1}{x \cdot (1 - \log x)^2} dx. \quad (10.107)$$

After the change of variables  $u = 1 - \log x$ , we have

$$\int \frac{1}{x \cdot (1 - \log x)^2} dx = \int -\frac{1}{u^2} du = \frac{1}{u} = \frac{1}{1 - \log x} + C. \quad (10.108)$$

Therefore,

$$\int_0^1 |f(x)|^2 dx = \left[ \frac{1}{1 - \log x} \right]_0^1 = 1. \quad (10.109)$$

So  $f \in L^2$ . Now, for  $p > 2$ ,

$$\int_0^1 |f(x)|^p dx = \int_0^1 \frac{1}{x^{\frac{p}{2}} \cdot (1 - \log x)^p} dx = \int_{-\infty}^0 \frac{e^{\frac{up}{2}}}{(1+u)^p} (-e^{-u} du), \quad (10.110)$$

for  $u = -\log x$ , so that  $x = e^{-u}$ , and  $dx = -e^{-u} du$ . After a bit of simplification, we get

$$\int_0^1 |f(x)|^p dx = \int_0^\infty \frac{e^{u(\frac{p}{2}-1)}}{(1+u)^p} du. \quad (10.111)$$

For  $p > 2$ ,  $e^{u(\frac{p}{2}-1)}$  grows exponentially while  $(1+u)^p$  just grows polynomially, i.e.

$$\lim_{u \rightarrow \infty} \frac{e^{u(\frac{p}{2}-1)}}{(1+u)^p} = \infty, \quad (10.112)$$

proving that  $\int_0^1 |f(x)|^p dx = \infty$ .

- (ii) Yes. In fact, we'll prove that the following function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is in  $L^p(\mathbb{R})$ , but not in any  $L^q(\mathbb{R})$  for  $q \neq p$ .

$$f(x) = \frac{1}{x^{\frac{1}{p}} (1 + (\log x)^2)} \text{ for } x > 0, \quad (10.113)$$

and we can define  $f(x) = 0$  for  $x \leq 0$ . Now,

$$\int_{\mathbb{R}} |f|^p \, dm = \int_0^\infty \frac{1}{x \left(1 + (\log x)^2\right)^p} \, dx. \quad (10.114)$$

With the change of variables  $u = \log x$ , we have

$$\int_{\mathbb{R}} |f|^p \, dm = \int_{-\infty}^\infty \frac{1}{(1+u^2)^p} \, du = 2 \int_0^\infty \frac{1}{(1+u^2)^p} \, du \quad (10.115)$$

$$= 2 \int_0^1 \frac{1}{(1+u^2)^p} \, du + 2 \int_1^\infty \frac{1}{(1+u^2)^p} \, du. \quad (10.116)$$

On  $[0, 1]$ ,  $\frac{1}{1+u^2} \leq 1$ , so the integral is finite on  $[0, 1]$ . Now, on  $[1, \infty)$ ,  $\frac{1}{1+u^2} \leq u^{-2}$ . Hence,

$$\int_1^\infty \frac{1}{(1+u^2)^p} \, du \leq \int_1^\infty u^{-2p} \, du = \left[ \frac{u^{1-2p}}{1-2p} \right]_1^\infty = \frac{1}{2p-1}, \quad (10.117)$$

since  $1-2p < 0$ , and hence  $\lim_{u \rightarrow \infty} u^{1-2p} = 0$ . Therefore,  $f \in L^p(\mathbb{R})$ . Now, we'll show that  $f \notin L^q(\mathbb{R})$  for  $q \neq p$ .

$$\int_{\mathbb{R}} |f|^q \, dm = \int_0^\infty \frac{1}{x^{\frac{q}{p}} \left(1 + (\log x)^2\right)^q} \, dx. \quad (10.118)$$

If  $q > p$ ,

$$\int_{\mathbb{R}} |f|^q \, dm \geq \int_0^{\frac{1}{e}} \frac{1}{x^{\frac{q}{p}} \left(1 + (\log x)^2\right)^q} \, dx = \int_\infty^1 \frac{e^{\frac{uq}{p}}}{(1+u^2)^q} (-e^{-u} \, du), \quad (10.119)$$

for  $u = -\log x$ , so that  $x = e^{-u}$ , and  $dx = -e^{-u} \, du$ . After a bit of simplification, we get

$$\int_{\mathbb{R}} |f|^q \, dm \geq \int_1^\infty \frac{e^{u\left(\frac{q}{p}-1\right)}}{(1+u^2)^q} \, du. \quad (10.120)$$

Since  $q > p$ ,  $\frac{q}{p} - 1 > 0$ . So  $e^{u\left(\frac{q}{p}-1\right)}$  grows exponentially while  $(1+u^2)^q$  just grows polynomially, i.e.

$$\lim_{u \rightarrow \infty} \frac{e^{u\left(\frac{q}{p}-1\right)}}{(1+u^2)^q} = \infty, \quad (10.121)$$

proving that  $\int_{\mathbb{R}} |f|^q \, dm = \infty$ . Now, consider the case  $q < p$ .

$$\int_{\mathbb{R}} |f|^q \, dm \geq \int_e^\infty \frac{1}{x^{\frac{q}{p}} \left(1 + (\log x)^2\right)^q} \, dx = \int_1^\infty \frac{e^{-\frac{uq}{p}}}{(1+u^2)^q} (e^u \, du), \quad (10.122)$$

for  $u = \log x$ , so that  $x = e^u$ , and  $dx = e^u \, du$ . After a bit of simplification, we get

$$\int_{\mathbb{R}} |f|^q \, dm \geq \int_1^\infty \frac{e^{u\left(1-\frac{q}{p}\right)}}{(1+u^2)^q} \, du. \quad (10.123)$$

Since  $q < p$ ,  $1 - \frac{q}{p} > 0$ . So  $e^{u(1-\frac{q}{p})}$  grows exponentially while  $(1+u^2)^q$  just grows polynomially, i.e.

$$\lim_{u \rightarrow \infty} \frac{e^{u(1-\frac{q}{p})}}{(1+u^2)^q} = \infty, \quad (10.124)$$

proving that  $\int_{\mathbb{R}} |f|^q \, dm = \infty$ .

(iii) Yes. First, consider the function

$$g(x) = \frac{1}{|x|(\log|x|)^2}, \quad \text{for } 0 < |x| < \frac{1}{2}, \quad (10.125)$$

and  $g(x) = 0$  elsewhere. First, note that  $g \in L^1(\mathbb{R})$ :

$$\int_{\mathbb{R}} |g| \, dm = 2 \int_0^{\frac{1}{2}} \frac{1}{x(\log x)^2} \, dx. \quad (10.126)$$

Now, take  $u = \log x$ , so that  $du = \frac{1}{x} \, dx$ .

$$\int \frac{1}{x(\log x)^2} \, dx = \int \frac{1}{u^2} \, du = -\frac{1}{u} = -\frac{1}{\log x} + C. \quad (10.127)$$

So,

$$\int_{\mathbb{R}} |g| \, dm = 2 \left[ -\frac{1}{\log x} \right]_0^{\frac{1}{2}} = \frac{2}{\log 2}. \quad (10.128)$$

Now we claim that  $g \notin L^p((-\varepsilon, \varepsilon))$  for  $0 < \varepsilon < \frac{1}{2}$  and  $p > 1$ .

$$\int_{-\varepsilon}^{\varepsilon} |g(x)|^p \, dx = 2 \int_0^{\varepsilon} \frac{1}{x^p (-\log x)^{2p}} \, dx = 2 \int_{\infty}^{-\log \varepsilon} \frac{e^{up}}{u^{2p}} (-e^{-u} \, du) \quad (10.129)$$

for  $u = -\log x$ , so that  $x = e^{-u}$ , and  $dx = -e^{-u} \, du$ . After a bit of simplification, we get

$$\int_{-\varepsilon}^{\varepsilon} |g(x)|^p \, dx = \int_{-\log \varepsilon}^{\infty} \frac{e^{u(p-1)}}{u^{2p}} \, du. \quad (10.130)$$

Since  $p-1 > 0$ ,  $e^{u(p-1)}$  grows exponentially while  $u^{2p}$  just grows polynomially, i.e.

$$\lim_{u \rightarrow \infty} \frac{e^{u(p-1)}}{u^{2p}} = \infty, \quad (10.131)$$

proving that  $\int_{-\varepsilon}^{\varepsilon} |g(x)|^p \, dx = \infty$ , i.e.  $g \notin L^p((-\varepsilon, \varepsilon))$ .

Now, let  $\{q_n\}_{n=1}^{\infty}$  be an enumeration of  $\mathbb{Q}$ . We define

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{2^n} g(x - q_n). \quad (10.132)$$

By applying [Proposition 2.8](#), we get

$$\int_{\mathbb{R}} |f| \, dm \leq \sum_{n=1}^{\infty} \frac{1}{2^n} \int_{-\infty}^{\infty} |g(x - q_n)| \, dx = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{2}{\log 2} = \frac{2}{\log 2}. \quad (10.133)$$

Therefore,  $f \in L^1(\mathbb{R})$ . Now, let  $I \subseteq \mathbb{R}$  be an arbitrary open interval. It contains some rational number, say  $q_k$ . Then there exists some  $\varepsilon > 0$  such that  $(q_k - \varepsilon, q_k + \varepsilon) \subseteq I$ . WLOG,  $\varepsilon < \frac{1}{2}$ . Now, on  $I$ ,

$$f(x) \geq \frac{1}{2^k} g(x - q_k), \quad (10.134)$$

so we have

$$\int_I |f|^p \, dm \geq \int_{q_k - \varepsilon}^{q_k + \varepsilon} \left| \frac{1}{2^k} g(x - q_k) \right|^p \, dx = \frac{1}{2^{kp}} \int_{-\varepsilon}^{\varepsilon} |g(u)|^p \, du, \quad (10.135)$$

for  $u = x - q_k$ . This integral diverges, as we have just seen. Therefore,  $f \notin L^p(I)$  for any  $I \subseteq \mathbb{R}$  and  $p > 1$ . ■

**Exercise 10.7.** Let  $g : \mathbb{R} \rightarrow \mathbb{C}$  be a bounded, measurable, periodic function with period 1, and let  $f$  be a function in  $L^1(\mathbb{R})$ . Show that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f(x) g(nx) \, dx = \left( \int_0^1 g(x) \, dx \right) \left( \int_{\mathbb{R}} f(x) \, dx \right). \quad (10.136)$$

*Solution.* Note that if (10.136) holds for  $f$ , then it holds for  $cf$ . Furthermore, if it holds for  $f = f_1$  and  $f = f_2$ , then it holds for  $f = f_1 + f_2$ .

First, we'll prove that (10.136) holds for  $\chi_{[a,b]}$ . Indeed,

$$\int_{\mathbb{R}} \chi_{[a,b]} g(nx) \, dx = \int_a^b g(nx) \, dx = \frac{1}{n} \int_{na}^{nb} g(u) \, du. \quad (10.137)$$

Since  $g$  is periodic with period 1, its integral over any interval with length 1 is the same,  $I := \int_0^1 g(x) \, dx$ . Therefore,

$$\int_{na}^{nb} g(u) \, du = I \times (\text{number of length-1 intervals in } [na, nb]) + \int_J g(x) \, dx, \quad (10.138)$$

where  $J$  is an interval of length less than 1. The number of length-1 intervals in  $[na, nb]$  is  $\lfloor nb - na \rfloor$ . Therefore, the RHS of (10.137) becomes

$$\frac{1}{n} \times I \times \lfloor n(b - a) \rfloor + \frac{1}{n} \int_J g(x) \, dx. \quad (10.139)$$

The second part is bounded by

$$\left| \frac{1}{n} \int_J g(x) \, dx \right| \leq \frac{1}{n} \int_{[0,1]} |g(x)| \, dx \leq \frac{M}{n} \rightarrow 0, \quad (10.140)$$

where  $M$  is the bound of  $g$ , i.e.  $|g| \leq M$ . For the first part, note that  $\lim_{n \rightarrow \infty} \frac{\lfloor nx \rfloor}{n} = x$  for any real  $x$  (this can be easily proven by squeeze theorem). Therefore,

$$\frac{1}{n} \times I \times \lfloor n(b-a) \rfloor \rightarrow I(b-a) = \left( \int_0^1 g(x) dx \right) \left( \int_{\mathbb{R}} \chi_{[a,b]}(x) dx \right). \quad (10.141)$$

Therefore, (10.136) holds for  $\chi_{[a,b]}$ . By linearity, it holds for any step function.

Now, let  $f \in L^1(\mathbb{R})$  be arbitrary, and fix  $\varepsilon > 0$ . Choose a step function  $\psi$  such that

$$\|f - \psi\|_1 \leq \frac{\varepsilon}{3M}. \quad (10.142)$$

Now,

$$\begin{aligned} & \left| \int_{\mathbb{R}} f(x)g(nx) dx - \left( \int_0^1 g(x) dx \right) \left( \int_{\mathbb{R}} f(x) dx \right) \right| \\ &= \left| \int_{\mathbb{R}} [f(x) - \psi(x)]g(nx) dx + \int_{\mathbb{R}} \psi(x)g(nx) dx - I \int_{\mathbb{R}} [f(x) - \psi(x)] dx - I \int_{\mathbb{R}} \psi(x) dx \right| \\ &\leq \int_{\mathbb{R}} |f(x) - \psi(x)| |g(nx)| dx + |I| \int_{\mathbb{R}} |f(x) - \psi(x)| dx + \left| \int_{\mathbb{R}} \psi(x)g(nx) dx - I \int_{\mathbb{R}} \psi(x) dx \right|. \end{aligned}$$

Since  $\int_{\mathbb{R}} \psi(x)g(nx) dx \rightarrow I \int_{\mathbb{R}} \psi(x) dx$ , we can find  $N$  such that for  $n \geq N$ ,

$$\left| \int_{\mathbb{R}} \psi(x)g(nx) dx - I \int_{\mathbb{R}} \psi(x) dx \right| \leq \frac{\varepsilon}{3}. \quad (10.143)$$

Furthermore,

$$|I| = \left| \int_0^1 g(x) dx \right| \leq \int_0^1 |g(x)| dx \leq \int_0^1 M dx = M. \quad (10.144)$$

Therefore, we have that for  $n \geq N$ ,

$$\begin{aligned} & \left| \int_{\mathbb{R}} f(x)g(nx) dx - \left( \int_0^1 g(x) dx \right) \left( \int_{\mathbb{R}} f(x) dx \right) \right| \\ &\leq \int_{\mathbb{R}} |f(x) - \psi(x)| |g(nx)| dx + M \int_{\mathbb{R}} |f(x) - \psi(x)| dx + \frac{\varepsilon}{3} \\ &\leq M \int_{\mathbb{R}} |f(x) - \psi(x)| dx + M \int_{\mathbb{R}} |f(x) - \psi(x)| dx + \frac{\varepsilon}{3} \\ &\leq M \cdot \frac{\varepsilon}{3M} + M \cdot \frac{\varepsilon}{3M} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned} \quad (10.145)$$

Therefore,  $\int_{\mathbb{R}} f(x)g(nx) dx \rightarrow \left( \int_0^1 g(x) dx \right) \left( \int_{\mathbb{R}} f(x) dx \right)$ . ■

**Exercise 10.8.** Let  $f \in L^1(\mathbb{R}^d)$ . For any  $\varepsilon > 0$ , show that there exists a  $\delta > 0$  such that for every Lebesgue measurable set  $E \subset \mathbb{R}^d$  with  $m(E) < \delta$  we have

$$\left| \int_E f(x) dx \right| < \varepsilon. \quad (10.146)$$

*Solution.* We define

$$f_n(x) = \begin{cases} f(x) & \text{if } |f(x)| \leq n, \\ 0 & \text{if } |f(x)| > n. \end{cases} \quad (10.147)$$

In other words,  $f_n = f\chi_{\{|f| \leq n\}}$ . Then we have  $f - f_n = f\chi_{\{|f| > n\}}$ . Since  $f$  is integrable, it's finite a.e. Therefore,  $f_n \rightarrow f$  pointwise a.e. Now,

$$|f - f_n| = |f|\chi_{\{|f| > n\}} \leq |f| \quad \text{pointwise everywhere.} \quad (10.148)$$

so  $f - f_n$  is dominated by the integrable function  $|f|$ . By [Dominated Convergence Theorem](#),

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^d} |f - f_n| \, dm = \int_{\mathbb{R}^d} \lim_{n \rightarrow \infty} |f - f_n| \, dm = 0. \quad (10.149)$$

Given  $\varepsilon > 0$ , we pick  $N$  such that

$$\int_{\mathbb{R}^d} |f - f_N| \, dm < \frac{\varepsilon}{2}. \quad (10.150)$$

We pick  $\delta = \frac{\varepsilon}{2N}$ . Now, given measurable  $E$  with  $m(E) < \delta = \frac{\varepsilon}{2N}$ ,

$$\left| \int_E f \, dm \right| \leq \int_E |f| \, dm \leq \int_E |f - f_N| \, dm + \int_E |f_N| \, dm. \quad (10.151)$$

Now, since  $|f_N(x)| \leq N$  for every  $x$ ,

$$\int_E |f_N| \, dm \leq \int_E N \, dm = N \cdot m(E) = N\delta = \frac{\varepsilon}{2}; \quad (10.152)$$

$$\int_E |f - f_N| \, dm \leq \int_{\mathbb{R}^d} |f - f_N| \, dm < \frac{\varepsilon}{2}. \quad (10.153)$$

Plugging these into (10.151), we have

$$\left| \int_E f \, dm \right| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad (10.154)$$

■

**Exercise 10.9.** Let  $\{f_n\} \subset L^1([0, 1])$  with

(i)  $f_n \rightarrow f$  a.e.,  $f \in L^1([0, 1])$ ;

(ii)  $\|f_n\|_1 \rightarrow \|f\|_1$ .

Show that  $f_n \rightarrow f$  in  $L^1([0, 1])$ .

*Solution.* We're gonna prove the following general result:

Let  $(\Omega, \Sigma, \mu)$  be a measure space, and  $f_n \in L^1(\Omega, \mu)$  be a sequence of integrable functions that converge pointwise a.e. to  $f \in L^1(\Omega, \mu)$ . Then

$$f_n \rightarrow f \text{ in } L^1(\Omega, \mu) \iff \int_{\Omega} |f_n| d\mu \rightarrow \int_{\Omega} |f| d\mu. \quad (10.155)$$

( $\Rightarrow$ ) By reverse triangle inequality in  $L^1$ ,

$$|\|f\|_1 - \|f_n\|_1| \leq \|f - f_n\|_1. \quad (10.156)$$

Since the RHS converges to 0 (by hypothesis  $f_n \rightarrow f$  in  $L^1$ ), therefore, the LHS also converges to 0, i.e.  $\int_{\Omega} |f_n| d\mu \rightarrow \int_{\Omega} |f| d\mu$ .

( $\Leftarrow$ ) We are given that  $\int_{\Omega} |f_n| d\mu \rightarrow \int_{\Omega} |f| d\mu$ . We need to show that  $\int_{\Omega} |f - f_n| d\mu \rightarrow 0$ . Note that,

$$|f(x)| + |f_n(x)| - |f(x) - f_n(x)| \geq 0, \quad (10.157)$$

by triangle inequality. Now, applying [Fatou's Lemma](#) on  $|f| + |f_n| - |f - f_n|$ ,

$$\int_{\Omega} \liminf_{n \rightarrow \infty} [|f| + |f_n| - |f - f_n|] d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} [|f| + |f_n| - |f - f_n|] d\mu. \quad (10.158)$$

$|f| + |f_n| - |f - f_n|$  converges pointwise a.e. to  $2|f|$ . So the LHS is  $\int_{\Omega} 2|f| d\mu$ . In the RHS, applying linearity and  $\int_{\Omega} |f_n| d\mu \rightarrow \int_{\Omega} |f| d\mu$ , we get

$$\text{RHS} = \liminf_{n \rightarrow \infty} \left[ \int_{\Omega} |f| d\mu + \int_{\Omega} |f_n| d\mu - \int_{\Omega} |f - f_n| d\mu \right] \quad (10.159)$$

$$= 2 \int_{\Omega} |f| d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} |f - f_n| d\mu. \quad (10.160)$$

Cancelling out  $2 \int_{\Omega} |f| d\mu$  from LHS  $\leq$  RHS, we get

$$\limsup_{n \rightarrow \infty} \int_{\Omega} |f - f_n| d\mu \leq 0. \quad (10.161)$$

Since this is a nonnegative sequence with  $\limsup 0$ , therefore, the limit must be 0, i.e.  $\|f - f_n\|_1 \rightarrow 0$ . ■

**Exercise 10.10.** Let  $B = \{f \in L^2(\mathbb{R}) : \|f\|_2 \leq 1\}$ . Let  $\{f_n\} \subset B$  be a sequence such that  $f_n \rightarrow f$  a.e. in  $\mathbb{R}$ . Show that  $f \in B$  and that

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n(x)g(x) dx = \int_{-\infty}^{\infty} f(x)g(x) dx \quad (10.162)$$

for all  $g \in L^2(\mathbb{R})$ .

*Solution.* The fact that  $f \in B$  follows from [Fatou's Lemma](#):

$$\|f\|_2^2 = \int_{\mathbb{R}} |f|^2 dm \leq \liminf_{n \rightarrow \infty} \int_{\mathbb{R}} |f_n|^2 dm \leq \liminf_{n \rightarrow \infty} 1 \leq 1. \quad (10.163)$$

Now, fix  $\varepsilon > 0$ . Since  $\int_{\mathbb{R}} |g|^2 \, dm$  is finite, we can choose  $M$  large enough so that

$$\int_{|x|>M} |g(x)|^2 \, dx \leq \frac{\varepsilon^2}{36}. \quad (10.164)$$

Now,

$$\begin{aligned} \left| \int_{\mathbb{R}} f_n g \, dm - \int_{\mathbb{R}} f g \, dm \right| &\leq \int_{\mathbb{R}} |(f_n - f)g| \, dm \\ &= \int_{[-M, M]} |(f_n - f)g| \, dm + \int_{|x|>M} |(f_n - f)g| \, dm. \end{aligned} \quad (10.165)$$

By [Hölder's inequality](#),

$$\begin{aligned} \int_{|x|>M} |(f_n - f)g| \, dm &\leq \|f_n - f\|_{L^2(\mathbb{R}-[-M, M])} \|g\|_{L^2(\mathbb{R}-[-M, M])} \\ &\leq \|f_n - f\|_{L^2(\mathbb{R})} \|g\|_{L^2(\mathbb{R}-[-M, M])} \\ &\leq \left[ \|f_n\|_{L^2(\mathbb{R})} + \|f\|_{L^2(\mathbb{R})} \right] \frac{\varepsilon}{6} \\ &\leq 2 \cdot \frac{\varepsilon}{6} = \frac{\varepsilon}{3}. \end{aligned} \quad (10.166)$$

Now, let's bound  $\int_{[-M, M]} |(f_n - f)g| \, dm$ . By [Exercise 10.8](#), since  $|g|^2 \in L^1(\mathbb{R})$ , we can find  $\delta > 0$  such that for every measurable set  $E$  with  $m(E) < \delta$ ,

$$\int_E |g|^2 \, dm \leq \frac{\varepsilon^2}{36}. \quad (10.167)$$

Since  $f_n \rightarrow f$  pointwise a.e. by [Egorov's theorem](#), there exists a measurable set  $A \subseteq [-M, M]$  such that  $m([-M, M] - A) < \delta$ , and  $f_n \rightarrow f$  uniformly on  $A$ .

$$\int_{[-M, M]} |(f_n - f)g| \, dm = \int_A |(f_n - f)g| \, dm + \int_{[-M, M]-A} |(f_n - f)g| \, dm. \quad (10.168)$$

Again by [Hölder's inequality](#),

$$\int_{[-M, M]-A} |(f_n - f)g| \, dm \leq 2 \left( \int_{[-M, M]-A} |g|^2 \, dm \right)^{\frac{1}{2}} \leq \frac{\varepsilon}{3}. \quad (10.169)$$

Since  $f_n \rightarrow f$  uniformly on  $A$ , we can choose  $N$  such that for every  $n \geq N$ ,

$$\sup_{x \in A} |f_n - f| \leq \frac{\varepsilon}{3\sqrt{2M} \|g\|_2}. \quad (10.170)$$

Now,

$$\int_A |(f_n - f)g| \, dm \leq \frac{\varepsilon}{3\sqrt{2M} \|g\|_2} \int_A |g| \, dm. \quad (10.171)$$

Again by Hölder's inequality,

$$\int_A |g| \, dm \leq \left( \int_A 1^2 \, dm \right)^{\frac{1}{2}} \left( \int_A |g|^2 \, dm \right)^{\frac{1}{2}} \leq \sqrt{m(A)} \|g\|_2 \leq \sqrt{2M} \|g\|_2. \quad (10.172)$$

Therefore,

$$\int_A |(f_n - f)g| \, dm \leq \frac{\varepsilon}{3\sqrt{2M} \|g\|_2} \cdot \sqrt{2M} \|g\|_2 = \frac{\varepsilon}{3}. \quad (10.173)$$

Now, for  $n \geq N$ ,

$$\begin{aligned} \left| \int_{\mathbb{R}} f_n g \, dm - \int_{\mathbb{R}} f g \, dm \right| &\leq \int_{[-M, M]} |(f_n - f)g| \, dm + \int_{|x| > M} |(f_n - f)g| \, dm \\ &= \int_A |(f_n - f)g| \, dm + \int_{[-M, M] - A} |(f_n - f)g| \, dm \\ &\quad + \int_{|x| > M} |(f_n - f)g| \, dm \\ &\leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned} \quad (10.174)$$

■

**Exercise 10.11.** Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  be a Lebesgue measurable function such that

$$m\{x : |f(x)| > \lambda\} \leq \frac{C}{\lambda^2} \quad (10.175)$$

for some fixed constant  $C > 0$  and all  $\lambda > 0$ . Prove that there exists a constant  $C_1$  such that for any Lebesgue measurable set  $E \subset \mathbb{R}^d$  of finite and positive measure we have

$$\int_E |f(x)| \, dx \leq C_1 \sqrt{m(E)}. \quad (10.176)$$

(Note: Here  $m$  denotes the Lebesgue measure.)

*Solution.* For  $\lambda > 0$  can divide  $\int_E |f| \, dm$  into

$$\int_E |f| \, dm = \int_{E \cap \{|f| \leq \lambda\}} |f| \, dm + \int_{E \cap \{|f| > \lambda\}} |f| \, dm. \quad (10.177)$$

In the first integral  $|f| \leq \lambda$ , where the integration is over a set with measure at most  $m(E)$ . Therefore,

$$\int_{E \cap \{|f| \leq \lambda\}} |f| \, dm \leq \lambda \, m(E). \quad (10.178)$$

In the second integral of (10.177), we rewrite the integral into

$$\int_{E \cap \{|f| > \lambda\}} |f| \, dm = \sum_{k=0}^{\infty} \int_{E \cap \{\lambda 2^k < |f| \leq \lambda 2^{k+1}\}} |f| \, dm. \quad (10.179)$$

On each summand,  $|f| \leq \lambda 2^{k+1}$ . Furthermore,

$$m\left(E \cap \left\{\lambda 2^k < |f| \leq \lambda 2^{k+1}\right\}\right) \leq m\left(\left\{\lambda 2^k < |f|\right\}\right) \leq \frac{C}{\lambda^2 2^{2k}}. \quad (10.180)$$

Hence,

$$\int_{E \cap \{\lambda 2^k < |f| \leq \lambda 2^{k+1}\}} |f| \, dm \leq \lambda 2^{k+1} \frac{C}{\lambda^2 2^{2k}} = \frac{C}{\lambda} 2^{-k+1}. \quad (10.181)$$

Summing up, we get

$$\int_{E \cap \{|f| > \lambda\}} |f| \, dm \leq \sum_{k=0}^{\infty} \frac{C}{\lambda} 2^{-k+1} = \frac{4C}{\lambda}. \quad (10.182)$$

Plugging everything into (10.177), we have

$$\int_E |f| \, dm \leq \lambda m(E) + \frac{4C}{\lambda}. \quad (10.183)$$

This is true for every  $\lambda > 0$ . So we minimize the RHS. The RHS is minimized when  $\lambda = 2\sqrt{\frac{C}{m(E)}}$ , and for this value, the RHS is  $4\sqrt{C}\sqrt{m(E)}$ . Therefore,

$$\int_E |f| \, dm \leq 4\sqrt{C}\sqrt{m(E)}, \quad (10.184)$$

as desired, where  $C_1 = 4\sqrt{C}$ . ■

**Exercise 10.12.** Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  be a measurable function in  $L^1_{\text{weak}}(\mathbb{R}^d)$ , i.e.

$$\theta = \sup_{\lambda > 0} \lambda \cdot m(\{x \in \mathbb{R}^d : |f(x)| > \lambda\}) < \infty. \quad (10.185)$$

(here  $m$  denotes the Lebesgue measure). Prove that for every  $\delta \in (0, 1)$ , and for every measurable set  $E \subset \mathbb{R}^d$ ,  $m(E) < \infty$ , we have

$$\int_E |f(x)|^\delta \, dx \leq C m(E)^{1-\delta} \theta^\delta, \quad (10.186)$$

with an appropriate constant  $C$  that is independent of  $f$  and  $E$ .

*Solution.* We proceed similarly as last problem. For  $\lambda > 0$ , we write

$$\int_E |f|^\delta \, dm = \int_{E \cap \{|f| \leq \lambda\}} |f|^\delta \, dm + \sum_{k=0}^{\infty} \int_{E \cap \{\lambda 2^k < |f| \leq \lambda 2^{k+1}\}} |f|^\delta \, dm. \quad (10.187)$$

In the first integral, the  $|f|^\delta \leq \lambda^\delta$ , and the integration is over a set of measure at most  $m(E)$ . Therefore,

$$\int_{E \cap \{|f| \leq \lambda\}} |f|^\delta \, dm \leq \lambda^\delta \cdot m(E). \quad (10.188)$$

For the second one, we'll use the weak-integrability condition. Given that for every  $\lambda > 0$ ,

$$\lambda \cdot m(\{x \in \mathbb{R}^d : |f(x)| > \lambda\}) \leq \theta \implies m(\{x \in \mathbb{R}^d : |f(x)| > \lambda\}) \leq \frac{\theta}{\lambda}. \quad (10.189)$$

In the second part of (10.187), on each summand,  $|f|^\delta \leq \lambda^\delta 2^{(k+1)\delta}$ . Furthermore,

$$m\left(E \cap \left\{\lambda 2^k < |f| \leq \lambda 2^{k+1}\right\}\right) \leq m\left(\left\{\lambda 2^k < |f|\right\}\right) \leq \frac{\theta}{\lambda 2^k}. \quad (10.190)$$

Hence,

$$\int_{E \cap \{\lambda 2^k < |f| \leq \lambda 2^{k+1}\}} |f|^\delta \, dm \leq \lambda^\delta 2^{(k+1)\delta} \frac{\theta}{\lambda 2^k} = \theta \lambda^{\delta-1} 2^\delta 2^{(\delta-1)k} \quad (10.191)$$

Plugging everything into (10.187), we get

$$\int_E |f|^\delta \, dm \leq \lambda^\delta \cdot m(E) + \theta \lambda^{\delta-1} 2^\delta \sum_{k=0}^{\infty} 2^{(\delta-1)k} = \lambda^\delta \cdot m(E) + \theta \lambda^{\delta-1} 2^\delta \frac{1}{1 - 2^{\delta-1}} \quad (10.192)$$

This is true for any  $\lambda > 0$ . So we put  $\lambda = \frac{\theta}{m(E)}$ . Then the RHS becomes

$$\left(\frac{\theta}{m(E)}\right)^\delta m(E) + \theta \left(\frac{\theta}{m(E)}\right)^{\delta-1} \frac{2^\delta}{1 - 2^{\delta-1}} = \left(1 + \frac{2^\delta}{1 - 2^{\delta-1}}\right) \theta^\delta m(E)^{1-\delta}. \quad (10.193)$$

Hence,

$$\int_E |f|^\delta \, dm \leq \left(1 + \frac{2^\delta}{1 - 2^{\delta-1}}\right) \theta^\delta m(E)^{1-\delta}, \quad (10.194)$$

as desired, with  $C = 1 + \frac{2^\delta}{1 - 2^{\delta-1}}$ , which is independent of  $f$  and  $E$ . ■

**Exercise 10.13.** We say that a sequence  $\{f_k\}$  of measurable functions in  $\mathbb{R}^d$  is *Cauchy in measure* if for every  $\varepsilon$  we have

$$m(\{x \in \mathbb{R}^d : |f_k(x) - f_\ell(x)| > \varepsilon\}) \rightarrow 0 \quad \text{as } k, \ell \rightarrow \infty. \quad (10.195)$$

We say that  $f_k$  *converges in measure* to a measurable function  $f$  if for every  $\varepsilon > 0$  we have

$$m(\{x \in \mathbb{R}^d : |f_k(x) - f(x)| > \varepsilon\}) \rightarrow 0 \quad \text{as } k \rightarrow \infty. \quad (10.196)$$

Prove the following:

- (i) If  $\{f_k\}$  is Cauchy in measure, then there is a measurable function  $f$  to which this sequence converges in measure.
- (ii) If  $f_k$  converges to  $f$  in measure, show that there exists a subsequence  $f_{n_k}$  that converges to  $f$  pointwise almost everywhere. Passing to a subsequence is really necessary here?
- (iii) If  $f_k \rightarrow f$  in  $L^p$ , then  $f_k$  converges to  $f$  in measure. Is the converse true? (say, assuming that all  $f_k$  and  $f$  are in  $L^p$ ).

*Solution.* (i) We construct a subsequence  $\{g_k\}$  such that

$$m\left(\left\{x : |g_{k+1}(x) - g_k(x)| > 2^{-k}\right\}\right) \leq 2^{-k}. \quad (10.197)$$

Suppose

$$E_k = \left\{x : |g_{k+1}(x) - g_k(x)| > 2^{-k}\right\}, \quad \text{then } \sum_{k=1}^{\infty} m(E_k) \leq \sum_{k=1}^{\infty} 2^{-k} < \infty. \quad (10.198)$$

By [Borel–Cantelli lemma](#), the set  $E$  of points which belong to infinitely many  $E_k$ 's has measure 0. Now we claim that  $\{g_k(x)\}_k$  is a Cauchy sequence for a fixed  $x \notin E$ .

Since  $x \notin E$ , there exists  $K$  such that  $x \notin E_k$  for  $k \geq K$ . Now, for  $n > m \geq K + 1$ ,

$$|g_n(x) - g_m(x)| \leq \sum_{i=m}^{n-1} |g_{i+1}(x) - g_i(x)| \leq \sum_{i=m}^{n-1} 2^{-i} \leq \sum_{i=m}^{\infty} 2^{-i} = 2^{-m+1} \leq 2^{-K}. \quad (10.199)$$

Therefore,  $\{g_k(x)\}_k$  is a Cauchy sequence. Hence, by the completeness of  $\mathbb{R}$  (or  $\mathbb{C}$ ), we can define

$$f(x) = \lim_{n \rightarrow \infty} g_n(x), \quad \text{for } x \notin E. \quad (10.200)$$

For  $x \in E$ , we can just define  $f(x) = 0$ . Now we'll prove that  $g_k \rightarrow f$  in measure. Indeed, for  $x \notin \bigcup_{n=k}^{\infty} E_n$ ,

$$|f(x) - g_k(x)| = \lim_{n \rightarrow \infty} |g_n(x) - g_k(x)| \leq 2^{-k+1}. \quad (10.201)$$

Therefore, if  $|f(x) - g_k(x)| > 2^{-k+1}$ , then  $x$  must belong to  $\bigcup_{n=k}^{\infty} E_n$ , or the nullset  $E$ . Hence,

$$\begin{aligned} m\left(\left\{x : |f(x) - g_k(x)| > 2^{-k+1}\right\}\right) &\leq m\left(A \cup \bigcup_{n=k}^{\infty} E_n\right) \\ &\leq \sum_{n=k}^{\infty} m(E_n) \\ &\leq \sum_{n=k}^{\infty} 2^{-n} \leq 2^{-k+1}. \end{aligned} \quad (10.202)$$

Hence,  $g_n \rightarrow f$  in measure. Now we claim that  $f_n \rightarrow f$  in measure. Notice that

$$|f(x) - f_n(x)| \leq |f(x) - g_n(x)| + |g_n(x) - f_n(x)|. \quad (10.203)$$

This gives

$$\{x : |f(x) - f_n(x)| \geq \varepsilon\} \subseteq \left\{x : |f(x) - g_n(x)| \geq \frac{\varepsilon}{2}\right\} \cup \left\{x : |g_n(x) - f_n(x)| \geq \frac{\varepsilon}{2}\right\}. \quad (10.204)$$

As  $n \rightarrow \infty$ , the first set's measure approaches 0, since  $g_n \rightarrow f$  in measure. As  $n \rightarrow \infty$ , the second set's measure also approaches 0, since  $f_n$  is a Cauchy sequence and  $g_n$  is a

subsequence of  $f_n$ . Therefore,

$$\lim_{n \rightarrow \infty} m(\{x : |f(x) - f_n(x)| \geq \varepsilon\}) = 0, \quad (10.205)$$

i.e.  $f_n \rightarrow f$  in measure.

- (ii) We proceed similarly as before. We construct a subsequence  $\{g_k\}$  as follows: for a fixed  $k$ , there exists  $N_k$  such that for all  $m \geq N_k$ ,

$$m\left(\left\{x : |f_m(x) - f(x)| > 2^{-k}\right\}\right) \leq 2^{-k}. \quad (10.206)$$

Then  $g_k = f_{N_k}$ . Suppose

$$E_k = \left\{x : |g_k(x) - f(x)| > 2^{-k}\right\}, \quad \text{then } \sum_{k=1}^{\infty} m(E_k) \leq \sum_{k=1}^{\infty} 2^{-k} < \infty. \quad (10.207)$$

By [Borel–Cantelli lemma](#), the set  $E$  of points which belong to infinitely many  $E_k$ ’s has measure 0. Now we claim that  $g_k \rightarrow f$  pointwise on  $\mathbb{R}^d - E$ .

For  $x \notin E$ , there exists  $K$  such that  $x \notin E_k$  for  $k \geq K$ . Then for all  $k \geq K$ ,

$$|g_k(x) - f(x)| \leq 2^{-k}. \quad (10.208)$$

Therefore,  $g_k \rightarrow f$  pointwise a.e.

Yes, passing to a subsequence is necessary. For example, consider the sequence of functions  $[0, 1] \rightarrow \mathbb{R}$  inspired from [Exercise 6.1](#):

$$\begin{aligned} f_1 &= \chi_{[0,1]} \\ f_2 &= \chi_{[0, \frac{1}{2}]}, \quad f_3 = \chi_{[\frac{1}{2}, 1]} \\ f_4 &= \chi_{[0, \frac{1}{4}]}, \quad f_5 = \chi_{[\frac{1}{4}, \frac{1}{2}]}, \quad f_6 = \chi_{[\frac{1}{2}, \frac{3}{4}]}, \quad f_7 = \chi_{[\frac{3}{4}, 1]} \\ &\dots \end{aligned}$$

In other words, for  $n = 2^k + m$ , with  $k$  being the unique number satisfying  $2^k \leq n < 2^{k+1}$ , and  $0 \leq m < 2^k$ , we define

$$f_n = \chi_{\left[\frac{m}{2^k}, \frac{m+1}{2^k}\right]}. \quad (10.209)$$

Then the pointwise sequence  $f_n(x)$  doesn’t converge for any  $x$ , since the sequence is a sequence of 0 and 1, which is not eventually constant, and hence it doesn’t converge. However, it converges to 0 in measure. Given  $\varepsilon > 0$

$$m(x : |f_n(x) - 0| > \varepsilon) = m\left(\left[\frac{m}{2^k}, \frac{m+1}{2^k}\right]\right) = \frac{1}{2^k} \rightarrow 0. \quad (10.210)$$

Therefore, a sequence of functions converging in measure may fail to converge pointwise.

(iii) Fix  $\varepsilon > 0$ . We need to show that

$$\lim_{n \rightarrow \infty} m(\{x : |f_n(x) - f(x)| > \varepsilon\}) = 0. \quad (10.211)$$

Now,

$$\begin{aligned} \|f_n - f\|_p^p &= \int_{\mathbb{R}^d} |f_n - f|^p \, dm \\ &\geq \int_{\{x : |f_n(x) - f(x)| > \varepsilon\}} |f_n - f|^p \, dm \\ &\geq \int_{\{x : |f_n(x) - f(x)| > \varepsilon\}} \varepsilon^p \, dm \\ &= \varepsilon^p m(\{x : |f_n(x) - f(x)| > \varepsilon\}). \end{aligned}$$

In other words,

$$m(\{x : |f_n(x) - f(x)| > \varepsilon\}) \leq \frac{\|f_n - f\|_p^p}{\varepsilon^p}. \quad (10.212)$$

$\|f_n - f\|_p^p \rightarrow 0$  as  $n \rightarrow \infty$ . Therefore,

$$\lim_{n \rightarrow \infty} m(\{x : |f_n(x) - f(x)| > \varepsilon\}) = 0, \quad (10.213)$$

i.e.  $f_n \rightarrow f$  in measure.

No, the converse is not true, in general. Suppose

$$f_n = n^{\frac{1}{p}} \chi_{(0, \frac{1}{n})}. \quad (10.214)$$

Then  $f_n \not\rightarrow 0$  in  $L^p$ , since

$$\|f_n - 0\|^p = \int_{\mathbb{R}} \left[ n^{\frac{1}{p}} \chi_{(0, \frac{1}{n})} \right]^p \, dm = n \cdot m\left(\left(0, \frac{1}{n}\right)\right) = 1. \quad (10.215)$$

However,  $f_n \rightarrow 0$  in measure. Because,

$$m(x : |f_n(x) - 0| > \varepsilon) = m\left(\left(0, \frac{1}{n}\right)\right) = \frac{1}{n} \rightarrow 0. \quad (10.216)$$

■

**Exercise 10.14.** Assume that  $(\Omega, \Sigma, \mu)$  is a measure space and  $f \in L^1(\Omega, \Sigma, \mu)$  satisfies  $\|f\|_1 = 1$ . Prove that

$$\int_E \log |f(x)| \, d\mu(x) \leq -\mu(E) \log \mu(E) \quad (10.217)$$

for any measurable subset  $E$  such that  $0 < \mu(E) < \infty$ .

*Solution.* If  $f = 0$  in a positive measure set, then  $\log |f(x)| = -\infty$  on a positive measure set, implying that the LHS is  $-\infty$ , in which case the inequality is satisfied trivially. So we can

assume  $|f| > 0$  a.e. We shall use the bound

$$\log x \leq x - 1, \quad \text{for } x > 0. \quad (10.218)$$

As a result,

$$\log(|f(x)|\mu(E)) \leq |f(x)|\mu(E) - 1 \implies \log|f(x)| \leq |f(x)|\mu(E) - 1 - \log(\mu(E)). \quad (10.219)$$

Integrating both sides, we get

$$\int_E \log|f| \, d\mu \leq \mu(E) \int_E |f| \, d\mu - \int_E d\mu - \log(\mu(E)) \int_E d\mu \quad (10.220)$$

$$\leq \mu(E) \int_E |f| \, d\mu - \mu(E) - \mu(E) \log(\mu(E)). \quad (10.221)$$

Now,

$$\int_E |f| \, d\mu \leq \int_\Omega |f| \, d\mu = 1, \quad (10.222)$$

so we have

$$\int_E \log|f| \, d\mu \leq \mu(E) - \mu(E) - \mu(E) \log(\mu(E)) = -\mu(E) \log(\mu(E)). \quad (10.223)$$

■

**Exercise 10.15.** Let  $\{f_n\}_{n=1}^\infty \subset L^2([0, 1])$ . Assume that  $\sup_n \|f_n\|_2 < \infty$  and let  $1 \leq p < 2$ . Let  $f : [0, 1] \rightarrow \mathbb{R}$  and assume that  $f_n \rightarrow f$  pointwise a.e. Prove that

$$\lim_{n \rightarrow \infty} \|f_n - f\|_p = 0. \quad (10.224)$$

Give an example to show that this is false for  $p = 2$ .

*Solution.* Suppose  $M := \sup_n \|f_n\|_2$ . By [Fatou's Lemma](#), since  $|f_n|^2 \rightarrow |f|^2$  pointwise a.e.

$$\int_{[0,1]} |f|^2 \, dm \leq \liminf_{n \rightarrow \infty} \int_{[0,1]} |f_n|^2 \, dm \leq \liminf_{n \rightarrow \infty} M^2 \leq M^2, \quad (10.225)$$

proving that  $\|f\|_2 \leq M$ . Therefore,

$$\|f_n - f\|_2 \leq \|f_n\|_2 + \|f\|_2 \leq 2M. \quad (10.226)$$

Now, fix  $\varepsilon > 0$ . By [Egorov's theorem](#), there exists a measurable subset  $E \subseteq [0, 1]$  such that  $f_n \rightarrow f$  uniformly on  $[0, 1] - E$ , with

$$m(E) \leq \delta := \left( \frac{\varepsilon^2}{4(2M)^{2p}} \right)^{\frac{1}{2-p}}. \quad (10.227)$$

Since  $f_n \rightarrow f$  uniformly on  $[0, 1] - E$ , there exists  $N$  such that for  $n \geq N$ ,

$$\sup_{x \in [0,1] - E} |f_n(x) - f(x)| \leq \left(\frac{\varepsilon}{2}\right)^{\frac{1}{p}}. \quad (10.228)$$

Now, for  $n \geq N$ ,

$$\|f_n - f\|_p^p = \int_{[0,1]} |f_n - f|^p \, dm = \int_{[0,1] - E} |f_n - f|^p \, dm + \int_E |f_n - f|^p \, dm. \quad (10.229)$$

On  $[0, 1] - E$ , the integrand  $|f_n - f|^p$  is at most  $\frac{\varepsilon}{2}$ , and the integration is over a set of measure at most 1. Therefore

$$\int_{[0,1] - E} |f_n - f|^p \, dm \leq \frac{\varepsilon}{2}. \quad (10.230)$$

By [Hölder's inequality](#) on  $\frac{2}{p}$  and its conjugate exponent  $s = \frac{2}{2-p}$ ,

$$\begin{aligned} \int_E |f_n - f|^p \, dm &\leq \left( \int_E (|f_n - f|^p)^{\frac{2}{p}} \, dm \right)^{\frac{p}{2}} \left( \int_E 1^{\frac{2}{2-p}} \, dm \right)^{\frac{2-p}{2}} \\ &\leq \left( \int_{[0,1]} |f_n - f|^2 \, dm \right)^{\frac{p}{2}} [m(E)]^{\frac{2-p}{2}} \\ &\leq ((2M)^2)^{\frac{p}{2}} \delta^{\frac{2-p}{2}} \\ &\leq (2M)^p \left( \frac{\varepsilon^2}{4(2M)^{2p}} \right)^{\frac{1}{2}} = \frac{\varepsilon}{2}. \end{aligned} \quad (10.231)$$

Plugging everything into (10.229), we have

$$\|f_n - f\|_p^p \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \quad \text{for } n \geq N. \quad (10.232)$$

Hence,  $\lim_{n \rightarrow \infty} \|f_n - f\|_p^p = 0$ .

For  $p = 2$ , consider the following example:

$$f_n = \sqrt{n} \chi_{(0, \frac{1}{n})}. \quad (10.233)$$

Then

$$\int_{[0,1]} |f_n|^2 \, dm = n \cdot m\left(\left(0, \frac{1}{n}\right)\right) = 1, \quad (10.234)$$

so that  $f_n \in L^2([0, 1])$  with  $\|f_n\|_2 = 1$ , in particular,  $\sup_n \|f_n\|_2 = 1 < \infty$ . Now,  $f_n \rightarrow 0$  pointwise, but  $\|f_n - 0\|_2 = 1$ , so  $\lim_{n \rightarrow \infty} \|f_n - 0\|_2 \neq 0$ . ■

## §11 28 September 2026

Throughout,  $(\Omega, \Sigma, \mu)$  is a measure space. For  $1 \leq p \leq \infty$ , the *conjugate exponent*  $p'$  is defined by

$$\frac{1}{p} + \frac{1}{p'} = 1, \quad \text{i.e.} \quad p' = \frac{p}{p-1}. \quad (11.1)$$

Recall that  $L^p$  is a Banach space. We assume that  $\mu$  is  $\sigma$ -finite, that is,

$$\Omega = \bigcup_{n=1}^{\infty} \Omega_n, \quad \mu(\Omega_n) < \infty \text{ for all } n. \quad (11.2)$$

For example, Lebesgue measure on  $\mathbb{R}^d$  is  $\sigma$ -finite. This hypothesis is needed in the case  $p = \infty$  below.

### Proposition 11.1 (Duality of the norm)

Let  $f \in L^p(\Omega)$  with  $1 \leq p \leq \infty$ . Then

$$\|f\|_p = \sup \left\{ \left| \int_{\Omega} fg \, d\mu \right| : g \in L^{p'}(\Omega), \|g\|_{p'} \leq 1 \right\}. \quad (11.3)$$

(For real-valued functions the absolute value may be dropped.) Moreover, if  $1 \leq p < \infty$ , the supremum is always attained, so it is a maximum.

This is the statement, in the language of norms, that  $L^{p'}$  is the dual space of  $L^p$ .

*Proof.* By Hölder's inequality, if  $g \in L^{p'}$  with  $\|g\|_{p'} \leq 1$ , then

$$\left| \int_{\Omega} fg \, d\mu \right| \leq \|f\|_p \|g\|_{p'} \leq \|f\|_p. \quad (11.4)$$

Hence the supremum is at most  $\|f\|_p$ .

Now suppose  $1 \leq p < \infty$ . We look for a good  $g$ , one that “resonates” with the sign of  $f$  and makes  $fg$  comparable to  $|f|^p$ . Since  $fg$  should carry one power of  $f$  and we want  $|f|^p$ , we take  $|f|^{p-1}$  times the sign of  $f$ . Define

$$g(x) = \frac{\operatorname{sgn}(f(x)) |f(x)|^{p-1}}{\|f\|_p^{p-1}}, \quad (11.5)$$

where

$$\operatorname{sgn}(t) = \begin{cases} 1 & \text{if } t > 0, \\ -1 & \text{if } t < 0, \\ 0 & \text{if } t = 0. \end{cases} \quad (11.6)$$

For complex case, we just define

$$\operatorname{sgn}(z) = \begin{cases} \frac{|z|}{z} & \text{if } z \neq 0, \\ 0 & \text{if } z = 0. \end{cases} \quad (11.7)$$

**Claim 11.1:**  $g \in L^{p'}$  and  $\|g\|_{p'} = 1$ .

*Proof.* Since  $p' = \frac{p}{p-1}$ , we have  $(p-1)p' = p$ , so

$$\int_{\Omega} |g|^{p'} d\mu = \frac{1}{\|f\|_p^{(p-1)p'}} \int_{\Omega} |f|^{(p-1)p'} d\mu = \frac{1}{\|f\|_p^p} \int_{\Omega} |f|^p d\mu = 1. \quad (11.8)$$

In particular the integral is finite because  $f \in L^p$ .  $\square$

**Claim 11.2:**  $g$  attains the supremum.

*Proof.* Since  $f \operatorname{sgn}(f) = |f|$ ,

$$\int_{\Omega} fg d\mu = \frac{1}{\|f\|_p^{p-1}} \int_{\Omega} |f| |f|^{p-1} d\mu = \frac{\|f\|_p^p}{\|f\|_p^{p-1}} = \|f\|_p. \quad (11.9)$$

$\square$

This proves the proposition for  $1 \leq p < \infty$ , and shows the supremum is attained.

*The case  $p = \infty$  (so  $p' = 1$ ).* Let  $\varepsilon > 0$  and set

$$X_{\varepsilon} = \{x \in \Omega : |f(x)| \geq \|f\|_{\infty} - \varepsilon\}. \quad (11.10)$$

By the definition of the essential supremum,  $\mu(X_{\varepsilon}) > 0$  (otherwise  $\|f\|_{\infty} - \varepsilon$  would be a smaller essential upper bound). However,  $\mu(X_{\varepsilon})$  could be  $+\infty$ . This is where  $\sigma$ -finiteness is used: writing

$$X_{\varepsilon} = \bigcup_n (X_{\varepsilon} \cap \Omega_n), \quad (11.11)$$

at least one piece has positive measure, otherwise  $\mu(X_{\varepsilon}) = 0$ . So there is a set  $Y_{\varepsilon} \subseteq X_{\varepsilon}$  with  $0 < \mu(Y_{\varepsilon}) < \infty$ . Define

$$g = \frac{\operatorname{sgn}(f) \chi_{Y_{\varepsilon}}}{\mu(Y_{\varepsilon})}. \quad (11.12)$$

Then  $\|g\|_1 = 1$ , since

$$\int_{\Omega} |g| d\mu = \frac{1}{\mu(Y_{\varepsilon})} \int_{\Omega} |\operatorname{sgn}(f)| \chi_{Y_{\varepsilon}} d\mu = \frac{1}{\mu(Y_{\varepsilon})} \int_{\Omega} \chi_{Y_{\varepsilon}} d\mu = 1. \quad (11.13)$$

Then

$$\int_{\Omega} fg d\mu = \frac{1}{\mu(Y_{\varepsilon})} \int_{Y_{\varepsilon}} |f| d\mu \geq \|f\|_{\infty} - \varepsilon. \quad (11.14)$$

Since  $\varepsilon > 0$  was arbitrary, the supremum equals  $\|f\|_\infty$ . ■

Note that, the whole proof works if we replace  $\|g\|_{p'} \leq 1$  with  $\|g\|_{p'} = 1$ . Hence,

$$\|f\|_p = \sup_{\substack{g \in L^{p'}(\Omega), \\ \|g\|_{p'}=1}} \left| \int_{\Omega} fg \, d\mu \right|. \quad (11.15)$$

**Remark 11.1.** The supremum need not be attained for  $p = \infty$ . If  $|f|$  has a plateau at height  $\|f\|_\infty$  (on a set of positive finite measure), one can take  $g$  supported on that plateau and the supremum is attained. The opposite situation is a function that reaches its maximum at a single point, for instance  $f(x) = (1 - |x|)^+$  on  $\mathbb{R}$  with  $\|f\|_\infty = 1$ .

To have the supremum attained, we need to have  $\int_{\mathbb{R}} fg \, dm = 1$ , with  $\|g\|_1 \leq 1$ . Now, notice that since  $f < 1$  a.e., we have

$$1 = \left| \int_{\mathbb{R}} fg \, dm \right| \leq \int_{\mathbb{R}} |f| |g| \, dm < \int_{\mathbb{R}} |g| \, dm = \|g\|_1 \leq 1, \quad (11.16)$$

which is a contradiction! Therefore, the supremum need not be attained for  $p = \infty$ .

**Remark 11.2** (Compactness of the unit ball; weak topologies). The unit ball in an infinite-dimensional normed space is not compact in the norm topology. For example, on  $\Omega = \mathbb{N}$  with counting measure,  $L^p = \ell^p$  is the space of sequences  $(a_n)$  with  $\sum |a_n|^p < \infty$ . Let  $e_k$  be the sequence with 1 in position  $k$  and 0 elsewhere. Then  $\|e_k\|_p = 1$  and for  $j \neq k$ ,

$$\|e_j - e_k\|_p = 2^{1/p} > 0, \quad (11.17)$$

so  $(e_k)$  has no convergent subsequence. Since  $\ell^p$  is a metric space, compactness is equivalent to sequential compactness, so the closed unit ball is not compact. (Compact means every open cover has a finite subcover; “closed and bounded” is a property of compact sets in  $\mathbb{R}^d$ , not the definition.)

This motivates *weak topologies*: a topology with fewer open sets than the norm (strong) topology, hence more compact sets, but still meaningful. This is useful, for example, for producing solutions of PDEs by minimizing a functional: one needs a minimizing sequence to have a convergent subsequence. This will be discussed in a functional analysis course.

**Definition 11.1.** Let  $f : \Omega \rightarrow \mathbb{R}$  (or  $\mathbb{C}$ ) be measurable. For  $\lambda > 0$  define the **super-level sets**

$$E_\lambda = \{x \in \Omega : |f(x)| > \lambda\}. \quad (11.18)$$

If  $\lambda_1 > \lambda_2$ , then  $E_{\lambda_1} \subseteq E_{\lambda_2}$ . The function  $\lambda \mapsto \mu(E_\lambda)$  is called the **distribution function** of  $f$ .

The distribution function encodes much of the information about  $f$ . For instance, let  $f \in L^p$  with  $1 \leq p < \infty$ . On  $E_\lambda$  we have  $|f| > \lambda$ , so

$$\int_{\Omega} |f|^p d\mu \geq \int_{E_\lambda} |f|^p d\mu \geq \lambda^p \mu(E_\lambda). \quad (11.19)$$

Hence,

$$\mu(E_\lambda) \leq \frac{\|f\|_p^p}{\lambda^p} \quad \text{for all } \lambda > 0. \quad (11.20)$$

So the super-level sets cannot be large when  $\|f\|_p$  is fixed. This is called Chebyshev's inequality. (This is the generalization of Markov's inequality in probability language.)

**Proposition 11.2** (Chebyshev's inequality)

Let  $f \in L^p$  with  $1 \leq p < \infty$ . Then

$$\mu(\{x \in \Omega : |f(x)| > \lambda\}) \leq \frac{\|f\|_p^p}{\lambda^p} \quad \text{for all } \lambda > 0. \quad (11.21)$$

This motivates the definition of weak  $L^p$  spaces.

**Definition 11.2** (Weak  $L^p$ ). Let  $1 \leq p < \infty$ . The space  $L_{\text{weak}}^p$  (also written  $L^{p,\infty}$ ) is the set of measurable functions  $f : \Omega \rightarrow \mathbb{R}$  (or  $\mathbb{C}$ ) for which there exists a constant  $C > 0$ , depending on  $f$  (and  $p$ ) but *not* on  $\lambda$ , such that

$$\mu(\{x \in \Omega : |f(x)| > \lambda\}) \leq \frac{C}{\lambda^p} \quad \text{for all } \lambda > 0. \quad (11.22)$$

**Remark 11.3.** By (11.20), every  $f \in L^p$  satisfies this with  $C = \|f\|_p^p$ , so

$$L^p \subset L_{\text{weak}}^p. \quad (11.23)$$

The inclusion is strict. Let  $f(x) = |x|^{-d}$  on  $\mathbb{R}^d$ . Then

$$E_\lambda = \{x : |x|^{-d} > \lambda\} = \{x : |x| < \lambda^{-1/d}\}, \quad (11.24)$$

a ball of radius  $\lambda^{-1/d}$ , so

$$\mu(E_\lambda) = \omega_d \left( \lambda^{-1/d} \right)^d = \frac{\omega_d}{\lambda}, \quad (11.25)$$

where  $\omega_d$  is the volume of the unit ball in  $\mathbb{R}^d$ . Hence  $f \in L_{\text{weak}}^1(\mathbb{R}^d)$ .

But  $f \notin L^1(\mathbb{R}^d)$ . In polar coordinates,

$$\int_{|x|<1} \frac{dx}{|x|^d} \sim \int_0^1 \frac{1}{r^d} r^{d-1} dr = \int_0^1 \frac{dr}{r} = \infty, \quad \int_{|x|>1} \frac{dx}{|x|^d} \sim \int_1^\infty \frac{dr}{r} = \infty. \quad (11.26)$$

So  $|x|^{-d}$  is exactly the threshold function for integrability: it fails to be integrable both near 0 and near  $\infty$ , but only barely.

For  $\alpha > 0$ , on  $\mathbb{R}^d$ ,

$$\int_{|x|<1} \frac{dx}{|x|^\alpha} < \infty \iff \alpha < d, \quad \int_{|x|>1} \frac{dx}{|x|^\alpha} < \infty \iff \alpha > d. \quad (11.27)$$

For a function that is continuous away from the origin, to decide whether  $f \in L^p$  one only needs its asymptotic behavior near 0 and near  $\infty$ .

**Example 11.1**

On  $\mathbb{R}^3$  let  $f(x) = \frac{1}{|x|(1+|x|)}$ . Near 0,  $f \sim |x|^{-1}$ , so  $f^p \sim |x|^{-p}$  is integrable near 0 iff  $p < 3$ . Near  $\infty$ ,  $f \sim |x|^{-2}$ , so  $f^p \sim |x|^{-2p}$  is integrable near  $\infty$  iff  $2p > 3$ . Hence

$$f \in L^p(\mathbb{R}^3) \iff \frac{3}{2} < p < 3. \quad (11.28)$$

**Remark 11.4.** Power functions are not the only threshold. On  $[2, \infty)$ ,

$$\int \frac{dx}{x \log x} = \log \log x \quad (11.29)$$

diverges, while

$$\int \frac{dx}{x(\log x)^{1+\varepsilon}} < \infty \quad (11.30)$$

**Proposition 11.3** (The layer-cake formula)

Let  $f \in L^p$ ,  $1 \leq p < \infty$ .

$$\|f\|_p^p = \int_0^\infty p t^{p-1} \mu(E_t) dt. \quad (11.31)$$

*Proof.* Let  $f \in L^p$ ,  $1 \leq p < \infty$ . Since

$$|f(x)|^p = \int_0^{|f(x)|} p t^{p-1} dt, \quad (11.32)$$

we can write

$$\|f\|_p^p = \int_\Omega |f|^p d\mu = \int_\Omega \int_0^\infty p t^{p-1} \chi_{\{t < |f(x)|\}} dt d\mu. \quad (11.33)$$

The integrand is nonnegative and measurable, so by Fubini's theorem (interchanging integrals of nonnegative functions is always allowed, though the value may be  $+\infty$ , we'll prove this result soon),

$$\|f\|_p^p = \int_\Omega |f|^p d\mu = \int_0^\infty p t^{p-1} \int_\Omega \chi_{\{t < |f(x)|\}} d\mu dt = \int_0^\infty p t^{p-1} \mu(E_t) dt. \quad (11.34)$$

■

Consequently, the norm of  $f$  is determined by its distribution function, and two functions with

the same distribution function have the same  $L^p$  norm. Similarly, for a measurable set  $E \subset \Omega$ ,

$$\int_E |f|^p d\mu = \int_0^\infty p t^{p-1} \mu(E_t \cap E) dt. \quad (11.35)$$

**Definition 11.3.** Let  $(a, b)$  be an interval (one may have  $a = -\infty$  and/or  $b = \infty$ ). A function  $\varphi : (a, b) \rightarrow \mathbb{R}$  is said to be **convex** if

$$\varphi((1 - \lambda)x + \lambda y) \leq (1 - \lambda)\varphi(x) + \lambda\varphi(y) \quad (11.36)$$

for all  $x, y \in (a, b)$  and  $0 \leq \lambda \leq 1$ .

Looking at the graph of the function, the convexity condition says that, if  $x < t < y$ , then the point  $(t, \varphi(t))$  lies below or on the line connecting the points  $(x, \varphi(x))$  and  $(y, \varphi(y))$  in the plane. This is equivalent to saying that

$$\frac{\varphi(t) - \varphi(s)}{t - s} \leq \frac{\varphi(u) - \varphi(t)}{u - t} \quad (11.37)$$

whenever  $a < s < t < u < b$ . By [Exercise 10.1](#), if  $\varphi : (a, b) \rightarrow \mathbb{R}$  is convex, then it is continuous.

**Theorem 11.4 (Jensen's inequality)**

Let  $(\Omega, \Sigma, \mu)$  be a measure space with  $\mu(\Omega) = 1$  (i.e.  $\mu$  is a probability measure). Let  $f \in L^1$  be a real-valued function such that  $a < f(x) < b$  for all  $x \in \Omega$  (here you may have  $a = -\infty$  and/or  $b = \infty$ ). If  $\varphi : (a, b) \rightarrow \mathbb{R}$  is convex, then

$$\varphi\left(\int_\Omega f d\mu\right) \leq \int_\Omega (\varphi \circ f) d\mu. \quad (11.38)$$

*Proof.* Let  $t = \int_\Omega f d\mu$ . Then since  $a < f(x) < b$  everywhere, we have

$$\int_\Omega a d\mu < \int_\Omega f d\mu < \int_\Omega b d\mu \implies a < t < b, \quad (11.39)$$

since  $\mu(\Omega) = 1$ . Let

$$\beta = \sup_{s \in (a, t)} \frac{\varphi(t) - \varphi(s)}{t - s}. \quad (11.40)$$

Then since  $\frac{\varphi(t) - \varphi(s)}{t - s} \leq \frac{\varphi(u) - \varphi(t)}{u - t}$  for every  $a < s < t < u < b$ , taking supremum on the LHS, we have

$$\beta \leq \frac{\varphi(u) - \varphi(t)}{u - t}, \quad \text{for every } u \in (t, b). \quad (11.41)$$

Now we claim that for any  $a < s < b$  we have

$$\varphi(s) \geq \varphi(t) + \beta(s - t). \quad (11.42)$$

Indeed, it's true for  $s = t$ . If  $s < t$ ,

$$\frac{\varphi(t) - \varphi(s)}{t - s} \leq \beta, \quad (11.43)$$

since  $\beta$  is the supremum. Simplifying gives

$$\varphi(t) - \varphi(s) \leq \beta(t - s) \implies \varphi(s) \geq \varphi(t) + \beta(s - t). \quad (11.44)$$

If  $s > t$ , then by (11.41),

$$\beta \leq \frac{\varphi(s) - \varphi(t)}{s - t} \implies \varphi(s) \geq \varphi(t) + \beta(s - t). \quad (11.45)$$

Hence (11.42) holds! Now, taking  $s = f(x)$ , we have

$$\varphi(f(x)) - \varphi(t) - \beta(f(x) - t) \geq 0 \quad (11.46)$$

for every  $x \in \Omega$ . Since  $\varphi$  is continuous,  $\varphi \circ f$  is measurable. Integrating both sides of (11.46),

$$\int_{\Omega} (\varphi \circ f) \, d\mu - \varphi(t) \int_{\Omega} 1 \, d\mu - \beta \left( \int_{\Omega} f \, d\mu - t \right) \geq 0. \quad (11.47)$$

Since  $t = \int_{\Omega} f \, d\mu$ , and  $\mu(\Omega) = 1$ , we have

$$\int_{\Omega} (\varphi \circ f) \, d\mu - \varphi \left( \int_{\Omega} f \, d\mu \right) \geq 0, \quad (11.48)$$

as desired. ■

## §12 30 September 2026

Today we review inner product spaces and Hilbert spaces and prove the existence of orthogonal projections onto closed subspaces. Throughout, vector spaces are over  $\mathbb{C}$ . (Over  $\mathbb{R}$  the same statements and proofs hold, with all complex conjugations removed and  $\text{Re}$  dropped.)

**Definition 12.1** (Inner product). Let  $H$  be a vector space over  $\mathbb{C}$ . An **inner product** on  $H$  is a function  $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{C}$  such that for all  $x, y, z \in H$  and  $\lambda \in \mathbb{C}$ :

1. (conjugate symmetry)  $\langle x, y \rangle = \overline{\langle y, x \rangle}$ ;
2. (linearity in the first variable)  $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$  and  $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$ ;
3. (positive definiteness)  $\langle x, x \rangle \geq 0$ , and  $\langle x, x \rangle = 0$  if and only if  $x = 0$ .

Note that  $\langle x, x \rangle$  is real by (1), so (3) makes sense. By (1) and (2), the inner product is conjugate-linear in the second variable:

$$\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle, \quad \langle x, \lambda y \rangle = \bar{\lambda} \langle x, y \rangle. \quad (12.1)$$

Thus a constant  $\lambda$  comes out of the first variable as  $\lambda$  and out of the second variable as  $\bar{\lambda}$ .

### Example 12.1

For  $f, g \in L^2(\Omega)$  (complex-valued), set

$$\langle f, g \rangle = \int_{\Omega} f \bar{g} \, d\mu. \quad (12.2)$$

This satisfies all the axioms. (Over  $\mathbb{R}$  one simply takes  $\int_{\Omega} f g \, d\mu$ .)

Every inner product induces a norm.

$$\|x\| = \sqrt{\langle x, x \rangle}, \quad x \in H. \quad (12.3)$$

### Theorem 12.1 (Cauchy–Schwarz inequality)

Let  $H$  be a vector space with an inner product. Then for all  $x, y \in H$ ,

$$|\langle x, y \rangle| \leq \|x\| \|y\|. \quad (12.4)$$

The inequality is named after Hermann Schwarz (no “t”), the German mathematician. The Frenchman Laurent Schwartz, with a “t”, is the one of Schwartz space and Schwartz distributions.

*Proof.* Choose  $\omega \in \mathbb{C}$  with  $|\omega| = 1$  such that

$$\omega \langle x, y \rangle = |\langle x, y \rangle|, \quad (12.5)$$

(if  $\langle x, y \rangle = 0$  any  $\omega$  works). Then

$$\langle \omega x, y \rangle = \omega \langle x, y \rangle = |\langle x, y \rangle| \in \mathbb{R}, \quad (12.6)$$

and  $\|\omega x\| = \|x\|$ . For every real  $t$ , positivity of the inner product gives

$$0 \leq \|\omega x + ty\|^2 = \langle \omega x, \omega x \rangle + t \langle \omega x, y \rangle + t \langle y, \omega x \rangle + t^2 \langle y, y \rangle \quad (12.7)$$

$$= \|x\|^2 + 2t \operatorname{Re} \langle \omega x, y \rangle + t^2 \|y\|^2 \quad (12.8)$$

$$= \|x\|^2 + 2t |\langle x, y \rangle| + t^2 \|y\|^2. \quad (12.9)$$

Here we used  $\langle y, \omega x \rangle = \overline{\langle \omega x, y \rangle}$ . The right-hand side is a quadratic polynomial in the real variable  $t$  which is nonnegative for all  $t \in \mathbb{R}$ . If  $y = 0$  the claim is trivial. If  $y \neq 0$ , the polynomial  $at^2 + bt + c$  with  $a = \|y\|^2 > 0$ ,  $b = 2|\langle x, y \rangle|$ ,  $c = \|x\|^2$  is nonnegative everywhere, so its discriminant is nonpositive:

$$4|\langle x, y \rangle|^2 - 4\|x\|^2\|y\|^2 \leq 0. \quad (12.10)$$

Hence  $|\langle x, y \rangle|^2 \leq \|x\|^2\|y\|^2$ , and taking square roots gives the claim.  $\blacksquare$

### Proposition 12.2

Let  $H$  be a vector space with an inner product. Then  $\|x\| = \sqrt{\langle x, x \rangle}$  is a norm on  $H$ .

*Proof. Positive Definiteness.*  $\|x\| = 0$  if and only if  $\langle x, x \rangle = 0$  if and only if  $x = 0$ .

*Homogeneity.* For  $\lambda \in \mathbb{C}$ ,

$$\|\lambda x\|^2 = \langle \lambda x, \lambda x \rangle = \lambda \bar{\lambda} \langle x, x \rangle = |\lambda|^2 \|x\|^2, \quad (12.11)$$

so  $\|\lambda x\| = |\lambda| \|x\|$ .

*Triangle inequality.* Expanding and using Cauchy–Schwarz,

$$\|x + y\|^2 = \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2 \quad (12.12)$$

$$= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \quad (12.13)$$

$$\leq \|x\|^2 + 2|\langle x, y \rangle| + \|y\|^2 \quad (12.14)$$

$$\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2. \quad (12.15)$$

Taking square roots gives  $\|x + y\| \leq \|x\| + \|y\|$ .  $\blacksquare$

**Definition 12.2** (Hilbert space). A **Hilbert space** is a vector space  $H$  over  $\mathbb{C}$  with an inner product which is complete with respect to the induced norm  $\|x\| = \sqrt{\langle x, x \rangle}$ . In other words, it is a Banach space whose norm comes from an inner product.

**Proposition 12.3** (Parallelogram law)

Let  $H$  be an inner product space. For all  $x, y \in H$ ,

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2. \quad (12.16)$$

*Proof.* Expanding as above,

$$\|x + y\|^2 = \|x\|^2 + 2\operatorname{Re} \langle x, y \rangle + \|y\|^2, \quad (12.17)$$

$$\|x - y\|^2 = \|x\|^2 - 2\operatorname{Re} \langle x, y \rangle + \|y\|^2. \quad (12.18)$$

Adding the two identities, the  $\pm 2\operatorname{Re} \langle x, y \rangle$  terms cancel. ■

**Remark 12.1.** Conversely, a norm satisfying the parallelogram law comes from an inner product, given by the polarization identity

$$\langle x, y \rangle = \frac{1}{4} \sum_{k=0}^3 i^k \|x + i^k y\|^2. \quad (12.19)$$

**Example 12.2**

For any measure space  $(\Omega, \Sigma, \mu)$ ,  $L^2(\Omega)$  with

$$\langle f, g \rangle = \int_{\Omega} f \bar{g} \, d\mu \quad (12.20)$$

is a Hilbert space: completeness was shown earlier, and the norm

$$\|f\|_2 = \left( \int_{\Omega} |f|^2 \, d\mu \right)^{\frac{1}{2}} \quad (12.21)$$

is the one induced by the inner product. In particular,  $\mathbb{C}^d$  is a Hilbert space.

Take  $\Omega = \mathbb{N}$ , with  $\Sigma$  the set of all subsets of  $\mathbb{N}$  and  $\mu$  the counting measure. Then  $L^2(\mathbb{N}) = \ell^2$  is the space of sequences  $a = (a_1, a_2, \dots)$  with  $\sum_n |a_n|^2 < \infty$  (a function  $\mathbb{N} \rightarrow \mathbb{C}$  is just a sequence), with

$$\langle a, b \rangle = \sum_{n=1}^{\infty} a_n \bar{b}_n, \quad \|a\| = \left( \sum_{n=1}^{\infty} |a_n|^2 \right)^{\frac{1}{2}}. \quad (12.22)$$

This is an infinite-dimensional analogue of  $\mathbb{C}^d$ . It is the prototype: every separable Hilbert space is isomorphic to  $\ell^2$ , as will be seen in a functional analysis course.

**Remark 12.2** (The unit ball is not compact). In  $\ell^2$ , let  $e_k$  be the sequence with 1 in position  $k$  and 0 elsewhere. Then  $\|e_k\| = 1$  and  $\|e_j - e_k\| = \sqrt{2}$  for  $j \neq k$ , so  $(e_k)$  has no convergent subsequence. Since  $\ell^2$  is a metric space, compactness is equivalent to sequential compactness, so the closed unit ball of  $\ell^2$  is not compact. The same holds in any infinite-dimensional Hilbert space.

A subspace  $M \subseteq H$  is a subset which is itself a vector space (closed under sums and scalar multiplication). It is **closed** if it is closed in the norm topology, i.e.  $M = \overline{M}$ .

In finite dimensions every subspace is closed. In infinite dimensions this fails.

### Example 12.3

Let  $c_{00} \subseteq \ell^2$  be the set of sequences with only finitely many nonzero entries. It is a subspace, and it contains every  $e_j$ . Any  $a \in \ell^2$  is the limit of its truncations

$$(a_1, \dots, a_N, 0, 0, \dots) \in c_{00}, \quad (12.23)$$

because

$$\sum_{n>N} |a_n|^2 \rightarrow 0. \quad (12.24)$$

So

$$\overline{c_{00}} = \ell^2 \neq c_{00}. \quad (12.25)$$

for instance the sequence  $a_n = \frac{1}{n}$  is in  $\ell^2$ , but not in  $c_{00}$ . Thus  $c_{00}$  is not closed.

Let  $\ell^1 \subseteq \ell^2$  be the summable sequences. Indeed, if  $\sum |a_n| < \infty$ , then  $|a_n| < 1$  for all large  $n$ , so  $|a_n|^2 \leq |a_n|$  eventually and

$$\sum_{n=1}^{\infty} |a_n|^2 < \infty. \quad (12.26)$$

It is a subspace containing all  $e_j$ , so it is dense in  $\ell^2$ , but it is not all of  $\ell^2$  since the sequence  $a_n = \frac{1}{n}$  is in  $\ell^2$ , but not in  $\ell^1$ . Thus  $\ell^1$  is not closed in  $\ell^2$ .

**Definition 12.3** (Orthogonality). Two vectors  $x, y \in H$  are **orthogonal**, written  $x \perp y$ , if  $\langle x, y \rangle = 0$ . For a subspace  $M \subseteq H$ , the **orthogonal complement** is

$$M^\perp = \{x \in H : \langle x, m \rangle = 0 \text{ for all } m \in M\}. \quad (12.27)$$

### Lemma 12.4

$M^\perp$  is a closed subspace of  $H$ , and  $M \cap M^\perp = \{0\}$ .

*Proof.* For fixed  $m \in M$ , the map  $x \mapsto \langle x, m \rangle$  is linear and continuous, since by [Cauchy–Schwarz inequality](#)

$$|\langle x, m \rangle - \langle x', m \rangle| \leq \|x - x'\| \|m\|. \quad (12.28)$$

Hence  $\{x : \langle x, m \rangle = 0\}$  is a closed subspace (a kernel of a continuous linear map), and

$$M^\perp = \bigcap_{m \in M} \{x : \langle x, m \rangle = 0\} \quad (12.29)$$

is an intersection of closed subspaces, hence a closed subspace. (Equivalently: if  $x_k \rightarrow x$  with  $x_k \in M^\perp$ , one can pass to the limit inside the inner product.) If  $z \in M \cap M^\perp$ , then  $\langle z, z \rangle = 0$ , so  $z = 0$ . ■

### Theorem 12.5 (Orthogonal projection)

Let  $H$  be a Hilbert space,  $M \subseteq H$  a closed subspace, and  $x \in H$ .

1. There exists a unique  $Px \in M$  such that

$$\|x - Px\| = \text{dist}(x, M) = \inf_{y \in M} \|x - y\|. \quad (12.30)$$

2. There is a unique decomposition  $x = y + w$  with  $y \in M$  and  $w \in M^\perp$ . Moreover  $y = Px$ ; we write  $w = Qx = x - Px$ .
3. (Pythagoras)  $\|x\|^2 = \|Px\|^2 + \|Qx\|^2$ .
4. The maps  $P : H \rightarrow H$  and  $Q : H \rightarrow H$  are linear.

Geometrically,  $Px$  is the foot of perpendicular from  $x$  to  $M$ , i.e. the point of  $M$  closest to  $x$ .

*Proof.* 1. Let  $\delta = \text{dist}(x, M)$  and choose  $z_n \in M$  with

$$\|x - z_n\| \leq \delta + \frac{1}{n}, \text{ or equivalently, } \|x - z_n\| \rightarrow \delta. \quad (12.31)$$

(If  $\delta = 0$  then  $x \in \overline{M} = M$  because  $M$  is closed, and we may simply take  $Px = x$ ; the argument below also covers this case.) We claim that  $(z_n)$  is Cauchy. Apply the [Parallelogram law](#) to  $a = x - z_n$  and  $b = x - z_m$ . Since  $a + b = 2(x - \frac{z_n + z_m}{2})$  and  $a - b = z_m - z_n$ , we get

$$4 \left\| x - \frac{z_n + z_m}{2} \right\|^2 + \|z_n - z_m\|^2 = 2 \|x - z_n\|^2 + 2 \|x - z_m\|^2. \quad (12.32)$$

Since  $M$  is a subspace,  $\frac{z_n + z_m}{2} \in M$ , hence  $\left\| x - \frac{z_n + z_m}{2} \right\| \geq \delta$ . Therefore

$$\|z_n - z_m\|^2 \leq 2 \|x - z_n\|^2 + 2 \|x - z_m\|^2 - 4\delta^2 \longrightarrow 0 \quad (12.33)$$

as  $n, m \rightarrow \infty$ . So  $(z_n)$  is Cauchy. As  $H$  is complete,  $z_n \rightarrow z$  for some  $z \in H$ , and  $z \in M$  because  $M$  is closed. By continuity of the norm,

$$\|x - z\| = \lim_{n \rightarrow \infty} \|x - z_n\| = \delta. \quad (12.34)$$

So we set  $Px = z$ . Now, about the uniqueness, suppose  $z, z' \in M$  both satisfy  $\|x - z\| = \|x - z'\| = \delta$ . We apply the [Parallelogram law](#) with  $a = x - z$ ,  $b = x - z'$

$$\|z - z'\|^2 = 2\|x - z\|^2 + 2\|x - z'\|^2 - 4\left\|x - \frac{z + z'}{2}\right\|^2 \leq 4\delta^2 - 4\delta^2 = 0, \quad (12.35)$$

so  $z = z'$ .

2. First, we show the uniqueness of this decomposition. If  $x = y + w = y' + w'$  with  $y, y' \in M$  and  $w, w' \in M^\perp$ , then using [Lemma 12.4](#),

$$y - y' = w' - w \in M \cap M^\perp = \{0\}. \quad (12.36)$$

So  $y = y'$  and  $w = w'$ . Now we'll show existence of this decomposition. Take  $y = Px \in M$  and  $w = x - Px$ . We must show  $w \in M^\perp$ . Note  $\|w\| = \delta$ . Let  $z \in M$  with  $\|z\| = 1$  and  $\lambda \in \mathbb{C}$ . Since  $Px + \lambda z \in M$ ,

$$\delta^2 \leq \|x - (Px + \lambda z)\|^2 = \|w - \lambda z\|^2 \quad (12.37)$$

$$= \|w\|^2 - \bar{\lambda} \langle w, z \rangle - \lambda \langle z, w \rangle + |\lambda|^2 \|z\|^2 \quad (12.38)$$

$$= \delta^2 - 2 \operatorname{Re}(\bar{\lambda} \langle w, z \rangle) + |\lambda|^2. \quad (12.39)$$

Hence

$$2 \operatorname{Re}(\bar{\lambda} \langle w, z \rangle) \leq |\lambda|^2 \quad \text{for all } \lambda \in \mathbb{C}. \quad (12.40)$$

Choose  $\lambda = t \langle w, z \rangle$  with  $t > 0$  real. Then

$$\bar{\lambda} \langle w, z \rangle = t |\langle w, z \rangle|^2 \quad (12.41)$$

and  $|\lambda|^2 = t^2 |\langle w, z \rangle|^2$ , so

$$2t |\langle w, z \rangle|^2 \leq t^2 |\langle w, z \rangle|^2 \implies 2 |\langle w, z \rangle|^2 \leq t |\langle w, z \rangle|^2. \quad (12.42)$$

Letting  $t \rightarrow 0^+$  gives  $\langle w, z \rangle = 0$ . This holds for every unit vector  $z \in M$ , hence (by scaling) for every  $z \in M$ . So  $w \in M^\perp$ .

3. Since  $Px \in M$  and  $Qx \in M^\perp$ , we have  $\langle Px, Qx \rangle = 0$ , so

$$\|x\|^2 = \|Px + Qx\|^2 = \|Px\|^2 + \|Qx\|^2 + 2 \operatorname{Re} \langle Px, Qx \rangle = \|Px\|^2 + \|Qx\|^2. \quad (12.43)$$

4. Let  $x, x' \in H$  and  $\alpha, \beta \in \mathbb{C}$ . Then

$$\alpha x + \beta x' = (\alpha Px + \beta Px') + (\alpha Qx + \beta Qx'), \quad (12.44)$$

where the first summand lies in  $M$  and the second in  $M^\perp$ , since both are subspaces. By uniqueness in (2),  $P(\alpha x + \beta x') = \alpha Px + \beta Px'$ . Hence  $P$  is linear. Finally  $Q = I - P$  is linear as well.

■

**Remark 12.3.** Existence and uniqueness in (1) used the completeness, so the argument does not work in a general normed space. For instance, in  $\mathbb{R}^2$  with the norm

$$\|(u, v)\|_\infty = \max(|u|, |v|), \quad (12.45)$$

let

$$M = \{(0, y) : y \in \mathbb{R}\} \quad (12.46)$$

and  $x = (1, 0)$ . Then

$$\|(1, 0) - (0, y)\|_\infty = \max(1, |y|) \quad (12.47)$$

which equals  $\text{dist}(x, M) = 1$  for every  $|y| \leq 1$ . Hence, the nearest point is not unique.

Tomorrow we will prove the Riesz representation theorem for Hilbert spaces.

## §13 1 October 2026

A *Hilbert space*  $H$  is a vector space over  $\mathbb{C}$  with an inner product  $\langle \cdot, \cdot \rangle$  (linear in the first variable, conjugate-linear in the second) which is complete for the induced norm  $\|x\| = \sqrt{\langle x, x \rangle}$ . The basic example is  $L^2(\Omega, \Sigma, \mu)$  with  $\langle f, g \rangle = \int_{\Omega} f \bar{g} d\mu$ .

Last time we proved the projection theorem: if  $M \subset H$  is a closed subspace and  $x \in H$ , there is a unique  $Px \in M$  realizing the distance from  $x$  to  $M$ , and  $Qx = x - Px$  belongs to  $M^\perp$ , the set of vectors orthogonal to every element of  $M$ .

**Remark 13.1.** A natural question arises: is  $L^2$  the only  $L^p$  which is a Hilbert space? Essentially yes, because the parallelogram law fails for  $p \neq 2$  for most of the measure spaces. The concrete statement is:

Suppose  $(\Omega, \Sigma, \mu)$  is a measure space that contains two disjoint sets  $A, B$  with  $0 < \mu(A), \mu(B) < \infty$ . Then  $L^p$  is **NOT** a Hilbert space for  $p \neq 2$ .

To prove this, we choose

$$f = \frac{1}{\mu(A)^{\frac{1}{p}}} \chi_A, \quad g = \frac{1}{\mu(B)^{\frac{1}{p}}} \chi_B. \quad (13.1)$$

Then for every  $p$ ,

$$\|f\|_p = \|g\|_p = 1. \quad (13.2)$$

Since  $f, g$  have disjoint supports,  $|f + g| = |f - g|$ , so that for  $1 \leq p < \infty$ ,

$$\|f + g\|_p^p = \|f - g\|_p^p = \int_{\Omega} \frac{1}{\mu(A)} \chi_A d\mu + \int_{\Omega} \frac{1}{\mu(B)} \chi_B d\mu = 2. \quad (13.3)$$

Hence,

$$\|f + g\|_p^2 + \|f - g\|_p^2 = 2 \cdot 2^{\frac{2}{p}} = 2^{1+\frac{2}{p}}, \quad 2\|f\|_p^2 + 2\|g\|_p^2 = 4, \quad (13.4)$$

and these two are equal only when  $p = 2$ . For  $p = \infty$ ,

$$\|f + g\|_{\infty}^2 + \|f - g\|_{\infty}^2 = 2, \quad 2\|f\|_{\infty}^2 + 2\|g\|_{\infty}^2 = 4. \quad (13.5)$$

Hence, for *well-behaved* measure spaces (i.e. measure spaces with two disjoint finite-measure sets),  $L^2$  is the only  $L^p$  that is a Hilbert space.

**Remark 13.2.** Having two disjoint finite-measure sets is not an unreasonable condition to ask. In fact, the only measure spaces where this doesn't hold are:

1. Measure spaces where every nonempty set has infinite measure. But in this case,  $L^p = \{0\}$ . Because if  $|f| > 0$  on a nonempty measurable set  $E$ , then  $\int_{\Omega} |f|^p d\mu = \infty$ .
2. There is exactly one “component” of finite measure. Suppose there is a measurable set  $A$  with  $0 < \mu(A) < \infty$ . Then any set  $B$  disjoint from  $A$  will have either  $\mu(B) = 0$  or  $\mu(B) = \infty$ . If we have  $B \subseteq A$  is measurable, then we must have  $\mu(B) = 0$  or  $\mu(B) = \mu(A)$ .

(otherwise  $B$  and  $A - B$  will be two disjoint finite-measure sets). In this case, we claim that

$$L^p = \{c \cdot \chi_A : c \in \mathbb{C}\}, \quad (13.6)$$

i.e.  $L^p$  is 1-dimensional. Indeed, for  $f \in L^p$ , first we show that  $f = 0$  a.e. on  $A^c$ . Indeed, if

$$B_n = \left\{x \in A^c : |f(x)| \geq \frac{1}{n}\right\}. \quad (13.7)$$

Since  $B_n$  is disjoint from  $A$ , its measure is either 0 or infinite. If it's infinite, then

$$\int_{\Omega} |f|^p d\mu \geq \int_{B_n} |f|^p d\mu \geq \frac{1}{n^p} \mu(B_n) = \infty, \quad (13.8)$$

which contradicts with  $f \in L^p$ . Hence,  $\mu(B_n) = 0$  for all  $n$ . Therefore,

$$\mu(\{x \in A^c : |f(x)| > 0\}) = \mu\left(\bigcup_{n=1}^{\infty} B_n\right) \leq \sum_{n=1}^{\infty} \mu(B_n) = 0. \quad (13.9)$$

We assume  $f$  take real values. For the complex case, we can just work with  $\operatorname{Re} f$  and  $\operatorname{Im} f$  separately. For  $t \in \mathbb{R}$ , define

$$E_t = \{x \in A : f(x) \leq t\}. \quad (13.10)$$

Each  $E_t \subseteq A$ , so  $\mu(E_t) = 0$  or  $\mu(E_t) = \mu(A)$ . Also, for  $t < t'$ ,  $E_t \subseteq E_{t'}$ , so  $\mu(E_t) \subseteq \mu(E_{t'})$ .

Now, as  $t \rightarrow -\infty$ ,  $\mu(E_t) \rightarrow 0$ , and as  $t \rightarrow \infty$ ,  $\mu(E_t) \rightarrow \mu(A)$ . We pick

$$c = \inf \{t \in \mathbb{R} : \mu(E_t) = \mu(A)\}. \quad (13.11)$$

The first order of business is to show that it's a finite real number. If it's not, suppose it's  $-\infty$ , then  $\mu(E_t) = \mu(A)$  holds for every  $t \in \mathbb{R}$ . If  $M = \int_{\Omega} |f|^p d\mu$ , choose

$$t = -\left(\frac{2M}{\mu(A)}\right)^{\frac{1}{p}}. \quad (13.12)$$

By hypothesis,  $f(x) \leq -\left(\frac{2M}{\mu(A)}\right)^{\frac{1}{p}}$  almost everywhere on  $A$  (since  $\mu(E_t) = \mu(A)$ ),

$$M = \int_{\Omega} |f|^p d\mu \geq \int_A \left(\frac{2M}{\mu(A)}\right) d\mu = 2M, \quad (13.13)$$

implies  $M$  must be 0. But this just means  $f = 0$ .

On the other hand, if  $c = \infty$ , that means there is no  $t$  for which  $\mu(E_t) = \mu(A)$  happens, i.e.  $\mu(E_t) = 0$  for every  $t$ . Therefore,  $\mu(A - E_t) = \mu(A)$ , i.e.

$$\mu(\{x \in A : f(x) > t\}) = \mu(A), \quad (13.14)$$

for all  $t$ . We choose

$$t = \left( \frac{2M}{\mu(A)} \right)^{\frac{1}{p}}. \quad (13.15)$$

Since  $f(x) > t$  almost everywhere on  $A$  (since  $\mu(E_t) = 0$ ),

$$M = \int_{\Omega} |f|^p d\mu \geq \int_A \left( \frac{2M}{\mu(A)} \right) d\mu = 2M, \quad (13.16)$$

implies  $M$  must be 0. But this just means  $f = 0$ .

So for nonzero  $f \in L^p$ ,  $c$  is a finite real number. We claim that  $f = c \cdot \chi_A$ . For  $t < c$ ,  $\mu(E_t) = 0$ . Therefore,

$$\mu(\{x \in A : f(x) < c\}) = \mu\left(\bigcup_{n=1}^{\infty} \left\{x \in A : f(x) \leq c - \frac{1}{n}\right\}\right) \leq \sum_{n=1}^{\infty} \mu(E_{c-\frac{1}{n}}) = 0. \quad (13.17)$$

On the other hand, for  $t > c$ ,  $\mu(E_t) = \mu(A)$ . Hence,

$$\mu(\{x \in A : f(x) > c\}) = \mu\left(\bigcup_{n=1}^{\infty} \left\{x \in A : f(x) > c + \frac{1}{n}\right\}\right) \leq \sum_{n=1}^{\infty} \mu(A - E_{c+\frac{1}{n}}) = 0. \quad (13.18)$$

Therefore,  $f = c$  almost everywhere on  $A$ , hence  $f = c \cdot \chi_A$ . Thus, we have proved that  $\dim L^p = 1$  in this case.

Of course, for 1-dimensional normed spaces, the norm is always induced by an inner product, so this is not particularly an interesting scenario.

**Definition 13.1.** Let  $E$  be a normed vector space over  $\mathbb{C}$  (not necessarily complete). A **functional** on  $E$  is a function  $L : E \rightarrow \mathbb{C}$ .

**Definition 13.2** (Linear and bounded functionals). A functional  $L : E \rightarrow \mathbb{C}$  is **linear** if

$$L(x + y) = L(x) + L(y) \text{ and } L(\lambda x) = \lambda L(x) \quad (13.19)$$

for all  $x, y \in E$  and  $\lambda \in \mathbb{C}$ . It is **bounded** if there exists a constant  $C \geq 0$  such that

$$|L(x)| \leq C \|x\| \quad \text{for all } x \in E. \quad (13.20)$$

Equivalently,  $L$  maps the unit ball to a bounded set, or  $\sup_{\|x\| \leq 1} |L(x)| < \infty$ . In that case we define

$$\|L\| = \sup_{\|x\| \leq 1} |L(x)| = \sup_{\|x\|=1} |L(x)|. \quad (13.21)$$

The two suprema agree by dilation: if  $0 < \|x\| < 1$ , then  $\frac{x}{\|x\|}$  has norm 1 and

$$\left| L\left(\frac{x}{\|x\|}\right) \right| = \frac{|L(x)|}{\|x\|} \geq |L(x)|, \quad (13.22)$$

so points in the interior of the ball never beat points on the sphere. Directly from the definition,

$$|L(x)| \leq \|L\| \|x\|, \quad (13.23)$$

for all  $x$ , and  $\|L\|$  is the smallest admissible constant  $C$ .

**Definition 13.3** (Dual space). The *dual space*  $E^*$  of  $E$  is the set of all bounded linear functionals on  $E$ .

**Remark 13.3.** In finite dimensions  $E^*$  is the usual dual space of linear algebra, but in infinite dimensions the space of all linear functionals is strictly larger than the space of bounded ones. The word “dual” here always refers to  $E^*$ .

$E^*$  is a vector space: for  $L_1, L_2 \in E^*$  and  $\lambda \in \mathbb{C}$ , set

$$(L_1 + L_2)(x) = L_1(x) + L_2(x) \quad (13.24)$$

$$(\lambda L)(x) = \lambda L(x). \quad (13.25)$$

These are again bounded linear functionals, as we’ll see in the next proposition.

**Proposition 13.1**

$\|\cdot\|$  is a norm on  $E^*$ .

*Proof.* If  $\|L\| = 0$ , then  $L = 0$  on the unit ball, hence everywhere by dilation, so  $L$  is the zero functional. For  $\lambda \in \mathbb{C}$ ,

$$\|\lambda L\| = \sup_{\|x\|=1} |\lambda L(x)| = |\lambda| \sup_{\|x\|=1} |L(x)| = |\lambda| \|L\|. \quad (13.26)$$

Finally, for  $\|x\| = 1$ ,

$$|(L_1 + L_2)(x)| \leq |L_1(x)| + |L_2(x)| \leq \|L_1\| + \|L_2\|, \quad (13.27)$$

and taking the supremum over  $x$  gives  $\|L_1 + L_2\| \leq \|L_1\| + \|L_2\|$ . ■

**Proposition 13.2**

$E^*$  is complete, even if  $E$  is not.

*Proof.* Let  $L_1, L_2, \dots$  be a Cauchy sequence in  $E^*$ , i.e.

$$\|L_m - L_n\| \rightarrow 0 \text{ as } m, n \rightarrow \infty. \quad (13.28)$$

Now, for every  $x \in E$ ,

$$|L_m(x) - L_n(x)| \leq \|L_m - L_n\| \|x\| \rightarrow 0 \text{ as } m, n \rightarrow \infty. \quad (13.29)$$

Therefore,  $L_n(x)$  is a Cauchy sequence of complex numbers. By the completeness of  $\mathbb{C}$ , it converges. So we can define a functional  $L : E \rightarrow \mathbb{C}$

$$L(x) = \lim_{n \rightarrow \infty} L_n(x) \text{ for } x \in E. \quad (13.30)$$

$L$  is linear by the linearity of limit and linearity of  $L_n$ 's:

$$L(\alpha x + y) = \lim_{n \rightarrow \infty} L_n(\alpha x + y) = \alpha \lim_{n \rightarrow \infty} L_n(x) + \lim_{n \rightarrow \infty} L_n(y) = \alpha L(x) + L(y). \quad (13.31)$$

Now, we need to show that  $L_n \rightarrow L$  in the operator norm. Given  $\varepsilon > 0$  choose  $N$  large enough so that  $\|L_m - L_n\| < \varepsilon$  for  $m, n \geq N$ .

$$|L(x) - L_n(x)| \leq |L(x) - L_m(x)| + |L_m(x) - L_n(x)| \quad (13.32)$$

Taking  $m \rightarrow \infty$ , we have

$$|L(x) - L_n(x)| \leq |L_m(x) - L_n(x)| \leq \varepsilon \|x\|. \quad (13.33)$$

Therefore,  $\|L - L_n\| \leq \varepsilon$ . Finally,  $L \in E^*$ , since

$$L = L_n + (L - L_n), \quad (13.34)$$

and  $L_n$  has a bounded norm, and so does  $L - L_n$ . ■

### Proposition 13.3

A linear functional  $L : E \rightarrow \mathbb{C}$  is bounded if and only if it is continuous.

*Proof.* Suppose  $L$  is bounded, say  $|L(x)| \leq C \|x\|$ . Then for all  $x, x_0$ , by linearity,

$$|L(x) - L(x_0)| = |L(x - x_0)| \leq C \|x - x_0\|, \quad (13.35)$$

so  $L$  is continuous (even Lipschitz).

Conversely, suppose  $L$  is continuous. Then it is continuous at 0, so there is  $\delta > 0$  such that

$$\|y\| < \delta \text{ implies } |L(y)| < 1 \quad (13.36)$$

If  $\|x\| \leq 1$ , then

$$\left\| \frac{\delta x}{2} \right\| < \delta \implies \left| L\left(\frac{\delta x}{2}\right) \right| < 1 \implies |L(x)| < \frac{2}{\delta}. \quad (13.37)$$

Hence  $L$  maps the unit ball into a bounded set, so  $L$  is bounded. ■

Let  $H$  be a Hilbert space. Every vector  $v \in H$  defines a functional

$$L_v(x) = \langle x, v \rangle, \quad x \in H. \quad (13.38)$$

It is linear since the inner product is linear in the first variable. It is bounded by the Cauchy–Schwarz inequality:

$$|L_v(x)| \leq \|x\| \|v\| \quad (13.39)$$

so  $\|L_v\| \leq \|v\|$ . In fact  $\|L_v\| = \|v\|$ : if  $v \neq 0$ , take  $x = \frac{v}{\|v\|}$ , which has norm 1, and

$$L_v(x) = \frac{\langle v, v \rangle}{\|v\|} = \|v\|. \quad (13.40)$$

Thus every  $v \in H$  gives an element  $L_v \in H^*$  with the same norm. The following theorem says that there are no others.

**Theorem 13.4 (Riesz representation theorem)**

Let  $H$  be a Hilbert space and  $L \in H^*$ . Then there exists a unique  $v \in H$  such that

$$L(x) = \langle x, v \rangle \quad \text{for all } x \in H. \quad (13.41)$$

Moreover  $\|L\| = \|v\|$ .

*Proof.* If  $L = 0$ , take  $v = 0$ . Assume  $L \neq 0$  and let

$$M = \text{Ker } L = \{x \in H : L(x) = 0\}. \quad (13.42)$$

$M$  is a subspace because  $L$  is linear, and it is closed because  $L$  is continuous (Proposition 13.3), since  $M = L^{-1}\{0\}$  and  $\{0\}$  is closed in  $\mathbb{C}$ . Since  $L \neq 0$ , there is  $y \in H - M$ . Let

$$z = \frac{y - Py}{\|y - Py\|}, \quad (13.43)$$

where  $P$  is the orthogonal projection onto  $M$ . Here  $y - Py \neq 0$  since otherwise  $y = Py \in M$ . By the projection theorem (Theorem 12.5)  $y - Py \in M^\perp$ , so  $z \in M^\perp$  and  $\|z\| = 1$ . Moreover  $L(z) \neq 0$ , since otherwise  $z \in M \cap M^\perp = \{0\}$ , contradicting  $\|z\| = 1$ .

For  $x \in H$ , define

$$u = L(x)z - L(z)x. \quad (13.44)$$

Then

$$L(u) = L(x)L(z) - L(z)L(x) = 0, \quad (13.45)$$

so  $u \in M$ . Since  $z \in M^\perp$ ,  $\langle u, z \rangle = 0$ . Expanding with linearity in the first variable and  $\langle z, z \rangle = 1$ ,

$$0 = \langle u, z \rangle = L(x)\langle z, z \rangle - L(z)\langle x, z \rangle = L(x) - L(z)\langle x, z \rangle, \quad (13.46)$$

that is,

$$L(x) = L(z)\langle x, z \rangle = \left\langle x, \overline{L(z)}z \right\rangle, \quad (13.47)$$

where we used  $\langle x, \lambda w \rangle = \bar{\lambda} \langle x, w \rangle$ . So  $v = \overline{L(z)}z$  works.

For the uniqueness, if  $\langle x, v \rangle = \langle x, v' \rangle$  for all  $x$ , then

$$\langle x, v - v' \rangle = 0, \quad (13.48)$$

for all  $x$ . Taking  $x = v - v'$  gives  $\|v - v'\|^2 = 0$ , so  $v = v'$ . ■

**Remark 13.4.** The map  $v \mapsto L_v$  is a bijective isometry from  $H$  onto  $H^*$ . Over  $\mathbb{C}$  it is conjugate-linear, since  $L_{\lambda v} = \bar{\lambda} L_v$ . In this sense Hilbert spaces are self-dual. Since  $H^*$  is complete, this also shows again that  $H^*$  is a Hilbert space.

Up to now measures were nonnegative. We now allow measures with positive and negative weights. This is the setting for the Radon–Nikodym theorem, to be proved next time.

**Definition 13.4** (Signed measure). Let  $(\Omega, \Sigma)$  be a measurable space. A *signed measure* on  $\Omega$  is a function  $\nu : \Sigma \rightarrow \mathbb{R}$  such that for every set  $E \in \Sigma$  and every partition of  $E$  into countably many disjoint sets  $E = \bigsqcup_{n=1}^{\infty} E_n$  with  $E_n \in \Sigma$ ,

$$\nu(E) = \sum_{n=1}^{\infty} \nu(E_n), \quad \text{and the series converges absolutely.} \quad (13.49)$$

(One can also consider complex-valued  $\nu$ , with small modifications.)

**Remark 13.5.** In this definition  $\nu$  takes only finite real values. Allowing  $\nu(E) = \pm\infty$  would conflict with absolute convergence in (13.49), and we want to avoid expressions such as  $\infty - \infty$ . (In particular, Lebesgue measure on  $\mathbb{R}$  is not a signed measure in this sense, since  $\nu(\mathbb{R}) = \infty$ .)  $\nu(\emptyset) = 0$ : writing  $\emptyset = \emptyset \sqcup \emptyset$  gives  $\nu(\emptyset) = 2\nu(\emptyset)$ .

### Example 13.1

Let  $\mu$  be a nonnegative measure and  $f \in L^1(\mu)$  real-valued. Then

$$\nu(E) = \int_E f \, d\mu \quad (13.50)$$

is a signed measure. Indeed, for a partition  $E = \bigsqcup_{n=1}^{\infty} E_n$ , by [Dominated Convergence Theorem](#),  $\nu(E) = \sum_{n=1}^{\infty} \nu(E_n)$ , and

$$\sum_n |\nu(E_n)| \leq \sum_n \int_{E_n} |f| \, d\mu = \int_E |f| \, d\mu < \infty. \quad (13.51)$$

We want to majorize a signed measure  $\nu$  by a nonnegative measure  $\mu$ , i.e.

$$|\nu(E)| \leq \mu(E) \quad (13.52)$$

for all  $E$ , in the *cheapest possible way*:  $\mu$  should be smaller than any other nonnegative measure  $\tilde{\mu}$  with  $|\nu(E)| \leq \tilde{\mu}(E)$ .

If  $E = \bigsqcup_{n=1}^{\infty} E_n$  is a partition and  $\mu$  is such a majorant, then

$$\mu(E) = \sum_{n=1}^{\infty} \mu(E_n) \geq \sum_{n=1}^{\infty} |\nu(E_n)|. \quad (13.53)$$

This suggests the following definition.

**Definition 13.5** (Total variation). For a signed measure  $\nu$  and  $E \in \Sigma$ , define  $|\nu| : \Sigma \rightarrow [0, \infty]$

$$|\nu|(E) = \sup \left\{ \sum_{n=1}^{\infty} |\nu(E_n)| : E = \bigsqcup_{n=1}^{\infty} E_n \text{ is a countable partition of } E \right\}. \quad (13.54)$$

A priori  $|\nu|$  could take the value  $+\infty$ . Taking the trivial partition  $E = E \sqcup \emptyset \sqcup \emptyset \sqcup \cdots$  shows

$$|\nu(E)| \leq |\nu|(E) \quad \text{for all } E \in \Sigma, \quad (13.55)$$

so  $|\nu|$  is a majorant. By the discussion above, any nonnegative measure majorizing  $\nu$  which is additive is at least  $|\nu|$  on every set. It remains to show that  $|\nu|$  is itself a measure.

For  $\nu(E) = \int_E f \, d\mu$ , one expects  $|\nu|(E) = \int_E |f| \, d\mu$ .

**Proposition 13.5**

$|\nu|$  is a nonnegative measure on  $(\Omega, \Sigma)$  (possibly with infinite values), called the **total variation measure** of  $\nu$ .

*Proof.* Clearly  $|\nu| \geq 0$ , and  $|\nu|(\emptyset) = 0$  since every partition of  $\emptyset$  consists of empty sets and  $\nu(\emptyset) = 0$ . We show countable additivity: let  $E = \bigsqcup_{n=1}^{\infty} E_n$ ; we prove

$$|\nu|(E) = \sum_{n=1}^{\infty} |\nu|(E_n). \quad (13.56)$$

Fix real numbers  $t_n$  with  $0 \leq t_n < |\nu|(E_n)$  (and  $t_n = 0$  if  $|\nu|(E_n) = 0$ ). By the definition of the supremum, for each  $n$  there is a partition  $E_n = \bigsqcup_j E_{n,j}$  with

$$\sum_j |\nu(E_{n,j})| \geq t_n \quad (13.57)$$

The family  $\{E_{n,j}\}_{n,j}$  is a countable partition of  $E$ , so

$$|\nu|(E) \geq \sum_n \sum_j |\nu(E_{n,j})| \geq \sum_n t_n. \quad (13.58)$$

Taking the supremum over all such choices of  $t_n$  gives  $|\nu|(E) \geq \sum_{n=1}^{\infty} |\nu|(E_n)$ , which holds also if the right-hand side is  $+\infty$ .

Now we need to show the reverse inequality. Let  $E = \bigsqcup_j A_j$  be any countable partition of  $E$ . For each  $j$ ,

$$A_j = \bigsqcup_n (A_j \cap E_n) \quad (13.59)$$

and for each  $n$ ,

$$E_n = \bigsqcup_j (A_j \cap E_n). \quad (13.60)$$

Using the additivity of  $\nu$  and exchanging sums of nonnegative terms,

$$\sum_j |\nu(A_j)| = \sum_j \left| \sum_n \nu(A_j \cap E_n) \right| \leq \sum_j \sum_n |\nu(A_j \cap E_n)| \quad (13.61)$$

$$= \sum_n \sum_j |\nu(A_j \cap E_n)| \leq \sum_n |\nu(E_n)|, \quad (13.62)$$

because  $\{A_j \cap E_n\}_j$  is a partition of  $E_n$ . Taking the supremum over partitions  $\{A_j\}$  gives  $|\nu|(E) \leq \sum_n |\nu|(E_n)$ . ■

### Lemma 13.6

Let  $\nu$  be a signed measure and  $E \in \Sigma$  with  $|\nu|(E) = \infty$ . Then  $E = A \sqcup B$  with  $A, B \in \Sigma$  such that

$$|\nu(A)| > 1, \quad |\nu(B)| > 1, \quad \text{and at least one of } |\nu(A)|, |\nu(B)| \text{ is } \infty.$$

*Proof.* Let  $t > 0$  be a number to be chosen. Since  $|\nu|(E) = \infty > t$ , there is a partition  $E = \bigsqcup_n E_n$  with

$$\sum_{n=1}^{\infty} |\nu(E_n)| > t \quad (13.63)$$

Let  $A_+$  be the union of those  $E_n$  with  $\nu(E_n) > 0$ , and  $A_-$  the union of those with  $\nu(E_n) < 0$ . By absolute convergence and additivity,

$$\nu(A_+) = \sum_{\nu(E_n) > 0} \nu(E_n), \quad -\nu(A_-) = \sum_{\nu(E_n) < 0} |\nu(E_n)|, \quad (13.64)$$

and these two sums add up to more than  $t$ . So one of them exceeds  $\frac{t}{2}$ ; let  $A$  be the corresponding set, so that  $|\nu(A)| > \frac{t}{2}$ . Let  $B = E - A$ . Then

$$|\nu(B)| = |\nu(E) - \nu(A)| \geq |\nu(A)| - |\nu(E)| > \frac{t}{2} - |\nu(E)|. \quad (13.65)$$

Here  $|\nu(E)|$  is a finite number, since  $\nu$  is real-valued. Choose  $t > 2(|\nu(E)| + 1)$ ; then  $|\nu(A)| > \frac{t}{2} > 1$  and  $|\nu(B)| > 1$ . Finally, by [Proposition 13.5](#),  $|\nu|(A) + |\nu|(B) = |\nu|(E) = \infty$ , so at least one of the two is infinite. ■

**Proposition 13.7**

If  $\nu$  is a signed measure, then  $|\nu|(\Omega) < \infty$ .

*Proof.* Suppose  $|\nu|(\Omega) = \infty$ . By Lemma 13.6,  $\Omega = A_1 \sqcup B_1$  with

$$|\nu(A_1)| > 1, \quad |\nu(B_1)| > 1, \quad |\nu|(B_1) = \infty \quad (13.66)$$

(We rename the sets so that  $|\nu|(B_1) = \infty$ .) Applying the lemma again to  $B_1$ , we obtain  $B_1 = A_2 \sqcup B_2$  with

$$|\nu(A_2)| > 1, \quad |\nu(B_2)| > 1, \quad |\nu|(B_2) = \infty \quad (13.67)$$

Repeating, we obtain pairwise disjoint sets  $A_1, A_2, A_3, \dots$  with  $|\nu(A_n)| > 1$  for all  $n$ . Let  $A = \bigsqcup_n A_n$ . By the definition of a signed measure,  $\sum_n \nu(A_n)$  must converge absolutely, but that's not possible here as  $|\nu(A_n)| > 1$ . Hence  $|\nu|(\Omega) < \infty$ . ■

**Remark 13.6.** For complex-valued  $\nu$ , apply the same results to the real signed measures  $\operatorname{Re} \nu$  and  $\operatorname{Im} \nu$ , using  $|\nu|(E) \leq |\operatorname{Re} \nu|(E) + |\operatorname{Im} \nu|(E)$ . Alternatively, in Lemma 13.6 one chooses among four sets (positive and negative real parts, positive and negative imaginary parts). The finiteness of  $|\nu|(\Omega)$  is the measure-theoretic analogue of the notion of a function of bounded variation, which will come up later.

Define

$$\nu^+ = \frac{|\nu| + \nu}{2}, \quad \nu^- = \frac{|\nu| - \nu}{2}. \quad (13.68)$$

Both are nonnegative, because  $|\nu(E)| \leq |\nu|(E)$ ; both are countably additive, being combinations of countably additive set functions; and both are finite by Proposition 13.7. Thus  $\nu^\pm$  are finite nonnegative measures and

$$\nu = \nu^+ - \nu^-, \quad |\nu| = \nu^+ + \nu^-. \quad (13.69)$$

This is the *Jordan decomposition* of  $\nu$ : every signed measure is the difference of two finite nonnegative measures.

For  $\nu(E) = \int_E f \, d\mu$ , it corresponds to writing  $f = f^+ - f^-$  with  $f^\pm \geq 0$ , so

$$\nu^\pm(E) = \int_E f^\pm \, d\mu. \quad (13.70)$$

The collection of signed measures on  $(\Omega, \Sigma)$  is a vector space: the sum of two signed measures and any scalar multiple of a signed measure (positive or negative) is again a signed measure. By Proposition 13.7, we set

$$\|\nu\| = |\nu|(\Omega) < \infty. \quad (13.71)$$

**Proposition 13.8**

(13.71) defines a norm on the space of signed measures.

*Proof.*  $\|\nu\| = 0$  implies  $|\nu|(\Omega) = 0$ , which implies  $|\nu|(E) = 0$  for every  $E \in \Sigma$ , in particular,  $\nu(E) = 0$  for every  $E \in \Sigma$ .

For  $\alpha \in \mathbb{R}$ , and partitions  $\Omega = \bigsqcup_{n=1}^{\infty} E_n$ ,

$$\|\alpha \nu\| = \sup_{\text{partitions}} \sum_{n=1}^{\infty} |\alpha \nu(E_n)| = |\alpha| \sup_{\text{partitions}} \sum_{n=1}^{\infty} |\nu(E_n)| = |\alpha| \|\nu\|. \quad (13.72)$$

Now, if  $\nu_1, \nu_2$  are two signed measures, then

$$|\nu_1(E) + \nu_2(E)| \leq |\nu_1(E)| + |\nu_2(E)|. \quad (13.73)$$

Therefore, for a partition  $\Omega = \bigsqcup_{n=1}^{\infty} E_n$ ,

$$\sum_{n=1}^{\infty} |\nu_1(E_n) + \nu_2(E_n)| \leq \sum_{n=1}^{\infty} |\nu_1(E_n)| + \sum_{n=1}^{\infty} |\nu_2(E_n)|. \quad (13.74)$$

Taking supremum over all partitions, we get

$$\|\nu_1 + \nu_2\| \leq \|\nu_1\| + \|\nu_2\|. \quad (13.75)$$

■